

Cosmic System Dynamics and Dimensional Relations under Generalized Relational State Evolution System

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Abstract

Starting from the generalized state-relation evolution system, this theory extends sequentially to inertial evolution, standard quantization and cosmic systems, and systematically reconstructs necessary conditions for stable cosmic evolution, emergent phenomena, dynamical correlations, dimensional relations and causal hierarchies. From this unified framework, classical mechanics, relativity, quantum mechanics and celestial mechanics are reconstructed under identical fundamental logical and mathematical relations, with core conclusions summarized as follows:

Fundamental Mechanics: All elementary cosmic entities are Momentum units defined as $p_0 = m_0 \cdot c$. Particles possess quasi-standard quantized internal configurations and form co-evolving bodies constrained by Internal Coupling from quantum entanglement; Orientation Ratio originating from internal Vector Composition generates interference, whose macroscopic statistical outcomes correspond to mass-energy transformation and Lagrangian quantity. Classical mechanics, relativity and the Standard Model are macroscopic integral effects emerging across diverse physical scenarios from particle Encapsulation, External Coupling interaction rules, three-dimensional Perceptual Cross-section attenuation and instantaneous Update of interactions.

Dynamical Relations: All forces are equivalent to motion potential (reverse evolution) and resultant velocity capacity (evolutionary frequency synthesis), and all forms of energy equal the interval integral of relevant forces.

Celestial Mechanics: Neither terrestrial weighing measurements nor torsion balance experiments require dark matter for data correction. The solar system is dominated by central gravitational attraction, while large-scale galaxies evolve under mesh-structured gravitation; such structural divergence eliminates the necessity of dark matter for dynamical revision. For uniformly distributed large-scale cosmic structures, gravitation behaves as local Attraction Potential and drives inward contraction toward respective centers at ultra-large cosmic scales. Reverse integral of photons produces apparent cosmic expansion purely from observational perspective, which rules out dark energy as a mandatory dynamical correction. Although dark matter and dark energy can mathematically fit cosmic evolutionary discrepancies among celestial bodies, they are physically redundant within the present theoretical framework.

Keywords: Evolution Theory; dynamics evolution system; physical dimensional relation; dynamical relation; dark matter problem; dark energy problem; origin of gravitation; internal particle structure; generalized equivalence principle

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1 Introduction

1.1 Current Challenges

For a long time, in our pursuit of a deeper understanding and application of the physical world, we have established an extensive array of physical disciplines. These have achieved remarkable precision in mathematical predictions and proven highly effective in engineering applications. However, it is crucial not to overlook the following emerging vulnerabilities:

1. Diminishing physical intuition in descriptions: Concepts such as operators, probability waves, matrices, fiber bundles, spacetime curvature, dark energy, and dark matter are becoming increasingly abstract. There is a growing lack of a unified physical description that is observable, evolvable, and capable of materialization.

2. Weak compatibility across domains: The logical connections between energy and momentum, spacetime backgrounds and entity evolution, fields and particles, as well as information transmission, perception mechanisms, and entity evolution, remain unresolved. There is a tendency to artificially bridge these gaps rather than fundamentally reconciling them.

3. Lack of microscopic origins for dimensional relationships: The interrelationships among fundamental dimensions—such as time, space, mass, energy, momentum, force, information, and entropy—lack a foundational connection and a unified descriptive framework.

4. Incomplete physical mechanisms: There is an absence of a comprehensive mechanistic chain describing the complete process of a particle, from its generation, origin of motion, causal interactions, and phenomenon emergence, to dimensional relationships, conservation laws, and macroscopic observations.

1.2 Theoretical Tasks

This theory does not address the origin of the cosmic system; rather, it focuses on constructing the core elements of the “Generalized Relational State Evolution System.” Based on this foundation, it further deduces the emergence mechanisms of fundamental physical quantities and their dynamical relationships—including mass, momentum, time, space, the speed of light, kinetic energy, energy, force, and the Lorentz factor—and validates the dynamical evolution relationships within the cosmic system. The ultimate goal of this framework is to identify the missing directions and the true physical essence in contemporary physics, providing a fundamentally unified form of compensation for these gaps. Furthermore, it aims to offer a unified conceptual direction for ongoing research and present specific theoretical predictions.

1.3 Theoretical Methodology

This paper primarily adopts the “Standard Three-Dimensional Orthogonal Inertial Relational State Evolution System” as the default reference perspective. This ensures that the derivation process remains comparable to and verifiable against the structures of classical physics. Simultaneously, the logical framework constructed herein is compatible with the more general “Generalized Relational State Evolution System.” This universality allows the derived conclusions to apply not only to the physical cosmic system but also to be generalized to any other system possessing state representations, relational interactions, and evolution rules. The compatibility of this theory is rigorously validated across

four dimensions: logical consistency, mathematical rigor, existing experimental data, and established traditional theories.

1.4 Statement of Position

The proposed framework remains compatible with conventional field theory and other effective descriptive systems while reinterpreting them through underlying relational evolution mechanisms, introducing necessary modeling hypotheses consistent with theoretical, mathematical and experimental constraints.

1.5 Notes on Expression

For dimensional representation, this framework exclusively deduces the underlying logic and statistical dependence between the generalized relational state evolution system and cosmic evolution, without adopting conventional dimensional units such as meter, second, kilogram, Newton and Joule; it does not accept the treatment of statistical outcomes from specific observational viewpoints (e.g., square second) as fundamental dimensional units. In physical description, all evolutionary rules and other physical depictions in this framework are formulated for intuitive interpretation from the generalized relational state evolution system. Namely, all physical phenomena are described via state evolution relations while conforming to observational results and conventional theories following a three-tier logic: state evolution $\rightarrow$ observational phenomenon $\rightarrow$ conventional theory compatibility. The proposed theory is abbreviated as Evolution Theory hereafter.

1.6 Framework Explanation

This theory re-examines the fundamental operating principles of physical systems from the perspective of evolutionary relations. Consequently, it differs fundamentally from traditional theories in both conceptual framework and mathematical formulation. The two approaches can only maintain compatibility, but cannot be directly mapped or cited item by item. Researchers with a background in traditional theories may inevitably harbor doubts about some of the conclusions and derivations upon their initial encounter with this framework. To this end, I hereby declare the following:

The core mechanisms of this framework—including the necessary conditions for evolution, the emergent relationships of dimensional quantities, the causal influences of dynamics, and the microscopic origins of various observational effects—can all be concretely demonstrated and verified through digital parametric simulations. By constructing corresponding computational models, simulations can be run based on both this framework and traditional theories separately, and the output results (such as orbital morphology, momentum distribution, interference patterns, etc.) can be systematically compared—including comparisons of complexity or even the feasibility of simulation itself—to intuitively demonstrate the correctness and compatibility of this framework.

Moreover, while maintaining the self-consistency of its underlying logic, this framework has already covered all major domains of physics (classical mechanics, relativity, quantum mechanics, celestial mechanics, etc.), and provides compatible mathematical formulations and testable quantitative predictions for each. Any further discussion or challenge to this theory is welcomed to be based on simulation verification and mathematical derivation, rather than merely on subjective judgments rooted in traditional conceptual systems.

1.7 Reading Guide

The progressive logic of the present theory is summarized as follows: system classification and static construction → relational evolution → free dimension and evolutionary motion modes → prerequisites for complex relation formation (standard quantization) → reference to the cosmic system, with cosmic physical phenomena deduced sequentially along this logical chain. Detailed elaboration is listed below:

Layer 1 – Static Characterization and Independent Evolution (System Foundation): Clarify diversified feasible forms of evolution systems and establish rules for state characterization and independent evolution, serving as the prerequisite for all subsequent derivations. Examples include independent evolution of RGB three primary colors, status updating on checkerboard lattices and sequential discrete state transition.

Layer 2 – Relational Evolution (Causal Interaction Establishment): Built upon Layer 1, individual evolution carriers establish causal interactions via perception and response mechanisms, laying the underlying foundation for mutually constrained systemic evolution.

Layer 3 – Inertial Evolution (Free Dimension and State Persistence): Free dimensions and state persistence rules are introduced atop relational evolution. Evolution carriers sustain uniform state expansion along free dimensions, from which inertial motion and classical mechanical phenomena emerge. Relevant inertial free dimension types include single-step progression and checkerboard jump-style displacement.

Layer 4 – Standard Quantization (Discrete Minimum Distributed Evolution): Minimum evolutionary events, baseline state characterization scales and discrete constraints are added onto inertial evolution. Quantum phenomena statistically emerge as inertial evolution confined by discrete restrictions.

Layer 5 – Cosmic System (Real-World Physical Universe): Rules from the preceding four layers are applied to cosmic scale. Relativistic effects and celestial mechanical phenomena are deduced starting from Momentum unit encapsulation, internal coupling entanglement and perceptual attenuation.

The five layers constitute an irreversible progressive chain ranging from abstract abstraction to concrete reality and from simplicity to complexity. Every upper-layer rule remains intact when constructing subsequent layers, and richer physical phenomena are derived via supplementary restrictive conditions. Instead of predefining pre-existing entities or concepts including forces, energy, spacetime, reference frames, relational groups and strings, spacetime, dimensional emergence and dynamic relations are deduced from systemic necessity alongside corresponding physical phenomena.

1.8 Content Declaration

No single subfield of the proposed theory can be separated for independent elaboration due to mutual interdependence across all modules, resulting in an extensive overall manuscript. The established Evolution Theory spans classical mechanics, relativity, quantum mechanics and celestial mechanics, matching their inherent logical and mathematical formulations with an intricate reasoning chain and abundant customized symbolic correlations. Restricted by proofreading resources and manuscript length, occasional imprecise wording, inconsistent symbol usage, abbreviated derivation steps and incomplete numerical verification details may exist within the text. Such minor imperfections do not compromise the validity of core axioms, primary logical threads and key conclusions; revisions and optimizations will be supplemented in subsequent versions, and academic corrections from peer scholars are welcomed.

2 Fundamental Features of the Generalized Relational State Evolution System

2.1 Definition of Generalized State Evolution System

The generalized relational state evolution system refers to a dynamic causal state evolution network assembled from discrete Evolution carrier units. Three irreducible fundamental components underpin the entire system: intrinsic Representational Quanton m_0 , Driving Capacity D , and Rule set $\mathcal{R}$ [1]. Evolution carriers correspond to Representational Quanton units; Driving Capacity determines the mode of state transition; Rule set imposes global constraints on all evolutionary pathways[2].

Constrained collectively by Driving Capacity and Rule set, each Evolution carrier expands its intrinsic state within exclusive free dimensions. Local causal interactions rather than spontaneous individual updates govern systemic state variation and inertial modification. Through continuous accumulation of localized interactions under competing and synergistic effects, stable adjacent causal structures gradually form, with macroscopic substantial configurations sequentially emerging following the evolutionary path: free dimension $\rightarrow$ evolutionary space $\rightarrow$ stable adjacent causal domain $\rightarrow$ substantial physical space[1], alongside derivative spatial order and sequential evolutionary events constituting temporal ordering[3].

Since information denotes interdependent relational states between interacting components, the entire system inherently constitutes an iterative dynamic Information-dynamics evolution system[4][5]. Its core underlying dynamic formulation reads:

$$\text{System} = (m_0, D, \mathcal{R}), \quad S_{t+1} = \mathcal{R}(S_t, D(S_t))$$

where S_t stands for the aggregate set of characterized states at moment t , $D(S_t)$ denotes driving output determined by instantaneous system configuration, and $\mathcal{R}$ represents the predefined Rule set[6].

2.2 Correlation between State Evolution and Information

Information here is not confined to digitized data applicable for electronic simulation, but refers to mutually influential causal connections formed among systemic constituent states. For example, a complete English dictionary carries conventional linguistic information for human readers, yet only static character states are recognizable to extraterrestrial observers lacking matching neural mapping frameworks; absent valid causal state correlation linking dictionary symbols to alien cognitive structures, the dictionary contains no interpretable linguistic information for such observers. By contrast, mutual causal state correlations among multiple atoms generate synergistic evolutionary behaviors and form molecular structures, where standalone discrete Representational Quanton aggregate into information architectures governing molecular evolution.

The above triple-layer structure decomposes inherent informational composition. Information follows the fundamental definition $\mathcal{I} = (S, E), E = \mathcal{C}(S)$. The constraint operator $\mathcal{C}$ generates relational architecture instead of acting as an independent constituent. Pure static states amount to isolated coding without systemic connection or evolutionary attribute; complete information sustaining systemic evolution only arises when static states dynamically develop structured correlations under predefined constraints. Hence, information embodied via inter-state relations fully characterizes overall systemic performance[5].

Distinction between State Information and Generalized Information

1. Generalized Information (Relational Information): Complete informational form defined by the ontology $\mathcal{J} = (S, E)$, integrating static foundations and dynamic interactive evolution, functioning as the core driver for systemic evolution and structural Emergence.
2. State Information: Baseline constituent of systemic evolution, consisting of inherent static Eigen-state of carriers plus incremental spatial and temporal variations generated throughout evolution, elaborated from multiple perspectives in subsequent chapters.

State information merely answers the descriptive question of “what a system is”, without reflecting inter-component interactions and derived structural formations[7]. After constraint operator $\mathcal{C}$ constructs relational structure $E = \mathcal{C}(S)$ from raw state data, static foundations evolve dynamically into complete generalized information[5].

Generalized Information = State Information + Relational Structure Generated via Constraints

2.3 Classification of State Evolution Systems

Under identical foundational constraints (State–Drive–Rule), the generalized relational state evolution system branches into multiple concrete realizations differentiated by free-dimensional layout and causal connection topology[8], representative cases listed as below:

Checkerboard-type System: Discrete lattice structure with local adjacent interactions (analogous to cellular automata);

Game-type System: Finite state machine governed by turn-based rule-driven evolution;

RGB State System: Multi-channel state coding supporting parallel independent evolution;

Standard Quantized System: Fixed minimum discrete unit (l_0, t_0) with unified fixed update step length;

Three-dimensional Relational Quantized System: Standard quantized evolution embedded within three-dimensional space, serving as the direct abstraction of the cosmic physical system.

All aforementioned systems comply with unified structural requirements including state characterization, free-dimensional expansion, causal interaction and rule restriction, while diverging across three core dimensions detailed below:

1. Driving Mode (Evolution Capacity Form): Evolution carriers adopt distinct state transition patterns, categorized into three core Driving Capacity types within this work:
 - (a) Inertial Drive (v): Continuous uniform state expansion along free dimensions generating classical inertial motion via Emergence;
 - (b) Single-step Drive (step): One discrete state transition per fundamental time unit with no sustained inertial progression[9];
 - (c) Skip Drive (skip): Non-local state transition crossing intermediate free-dimensional coordinates, exemplified by pieces jumping over intermediate checkerboard grids under preset game rules.
2. Free-dimensional Configuration: Corresponding free-dimensional expansion characteristics vary across driving modes:

- (a) Under inertial drive, carriers expand synchronously and proportionally alongside free dimensions, yielding concurrent emergent space and time structures[9];
 - (b) Under single-step drive, free dimensions refresh independently at discrete time nodes without continuous spatial extension;
 - (c) Under skip drive, intermediate free-dimensional coordinates are bypassed to generate non-local correlations such as checkerboard cross-grid movement.
3. Causal Connection Topology: Driving mode plus dimensional layout jointly determine interaction topology between Evolution carriers, categorized as local adjacent coupling, instantaneous global linkage and conditional triggering.

Despite structural discrepancies, all state evolution systems admit unified formal expression. To eliminate symbolic ambiguity, symbol Σ (distinct from integral notation) is defined as the complete set characterizing full generalized relational state evolution systems, with the canonical expression:

$$\Sigma = (\Omega, T, S, \mathcal{K})$$

where:

- Ω : Free-dimensional space;
- T : Causal topology;
- S : State space;
- $\mathcal{K}$: Evolution constraints.

Logical deduction confirms all evolution systems originate from three irreducible bottom-level components: baseline Representational Quanton m_0 , Driving Capacity (v , step or skip), and Rule set $\mathcal{R}$. Systematic disparities fundamentally derive from alternative driving mode selection and customized free-dimensional constraint configuration.

2.4 State Characterization (Evolution Carrier)

State Characterization, namely Evolution carrier, denotes the fundamental systemic unit exclusively occupying independent evolutionary free dimensions and the sole tangible participant in dynamic evolution and causal interaction, defined by intrinsic state characterization plus evolutionary capacity[1]. Its evolutionary behavior is jointly determined by Driving Capacity and Rule set constraints[10], with unified mathematical formulation:

$$S_{t+1} = \mathcal{R}(S_t, D)$$

where: S stands for state characterization (the fundamental basis of existential coding), D denotes Driving Capacity (state transition capacity), $\mathcal{R}$ represents Rule set (system constraint specification). The evolutionary function in Subsection 3.1 is a unit-specific instantiation of Rule set $\mathcal{R}$, fully equivalent to the constraint formula in Section 2.7; symbol $\mathcal{R}$ uniformly represents Rule set throughout the full manuscript.

Systematic differences emerge regarding characterization format, baseline characterization scale, free-dimensional layout and causal connection configuration.

Minimum Characterization Scale: A baseline spatial threshold Λ exists within generalized state evolution systems, representing the minimum static or dynamic Representational Quanton footprint of single Evolution carrier as well as the shortest feasible distance for valid causal coupling[9]. Mutual

causal interaction between two Evolution carriers is permitted only for separation $r \geq \Lambda$; carriers with $r < \Lambda$ merge into an identical indistinguishable state node.

Remark: Representational Quanton quantifies existential attributes inside arbitrary state evolution systems, functioning as the sole tangible carrier for observable informational states. Static Representational Quanton generates emergent space; dynamic evolution yields sequential temporal events plus evolving dynamic space. For instance, three-dimensional systems require Representational Quanton with intrinsic three-dimensional emergent attributes, which describe dimensional properties rather than pure numerical magnitudes.

2.5 Driving Capacity

Generic Driving Capacity is abbreviated as D . Subject to specific system classification, D splits into three concrete variants: single-step drive (step), inertial drive (v_0), skip drive (skip). Inertial drive v_0 originates from cumulative continuous integration of generic Driving Capacity D within continuous free dimensions, with its baseline Dimension denoted by v .

Driving Capacity constitutes the core power source enabling Evolution carriers to complete state transition and free-dimensional state expansion. Distinct systemic frameworks correspond to differentiated free-dimensional unfolding pathways alongside customized transition modes: single-step drive matches discrete low-continuity evolution systems; inertial drive applies to three-dimensional orthogonal frameworks featuring inherent state persistence; skip drive fits non-local cross-lattice expansion within checkerboard topological systems.

Driving Capacity undergoes aggregation and coupling during multi-carrier interaction:

$$D_{\text{total}} = \sum_i D_i + \mathcal{F}(T)$$

where T denotes causal topological structure, and $\mathcal{F}(T)$ quantifies nonlinear coupling corrections induced by topological layout, whose macroscopic statistical manifestations correspond to emergent bulk physical quantities including momentum, force and energy.

2.6 Rule set

Rule set $\mathcal{R}$ is defined as the aggregate of constraints governing valid state evolution and causal interaction[1][9].

Its regulatory scope includes legality of state transition, causal interaction pathways, information exchange protocols, perception-coupling modes and free-dimensional accessibility.

Unified constraint formulation:

$$S_{t+1} = \mathcal{R}(S_t, D)$$

2.7 Chapter Summary

This chapter establishes the core axiom of generalized relational state evolution systems: all evolution systems consist of three irreducible building blocks: state characterization (Evolution carrier), Driving Capacity and Rule set, complying with the canonical expression:

$$S_{t+1} = \mathcal{R}(S_t, D)$$

Derived from this foundational axiom, the relational essence of information is formulated as $\mathcal{J} = (S, E), E = \mathcal{C}(S)$. System classifications including checkerboard-type, standard quantized and three-dimensional orthogonal inertial systems are distinguished; the minimum interactive characterization threshold Λ of state carriers, coupling mechanism of Driving Capacity and regulatory scope of Rule set are formally defined.

All axioms and definitions above serve as the logical starting point for subsequent chapters. Following the sequential reasoning path: relational evolution $\rightarrow$ inertial evolution $\rightarrow$ standard quantization $\rightarrow$ cosmic system, follow-up content deduces stability prerequisites for cosmic-scale evolution from systemic necessity, elaborates how conventional physical theories emerge as macroscopic integral statistical effects, and systematically derives emergent rules of fundamental physical Dimensions and dynamic relations exclusively from the Emergence perspective via logical and mathematical verification, with momentum (inertial state magnitude encapsulating substance and evolutionary attribute) treated as the core baseline physical variable.

2.8 Chapter Reference Commentary

[1] Wong, M. L., Cleland, C. E., et al. (2023). On the roles of function and selection in evolving systems. *Proceedings of the National Academy of Sciences (PNAS)*, 120(43), e2310223120. <https://doi.org/10.1073/pnas.2310223120>

This research summarizes three universal traits shared across all evolving systems: assembly from massive constituent components (evolution subjects), configurational generation (complexity construction), and configuration screening driven by functional selection (mutual evolution and constraint). It proposes the law of functional information growth: functional systemic information (orderliness and complexity) rises under selective environmental pressure. The conclusion aligns with this work's objective of constructing a unified underlying framework compatible with established physical theories and confirming evolutionary universality.

[2] Liang, J., Ban, X., Yu, K., Qu, B., Qiao, K., Yue, C., Chen, K., & Tan, K. C. (2023). A Survey on Evolutionary Constrained Multiobjective Optimization. *IEEE Transactions on Evolutionary Computation*, 27(2), 201–221. <https://doi.org/10.1109/TEVC.2022.3155533>

This review systematically sorts constraint handling methodologies within evolutionary optimization and confirms constraint functions directly define feasible searching domains. From computational mathematics, such constraints correspond to Rule set $\mathcal{R}$ in this manuscript acting as filtering operators determining validity and reachability of subsequent systemic states.

[3] Eddington, A. S. (1928). *The Nature of the Physical World*. Cambridge University Press.

Time is interpreted as irreversible globally entropic sequential evolutionary events, with the arrow of time identical to the direction of entropy increment, defining time as ordered evolutionary sequences and supporting the temporal definition adopted in this work; the global scope implies cross-system mutual correlation of full evolutionary sequences.

[4] van der Wijngaart, W. (2024). A Physical Basis for Information. *arXiv preprint*, arXiv:2407.09567v5. <https://doi.org/10.48550/arXiv.2407.09567>

The paper defines information as generative influential relational sets (state characterization) among topological nodes and proves ontological interconnection between information and evolution, with information acting as causal evolutionary motivation. Its core logic matches this theory's definition of information as mutually influential relational states of systemic components and underpins framing evolution

rooted in informational first principles.

[5] Gao, J. P., & Qi, Z. Y. (2022). *Xitong zhexue sixiangshi yanjiu* [A Study on the Intellectual History of Systems Philosophy]. Beijing: Science Press. pp. 443-456.

This monograph reviews the paradigm shift from substance-oriented thinking to relation-oriented thinking within systems philosophy and elaborates holistic, nonlinear and generative evolutionary attributes of relations. Its arguments regarding Reality, physicality, relationality and informational nature match the core definition framing the universe as a relational state evolution system in this paper.

[6] Maynard Smith, J., et al. (1985). *The Costs of Morphogenesis*. Princeton University Press.

A foundational monograph on evolutionary constraint theory demonstrating physical laws and developmental restrictions limit viable biological morphological space. Its core viewpoint that systemic states cannot randomly traverse unrestricted dimensions but are rigorously bounded by underlying rules directly corroborates the core mechanism where Rule set $\mathcal{R}$ constrains free-dimensional expansion capacity within this framework.

[7] Nicolis, G., & Prigogine, I. (1977). *Self-Organization in Nonequilibrium Systems: From Dissipative Structures to Order through Fluctuations*. Wiley.

This canonical work elaborates spontaneous ordered structural formation via fluctuation and dissipation within nonequilibrium open systems, providing theoretical grounding for the manuscript's deduction of systemic self-organization from disorder to order regulated by Rule set $\mathcal{R}$.

[8] Waldrop, M. M. (1992). *Complexity: The Emerging Science at the Edge of Order and Chaos*. Simon & Schuster.

This book documents core outcomes from Santa Fe Institute complexity research: localized interactions between adaptive subjects produce global emergent structural patterns (hierarchical informational relations), and systems develop hybrid ordered-chaotic properties near the edge of chaos. It supplies computational case evidence supporting the system classification and causal topology discussion in this paper.

[9] Whitehead, A. N. (1929). *Process and Reality: An Essay in Cosmology*. New York: Macmillan.

Whitehead's process philosophy proposes all actual entities exist as discrete generative events instead of static preset spacetime, denying infinitesimal instantaneous time points; all temporal instants correspond to finite fundamental interval t_0 consistent with identical logical settings within the present Evolution Theory.

[10] Holland, J. H. (1998). *Emergence: From Chaos to Order*. Reading, MA: Addison-Wesley (Basic Books).

The monograph elaborates complex systemic behavior Emergence originating from simple fixed evolutionary rules, validating the core logical framework of this work: complicated macroscopic phenomena derive from basic underlying evolutionary regulations.

3 Fundamental Emergence of Generalized State Evolution System

3.1 Definition and Theoretical Relevance of Emergence

Emergence denotes the generated **informational influence quantity** emerging across varied scales, viewpoints, influencing factors and relational configurations within a state or relational state

evolution system built on foundational system components: Representational Quanton, Driving Capacity and Rule set. All statistically describable macroscopic phenomena including conventionally counted mechanical quantities are essentially quantified expressions of influence quantities (informational quantities) originating from state variations. Mathematically, emergence directly corresponds to the information definition $\mathcal{I} = (S, E)$, $E = \mathcal{C}(S)$ within this framework: underlying states S generate relational structure E under constraint operator $\mathcal{C}$, which constitutes the core generation mechanism of emergence. As the relational framework of mutual causal influences, E concretely embodies informational influence quantities and characterizes new correlations built via causal interactions between Evolution carriers across differing observational scales and perspectives. Accordingly, **Emergence is the generation process of information**, while **information is the resultant structural outcome of emergence**; mechanical quantities such as force, energy and momentum are quantified forms of these informational influence quantities under specific statistical constraints.

Theoretical Methodology: Following the above emergence logic, this framework deduces essential correlations among complex objects starting from underlying axioms and logical necessities with minimal requisite assumptions. Subsequent chapters sequentially derive emergence mechanisms of momentum, force, energy, spacetime, relativistic effects, classical mechanics, quantum effects and cosmological phenomena, and support further extension toward phenomena across other academic disciplines[10].

3.2 Mathematical Predictability

When the total count of Evolution carriers and Driving Capacity remain conserved and Rule set stays stable inside a state evolution system, mathematical predictability emerges. Evolution carriers and Driving Capacity serve as input variables mathematically, and Rule set corresponds to mathematical mapping relations, operators or functions. Predictability $\leftarrow$ stable rules plus definite inputs $\Rightarrow$ computable outputs[7].

$$\text{Predictability} \iff \text{Stable Rule} + \text{Determined Input} \Rightarrow \text{Computable Output}$$

Supplementary Relation of Mathematical Composition Any mathematical computation process possesses the same three core elements as a state evolution system: 1. At least one relational mapping rule must be established; for instance, addition depends on the accumulation rule; 2. Quantized standard variables act as input participating factors; 3. A complete computational procedure exists, representing the inherent calculation-driving relationship.

3.3 Unidirectionality of Time

Evolution carriers cannot independently alter their causal or inertial states, which gives rise to the unidirectionality of time[9]. The causal cost consumed to construct a steady evolutionary sequence far exceeds the cost required to destroy such structure, leading to asymmetric causal influences and concretely manifested temporal unidirectionality.

$$\text{Unidirectionality of Time} \iff \text{No Autonomous Modification of Causal State} + \text{Construction Cost} \gg \text{Destruction Cost}$$

3.4 CPT Symmetry (Invariance of Causal Generation Structure)

CPT symmetry emerges as the invariance of causal generation structure provided the system satisfies conserved Evolution resource, isomorphic Rule set, equivalent Driving Capacity and stable evolu-

tionary regulations[11][12].

$$\text{CPT Symmetry} \iff \text{Invariance of Causal Generation Structure}$$

The constraint conditions expand into the following system requirements:

$$\left\{ \begin{array}{l} \text{Resource Conservation (fixed total number } N \text{ of Evolution carriers)} \\ \text{Rule Stability (invariant generation mapping } \mathcal{R}) \\ \text{Rule Isomorphism (uniform constraint applied on all carriers)} \\ \text{Driving Capacity Equivalence (elimination of generation bias)} \\ \text{Reversible Free Dimension Generation (topological symmetric mapping)} \end{array} \right.$$

3.5 Cooperative Evolution

Multiple Evolution carriers build mutually influential cooperative evolutionary relations through the Perceptual Cross-section based perception mechanism, forming consistent or complementary macroscopic evolutionary trends via localized causal interactions[7].

1. **Dynamics-Constrained Cooperative Evolution** Cooperative relations originating from vector Cancellation of momentum or state and combined velocity are determined by internal system state distribution.

$$\sum_{i \in \text{Domain}} \vec{p}_i = M\vec{v}_{\text{total}}$$

Core Features:

- Vector Composition yields mutual cancellation or convergent motion
- Spontaneously generated via carrier causal interactions instead of compulsory rule provisions
- Cooperative state breaks upon unbalanced momentum input disrupting vector equilibrium

Examples: full internal momentum Cancellation inside stationary particles, combined bulk motion of multi-body systems and orbital motion of celestial bodies.

2. **Rule-Constrained Cooperative Evolution** Cooperative relations directly defined by Rule set $\mathcal{R}$, independent from momentum or directional vector Cancellation.

$$\mathcal{R}_{\text{sync}}(S_i, S_j) = \text{Cooperative Relation}$$

Subcategories:

- Quantum Entanglement: Internal Coupling rules enforce angular momentum conservation ($\vec{L}_A + \vec{L}_B = 0$)[13]
- Encapsulation Rule ($\hat{R}_2$): multi-layer statistical convergence of fundamental Momentum unit into stable structures from elementary particles to atoms and molecules
- Rule Coupling: checkerboard connected pieces, synchronous RGB color variation

Core Features:

- System must satisfy coupling constraints predefined by evolutionary rules
- Cooperative status persists even in absence of momentum exchange between carriers

- Cooperative relation terminates once governing rules are modified (e.g., entanglement collapse after measurement)

Examples: quantum entanglement, proton-electron structural Encapsulation, checkerboard linkage rules and synchronous RGB color evolution.

3.6 Steady State (Invariant Informational Topology)

Within generalized state evolution systems, a steady state refers to dynamic equilibrium achieved when localized Evolution carriers collectively share causal interaction costs in Cooperative Evolution, manifested as disturbance-resistant invariant informational topology[14][12].

$$\text{Steady State} = \underbrace{\text{Invariant Informational Topology}}_{\text{Resultant Property}} + \underbrace{\text{Local Perturbation Allocation Mechanism}}_{\text{Evolution Process}}$$

Mathematical formulation:

$$T(S) \approx T(S + \delta S_{\text{local}})$$

Where:

S : aggregate set of instantaneous system states

δS_{local} : localized perturbation or partial state shift

$T(S)$: causal topological framework (informational invariant) corresponding to state S

Physical Implication: A steady state does not require unchanged individual states; instead its overall informational topological structure remains invariant against minor perturbations. Localized disturbances are absorbed and redistributed inside the system without altering global informational configuration.

3.7 Breakdown (Irreversible Remolding of Informational Topology)

Definition: Breakdown stands for irreversible restructuring of causal relational topology (informational topology). The system cannot spontaneously revert back to its original topological configuration even after external perturbation is entirely eliminated.

$$\text{Breakdown} \iff \exists \delta S_{\text{local}} : \rho_{\text{recover}}(S, \delta S_{\text{local}}) < 1$$

Recovery coefficient is defined as:

$$\rho_{\text{recover}}(S, \delta S_{\text{local}}) = \max_k \left\{ \frac{|T(S_{\text{after}}) \cap T(S_{\text{before}})|}{|T(S_{\text{before}})|} \right\}$$

$T(S)$: causal informational topology of state S

$S_{\text{after}} = \mathcal{R}^k(S + \delta S_{\text{local}})$ (system state after k Update steps of evolution)

$\rho = 1$: complete topological recovery without breakdown

$\rho < 1$: irreversible topological modification representing system breakdown

3.8 Energy Storage (Asymmetric Evolution Cost)

Energy Storage originates from accumulation and retention of asymmetric evolution cost formed via localized causal interactions among multiple Evolution carriers constrained under shared relational

bindings.

$$\Delta C_{\text{asym}} \Rightarrow E_{\text{stor}}(\Delta C_{\text{asym}}) \Rightarrow T_{\text{encap}} \Rightarrow \Delta S \neq 0 \Rightarrow T_{\text{release}}$$

Asymmetric evolution cost ΔC_{asym} accumulates into stored evolutionary tendency E_{stor} , which maintains steady confinement via Encapsulation mechanism T_{encap} and releases reserved evolutionary trend T_{release} once triggering condition $\Delta S \neq 0$ is satisfied.

3.9 Covariance (Superposition of Spacetime Sequence from Observer Viewpoint Selection)

Covariance emerges from macroscopic geometric causal sampling integration formed when Evolution carriers interact across spatial distribution and temporal sequences under alternative observational viewpoints. Sampling only reflects primitive raw distribution without accumulated covariance effects at a single instantaneous moment under localized constraints.

Mathematical expressions:

$$\bar{\mathcal{P}} = \frac{1}{n} \sum_{k=1}^n \mathcal{P}(\theta_k, \{S(t_i)\}), \quad \bar{\mathcal{T}} = \frac{1}{n} \sum_{k=1}^n \mathcal{T}(\theta_k, \{S(t_i)\})$$

Where $\{S(t_i)\}$ denotes sequential carrier states embedding integral effects from causal interactions.

$$\bar{\mathcal{P}}, \bar{\mathcal{T}} \neq \mathcal{P}(\theta_{\text{true}}, \{S(t_i)\}), \mathcal{T}(\theta_{\text{true}}, \{S(t_i)\})$$

θ_{true} does not correspond to an exclusive global absolute perspective. Each viewpoint provides one projection slice of causal integral, and macroscopic Covariant structure arises from statistical superposition of multi-angle sampling slices rather than background absolute motion.

3.10 Domain

Definition: A Domain denotes the full set of Evolution carriers capable of establishing direct causal connections with a designated carrier within the Perceptual Cross-section Σ .

$$\text{Domain}(A) = \{B \mid |\vec{r}_B - \vec{r}_A| \leq r_\Sigma\}$$

The Domain forms the boundary of localized causal interaction; no immediate causal linkage exists between carriers separated beyond this boundary range.

3.11 Causal Chain

Definition: A Causal Chain represents sequential causal dependency constructed from discrete Update events $n \rightarrow n+1 \rightarrow n+2 \rightarrow \dots$ between Evolution carriers.

$$A \xrightarrow[\text{causal}]{n} B \xrightarrow[\text{causal}]{n+1} C \Rightarrow A \prec C$$

Causal Chains satisfy transitivity, and propagation direction strictly follows ascending incremental event index n .

3.12 Action-at-a-distance

Definition: Within relational state evolution systems, action-at-a-distance describes direct causal coupling between two Evolution carriers independent of their physical spatial separation, restricted only by valid connection status inside the Perceptual Cross-section.

Action-at-a-distance $\iff |\vec{r}_A - \vec{r}_B| \ll r_\Sigma \Rightarrow$ Causal Connection Insensitive to Intermediate Spatial Separation

Classification of distance definitions:

- Physical spatial distance $|\vec{r}_A - \vec{r}_B|$: geometric separation between carriers
- Perceptual distance $\leq r_\Sigma$: effective range enabling causal accessibility
- Indirect linkage across Causal Chains: mediated connection via intermediate Evolution carriers

Action-at-a-distance permits direct causal interaction bypassing physical spacing but never breaks sequential Causal Chain constraints of event progression.

3.13 State Distance

Definition: State Distance quantifies the overall discrepancy between two carrier states S_i, S_j within state space.

$$d(S_i, S_j) = \alpha |\vec{r}_i - \vec{r}_j| + \beta |\vec{p}_i - \vec{p}_j| + \gamma \cdot \delta(\sigma_i, \sigma_j)$$

α, β, γ are constant weighting coefficients, and $\delta(\sigma_i, \sigma_j)$ quantifies internal topological discrepancy between configurations.

State Distance determines whether paired carriers fall inside mutual Perceptual Cross-section and succeed in establishing causal coupling.

3.14 State Space

State space splits into two categories corresponding to single-step Update and inertial evolution modes:

1. Instantaneous State Space $\mathcal{S}_{\text{inst}}$ The set of static carrier states captured at individual discrete Update moments, including position $\vec{r}$, total momentum $\vec{p} = \sum \vec{p}_0$ and internal topological configuration σ :

$$\mathcal{S}_{\text{inst}} = \left\{ (\vec{r}, \vec{p}, \sigma) \mid \vec{r} \in \mathbb{R}^3, \vec{p} = \sum \vec{p}_0, \sigma \in \text{Topology} \right\}$$

Instantaneous State Space is discrete, distinguishable and constrained to configurations reachable under Rule set $\mathcal{R}$.

2. Evolution Trend Space $\mathcal{S}_{\text{trend}}$ (Hilbert-space structured) A linear Hilbert space collecting all feasible state variation tendencies constrained by current instantaneous configuration and Rule set $\mathcal{R}$:

$$\mathcal{S}_{\text{trend}}(\mathcal{S}(n)) = \left\{ (\Delta \vec{r}, \Delta \vec{p}, \Delta \sigma) \mid \mathcal{S}(n+1) = \mathcal{R}(\mathcal{S}(n)) \right\}$$

Core mathematical features embedded within Hilbert architecture:

- Superposition Principle (combined bulk velocity): linear Vector Composition of multiple evolutionary trends corresponds to macroscopic inertial motion

$$\sum \vec{p}_i = M\vec{v}_{\text{total}} \in \mathcal{H}_{\text{trend}}$$

- Orthogonality (opposing momentum Cancellation): inner product of reversed trending vectors yields negative value matching momentum offset

$$\langle \Delta \vec{p}, -\Delta \vec{p} \rangle = -|\Delta \vec{p}|^2$$

- Inner-product framework enabling angular calculation and orthogonal projection among evolutionary trends

$$\langle \Delta \vec{p}_1, \Delta \vec{p}_2 \rangle = |\Delta \vec{p}_1| |\Delta \vec{p}_2| \cos \theta$$

Physical interpretation:

- Macroscopic inertial motion arises from coherent superposition of trend vectors
- Observer viewpoint selection equals geometric projection inside trend space
- Steady states correspond to Eigenstate directional axes within trend space
- Quantum interference emerges macroscopically from overlapping superposition of evolutionary trend vectors

3.15 Hierarchical Emergence

Definition: Multi-layer Encapsulation Rule $\hat{R}_2$ drives stepwise statistical convergence from fundamental Momentum unit toward hierarchical structures including elementary particles, atoms and macroscopic condensed matter[10].

$$\boxed{\text{Level 0 : } \{\vec{p}_0\} \xrightarrow{\hat{R}_2} \text{Level 1 : Particle} \xrightarrow{\hat{R}_2} \text{Level 2 : Atom} \xrightarrow{\hat{R}_2} \dots}$$

Each hierarchical tier constitutes aggregated statistical ensemble of lower-layer basic units, generating newly emergent causal properties such as mass, electric charge and chemical valence.

3.16 Rule Set Hierarchical Emergence

Parallel to layered structural Encapsulation of material bodies, Rule set $\mathcal{R}$ develops novel effective constraint forms alongside ascending system hierarchy.

Core Mechanism: Fundamental interaction rules governing raw Momentum unit generate tier-specific effective Rule sets via successive topological Encapsulation:

$$\boxed{\mathcal{R}_0 \xrightarrow{\text{Encapsulation}} \mathcal{R}_1 \xrightarrow{\text{Encapsulation}} \mathcal{R}_2 \xrightarrow{\text{Encapsulation}} \dots}$$

Where $\mathcal{R}_0$ denotes baseline rules for Momentum unit interactions, $\mathcal{R}_1$ particle-scale physical laws, $\mathcal{R}_2$ chemical interaction rules and $\mathcal{R}_3$ biological evolutionary regulations, extending sequentially upward.

- **Hierarchical Rule Property:** Low-level original rules evolve into intermediate then high-level effective constraints via Encapsulation; each tiered rule represents statistical projection of underlying regulations mapped onto higher topological dimension.

- **Observer Illusion:** Observers confined within one hierarchical tier mistake locally emergent effective rules for universal fundamental laws ignoring their underlying generative origin.
- **Hierarchy Dependence of Mathematical Prediction:** Models formulated within fixed hierarchical scope remain valid only for corresponding tier. Representative cases:
 - Uniform gravitational field on Earth creates apparent unified periodic directional constraints misrecognized as inherent fundamental inertial rules: all evolution on Earth shares identical gravitational effects, identical periodicities, directional constraints, and identical boiling points, forming the illusion of a "weak inertia environment" as a rule of thumb.
 - Variable coupling between electrons and protons generates distinct elemental chemical regulations across atomic tiers
 - Molecular-atomic Encapsulation Rule serves as intrinsic evolutionary constraint within condensed matter tier

Define $\mathcal{R}_0$ as basal underlying rule set and $\hat{R}_2$ as Encapsulation operator; the k -th tier emergent Rule set satisfies:

$$\mathcal{R}_k = \hat{R}_2^{(k)}(\mathcal{R}_{k-1}), \quad \mathcal{R}_k \neq \mathcal{R}_{k-1}$$

Therefore, R is not merely a set of constraints but also embodies the hierarchical emergence of influence modes—the collective relations among multiple stable rules at the bottom level recursively combine upward into new composite evolutionary rules, while the relational network of bottom-level evolutionary rules forms a chain of influence factors. High-level rules are not absolutely invariant; rules themselves emerge as system complexity increases.

This explains why the laws of cosmic evolution appear inconsistent at different scales yet remain consistent at the fundamental level, and also reveals the potential cause of disciplinary fragmentation (physics, chemistry, biology) that hinders unification: the effective rules emerging at each hierarchical level differ from one another, and observers remain confined to their own level, unable to penetrate the Encapsulation to perceive the underlying unity. This also implies that at high levels, decay processes may violate quantum number conservation, while at the fundamental level, momentum is strictly conserved. It also suggests that at the fundamental level, all life on Earth shares the same underlying influence mode—the genetic code—while at higher levels, identical composite influence factors exist only among individuals of the same species.

Differentiation from Subsection 3.14:

- Hierarchical Emergence (3.14): Statistical convergence of material configuration ($\{\vec{p}_0\} \rightarrow \text{Particle} \rightarrow \text{Atom} \rightarrow \dots$) generating novel inherent causal attributes.
- Rule Set Hierarchical Emergence (3.15): Evolution of rule interactive modes ($\mathcal{R}_0 \rightarrow \mathcal{R}_1 \rightarrow \mathcal{R}_2 \rightarrow \dots$) producing tier-dependent effective constraint frameworks. Both forms of hierarchical emergence originate from the same underlying physical entities.

3.17 Chapter Summary

Starting from core axioms of generalized relational state evolution system compatible with inertial evolution framework, this chapter systematically derives fundamental emergent phenomena and intrinsic dynamic relations: mathematical predictability, temporal unidirectionality, CPT symmetry, dual-mode

Cooperative Evolution[15], steady state and breakdown mechanism, energy storage and Covariance. Definitions of Domain, Causal Chain, action-at-a-distance, State Distance, dual-category State Space and structural hierarchical Emergence are also established. All above contents compose complete foundational behavioral framework for systemic evolution and lay solid logical groundwork for progressive deduction in subsequent chapters.

3.18 Chapter Reference Commentary

[7] Nicolis, G., & Prigogine, I. (1977). *Self-Organization in Nonequilibrium Systems: From Dissipative Structures to Order through Fluctuations*. Wiley.

Dissipative structure theory elaborates how open systems sustain ordered configuration via energy exchange and restates the essential nature of entropy increase; theories of dissipative structure, nonequilibrium self-organization and fluctuation-induced ordering support structural evolution of life as nonequilibrium systems and asymmetric utilization of evolutionary constraints.

[9] Whitehead, A. N. (1929). *Process and Reality: An Essay in Cosmology*. New York: Macmillan.

Whitehead's process philosophy argues reality consists of discrete actual entity events instead of preset static spacetime and rejects infinitesimal zero-duration time points; all so-called instantaneous moments correspond to finite complete fundamental time interval t_0 , matching identical core logic adopted within this Evolution Theory.

[10] Holland, J. H. (1998). *Emergence: From Chaos to Order*. Reading, MA: Addison-Wesley (Basic Books).

The monograph establishes core Emergence logic that complex macroscopic behaviors originate from simple fixed underlying evolutionary rules.

[11] Anderson, P. W. (1972). More is different. *Science*, 177(4047), 393–396. <https://doi.org/10.1126/science.177.4047.393>

This paper demonstrates macroscopic complex order and symmetry Breaking develop from underlying stable Evolution resource, uniform rules and equivalent Driving Capacity rather than pre-existing fundamental universal symmetry laws. Symmetry properties within evolution systems emerge dynamically instead of being preset fundamental physical axioms.

[12] Roy, R. (2009). Topological phases and the quantum spin Hall effect in three dimensions. *Physical Review B*, 79(19), 195322. <https://doi.org/10.1103/PhysRevB.79.195322>

Topological order is confirmed as fundamental state classification criterion independent from density or geometric shape, defined via intrinsic topological invariant consistent with steady-state definition in this text.

[13] Schumacher, B., & Westmoreland, M. D. (2010). Quantum mutual information and the one-time pad. *Physical Review A*, 81(6), 062348. <https://doi.org/10.1103/PhysRevA.81.062348>

Entanglement is treated as quantum informational resource in existing research and materialized as Momentum unit reservoir carrying fixed angular momentum property within this framework.

[14] von Neumann, J. (1951). The general and logical theory of automata. In L. A. Jeffress (Ed.), *Cerebral mechanisms in behavior: The Hixon symposium* (pp. 1-31). John Wiley & Sons.

Von Neumann's automata theory first connects Evolution carrier to mathematical variables, Rule set to functional operators and computability to systemic predictability; fixed transition regulations plus definite input guarantee calculable output, forming original theoretical basis for the predictability relation derived

in this chapter.

[15] Sakharov, A. D. (1967). Violation of CP invariance, C asymmetry, and baryon asymmetry of the universe. *JETP Letters*, 5, 24–27. <https://doi.org/10.1134/S0021364067010029>

Sakharov' s conditions prove cosmic matter-antimatter asymmetry requires baryon-number nonconservation, CP Breaking and nonequilibrium phase transition, which reversely verifies CPT symmetry is emergent statistical conservation after partial dynamic Breaking instead of absolute pre-existing background law under conserved basal Evolution resource, Driving Capacity and Rule set $\mathcal{R}$.

4 Dynamical Emergence of Generalized Relational State Evolution System

Dynamics emerges from statistics of state evolution quantities and causal interaction quantities generated during interactions among Evolution carriers, yielding macroscopic mutual mechanical relations[16].

4.1 Mechanical Influence (Spatio-temporal State Count)

Direct mechanical influence refers to state variation quantities within the generalized relational state evolution system (quantities and rates inside inertial state evolution systems), including the count of Evolution carriers, magnitude of Driving Capacity, and variation magnitude of temporal or spatial states. These directly correspond to Breakdown, relational interactions and intrinsic carrier states; indirect causal components cover spatio-temporal state quantities, variations from Cooperative Evolution, steady-state evolution shifts and modifications of informational structures.

1. Spatio-temporal State Quantity

Static Representational Quanton (m_0) $\times$ motion capacity (step or v) $\times$ number of evolutionary events (t) equals the count of states generated via Space-time state shaping. The emergence rate of spatio-temporal states inside inertial systems reads:

$$S_{\text{state}} = m_0 \cdot v \cdot v = m_0 v^2$$

This quantity acts as an indirect mechanical causal term.

2. Inertial State Quantity

It describes the capacity of Evolution carriers to sustain persistent causal states expanding along free dimensions. The single-dimensional motion capacity for one expansion step is denoted v , meaning inertial velocity equals Evolution frequency v defined along a single spatial axis. Macroscopically, continuous state evolution based on intrinsic characteristic scale writes $l_0(\text{dimension } x, y, z \dots) * N/t_0$.

3. Frequency State Quantity

All categories of frequencies including Evolution frequency and interaction frequency are collectively defined as fixed emergent occurrence count v within a single macroscopic interval.

4. Spatial State Quantity

Topological configurations formed from free-dimensional expansion of multiple Evolution carriers throughout state evolution. For inertial- v state evolution systems under dynamic evolution, space emergence rate is v with statistical magnitude $m_0 * v$, serving as direct mechanical causal quantity from spatial perspective, namely momentum. Macroscopic realizations include volume V with finite density, surface area S , length L , and single-dimensional measure $l_0 * L$.

5. Temporal State Quantity

Total count of independent evolutionary events T of Evolution carriers inside state evolution systems. Within inertial- v state evolution systems under dynamic evolution, time emergence rate equals v with statistical value $m_0 * v$, functioning as direct mechanical causal quantity from temporal perspective, i.e., momentum. Its macroscopic form is $t_0 * T$ [17].

4.2 Perceptual Interaction

Within relational state evolution systems, causal link establishment between Evolution carriers is governed by attributes including perceptual attenuation law, perceptual period and Perceptual window frequency.

1. Perceptual Mechanism

Causal interactions between Evolution carriers are realized exclusively via the perceptual mechanism. The upper bound of perceptual capacity generally coincides with the maximum free-dimensional expansion frequency of Evolution carriers:

$$\text{Upper Limit of State Evolution Rate} = \text{Upper Limit of Perception Rate}$$

For inertial state evolution systems:

$$\text{Perception } v_0 = \text{Motion } v_0 = c \quad (\text{cosmic constant})$$

2. Perception Rate

Perception Rate denotes the lockable proportion constrained by three independent factors:

- (a) Rule-driven evolution (existence or absence of perceptual attenuation)
- (b) Spatial distribution (free-dimensional topological structure)
- (c) Kinematic status (intrinsic spatio-temporal expansion)

$$\text{Perception Rate} = \frac{\text{Count of lockable static spatio-temporal states}}{\text{Upper bound of perceptual capacity}} = \frac{\text{Number of evolved states}}{\text{Upper bound of perceptual capacity}}$$

3. Perceptual Attenuation in Three-dimensional State Space

Spherical radiative perceptual decay emerges for three-dimensional state space with finite perceptual attenuation:

$$\Sigma(r) = \frac{1}{4\pi r^2}$$

Spatio-temporal magnitude generated from carrier motion equals $T \times L$ (temporal accumulation $\times$ spatial expansion). Inside inertial state evolution systems, space and time synchronously emerge in strict direct proportionality:

$$T \cdot L = v \cdot v = v^2$$

Perception Rate inside Inertial State Evolution Systems

Defined parameters:

Upper limit of inertial evolution capacity: $v_0 = c$

Upper bound of perceptual capacity: c (identical to maximal motion capacity)

Spatio-temporal emergence at moving speed v : v^2

Total lockable perceptual capacity: c^2 (from synchronous proportional spatio-temporal generation)

4. Conservation of Perceptual Capacity:

$$c^2 = v_{\text{perception}}^2 + v^2$$

Perception rate = accessible perceptual fraction / total available capacity:

$$\eta = \frac{v_{\text{perception}}^2}{c^2} = \frac{c^2 - v^2}{c^2} = 1 - \frac{v^2}{c^2}$$

Incorporating spherical spatial attenuation factor $1/4\pi r^2$:

$$\eta = \left(1 - \frac{v^2}{c^2}\right) \cdot \frac{1}{4\pi r^2}$$

Under normalized radial coordinate r :

$$\eta = 1 - \frac{v^2}{c^2}$$

5. Covariance of Perception and Relation with Lorentz Factor

$$\gamma = \frac{1}{\sqrt{\eta}} = \frac{1}{\sqrt{1 - v^2/c^2}}$$

The Lorentz factor is the reciprocal square root of perception rate. At ultra-high velocity, $\eta \rightarrow 0$ and $\gamma \rightarrow \infty$. Macroscopic causal interactions produce covariant path integrals over spacetime; this relation neither implies separate spatial/temporal partitioning of evolution capacity nor orthogonal spacetime geometry, nor corresponds to intrinsic energy. It does not represent physical spacetime dilation or length contraction, but exclusively describes momentum integrals originating from causal interactions.

State Evolution Quantity & Evolution Event $\rightarrow$ Inertia $\rightarrow$ Space and Time Emergence Rate $\rightarrow$
Perceptual Mechanism $\rightarrow$ Perception Rate $\rightarrow$ Lorentz Factor

4.3 Causal Interaction Quantity (Action Quantity)

Generalized mechanics describes two core capacities of Evolution carriers (Representational Quantum) within free dimensions of relational state evolution systems: autonomous state Transition via intrinsic evolution, and mutual influence on counterpart Transition through Vector Composition of resultant velocity. Consequential observables include spatio-temporal distribution of Evolution carriers, total carrier population, Transition capacity such as inertial Evolution frequency, and directional or state variations during evolution.

Direct causal influences correspond to magnitude or rate of state shift along spatial/temporal axes and variation direction; indirect mechanical causal terms refer to interaction quantities, namely unitary or persistent causal interaction magnitudes.

1. Generalized Momentum

Inside inertial state evolution systems, Evolution carriers execute free-dimensional state evolution equipped with capacities for composite causal resultant motion and reverse-direction evolution. Here state characterization and causal influence unify into momentum, with no independently existent entities of force or energy[18][16].

In non-inertial state evolution systems, only discrete single-interaction driven state variations exist. Hence defining causal influence as force or energy is physically appropriate for non-inertial frameworks requiring successive causal excitations to sustain continuous evolution; causal influence fundamentally reduces to momentum within inertial systems.

Essential definition of generalized momentum:

$$\text{Count of Evolution carriers} \times \text{Driving Capacity (evolution/Transition frequency)} \Leftrightarrow \vec{p} = \sum \vec{p}_0$$

where $\vec{p}_0 = m_0 \cdot c$ denotes base momentum per elementary Evolution carrier, m_0 stands for unit Representational Quanton, and c is Driving Capacity (baseline state Evolution frequency). Macroscopic momentum $m\vec{v}$ is the statistical outcome of $\sum \vec{p}_0$ via Vector Composition.

Distinguishing Remark: No independent force or energy exists within inertial state evolution systems; precisely, momentum encapsulates instantaneous force and energy effects. Force and energy definitions rely upon successive discrete stepwise integration valid for step or skip-type evolution systems exclusively. All non-inertial states including rotation, discrete stepping motion and accelerated movement constitute **composite integrated motion**.

2. Generalized Force

Generalized Force denotes single-interaction momentum integral induced by causal coupling between Evolution carriers:

$$F = \Delta p \quad (\text{Inertial System: momentum variation after single interaction})$$

$$F = \Delta S \quad (\text{Non-inertial System: state shift after single interaction})$$

Classical mechanics averages acceleration over infinitesimal intervals to solve for force; such derivation yields identical dynamical relations as formulated in the present Evolution Theory. However, the two frameworks differ fundamentally in their physical descriptions. The subsequent sections will provide a detailed derivation of the formation of force and energy within classical mechanics, and identify certain conceptual issues inherent in the conventional approach. This discussion will not be elaborated further here.

3. Generalized Energy

Generalized Energy equals cumulative multiple-step momentum integrals generated from successive causal carrier interactions:

$$E = \sum_n \Delta p_n \quad (\text{Inertial System})$$

$$E = \sum_n \Delta S_n \quad (\text{Non-inertial System})$$

Classical mechanics evaluates energy by integrating force over finite large intervals to quantify ultimate evolutionary cost, applicable only under specific boundary conditions and invalid for inertial or Cancellation state non-interacting processes[19]. At any instantaneous moment, all existing

evolution follows inertial laws, meaning force and energy are merely integral constructions over mass and velocity[20].

Summary: All statistical dynamical quantities originate from projection of Evolution carrier state evolution and mutual causal interactions[18].

4. Dissipative and Non-dissipative Effects:

a. Dissipative Force

Relative spatial motion produces composite or split carrier velocities to realize shared or redistributed Evolution resource:

$$\{\text{Dissipative Force} \iff \text{Spatial Motion} \Rightarrow \text{Sharing or Redistribution of Evolution Capacity}\}$$

b. Non-dissipative Force

Without breaking pre-existing Cooperative Evolution relations, superluminal-range allocation of opposite Momentum Deviation proceeds via perceptual mechanism, generating paired action-reaction forces simultaneously:

$$\text{Non-dissipative Force} \iff \Sigma(r) \text{ Non-local Interaction} \Rightarrow (p, -p) \text{ Opposite Allocation}$$

Thus, dissipation manifests in cosmic systems as changes in particle species, intrinsic carrier properties, and evolutionary direction. Non-dissipative interactions conserve particle number, with only the directional distribution of internal Momentum units undergoing interference-driven rearrangement while Evolution carrier identity remains unchanged. State variations increase or decrease gradually according to the spatial distance attenuation relation $\Sigma(r) = 1/(4\pi r^2)$. A detailed derivation based on internal particle mechanisms will be provided in subsequent chapters.

4.4 Causal Influence Hierarchy

m and $v \rightarrow$ Resultant velocity and reverse evolution $\rightarrow$ Interaction quantity $\rightarrow$ Buffering relation $\rightarrow$ Cooperative Evolution capacity $\rightarrow$ Informational structure $\Rightarrow$ Spatio-temporal relation. All hierarchical causal influences finally reduce to modifications of spatio-temporal relations such as displacement magnitude, displacement frequency, Representational Quanton variation and structural restoration relations.

Causal Hierarchy (From Fundamental to Macroscopic)

Hierarchy Level	Content	Quantified Variation	Explanation
1	Fundamental Entity	m, v	$\Delta m = 0, \Delta v = \Delta p/m$
2	Resultant Motion and Reverse Evolution	Composite: $\sum m_i v_i = m_{\text{total}} v_{\text{total}};$ Reverse: $\Delta p_1 = -\Delta p_2$	Two macroscopic manifestations of momentum Vector Composition
3	Interaction Quantity	ΔP	Momentum transfer per single causal interaction
4	Buffering Relation	$e, \alpha, 1/4\pi r^2$	Restoration coefficient and dissipation control

5	Cooperative Evolution Capacity	$\sum p_i = \text{constant},$ $v_{\text{sync}} = \frac{\sum m_i v_i}{\sum m_i}$	Global coherent systemic evolution
6	Informational Structure	$\sum L_i = 0, I = (S, C, E)$	Long-range non-local correlation regulations

4.5 Chapter Summary

From the inertial state evolution system perspective, this chapter systematically elaborates direct and indirect causal dynamical relations: it defines spatio-temporal, inertial, spatial and temporal state quantities (momentum); derives the quantitative link between perception rate and Lorentz factor; establishes hierarchical definitions of generalized momentum, generalized force (single-step momentum integral) and generalized energy (multi-step cumulative momentum integral); distinguishes Dissipative Force from Non-dissipative Force; and constructs the complete causal influence ladder spanning from fundamental entities to final spatio-temporal relations. Core conclusion: momentum is the unique fundamental mechanical quantity inside inertial evolution frameworks, while force and energy are statistical derived quantities.

4.6 Chapter Reference Commentary

[16] Liu, L. (2026). Outline of intrinsic mechanics. *ScienceNet Blog*, 2026-03-17; Three core axioms of intrinsic mechanics, 2026-03-24.

In his intrinsic mechanics, Liu Lintao (2026) states force is not the source of acceleration but a metric quantifying object deviation from intrinsic motion state ($F = m(a - a_n)$). However several intermediate logical steps between state deviation and force are omitted in his formulation. This manuscript further specifies:

1. State deviation equals state variation, which itself corresponds to motion; 2. Acceleration is not elementary state change but statistical aggregation of multiple discrete inertial increments (Δp_0) per unit time; 3. Force originates from momentum shift after single causal interaction instead of generating acceleration (Inertial System: $F = \Delta p$); 4. Acceleration is a statistical manifestation rather than a definitional prerequisite of force.

Liu's "deviation-induced force" is incorporated as a limiting case within this framework: under sufficiently high interaction frequency with negligible statistical fluctuation, averaged $\sum \Delta p / \Delta t$ yields macroscopic acceleration such that $F = m(a - a_n)$ holds as valid macroscopic approximation.

[17] Rovelli, C. (2018). *The Order of Time*. New York: Riverhead Books.

The emergent-time theory supports the core standpoint that fundamental time does not exist and accumulates sequentially from discrete evolutionary events.

[18] Tho, T. (2017). *Vis Vim Vi: Declinations of Force in Leibniz's Dynamics*. Dordrecht: Springer. (Studies in History and Philosophy of Science, Vol. 46)

This monograph systematically reconstructs Leibniz's dynamical program, demonstrating Leibniz rejected Newton's exogenous force hypothesis and regarded force as intrinsic evolutionary tendency of object configuration (vis insita). In mature formulation, Leibniz's force is no longer defined as inter-object motion excitation but structural causality bound to whole-system arrangement; force precedes motion as intrinsic origin instead of external perturbation. This supplies the 17th-century philosophical foundation

for this theory' s statement that force is intrinsic dynamic relation of evolution itself.

Brief commentary on core inertial, momentum and energy viewpoints from Newton, Leibniz, Descartes and Huygens grounded in Evolution Theory: Newton correctly formulated inertia yet misdefined force as exogenous action; Leibniz properly denied external-origin force while erroneously rejecting momentum conservation; Descartes partially pioneered inertial and momentum concepts but discarded vector properties; Huygens successfully established momentum conservation yet incorrectly introduced scalar energy formulation.

[19] Zdesenko, Y. G., & Tretyak, V. I. (2001). Experimental tests of nucleon stability and conservation laws. *Physics of Particles and Nuclei*, 32(6), 469-494. <https://doi.org/10.1007/1027-4510-32-6-469>
Experimental data verifies invariant conserved mass, indirectly proving mass-energy conversion originates from Cancellation-Breaking of Evolution carriers, and mass remains conserved under non-decay Long-range Force interactions.

[20] Cabaret, D.-M., Grandou, T., & Perrier, E. (2024). Neither a field nor a matter: The case of energy. *arXiv preprint*, arXiv:2412.17858.

Its core conclusion that energy is neither fundamental field nor material substance but derived relational quantity describing system interaction and state variation supports this theory' s classification of energy as non-fundamental emergent relational concept rather than primitive cosmic ontology.

5 Core Conditions for Stable Evolution of Complex Relational State Evolution Systems

To resolve the longstanding complications in conventional physics concerning statistical treatments, conceptual definitions, and relational descriptions, this paper decisively abandons ambiguous causal boundaries and descriptive frameworks rooted in traditional approaches. Instead, it adopts a unified generalized relational state evolution system characterized by universal applicability, conciseness, physical realizability, dimensional quantifiability, derivability, and mathematical formalizability. Based on this foundation, derivations proceed sequentially—from state representation, evolutionary driving, causal interaction, to mechanical statistics—thereby recovering the core prerequisites necessary for stable evolution analogous to that of cosmic systems. Combined with the concepts established in preceding sections, the relevant constituent elements are elaborated as follows.

5.1 Spatial State Representation and Standard Quantization

Within complex relational state evolution systems, the minimal state quantities satisfy the following features: the magnitude of causal state quantities matches that of Driving Capacity, the magnitude of spatial relational quantities matches that of Representational Quanton for Evolution carriers, and the magnitude of temporal quantities corresponds to complete evolutionary events at matching scales. To characterize Planck-scale temporal and spatial states, the system requires complete intervals for perception, interaction and state Update at identical scales as elementary time units[21]. Meanwhile the system must possess fundamental minimal material constituents whose single-step motion quantity or static Representational Quanton matches Planck spatial magnitude. If the system features minimal inertial Driving Capacity yielding propagation speed equal to light speed c , every unit of matter holds conserved evolutionary capacity propagating at c . Derivations in this section target cosmic systems.

1. Standard Quantization

Four fundamental minimal quantities are defined to realize atomic discretization of causal interactions, guarantee spatial information storage and conserve Evolution resource[22].

Minimal Quantity	Symbol	Definition	Physical Implication
Minimal Causality	$\mathcal{C}_{\min}$	Unit counting for one full causal interaction	Dimensionless event counting unit, marking causal events where a Momentum unit finishes full momentum transfer or complete directional reversal; such events either fully occur or do not take place
Minimal Scale	l_0, t_0	Spatial and temporal scale for single evolutionary step	Minimal state quantity defined over free dimensions
Minimal Driving Capacity	$p_0 = m_0 c$	Minimal momentum variation generated from one single causal interaction	Elementary Momentum unit[23]
Minimal State Representation	m_0	Intrinsic mass of Evolution carrier	Fundamental discrete unit of material state

Naturally locked via light speed relation $c = l_0/t_0$, the four minimal quantities correlate through direct proportionality. Normalization under natural unit system is available by setting $\mathcal{C}_{\min} = 1, m_0 = 1, c = 1, l_0 = 1$, with numerical values mapped to practical physical magnitudes for specific application scenarios. Constraints among the four minimal quantities are intrinsic to the system rather than fixedly pegged to Planck constants.

2. Conservation Constraints

The core of standard quantization lies in Evolution resource conservation, summarized as the fundamental rule: "one unit of mass for self-existence, one unit of motional capacity for intrinsic evolution, one unit of interaction for relational establishment". Detailed specifications read:

Each minimal Representational Quanton m_0 carries fixed Intrinsic motional capacity c , forming elementary Momentum unit $p_0 = m_0 c$.

Macroscopic total mass $m = N m_0$, total systemic motional capacity $P_{\text{total}} = N m_0 c$.

Inertia emerges statistically from unit population instead of existing independently from mass; inertial magnitude is directly proportional to the count N of participating Momentum units.

Representational Quanton and evolutionary capacity satisfy two fundamental conservation laws inside the system:

(1) Conservation of total constituent count

$$\sum_i N_i = N_{\text{total}} = \text{constant}$$

N_i denotes the number of Momentum units moving along the i -th directional orientation, N_{total} stands for overall minimal Representational Quanton count of the whole system.

(2) Conservation of Driving Quantity

The system possesses a unified dimensionless driving frequency c at the fundamental level. It defines the minimum time interval t_0 for a single complete state evolution and causal interaction of the elementary Representational Quanton m_0 , as well as the emergent capacity for continuous time, c . The quantity c itself carries no spatial dimension—space is determined either by the static state magnitude characterized by m_0 or by the evolutionary scale per unit t_0 .

The speed of light emerges from the state evolution capacity of m_0 under continuous driving by c . The detailed derivation of the interrelations among matter, the speed of light, and spacetime is presented in Chapter 9.

(3) Conservation of total momentum

$$\sum_i N_i m_0 c \hat{n}_i = \vec{P}_{\text{total}} = \text{constant vector}$$

$\hat{n}_i$ is the unit vector along the i -th moving direction with $|\hat{n}_i| = 1$.

Inertial mass $m = N_{\text{total}} m_0$ is uniquely determined by the total unit count, irrespective of directional distribution.

The physical implication of the above derivations is that all physical observables—including mass, momentum, force, and energy—originate from countable discrete fundamental portions at the microscopic level; no infinitely divisible continuous fundamental entity exists. Moreover, these quantities emerge intrinsically from (1) and (2).

3. Spatial Causal Capacity and Constraints

(a) Definition of Spatial Causal Capacity

The **spatial causal capacity** is uniformly defined as: $\Omega_{\text{causal}} = l_0^3$.

Spatial causal capacity refers to the maximum number of complete causal units that any Evolution carrier can simultaneously maintain within three-dimensional space, determined by the intrinsic three-dimensional spatial scale of the carrier. $d = 3$ corresponds to the intrinsic volumetric state of the system, serving as the fundamental definition; $d = 2$ is merely a state quantity selected from a specific observational perspective and does not belong to the intrinsic state relation.

This quantity reflects the maximum causal information capacity that an Evolution carrier can bear, corresponding to the fundamental condition for causal integrity.

(b) Constraint Relations

For an Evolution carrier to maintain a causal state quantity of magnitude l_0^3 , two core constraints must be satisfied:

- All Evolution carriers possess identical spatial occupancy scale;
- The spatial displacement generated per single driving action matches this characteristic scale.

From the synchronous locking relation $c = l_0/t_0$, we obtain:

$$l_0 \cdot t_0 \cdot c = l_0 \cdot l_0 = l_0^2$$

4. Spatial Causal Capacity and Relevant Constraints

(a) **Definition of Spatial Causal Capacity**

Spatial causal capacity is uniformly defined as $\Omega_{\text{causal}} = l_0^3$.

Spatial causal capacity describes the maximum count of intact causal units an Evolution carrier can simultaneously sustain within three-dimensional space, constrained by intrinsic three-dimensional bulk scale of the carrier. Dimension $d = 3$ corresponds to intrinsic bulk state volume as foundational definition; $d = 2$ merely represents state quantity selected from specific observation perspective rather than intrinsic relational property.

This parameter quantifies the upper limit of causal information storage capacity for Evolution carriers and constitutes foundational prerequisite for causal integrity.

(b) **Restrictive Relations**

An Evolution carrier maintaining a causal state volume of magnitude l_0^3 must satisfy two core constraints:

- a. All Evolution carriers possess identical spatial occupancy scale;
- b. Spatial displacement generated per single driving action matches the characteristic scale.

From synchronous locking relation $c = l_0/t_0$:

$$l_0 \cdot t_0 \cdot c = l_0 \cdot l_0 = l_0^2$$

(c) **Necessity for Standard Quantization**

Conservation requirement: Evolution carriers hold quantifiable Intrinsic evolutionary capacity unified into Momentum unit $p_0 = m_0 c$, preventing spontaneous generation or vanishing of Evolution resource.

Elimination of point-particle hypothesis: Dimensionless point particles cannot accommodate finite spatial causal capacity and lead to ambiguous incomplete causal interactions.

Atomic causal discretization: each causal interaction either completes fully or fails entirely with no intermediate transitional states, ensuring complete and quantifiable causal behavior.

Finite information bound: spatial causal capacity caps the maximum processable causal information for Evolution carriers and eliminates undefined ambiguous intermediate states.

5. **Concluding Remark**

Removal of dimensionless point-particle formulation eliminates ambiguous partial causal interactions at local domains. Fundamental physics requires physically substantial material definition plus standard quantization specifying minimal interval and minimal action quantity. Contrasted with point particles limited to single-point state characterization[24], the standard quantization framework satisfies stable evolution requirements of generalized relational state evolution systems by realizing Evolution resource conservation and causal integrity, fulfilling the underlying postulate of "one unit mass, one unit motion capacity, one unit interaction". Fundamental physical quantities including momentum, force and energy emerge hierarchically from such constraints with mutually equivalent causal influences. For instance, transferring one Representational Quanton from carrier A to B respectively subtracts and adds identical Driving Capacity and state quantity for both sides, avoiding asymmetric causal behaviors such as infinite mass growth under relativistic particle acceleration or input-output imbalance.

Within any cosmic material system of total mass m , overall Driving Capacity permanently equals mc ; rest state or finite macroscopic bulk velocity arises from internal Cancellation state of constituent Momentum units.

5.2 Mechanical Representation and Generalized Equivalence Principle

1. Nature of Mechanics: From State Evolution to Causal Emergence

Mechanical representation serves as an indispensable component of state evolution systems, precisely describing state maintenance rules and evolutionary trends of Evolution carriers during spontaneous evolution or mutual causal interactions. In generalized relational state evolution systems, elementary constituents are structured Evolution carriers instead of isolated geometric mass points. Macroscopic matter-wave characteristics of particles arise statistically from Cooperative Evolution of internal Momentum units.

From this perspective, "force" is no standalone entity outside material bodies with invisible pulling effects; instead it represents integrated Momentum Deviation induced by environmental interactions following internal rearrangement of carrier momentum distribution. This internal structural mechanism uniformly accounts for peculiar mechanical phenomena including non-local action, penetration, instantaneous interaction, Vector Composition and Cancellation, and non-attenuating interactions, which collectively Emerge as observable macroscopic force and energy via multi-layer Encapsulation.

2. Generalized Equivalence Principle: Evolution Trend Identical to Mechanical Trend

Within arbitrary state evolution systems, causal effects directly act upon state evolution itself instead of intermediate interaction processes, which indirectly modify evolution via regulatory constraints. Formulation of mechanical relations relies on characterization of evolutionary states and their variations, ultimately manifested as spatial distribution trends of systemic state evolution. The generalized equivalence principle originates collectively from three core components: motion, relational coupling and state representation, from which mechanical observables including velocity, momentum, force, energy, structural configuration and intrinsic state Emerge.

1. Source of Causal Influence

State variation constitutes the fundamental implementation of causal action rather than standalone interaction quantity, which only reflects participation magnitude. When two Evolution carriers share identical interaction magnitude, they impose equivalent causal influences on one another. Therefore evolutionary trend equals mechanical trend, namely **equal participation** → **equivalent causal impact**[25].

2. Formation Mechanism of Evolutionary Relations

Causal relations in inertial state evolution systems are characterized via inertial velocity, combined bulk motion and reverse directional evolution; systemic Evolution frequency or velocity directly quantifies intensity and duration of causal action.

Critical Distinction: Scalar Energy Cannot Replace Vector Momentum as Fundamental Driving Capacity

Mechanical relations for complex relational state evolution systems must adopt vector-form momentum instead of scalar-form energy as elementary Driving Capacity. Detailed decomposition exposes inherent drawbacks of scalar descriptors:

a. Discrete single-step motion: Each causal transition updates system state via scalar energy increment $e \cdot \text{step} \cdot l_0$. Such formalism hinders free-dimensional expansion, permitting only short-range adjacent coupling and limited evolutionary capacity. Scalars lack directional information and cannot realize multi-body Vector Composition and combined bulk motion.

b. Linear motion with pseudo-scalar treatment: State Update governed by scalar increment $e \cdot v$ approximates momentum formalism yet discards directional causal records as pseudo-scalarization. The critical flaw lies in incapability of establishing combined vector motion with external carriers and missing relational tracking, hence failing to sustain cooperative or competitive coupling within complex evolutionary frameworks.

In summary, scalar energy only quantifies magnitude without directional and interactive information required for Vector Composition and causal registration in sophisticated evolutionary systems. Discrete-step and scalar-based formulations suit externally driven analog simulation systems exclusively; inertial and complex relational evolution systems necessitate vector momentum as basic Driving Capacity.

3. Functional Position of Force and Encapsulation Rule

The four fundamental interactions (electromagnetic, strong, weak and gravitational force) are not primary causal sources but **Encapsulation Rules** governing interaction magnitude, propagation mode and confinement criteria. Direct state modification still complies with equivalence principle: equal interaction magnitude yields equal state variation.

Intuitive Analogy: Human consumption of cooked rice acquires metabolic energy from biochemical state transformation inside rice rather than cooking personnel as intermediate procedural provider. Cooking only sets transfer pathway and regulation analogous to fundamental interactions providing interactive channels; actual causal state change originates from internal momentum rearrangement of constituent particles. Force fields define interactive pathways and action magnitude without supplying intrinsic evolutionary Driving Capacity.

3. Deduction for Validity of Generalized Equivalence Principle

1. Non-local action, penetration and instantaneous response: The intrinsic perceptual mechanism permits non-local penetrating instantaneous sensing, updating internal momentum distribution of involved particles toward opposite evolutionary trends after coupling.

2. Vector Composition and mutual Cancellation: Multiple Evolution carriers assemble into Cooperative Evolution structures constrained by quantum Internal Coupling or localized confinement; internal Momentum units superpose via Vector Composition to form unified macroscopic inertial motion, further combining with external particles to generate updated bulk inertial states and explaining vector superposition and offset phenomena.

3. Non-attenuating property: Despite perceptual attenuation, carrier Driving Capacity fixed at c by conservation forbids gradual fading of directional evolutionary capacity after interaction, generating non-decaying macroscopic mechanical behaviors. Distinction exists between perceptual attenuation and force attenuation: conserved momentum forbids spontaneous decay of previously acquired Momentum Deviation, which only redistributes or dissipates via structural rearrangement instead of continuous attenuation.

4. Emergence of acceleration: Macroscopic acceleration is not induced directly by external force but determined jointly by perceptual attenuation and spatial distribution: shorter inter-particle separation correlates larger exchanged Momentum Deviation and emergent bulk acceleration.

5. Emergence of Inertial Resistance from Rest Mass: In an ideal environment without gravity or friction, consider pushing two stationary objects of masses m_1 and m_2 with $m_2 = 2m_1$. To achieve the same final velocity v for both, the Momentum Deviation required for m_2 must satisfy $p_2 = 2p_1$. Therefore, the difference in inertial resistance between the two objects is exactly the difference in their required Momentum Deviations. Extending this to a thought experiment, if we

treat m_1 and m_2 as a single composite whole m_3 undergoing local Vector Composition and resultant velocity, this implies that within m_3 , two distinct velocities are simultaneously evolving. By the same principle, even when m_1 and m_2 are macroscopically stationary, their underlying internal dynamics still exhibit Vector Composition with mutual cancellation. Hence, the Generalized Equivalence Principle is indispensable for explaining why matter possesses rest mass (inertial resistance) and why the Momentum Deviation required for a mass m to reach velocity v is proportional to its mass.

Summary: All cosmic observables of force, energy, tensile rigidity and compressive hardness stem from kinematic relations assembled via massive internal Momentum unit coupling. Compression of iron triggers repulsive collective upward motion of inner constituents; steel cable load-bearing capacity originates from inward contraction governed by internal electromagnetic evolutionary capacity. All instantaneous upper and lower bounds of interactive force ranging from black hole to porous sponge fall within 0 to mc , corresponding to resistance against combined bulk motion.

4. Mathematical Formalism of Mechanical Quantities

All mechanical formulations follow integral rules of momentum variation. Single-interaction derived "force" equals total momentum shift; accumulated multi-interaction "energy" sums sequential incremental momentum variations over time.

Force generated by single interaction:

$$F = \Delta P = \Delta \left(\sum_{k=1}^N \vec{p}_{0k} \right)$$

Energy accumulated over successive multiple interactions:

$$E = \sum_{n=1}^M \Delta P_n = \sum_{n=1}^M \Delta \left(\sum_{k=1}^N \vec{p}_{0k} \right)_n$$

$\vec{p}_{0k}$ denotes the k -th internal Momentum unit, N total unit count and M cumulative interaction rounds. Macroscopic mechanical manifestations represent temporal integral projection of underlying microscopic momentum flux.

5. Theoretical Prediction and Verification Routes

Three testable predictions are proposed to verify physical validity of the framework:

A.Existence of intrinsic internal structure: Particles are non-pointlike with internal degrees of freedom participating in causal interactions, and internal momentum rearrangement correlates with de Broglie matter-wave wavelength.

B.Light-speed evolutionary bound: All internal Momentum unit evolution is capped absolutely at c , defining causal locality benchmark and guaranteeing equation Covariance.

C. Quantum Entanglement Internal Coupling Rule: The essential cooperative evolution mechanism among internal Momentum units that ensures the particle behaves as a unified causal entity during interactions. This guarantees coherent internal evolution and preserves mechanical causality integrity.

5.3 Relational Characterization: Multi-carrier Interactive Correlation and Cooperative Evolution Rules

Core definition: dual-layered internal intra-carrier linkage and external inter-carrier interaction governed by paired Internal Coupling and External Coupling rules (corresponding to quantum entanglement logic in cosmic systems), elaborating fundamental binding and interactive laws of multi-body evolution. Core research targets: how intra-particle constituent units assemble into bulk entities under Internal Coupling constraints, and how inter-particle External Coupling generates directional evolutionary trends including synchronized combined motion, collective inward contraction and collective outward expansion. Independent from single-body state and mechanical formalism, such relational rules underpin complex systemic coupling and macroscopic matter agglomeration.

1. Dual-layer Fundamental Mechanism: Internal Coupling Rule + External Coupling Rule

All correlative relations among Evolution carriers are implemented by mutually complementary independent Internal and External Coupling matching structural intra-particle composition and Vector Composition mathematical derivation.

A. Internal Coupling Rule (Intra-particle intrinsic binding)

Internal Coupling confines multiple Evolution carriers into unified Cooperative Evolution aggregates, enabling constituent units (elementary particles in cosmic systems) to construct complex one-to-one, one-to-many and many-to-many relational networks via distributed independent evolution complying with:

$$\vec{P}_{\text{total}} = N \cdot m_0 c$$

This rule realizes full synchronous constraint over all inner microscopic Momentum units inside individual particles following quantum entanglement cooperative logic of cosmic systems.

Inner constituent Momentum units $\vec{p}_{0k}$ superimpose vectorially to cancel random intrinsic trends and Emerge unified macroscopic velocity $\vec{v}$ plus stable inertial state with complete expression:

$$\vec{P}_{\text{total}} = \sum_{k=1}^N \vec{p}_{0k} = N m_0 \vec{v}, \quad \vec{v} = \frac{\sum_{k=1}^N \vec{p}_{0k}}{N m_0}$$

Internal Coupling works without external interaction or applied force, merely securing intrinsic wholeness and structural stability for standalone particles before any external mechanical evolution or coupling.

B. External Coupling Rule (Inter-particle cross-space causal interaction)

External Coupling governs long-distance causal interaction between discrete evolutionary units (cosmic particles) paired with spherical-space Perceptual Cross-section attenuation, realizing remote state sensing, bidirectional causal coupling, correlated state variation and quantitative interaction definition. Underlying formulations of four fundamental cosmic interactions Emerge from External Coupling fulfilling dual core functions: quantitative interaction magnitude restriction and directional evolutionary confinement.

$\begin{aligned} \text{External Coupling} &\iff a(\text{Perception}) + b(\text{Interaction Quantification}) + d(\text{State Update}) \\ &\Rightarrow \text{Evolution Relation} = N \cdot m_0 \cdot \vec{c} = \Delta P \end{aligned}$
--

Perception: sensing and perceptual attenuation mechanism between evolutionary units (determines interactive partners and cross-section)

Interaction Quantification: e.g., the four fundamental forces, determines interaction magnitude

State Update: quantum fluctuation or propagator-mediated transfer alters the evolutionary direction of internal $N \cdot m_0$

Evolution Relation: emergent new Momentum Deviation ΔP

Supplementary Remark: The aforementioned propagator is employed solely as an auxiliary conceptual tool for state update within this stage of hypothesis. Subsequent derivations will demonstrate that propagators lack the capacity to mediate Long-range Force transmission —long-range force establishment relies on the global synchronous update mechanism (see Chapter 11) and direct coupling via Perceptual Cross-section, requiring no intermediate propagator as a force transmission carrier.

2. Three Fundamental Evolution Modes under External Coupling

Constrained by quantitative and directional External Coupling limits, inter-particle interactions produce three primary evolutionary configurations dominating mainstream cosmic interactive phenomena; rotational and exotic modes may exist within alternative customized evolutionary systems.

Relation Category	Evolution Feature	Momentum Rearrangement Direction	Macroscopic Manifestation
Cooperative Evolution (Combined Motion)	Identical evolutionary trend without mutual approaching or separation	Shared evolutionary capacity	Bulk synchronized motion
Collective Inward Evolution	Attractive interaction	Mutually oriented toward counterpart	Cosmic gravitational and electromagnetic attractive phenomena
Collective Outward Evolution	Repulsive interaction	Mutually deviated away from counterpart	Cosmic strong-interaction and electromagnetic repulsive phenomena

“

5.4 Inertial State Characterization

Within inertial state evolution systems, Evolution carriers are required to characterize inertial states equipped with dissociation mechanisms for environmental interactions at the fundamental level. An Evolution carrier establishes complete, magnitude-matched causal relations with surroundings via the perceptual mechanism exclusively within the minimal elementary interaction interval t_0 .

The inertial state of Evolution carriers originates from discrete cyclic iterations of "causal interaction – state Update –inertial continuation". To prevent unbounded growth of systemic complexity and guarantee evolutionary stability, **an instant interaction decoupling mechanism** is introduced: immediately after finishing one complete causal interaction, an Evolution carrier decouples relational coupling from ambient environments and retains only post-interaction state for subsequent inertial evolution. New causal interactions can only proceed upon emergence of fresh perceptual and temporal interaction windows.

The decoupling mechanism dominates systemic complexity control. Without such a mechanism, Evolution carriers must permanently preserve all historical interaction correlations, leading to exponentially rising complexity and eventual collapse of stable evolutionary capacity. After implementation of decoupling, only instantaneous states and local interactions require maintenance, confining complexity within feasible evolutionary bounds.

Mathematical formulation: discrete temporal evolution governs state Update of Evolution carriers, with Planck time t_p denoted as t_0 within this evolutionary framework serving as the cycle for single interaction and state renewal. The state evolution formula with decoupling reads:

$$S(t + t_p) = F(S(t), E_{\text{local}})$$

$S(t)$ denotes systemic state at instant t , E_{local} stands for localized environmental interaction, and F represents the state evolution function. This expression indicates state renewal depends solely on instantaneous configuration and nearby ambient conditions, independent of historical relational records. If decoupling is absent, the evolution equation transforms into:

$$S(t + t_p) = F(S(t), E_{\text{local}}, H_{\text{all}})$$

H_{all} collects all historical interaction relations, yielding systemic complexity scaling as:

$$O(T \cdot N^2)$$

or even factorial-order complexity:

$$O\left(\left(\sum_{t=0}^T N_t^2\right)!\right)$$

T counts total evolutionary time steps, N_t gives the quantity of Evolution carriers at step t . After introducing decoupling, complexity reduces to:

$$O(N^2)$$

Standard evolution cycle: Interaction $\rightarrow$ State variation $\rightarrow$ Inertial continuation (decoupling) $\rightarrow$ New interaction $\rightarrow$ Cooperative steady state. The decoupling mechanism frees Evolution carriers from accumulated historical constraints to avoid systemic disorder and imposes rigorous complexity restrictions for persistent stable evolution. This also demonstrates that infinitely and continuously subdivided "fields" and "spacetime" cannot support their own stable evolution.

5.5 Isomorphism and Conservation of Evolutionary Mechanisms

1. Fundamental Principle of Isomorphism

Conservation of Evolution carriers, Driving Capacity and evolutionary rules constitutes the core prerequisite for complex relational state evolution systems; no spontaneous modification of carrier configuration or population occurs throughout state evolution. Complete underlying Isomorphism across the three aforementioned components supports steady evolution albeit not acting as an indispensable core constraint.

Conservation and Isomorphism satisfy three concrete criteria:

- No arbitrary skipping of predefined evolutionary states
- Spontaneous generation or vanishing of Evolution carriers and Driving Capacity is forbidden

- **Identical evolutionary boundary conditions yield identical evolutionary outcomes (isomorphism, i.e., possessing a unified underlying mechanism).**

2. Isomorphic Configuration of Three Core Constituents

Constituent	Description	Isomorphism Status
Representational Quanton of Evolution carrier	Minimal elemental unit (m_0) for causal information storage	Already isomorphic
Driving Capacity	Single-direction expansion frequency (c) over intrinsic free dimensions regulating evolution rate	Already isomorphic
Momentum unit	$p_0 = m_0 c$, elementary causal interaction unit combining state representation and Driving Capacity	Already isomorphic
Evolutionary Rule	Governs state Update, causal interaction, Encapsulation and perception	Pending isomorphism (core objective at this layer)

The first three items achieve inherent Isomorphism derived from the Generalized Equivalence Principle:

- m_0 fixes the minimal Representational Quanton
- c defines evolution speed and frequency of causal interaction.
- $p_0 = m_0 c$ quantifies minimal Momentum Deviation and elementary causal unit per single interaction

Core Conjecture: all underlying evolutionary rules are fully isomorphic; macroscopic discrepancies between particles of distinct Encapsulation tiers originate from projective transformation of uniform fundamental rules rather than divergent primitive laws.

3. Detailed Justification for Isomorphism Necessity

(a) Necessity of uniform isomorphic fundamental rules

Non-uniform underlying rules lead to three critical drawbacks:

- Inconsistent physical interpretation of macroscopic phenomena including particle generation, decay and scattering across Encapsulation hierarchies
- Breakdown of cross-tier conservation for total momentum and systemic state quantities
- Emergence of non-rigid composite force, tidal distortion and bent orbital trajectories

Conclusion: isomorphic underlying rules serve as foundational prerequisite for consistent evolution and Causal Continuity, underpinning particle indistinguishability and evolutionary conservation laws.

(b) Necessity of macroscopic Encapsulation equivalence

Macroscopic particles form as clustered Cooperative Evolution aggregates of intrinsic Momentum units $p_0 = m_0 c$, whose observable bulk properties must remain isomorphic to fundamental governing rules.

- New particle replication from Potential Space occurs when transient injected momentum equals p_0 of the parent particle; replicated particles inherently inherit all evolutionary rules and intrinsic state characterization of their progenitor.
- Rule non-isomorphism breaks uniform causal and mechanical responses between daughter and parent particles, violating global conservation and stable systemic evolution.

4. Isomorphism Principle of Evolutionary Rules

Definition: indistinguishability of particles with disparate Encapsulation architectures (elementary particles, atoms, molecules etc.) fundamentally arises from underlying Isomorphism shared among Evolution carriers, Driving Capacity and evolutionary regulations. Evolutionary conservation prohibits unprompted origination or annihilation of core physical quantities throughout state transition, and Isomorphism sustains both indistinguishability and global conservation constraints.

Mathematical Expression:

Let $\mathcal{R}_A(S)$ and $\mathcal{R}_B(S)$ denote respective evolution rules for distinct encapsulated Evolution carriers A and B . An isomorphic mapping ϕ satisfies the relation below:

$$\mathcal{R}_A(S) \cong \mathcal{R}_B(S) \iff \exists \phi : \mathcal{R}_B(S) = \phi \circ \mathcal{R}_A \circ \phi^{-1}(S)$$

Conservation Constraints:

$$\sum_i \vec{p}_i = \text{constant} \quad (\text{Momentum Conservation})$$

$$\sum_i N_i = \text{constant} \quad (\text{Conservation of total Evolution carrier count})$$

Global Conservation Formalism

$$\prod_i \mathcal{R}_i(S_i) = \mathcal{R}_{\text{global}}(S_{\text{total}})$$

Definitions of parameters:

S_{total} : overall systemic state

$\mathcal{R}_{\text{global}}$: unified global evolutionary regulation

Global conservation denotes invariant total evolutionary mode and preserved overall driving capacity, decomposed into conserved fundamental state representation, intrinsic driving power and invariant base-level evolutionary rules. In ideal configuration, all higher-tier rules emerge entirely from the same primitive underlying regulations, differing only in encapsulation schemes.

5. Particle Decay and Structural Reconfiguration

Conservation across multi-layer decay and reconstruction

Evolutionary conservation and fundamental Isomorphism persist intact regardless of sequential decay or repeated Encapsulation rearrangement of constituent units, constituting the core manifestation of particle indistinguishability:

- Total count, intrinsic Representational Quanton (m_0) and Driving Capacity (c) of elemental units experience no net loss across transformations
- Macroscopic composite particles or restructured assemblies fully inherit ancestral evolutionary rules after recombination or re-Encapsulation

- Global conservation holds for total causal interaction capacity, mechanical response and net momentum flux
- Varied Encapsulation configuration or changing particle population only modifies macroscopic observable traits without altering underlying causal logic and isomorphic primitive rules

Mathematical Formulation:

S_0 marks initial systemic state, S' represents post-decay/reconfiguration configuration. Conservation and Isomorphism impose:

$$\sum_i p_i(S_0) = \sum_j p_j(S') \quad \text{and} \quad \mathcal{R}_{\text{base}}(S_0) \cong \mathcal{R}_{\text{base}}(S')$$

$p_i = m_0 c$ corresponds to discrete Momentum unit, $\mathcal{R}_{\text{base}}$ restricts invariant fundamental evolutionary laws.

Key Remark: fundamental Momentum units and primitive evolutionary rules maintain strict Isomorphism; decay and restructuring merely rearrange macroscopic Encapsulation and population without compromising evolutionary conservation or intrinsic particle indistinguishability.

6. Theoretical Predictions

Prediction Category	Content	Verification Scheme
Encapsulation Layer Prediction	Particles are non-pointlike; clustered internal Momentum units ($p_0 = m_0 c$) generate matter waves[26]	High-energy scattering experiments
Light-speed Evolution Bound	Intrinsic Momentum unit evolution capped at upper limit c to guarantee causal locality and General Covariance	Relativistic effect measurements
Quantum Entanglement Internal Coupling Rule	Non-local synchronization among inner Momentum units yields isomorphic macroscopic mechanical responses	Entanglement characterization experiments
Particle Generation Mechanism	New particles duplicate from Potential Space upon ultra-short pulse momentum injection equal to parent p_0 , inheriting isomorphic governing rules	High-energy collision tests
Four Fundamental Interaction Isomorphism	Electromagnetic, weak, strong and gravitational interactions represent distinct projections of identical base rule across different Encapsulation tiers[26]	Isomorphism calibration experiments
Perceptual Mechanism Prediction	Minimal temporal window for non-local synchronized interaction determines macroscopic mechanical stability bounds	Ultrafast optical measurements

The Standard Model of quantum mechanics can be reformulated based on momentum evolution under the present evolutionary framework to validate listed predictions; physical forecasts remain valid even if the global Isomorphism postulate is discarded.

Conservation and Isomorphism collectively signify invariant fundamental elemental composition, fixed magnitude and mode of Driving Capacity, plus immutable primitive evolutionary laws through-

out state evolution. Indistinguishability at each hierarchical level derives from topological Encapsulation invariance or faithful replication, inevitably generating uniform causal behaviors and evolutionary configurations.

5.6 Global Synchronization Mechanism

Within three-dimensional complex relational state evolution systems, absent global synchronized Global Update protocols, Long-range Non-dissipative Force such as celestial non-local interactions induce causal paradoxes, irregular composite forces, tidal distortion and curved orbital paths analogous to seismic wave propagation. A system-wide minimal unified instantaneous interaction interval is introduced to resolve such inconsistencies. Independent of intermediate propagator-mediated transmission, localized quantum fluctuation borrows virtual momentum pairs $(p, -p)$ from Potential Space to toggle intrinsic momentum distribution inside particles via toggle effect, generating macroscopic attractive or repulsive interactive relations between paired objects. Long-distance causal linkage establishment relies on unified full-system minimal action cycles instead of third-party spatial mediator propagation[23].

1. Causal Contradiction Analysis

Consider mutual non-local interactions among three celestial bodies A, B, C . Conventional propagator-based delayed transmission without global synchronization introduces asynchronous interaction timestamps for resultant forces and prevents formation of unified instantaneous systemic Snapshot. Corresponding interaction formulas read:

$$\begin{aligned} F_A(t) &= f(S_B(t - \tau_{AB}), S_C(t - \tau_{AC})) \\ F_B(t) &= f(S_A(t - \tau_{BA}), S_C(t - \tau_{BC})) \\ F_C(t) &= f(S_A(t - \tau_{CA}), S_B(t - \tau_{CB})) \end{aligned}$$

Disparate delay constants τ_{XY} for pairwise couplings produce reverse causal ordering where future configuration t_3 determines present state t_2 . Delayed signal transmission breaks simultaneous matching of action-reaction pairs, yielding non-rigid composite force and destabilizing whole celestial systems[29].

Causal inversion cannot physically exist in standalone form, defined as present entities being altered by nonexistent future configurations. Such anomalous effect Emerges exclusively from mismatched locked evolutionary correlations within constrained systemic evolution.

2. Derivation of Composite Force Degradation and Mathematical Formalism

Without unified global synchronization, total aggregate force acting across A, B, C is formally defined as:

$$\vec{F}_{\text{total}}(t) = \sum_{i=A,B,C} \vec{F}_i(t)$$

Net interactive load on individual celestial body depends on time-delayed ambient configuration of remaining objects following general relation:

$$\vec{F}_i(t) = f(S_j(t - \tau_{ij}), S_k(t - \tau_{ik})), i \neq j \neq k$$

Component-wise interaction specifications:

$$F_A(t) = \sum_{j=B,C} F_{jA}(t - \tau_A)$$

$$F_B(t) = \sum_{j=A,C} F_{jB}(t - \tau_B)$$

$$F_C(t) = \sum_{j=A,B} F_{jC}(t - \tau_C)$$

Unequal pairwise delay parameters $\tau_{ij} \neq \tau_{ik}$ invalidate constant stable bulk force:

$$\vec{F}_{\text{total}}(t) \neq \vec{F}_{\text{stable}}$$

For long-range Non-dissipative Force including universal gravitation, propagator-dependent local coupling not only destabilizes total composite force but spawns spurious phenomena resembling seismic wave dispersion, tidal deformation and curved celestial trajectories. System evolution alternates between locked waiting phases and chaotic configurations, sustaining only repulsive interaction while prohibiting steady accumulation of attractive binding effects[31].

3. Global Synchronization Mechanism

A unified minimal collective interaction timescale $t_{\text{min,collective}}$ spanning the full system eliminates all transmission lag by enforcing simultaneous Global Update of all Evolution carriers within one fundamental cycle and constructing consistent global causal Snapshot satisfying:

$$S_i^{\text{global}}(t) = S_i(t), \forall i \in \text{whole system}$$

Four core attributes define global synchronization: propagator-independent operation, chronological order insensitivity, stable total composite force maintenance and consistent non-local Non-dissipative Force formation. Stable global aggregate force after synchronization reads:

$$\vec{F}_{\text{total}}^{\text{stable}}(t) = \sum_i f(S_j^{\text{global}}(t), S_k^{\text{global}}(t))$$

Complete removal of all propagation delay via universal synchronous Update restores rigid invariant bulk magnitude for all Long-range Force[32].

Synthetic Logical Summary

- Valid non-local interaction establishment depends on intrinsic causal accessibility rather than conventional field-mediated propagation medium
- Systems lacking global synchronization develop causal inversion, non-rigid resultant force, tidal deformation and curved orbital anomalies leading to structural instability
- Introduction of unified system-wide minimal collective interaction interval enables full-dimensional synchronous Global Update and stabilizes long-range Non-dissipative composite interaction

5.7 Perception and Perception Attenuation Mechanism

Causal interactions between Evolution carriers within relational state evolution systems require inherent perceptual interaction capability independent from spatial locomotion; meanwhile steady evolutionary systems must incorporate a perception attenuation mechanism.

1. Consequences in the Absence of Perception Attenuation

If no perception attenuation exists inside a state evolution system, every Evolution carrier (causal node) establishes fully equivalent causal coupling with all remaining nodes across state space, with interaction strength independent of separation distance. Global delocalization of cooperative evolutionary relations follows, eventually driving all carriers to collapse toward the dominant causal center.

Logical Deduction Chain

No perception attenuation $\Rightarrow$ interaction strength distance-independent $\Rightarrow$ globally extended relations $\Rightarrow$ collapse

$$\Sigma(r) = \text{constant}$$

$$\Delta P_{ij} \propto \Sigma(r) \cdot \mathcal{R}(S_i, S_j) \quad (\text{independent of } r)$$

where $\Sigma(r)$ denotes perception intensity function, and $\mathcal{R}(S_i, S_j)$ stands for coupling rule between paired Evolution carriers.

Collapse Mechanism

All Evolution carriers achieve peak interaction magnitude with the dominant coupling center $C_{\max} = \arg \max(\text{coupling constant})$, inducing bulk systemic collapse toward this central site and prohibiting formation of multi-center complicated structures.

2. System Behavior with Valid Perception Attenuation

A functioning perception attenuation mechanism prioritizes short-range adjacent interactions and localizes relational bonding, enabling stable multi-center structural configurations.

Perception Attenuation inside Three-dimensional Space

$$\Sigma(r) = \frac{1}{4\pi r^2}$$

$$\Delta P_{ij} \propto \frac{1}{4\pi r_{ij}^2} \cdot \mathcal{R}(S_i, S_j)$$

Physical Correspondence

The term $1/(4\pi r^2)$ naturally describes flux density distribution of the spherical symmetric perceptual field in three dimensions: source perception uniformly distributes over the spherical surface of radius r , with unit-area flux inversely proportional to total spherical area $4\pi r^2$. Terrestrial Newtonian gravitational acceleration cannot spontaneously emerge without this attenuation law[33].

Comparison of Collapse Conditions

Condition	Interaction Weight	Outcome
No attenuation	$w_{iC} \propto \text{coupling constant}_C$	All component i collapse toward the center with maximum coupling constant
With attenuation	$w_{iC} \propto \frac{\text{coupling constant}_C}{r_{iC}^2}$	Nearby carriers governed by adjacent bodies; remote objects weakly influenced by central core

Conclusion

The $1/(4\pi r^2)$ perception attenuation constitutes the geometric prerequisite for complex structural evolution within relational state evolution systems. Elimination of such attenuation equates interaction strength across infinite and infinitesimal separations, triggering full-system collapse onto a single dominant causal center and precluding stable localized multi-center architectures. Global collapse occurs without attenuation; localization and coexisting multiple stable centers are permitted once attenuation is introduced[34].

This spherical omnidirectional flux attenuation scales with r^2 growth, consistent with Internal Coupling constraints of quantum entanglement: any valid perceptual link between points A and B completes full momentum interchange, whereby perception behaves as a binary quantity (present or absent) with no intermediate half-activated perceptual states.

3. Perception Attenuation and Minimal Characteristic Scale

Implementation of perception attenuation defines a fundamental minimal spatial cutoff Λ , named the **minimum interactive scale**. Physically, any carrier compressed below this threshold redirects all intrinsic self-perception inward and loses capacity for external causal coupling; hence Λ marks the smallest stable elementary material constituent of the system[35].

Formal Definition

Perception saturation threshold $\rightarrow$ minimum material scale Λ

$$\int_0^\Lambda \Sigma(r) \cdot 4\pi r^2 dr = \int_0^\Lambda dr = \Lambda$$

Substitute three-dimensional attenuation relation $\Sigma(r) = 1/(4\pi r^2)$ for spherical-coordinate integration:

$$\int_0^\Lambda \frac{1}{4\pi r^2} \cdot 4\pi r^2 dr = \int_0^\Lambda dr = \Lambda$$

Physical Picture

- For integral upper bound $r < \Lambda$: accumulated perceptual flux is subthreshold and the perceptual field remains unsaturated
- For $r = \Lambda$: integrated perception equals Λ at saturation critical point
- For $r > \Lambda$: standard $1/r^2$ perception attenuation remains fully operative

Core Inference

- Λ marks the transition threshold where perception switches from outward distinguishable coupling to fully inward self-containment
- Domain $r < \Lambda$ (carrier interior) lacks well-defined "external space", with all causal interactions confined to self-interaction
- Perceptual saturation occurs exactly at the bounding surface $r = \Lambda$
- Normal inverse-square attenuation governs all exterior region $r > \Lambda$

Accordingly, Λ represents the minimal physically permissible material scale of the system, whose numerical value is fixed by intrinsic free-dimensional geometry and fundamental evolutionary rules. Matching against established physical frameworks via quantum and gravitational constraints inside three-dimensional orthogonal inertial cosmic systems yields $\Lambda = \sqrt{G\hbar/c^3} \approx l_0$ (refer to Section 9.3 for detailed derivation).

4. Emergence of Three-Dimensional Space

The state evolution of a particle is always one-dimensional, meaning that composite motion consists of multiple linear inertial motions. Three-dimensional space emerges from:

- (a) The static characterization of the Momentum unit m_0 as $l_0(x, y, z)$;
- (b) The internal structural distribution of particles (e.g., the electron's spherically symmetric three-dimensional distribution);
- (c) The selection of evolutionary directions along free dimensions.

Macroscopic three-dimensional material structures and the selection of free dimensions primarily emerge from three-dimensional spatial perceptual attenuation. Specifically, two particles A and B establish a causal Perceptual Cross-section through their three-dimensional relation, exchanging a Momentum Deviation $\Delta \vec{p}$ along that direction. Ultimately, through internal Vector Composition within each particle, a single inertial direction is selected among the free dimensions.

5. Section Summary

Mechanism	Mathematical Expression	Systemic Behavior
No perception attenuation	$\Sigma(r) = \text{constant}$	Global gravitational collapse; complex structural formation prohibited
With perception attenuation	$\Sigma(r) = 1/(4\pi r^2)$	Spatial localization; stable multi-center structures develop
Perception saturation cutoff	$\int_0^\Lambda \Sigma \cdot 4\pi r^2 dr = \Lambda$	Minimal fundamental scale Λ is well-defined

Central Conclusion

The $1/(4\pi r^2)$ perception attenuation provides essential geometric foundation for complex structural evolution within three-dimensional space. Extension of this mechanism naturally sets the saturation threshold Λ , the minimum permissible elementary material dimension of the evolutionary system.

5.8 State-Interaction Synchronization

Sustained stable evolution requires matching frequency between intrinsic state update and causal interaction. When state evolution proceeds faster than perceptual refresh rate, stable environmental coupling cannot be established; redundant unnecessary causal interactions emerge if perception cycles outpace intrinsic state renewal.

Synchronization condition equates state evolution frequency f_s and causal interaction frequency f_c :

Insufficient coupling ($f_c < f_s$): incomplete causal state locking produces permanent causal information loss.

Excessive coupling ($f_c > f_s$): superfluous interactions inflate total systemic causal complexity.

Total causal complexity C_{causal} directly quantifies imbalance between state progression and interactive refresh:

$$\begin{cases} f_c < f_s & \Rightarrow \text{causal information loss} \\ f_c > f_s & \Rightarrow C_{\text{causal}} \uparrow \end{cases}$$

1. Spatial Causal Expansion Rate

Per single evolutionary cycle, spatial causal expansion of Evolution carriers maintains synchronization with interactive frequency:

$$\Delta C_x = k \cdot \mathcal{C}_{\min}, \quad 0 < k \leq 1$$

Definitions:

$\mathcal{C}_{\min}$: minimal elementary causal unit

k : dimensionless spatial expansion coefficient

Interpretation: spatial causal expansion matches maximal interactive capacity to guarantee complete spatial state characterization per individual coupling event.

2. Temporal Causal Persistence Rate

Per-cycle temporal causal continuation synchronizes with state refresh:

$$\Delta C_t = \mathcal{C}_{\min}$$

Interpretation: one full causal update completes within each Planck-scale time interval to eliminate missing or redundant causal accumulation.

3. Causal Interaction Rate

Maximum permissible interactive count per unit evolutionary cycle:

$$N_{\max} = \text{total feasible causal rearrangement iterations within single } t_0$$

Synchronizes spatial extension, temporal persistence and overall interactive frequency.

4. Unified Synchronization Constraint

Consistent causal scaling across spatial extension, temporal evolution and interactive count:

$$\Delta C_x : \Delta C_t : N_{\max} \equiv k \cdot \mathcal{C}_{\min} : \mathcal{C}_{\min} : \mathcal{C}_{\min}$$

Physical meaning: state renewal, spatial spreading and temporal persistence evolve at identical pace for complete causal locking and stable systemic progression.

5. Causal Complexity Boundary Conditions

Excess interactions raise overall causal complexity:

$$C_{\text{causal}}(t) = C_0 + \sum_{i=1}^{N_{\text{redundant}}} \delta C_i$$

Deficient coupling induces irreversible causal loss:

$$C_{\text{causal}}(t) = C_0 - \sum_{j=1}^{N_{\text{lost}}} \delta C_j$$

Perfect synchronization criterion:

$$f_s = f_c \Rightarrow C_{\text{causal}} = C_0$$

5.9 Chapter Summary

Built upon generalized relational state evolution formalism and focused primarily on cosmic-type systems, this chapter systematically enumerates indispensable prerequisites enabling steady cosmic-analog evolution:

- **Standard Quantization and Discreteness:** State representation, intrinsic driving capacity, causal quantification and mechanical characterization all follow standardized discrete quantization constraints as foundational stability prerequisites.
- **Necessity of Internal Particle Architecture:** Derivation from mechanical and causal coupling rules rigorously proves mandatory internal substructure within particles and theoretically justifies generalization of the original equivalence principle into the generalized equivalence principle.
- **Global Perceptual Coupling Frequency:** Cosmic systems adopt uniform synchronized global perceptual frequency locked to intrinsic Driving Capacity oscillation frequency.
- **Implementation of Inertial States:** Inertial configuration cannot be modeled via continuous field theory and instead relies upon discretized quantized microstructure and localized causal interactions.
- **Conservation and Isomorphism of Evolutionary Rules:** Long-term system stability demands invariant and isomorphic evolutionary laws; identical particles satisfy indistinguishability constraints across evolutionary mechanics, state parameters, interaction laws and Encapsulation regulations.
- **Distance-dependent Perception Attenuation:** Radially decaying perceptual coupling prevents bulk global collapse onto a single causal core and sustains coexistence of multi-layer hierarchical structures.

Collectively these constraints constitute the complete foundational framework for stable cosmic-analog evolution and establish rigorous logical groundwork for subsequent derivation of cosmic dynamical relations and dimensional scaling.

5.10 Commentary on Chapter References

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Supports finite minimum elemental material dimension prohibiting unconfined infinite gravitational collapse inside black hole interiors.

6 Comparison of Four Evolution Modes for Three-Dimensional State Evolution Systems

Based on the derivations in Chapter 5 concerning the necessity of constructing complex three-dimensional relational state evolution systems, four relational models are summarized and contrasted below:

- A. One-to-one plus central-CPU architecture;
- B. Many-to-many configuration;
- C. Point-particle with hidden continuous-field centralized information processing;
- D. The evolutionary framework proposed herein (structured internal particle architecture establishing generalized mechanical relations with ambient surroundings).

6.1 Continuous Field and Background Processing Scheme (Equivalent to One-to-One plus Central-CPU Mode)

Both the discrete one-to-one CPU-type interaction model and continuous-field centralized information processing framework share identical fundamental limitations: neither can explicitly express full causal, mechanical and state-coupling relations of complicated systems via direct point-to-point linkage between point particles, mandating centralized information processing to acquire ambient environmental status, perform computation and export interactive outcomes within a single evolution cycle. Therefore, these two types of models are equivalent.

6.2 Discrete Many-to-Many Framework (Equivalent to Structured Particles plus Generalized Equivalence Principle)

Discarding the point-particle postulate, the present evolutionary theory introduces standard quantized minimal fundamental units (l_0, t_0, p_0, m_0) and treats every particle as a clustered structured aggregate composed of abundant internal Momentum units $p_0 = m_0 c$ at or near the fundamental scale,

endowed with intrinsic conserved evolutionary velocity c [37]. Particles directly build environmental relations via inherent Perceptual Cross-section $\Sigma(r) = \frac{1}{4\pi r^2}$, eliminating demand for intermediate central processing units or background fields. This will be verified in the subsequent chapters.

During momentum exchanges such as scattering and repulsive interaction, particles carry intrinsic descriptors including inherent effective interaction cross-section, oriented momentum configuration (internal Momentum unit distribution), positional trajectory memory and spin (cyclic manifold flow of inner Momentum units). Interactions are directly finalized following the Generalized Equivalence Principle via vector superposition and cancellation of constituent Momentum units:

$$\Delta \vec{p}_i = \sum_{j \in \Sigma} \mathcal{R}(\vec{r}_i, \vec{p}_i, \vec{r}_j, \vec{p}_j)$$

Global synchronized Global Update and decoupling detachment proceed per discrete t_0 timestep. No continuous differential sampling, static background metric or explicit precomputation of cross-section/spin parameters is required; all relational characteristics are intrinsically embedded within individual particles.

From multi-scale perspectives, the framework manifests distinct evolutionary classifications:

1. Macroscopic System View: Many-to-Many Evolution Mode

Globally, each particle hosts numerous internal Evolution carriers with complete mechanical and causal attributes and establishes extensive many-to-many relational networks with surrounding constituents. Consistent with structured-particle construction and the Generalized Equivalence Principle, internal compartments accommodate multiple independent speed components c without centralized computational mediation for global evolution[38].

2. Single-Particle View: One-to-Many Evolution Mode

When isolating an individual particle as a unified source of macroscopic effective force output, it develops one-to-many evolutionary couplings against multiple environmental bodies. A singular bulk macroscopic velocity v emerges at the particle level, whereby inner Momentum units and Perceptual Cross-section enable multi-directional environmental coupling. Structured internal anatomy effectively projects global many-to-many macroscopic connectivity into localized one-to-many interfacing originating from a single particle entity.

3. Pairwise Particle View: One-to-One Evolution Mode

Examining isolated binary particle encounters, each entire particle functions as an integrated macroscopic causal agent generating strictly one-to-one interactive evolution. Internal Evolution carriers within paired bodies compose combined bulk velocity relations, embedding local pairwise coupling into the broader global many-to-many network.

6.3 Emergence Relation between Discreteness and Continuity: Apparent Continuous Fields Emerging from Discrete Fundamental Constituents

The four preceding paradigms reduce to two core categories for comparative analysis: structured-particle plus equivalence principle versus point-particle plus continuous background field. Comparison demonstrates the point-particle continuous-field formalism is poorly suited for complex relational state evolution.

The following elaborates how discrete relational evolution systems produce macroscopically observable apparent continuous manifestations such as fields, forces, energy and space-time:

1. Discrete Constituent Carriers

All Evolution carriers inside relational evolution systems remain discretized with distinguishable states and independent well-defined events. Minimal fundamental causal and state quantization fixes discrete magnitudes for causality, configuration and kinematic motion; all physical entities, interactive events and Driving Capacity magnitudes follow discrete constraints.

2. Evolution Manifesting Apparent Continuity

Observed continuity does not imply infinite infinitesimal divisibility. All calculations in any mathematical framework inherently adopt standard quantized variables as baseline inputs, rather than being a unique assumption of this work. Unlimited subdivision corresponds to unbounded dimensional expansion, which would require matching discrete state encoding to generate corresponding Energomomentum and informational quantities. For instance, the arithmetic identity $1/3 \times 3 = 1$ skips exhaustive enumeration of all discrete states corresponding to $1/3$ via predefined mapping rules, proving that evolutionary continuity is not synonymous with infinite subdivision.

Three distinct origins generate continuous macroscopic behavior across cosmic evolution:

- (a) Perceptual mechanics and radial distance-dependent decay law: diminishing separation between particle A and B expands three-dimensional Perceptual Cross-section $\frac{1}{4\pi r^2}$, yielding larger inertial shifts Δv or ΔP within one t_0 interaction interval; progressive cross-section enlargement over successive evolution cycles spawns integral-style continuous descriptions of force and energy from discrete microscopic origins.
- (b) Continuous evolution of discrete states: Evolution carriers propagate over intrinsic fundamental length scale l_0 without arbitrary discrete spatial jumps, generating extensive spatial extent $L \cdot l_0$ and temporal duration $T \cdot t_0$. Dynamically synchronized proportional scaling of space and time constructs emergent discrete-continuous space-time geometry[36][38].
- (c) Condensation and Diffusion of Particle Architecture: Following intrinsic Encapsulation rules and dynamic interactions, particles reversibly toggle between compact minimal-volume condensed configuration and extended Diffuse State without abrupt state jumps, producing the macroscopic illusion of particle generation originating from continuous field excitation[39].

3. Continuous Observed Interaction $\neq$ Unbounded Unlimited Interaction Capacity

Phenomenologically continuous interaction does not correspond to unbounded coupling potential. Saturation occurs once all inner Momentum units align along a single uniform propagation direction preventing further causal superposition. Although Perceptual Cross-section and Momentum unit projection produce apparent continuous macroscopic response, total internal constituent count imposes an upper bound on aggregate particle momentum:

$$P_{\text{total}} = \sum_{i=1}^N p_0 = mc$$

where N denotes total internal Momentum unit population and $p_0 = m_0 c$.

At saturation limit with fully aligned constituent momenta, the particle attains maximum directional evolutionary capacity:

$$\Delta P_{\text{max}} = mc$$

Constraint Interpretation: Macroscopically continuous interactive magnitude carries inherent physical ceiling bounded by total internal particle momentum, securing systemic evolutionary stability and physical conservation laws. While macroscopic measurements display smoothly varying continuous response, underlying microscopic interaction sums remain confined by internal particle architecture yielding globally "continuous yet finite" apparent mechanical behavior.

System-Level Implication: Finite upper bounds on interactive intensity prevent unregulated unbounded force growth, safeguarding evolutionary stability, global state conservation and core constraints governing bounded causality, state representation and mechanical relations[40].

Macroscopic Correspondence: Continuous-looking observational variations are fundamentally restricted by microscopic internal particle composition to yield finite continuous bulk mechanical phenomena.

6.4 Chapter Summary

Systematic comparison across four relational models yields the core conclusions below:

- The one-to-one central-CPU and point-particle continuous-field frameworks share foundational flaws rooted in mandatory centralized data processing and absent intrinsic particle descriptors, prohibiting explicit full causal coupling characterization for complex systems.
- The proposed evolutionary theory abandons point-particle assumption in favor of structured constituents governed by the Generalized Equivalence Principle; particles natively embed cross-section, oriented momentum configuration, trajectory memory and spin, establishing direct environmental linkage via inherent Perceptual Cross-section independent of background field intermediaries.
- A unified underlying mechanism simultaneously produces many-to-many macroscopic coupling, one-to-many single-particle projection and pairwise one-to-one binary interaction across distinct observational scales.
- Apparent continuous macroscopic observables emerge via discrete-continuous genesis originating from radial decaying Perceptual Cross-section scaling, synchronously proportional discrete space-time formation and reversible particle transition between condensed and Diffuse State.
- Continuous-looking interactive strength obeys finite physical saturation bound $P_{\text{total}} = mc$ enforcing systemic stability and conservation principles.

Core Summary: The reason the point-particle plus central-field paradigm fails to meet the requirements of complex relational state evolution lies in its lack of the fundamental representational capacities derived in Chapter 5—namely, state representation capacity, generalized equivalence representation capacity, causal continuity representation capacity, etc. The central field can only delegate specific "energy" and "new inertial directions" to particles.

However, the field cannot access the many-to-many complex relationships inherent to particles, such as the instantaneous spin state, the dynamic relationship between a particle and each direction, and the fact that all state information and interaction quantities must be explicitly simulated, predicted, and computed through the central field (central processing unit).

Consequently, the assumption of point particles plus fields is not physically valid. Instead, fields emerge from discrete-continuous evolution, continuous integration, and the state-transition switching

capabilities of internal particle structures, giving rise to the appearance of a background-independent field and point particles. The foundation of all mathematical computations by humans is not infinite continuity, but rather the input of standard quantized information variables. All non-discrete computations employ mapping relations that bypass the complete representation of intermediate steps.

6.5 Reference Commentary for Chapter 6

[36] arXiv. (2009). Emergence of spatial structure from causal sets. *arXiv preprint*, arXiv:0905.0017. <https://doi.org/10.48550/arXiv.0905.0017>

Discrete causal ordering fully reconstructs emergent continuous space-time geometry without preexisting background manifold.

[37] PMC. (2025). Conceiving Particles as Undulating Granular Systems. *PMC8534518*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8534518/>

Supports the core postulate of composite particles assembled from clustered subconstituent granular Momentum units.

[38] Laughlin, R. B. (2005). *A different universe: Reinventing physics from the bottom down*. Basic Books.

Nobel laureate Laughlin establishes macroscopic physical laws including space-time geometry as collective emergent properties of underlying many-body microscopic systems, consistent with the present framework's discrete-micro to continuous-macro duality and space-time originating statistically from encapsulated Momentum units.

[39] Faddeev, L., & Niemi, A. (1999). Stable knot-like structures in classical field theories. *Nature*, 387(6628), 58–61. <https://doi.org/10.1038/387058a0>

Topological knot-soliton excitations within classical continuous fields behave as massive particle-like localized objects, substantiating particles as confined topological locking of continuous relational fields rather than independent primitive fundamental entities.

[40] Bekenstein, J. D. (1973). Black Holes and Entropy. *Physical Review D*, 7, 2333–2346. <https://doi.org/10.1103/PhysRevD.7.2333>

Validates information-based space-time entropy and state-quantity conservation, forbidding complete information erasure and unphysical unlimited black-hole collapse into unrepresentable singular states.

7 Fundamental Constituents of the Cosmic System

Synthesizing preceding derivations, the cosmos constitutes a **three-dimensional orthogonal free-dimensional inertial state evolution system** built upon three irreducible core components: elementary Representational Quanton m_0 , intrinsic Driving Capacity v_0 (light-speed evolutionary rate c), and global Rule set $\mathcal{R}$. System description below aligns with conventional physical formalism and dimensional scaling characteristics.

7.1 Cosmic System Description

The cosmos features three mutually orthogonal spatial degrees of freedom (x, y, z) ; independent temporal dimension t emerges synchronously with space following fixed proportionality:

$$c = \frac{l_0}{t_0}$$

Orthogonality underpins macroscopic spatial isotropy and causal locality, forming the geometric prerequisite for spherically symmetric causal interaction from radially decaying Perceptual Cross-section:

$$\Sigma(r) = \frac{1}{4\pi r^2}$$

The framework reduces to conventional 3+1 dimensional space-time at macroscopic limit consistent with established observational physics. Spatial geometry emerges from static Representational Quanton occupation plus dynamic evolutionary motion, while time arises purely from state evolution; synchronous proportional scaling eliminates requirement for predefined fixed background space-time manifold.

Core systemic attributes are summarized as:

- **Three-Dimensional Orthogonality:** Mutually orthogonal spatial axes (x, y, z) with synchronously emergent temporal coordinate t ;
- **Causal Locality:** Distance-dependent Perceptual Cross-section $\Sigma(r) = \frac{1}{4\pi r^2}$ enforces radially attenuated causal interaction[41];
- **Relational Emergence:** Causal coupling generates interparticle forces, energetic cost via integrated momentum variation, topological informational linkage and geometric spatial projection[42];
- **Discrete State Quantization:** Exists irreducible minimal Representational Quanton m_0 and elementary space-time quanta (l_0, t_0) ;
- **Apparent Continuous Evolution:** Statistical averaging over discrete microscopic constituents yields smooth continuous macroscopic kinematics;
- **Global System Coherence:** Full-system Global Update per fundamental t_0 timestep combined with global momentum conservation:

$$\sum \vec{p}_i = \text{constant}$$

securing internal theoretical self-consistency.

7.2 Representational Quanton m_0

Representational Quanton constitutes the singular fundamental Evolution carrier and material foundation of the cosmic system possessing defining properties:

- **Exclusive Free-Dimensional Occupancy:** Each elementary Representational Quanton occupies non-overlapping exclusive spatial volume within its available degrees of freedom.
- **Causal Coupling Capacity:** Enables cross-body causal interaction mediated through Perceptual Cross-section.

- **Intrinsic Evolutionary Motion:** Embedded inherent Driving Capacity c sustains discrete-continuous state transition throughout evolution.

All macroscopic bulk mass assemblies statistically from identical minimal irreducible constituent m_0 with elemental spatial footprint $l_0(x, y, z)$ and temporal quantum t_0 following:

$$m = Nm_0$$

m_0 denotes the indivisible smallest Evolution carrier of the cosmos lacking inherent macroscopic mass, energy or force attributes; such bulk observables exclusively Emergence from collective statistical aggregation, Encapsulation, interference, bulk vector composition and causal interaction among vast ensembles of fundamental quanta. Multiple m_0 constituents undergo topological Encapsulation governed by quantum entanglement and vector composition rules to form composite particles; interference and combined bulk motion generate emergent collective inertial velocity v . All causal linkage, measurable physical states and material corporeality originate from these primitive quanta manifesting macroscopically as condensed matter, spatial geometry and matter-wave kinematics[43][44].

7.3 Driving Capacity c

c acts as the universal sole conserved intrinsic Driving Capacity embedded within every elementary Representational Quanton m_0 characterized by:

- **No Direct Force or Energy Content:** c is neither force nor energy, but the **intrinsic frequency** of state evolution. It is essentially the **natural growth rate in space or time** of an Evolution carrier along its inertial evolutionary direction, constituting a **dimensionless emergent frequency**. That is, over N causal interactions (evolution cycles), the Evolution carrier performs state transitions at a fixed frequency, whose macroscopic statistical manifestation is the observed speed of light c .

Spatial Natural Growth Rate: the spatial extension l_0 per unit evolution cycle (causal interaction);

Temporal Natural Growth Rate: the increment of one elementary event count t_0 per unit evolution cycle (causal interaction).

- **Emergent Force and Energy:** Macroscopic force and energy originate from net momentum variation $\Delta P = Nm_0c$ induced via interparticle causal exchange rather than direct inherent contribution from c itself.
- **Vector Composition Kinematics:** Individual Evolution carriers follow vector superposition rules for coordinated collective evolutionary motion.

Emergent Potential and Inertial Behavior: Local Dissipative Force generates Resistance Potential, Thrust Potential and Inertial Potential via bulk vector composition effects; long-range Non-dissipative Force leverages quantum fluctuation mediated virtual momentum pair borrowing from Potential Space $(\Delta P, -\Delta P)$ modifying internal Momentum unit distribution to spawn Attraction Potential and Repulsion Potential.

Fundamentally the cosmos contains exclusively Momentum; force and energy emerge as secondary macroscopic phenomena stemming from coordinated forward/opposite directional evolution of inner constituent Momentum units, establishing the Generalized Equivalence Principle equating inertial motion and applied force.

7.4 Evolution Rule Set (Fundamental Constraining Laws)

Rule set $\mathcal{R}$ encapsulates all objective physical constraints governing coordinated Evolution carrier dynamics detailed in the tabulation below:

Sub-Rule	Detailed Content	Relevant Section
Encapsulation Rule	Governs aggregation of Momentum units into composite particles including electrons, quarks and photons	5.1
Perception Rule	Fixed Perceptual operating frequency c , inverse-square decaying Perceptual Cross-section $\Sigma(r) = \frac{1}{4\pi r^2}$	5.7, 5.8
Interaction Rule	Momentum lending-restoring mechanism via virtual momentum pairs $(p_0, -p_0)$ sourced from Potential Space	5.7
Replication Rule	New particle generation upon injected input momentum matching parent bulk momentum mc ; excess surplus momentum triggers particle-antiparticle pair production[46][47]	5.5
State Update Rule	Discrete evolution update relation $\vec{p}_i(n+1) = \vec{p}_i(n) + \sum \mathcal{R}_{\text{int}} \hat{r}$	5.3, 5.4
Internal Coupling Entanglement Rule	Global angular momentum conservation constraint $\sum \vec{L}_i = 0$ [48]	5.2, 5.3
External Coupling Perception Rule	Perceptual window relativistic contraction $v_{\text{Perception}} = \frac{c}{\gamma}$	5.2, 5.7
Global Update Rule	Full-system parallel synchronous snapshot refresh once per elementary t_0	5.6, 5.8
Detachment Rule	Instant decoupling post-interaction eliminating persistent historical relational binding	5.4

$\mathcal{R}$ exhaustively defines all cosmic evolutionary constraints covering constituent aggregation, interaction and dynamic state renewal.

7.5 Chapter Summary

Cosmic System = Representational Quanton m_0 + Driving Capacity c + Evolution Rule Set $\mathcal{R}$

- Foundational prerequisites rest upon three mutually orthogonal spatial axes (x, y, z) with core building blocks $m_0, c, \mathcal{R}$ anchored under universal momentum conservation;
- All macroscopic observables including bulk matter, force, energy, space-time, relativistic kinematics and quantum effects Emergence from statistical aggregation, geometric projection and hierarchical Encapsulation of the three core constituents within orthogonal free-dimensional geometry: Representational Quanton supplies material substrate, Driving Capacity provides evolutionary impetus and Rule set defines organizing physical constraints[42].

7.6 Reference Commentary for Chapter 7

[41] Jackson, J. D. (1999). *Classical Electrodynamics* (3rd ed.). Wiley.

Validates inverse-square $1/r^2$ flux attenuation within three-dimensional Euclidean space from Gauss' s theorem supporting radially decaying momentum flux core hypothesis.

[42] Crutchfield, J. P. (1994). *The Calculi of Emergence: Computation, Dynamics, and Induction*. Physica D: Nonlinear Phenomena.

Inter-agent boundary detection via flux gradient variations consistent with causal coupling strength fixed by decaying momentum flux gradient between Evolution carriers.

[43] Lamb, W. E., Jr., & Retherford, R. C. (1947). Fine structure of the hydrogen atom by a microwave method. *Physical Review*, 72(3), 241–243. (DOI: 10.1103/PhysRev.72.241).

Lamb-Retherford experiment confirms electron-vacuum fluctuation coupling, providing microscopic evidence for momentum shift during electronic level transition, conserved total topological state count and photon generation via encoded Momentum unit transformation.

[44] Schrödinger, E. (1930). Über die kräftefreie Bewegung in der relativistischen Quantenmechanik. *Sitzungsberichte der Preußischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse*, 24, 418–428. PDF

Predicts intrinsic Zitterbewegung frequency $2mc^2/h$ at propagation speed c for free electrons consistent with conserved internal constituent evolutionary velocity and fixed total Driving Capacity.

[45] Newton, I. (1687). *Philosophiæ Naturalis Principia Mathematica*.

Newton' s first law formalizes inertial existence reinterpreted within this framework: inertial momentum accumulates as integrated net historical causal interaction rather than primordial default configuration, all instantaneous inertial status equals cumulative prior coupling integral.

[46] Podgorsak, E. B. (2014). *Compendium to Radiation Physics for Medical Physicists*. Berlin: Springer. Free-space photon pair production requires adjacent massive absorber to dissipate surplus recoil momentum forbidding isolated unassisted pair generation.

[47] Bethe, H., & Heitler, W. (1934). On the stopping of fast particles and on the creation of positive electrons. *Proceedings of the Royal Society A*, 146(856), 83–112.

Foundational theoretical derivation for nuclear-field mediated photon-induced pair production requiring nucleus to absorb excess momentum.

[48] Bohm, D. (1951). *Quantum Theory*. New York: Prentice-Hall.

Reformulates EPR paradox within spin-singlet formalism establishing angular momentum conservation as core invariant constraint governing entangled Internal Coupling.

8 On Emergence Relations of Fundamental Physical Dimensions within the Cosmic System

From derivations in preceding chapters, the cosmos constitutes an inertial relational state evolution system. Only three foundational constituents, namely the Representational Quanton m_0 , Driving Capacity c , and evolution Rule set $\mathcal{R}$, suffice to produce material substrates, emergent spacetime, causal linkage and state variation. Accordingly, all statistically definable dimensional relations emerge endoge-

nously from these foundations; time, space, energy, force, field and particles possess no standalone ontological existence and differ merely by observational perspective. Relying on the three core components, mechanical interaction reduces exclusively to vector composition of bulk motion and counter-directional evolutionary motion, upon which a complete systematic relational catalogue is constructed below.

8.1 Core Fundamental Dimensions of the Systematic Ledger: Bottom-Up Definition, Quantum Mechanism and Macroscopic Extension

Remark: This dimensional table illustrates logical emergence and quantitative formulation of evolutionary relations. Symbols v and c do not denote conventional kinematic velocity but directional emergent growth rates (elementary transitional frequency of material evolution) defined under specific observational standpoints; decomposition into conventional metre, second or second-squared base units is inappropriate. Traditional physical units are statistical constructs tied to distinct measurement viewpoints rather than primitive fundamental dimensions[49]. For instance, the "second squared" term within energy expressions merely denotes statistical estimation of terminal momentum under bulk macroscopic interaction and bears no independent physical counterpart in reality. This framework targets underlying evolutionary logic, and no conventional SI units are marked throughout the table.

Table 8.1: Global Summary of Fundamental System Components and Dimensional Emergence Relations

Dimension / Component	Symbol	Microscopic Quantum Mechanism	Physical Implication of State Evolution	Mathematical Form
Representational Quanton (fundamental material unit) M_0	M_0	Generated via replication from Potential Space; enhanced perceptual capability under Cancellation State yields emergent rest mass[50]	Irreducible fundamental unit for matter, state representation, causal parameters and mechanical properties, also termed Evolution carrier	$l_0(x, y, z)$
Driving Capacity	c	One full l_0 state transition per elementary t_0 at microscopic scale; bulk macroscopic evolutionary range $0-c$ arises from Vector Composition	Conserved intrinsic Driving Capacity governing state evolution of the system; the Generalized Equivalence Principle establishes equivalence between velocity and force. Each Representational Quanton evolves at its intrinsic limiting transition frequency without full cancellation	c

Evolution Rule Set	$\mathcal{R}$	Emerged from primitive elemental rules or multi-layer Encapsulation, corresponding to classical physical fields; topological constraints regulate particle formation via Encapsulation[27]	Collective objective physical constraints governing state evolution, containing Encapsulation laws, perceptual rules, momentum control and interaction prescriptions	$\mathcal{P} = (\{p_{0k}\}, \mathcal{R}_{\text{encaps}}, \vec{x}, \vec{p})$
Elementary Spatial Unit	l_0	Originates from minimal resolvable evolutionary cutoff of the state system, alternative non-standard multi-scale minimal quanta and canonical fundamental discrete unit	Smallest distinguishable spatial increment of the system, emerging from static spatial occupation of primitive Representational Quanton across free dimensions	$l_0(x, y, z)$
Elementary Temporal Unit	t_0	Inversely derived from intrinsic constituent scale and Driving Capacity at microscopic level, matching intrinsic oscillatory phase inside composite particles[51]	Shortest complete event interval accommodating full state evolution, perception and interactive procedure	$\frac{l_0}{c}$
Interactive Capacity	c	Full environmental perception and coupling completed per discrete t_0 , persistently anchored to inherent rate c	Perceptual interaction frequency synchronised identically with fundamental state-evolution Driving Capacity	c
Particle	$[r = \lambda]$	Formed by multi-tier Encapsulation of Representational Quanton; dynamic equilibrium radius constrained by evolutionary laws equals observable de Broglie wavelength	Independently evolvable encapsulated aggregate; internally numerous Evolution carriers jointly realise holistic one-to-one, one-to-many and many-to-many mechanical and causal coupling with surroundings	$r = \lambda, \mathcal{P} = (\{p_{0k}\}, \mathcal{R})$
External Coupling Perception	$\mathcal{L}_{\text{out}}$	Underlies long-range Non-dissipative Force such as gravitation and electromagnetism via distance-dependent perceptual attenuation	Distance-dependent decaying causal linkage between particles; single-sided scaling as $1/r$, bidirectional spherical decay follows $1/4\pi r^2$	$\Sigma(r) = \frac{1}{4\pi r^2}, \Delta t = t_0$

Internal Coupling Entanglement	$\mathcal{L}_{\text{in}}$	Hierarchical entangled constraints govern coordinated evolution spanning Momentum unit, elementary particle and atomic scales; internal angular restrictions remain distance-insensitive	Intraparticle angular momentum conservation enforcing internal topological invariant amid external mechanical coupling	$\left[\sum_{i=1}^n \vec{L}_i = 0 \right],$ independent of r
Momentum Unit	P_0	Conserved intrinsic evolutionary capacity of a single elementary Representational Quanton	Unified intrinsic property of elementary material quanta denoting minimal causal and interactive potential, the sole globally conserved primitive entity of the system	$m_0 c$
Bulk Mass	m	Quantum entanglement and Cancellation State lock constituents into encapsulated particle configurations; total mass quantifies enclosed count of internal Momentum units[50]	Macroscopic bulk mass m , total internal Momentum unit population N and aggregate Representational Quanton count m_0 share identical physical essence via additive statistical summation	$m = N m_0$
Time $\mathcal{T}$	$\mathcal{T}$	Emerges from cyclic layered evolutionary periods spanning Planck scale, elementary particles, molecules and macroscopic bodies; time is a derivative parameter tracking relative material variation without standalone substance[52]	Derived from countable periodic evolutionary cycles (calendar-scale periods such as day/month/year) or dynamic emergent rate mv under inertial propagation; static configuration reduces to pure mass m	Macroscopic period or $N t_0$; emergence rate mv , stationary limit m
Space $\mathcal{V}$	$\mathcal{V}$	Macroscopic geometric extent generated from mutual evolutionary relations among massive aggregates; directional spatial growth arising under inertial motion	Structured spatial volume L, S, V defined by static material configuration or dynamic evolutionary generation rate mv ; stationary limit reduces to bare mass m	Macroscopic dimension or $N l_0(x, y, z)$; emergence rate mv , stationary limit m

Spacetime State	S_{state}	The matter-wave state of Momentum units continuously undergoes vector superposition and motion based on microscopic evolution.	Emergent count rate of comprehensive spacetime configurations from dynamic material evolution	Stationary: $Nl_0t_0 = \frac{ml_0t_0}{m_0}$; inertial motion: mvv or mcc
Velocity V	V	Bulk evolutionary rate generated via Vector Composition (Orientation Ratio s) across multiple constituent Representational Quanton	Directional propagation rate along free spatial axes aggregated from coordinated constituent motion	Nsc
Momentum p	p	Vector superposition and phase coherence of discrete Momentum units producing net oriented bulk motion	Bulk kinematic magnitude equivalent to temporal and spatial emergence rate directly quantifying net evolutionary capacity	Conserved rest limit mc , macroscopic kinematic form $mv, p = m\lambda v$
Macroscopic Force F	F	Unified coupling of four fundamental interactions; Dissipative Force splits combined bulk velocity into partial components, while Non-dissipative Force modifies internal Momentum unit distribution via quantum fluctuation. Force characterises transient momentum shift rather than independent entity[53]	Net momentum variation from single discrete causal exchange or averaged per-unit-interval momentum increment	$F = \frac{\Delta P}{\Delta t_0}$ or $F = \frac{\Delta P}{\Delta t_0}$, terminal kinematic value mv
Energy E	E	Integrated cumulative momentum change over stable acceleration intervals constrained by three-dimensional spherical perceptual decay; equivalent integral via Perceptual Cross-section, separation distance, momentum and inertial progression. Energy denotes statistical integral over evolutionary history[53]	Integrated aggregate of incremental interactive momentum or net force across finite duration; total terminal energy matches final bulk momentum	$\frac{1}{2}mv^2 = \sum \Delta P_n v_n$; terminal value mv , conserved intrinsic bound mc

Intrinsic Information I	I	Total Evolution carrier population eligible for causal coupling and state rearrangement	Count of discrete constituent bodies participating in systemic evolution	m
Composite Information I_s	I_s	Aggregate information content emerging from diversified relational combinations across alternative observational frames	Cumulative configuration count from varied inter-carrier interaction topologies and measurement perspectives	$I_s = (\mathcal{S}, \mathcal{C}, E = \mathcal{C}(\mathcal{S}))$
Entropy S	S	Statistical randomness of internal Momentum unit directional distribution; observational ignorance of inner microscopic arrangement bounded by $S \leq k_B \ln N$	Degree of configurational disorder induced upon constituent rearrangement during single interaction, positively correlated with available microscopic state count	$S = k \log W$ (W = number of microscopic configurations)
Composite Bulk Evolution	Composite velocity $\vec{v}_{\text{comb}}$	Vector Composition and statistical averaging over multiple discrete constituent momenta: $\vec{v}_{\text{comb}} = (\sum \vec{p}_{0k}) / (Nm_0)$	Net oriented superposition of constituent momenta governed by vector addition rules	$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_{\text{comb}}$
Counter-directional Evolution	Reaction Force	Virtual momentum pair exchange borrowed from Potential Space constrained by spacetime shaping conservation and finite Perceptual Cross-section	Long-range Non-dissipative interaction inducing opposing momentum shift $\Delta p_1 = -\Delta p_2$ per pairwise exchange	$F_{12} = -F_{21}$
Decay Evolution	Decay partial velocity v_{emergent}	Internal topological phase transition splitting stable cyclic manifold (parent particle) into two distinct daughter topological structures (decay products)	Separation velocity of fragmented constituents following global vector momentum conservation after perturbative interaction	$v_{\text{emergent}} = \frac{\Delta N}{N_{\text{total}}} V_{\text{bandwidth}}, \sum m_i \vec{v}_i = 0$

Summary: $P = F_{\text{final}} = E_{\text{final}} = mv$. That is, force corresponds to the magnitude of momentum exchanged in a single interaction, while energy corresponds to the integral of momentum over an interval. Neither force nor energy is an independent physical quantity; all three are fundamentally unified as momentum [53].

Note that this relation holds within the framework of mechanical action: the final-state action is the resultant-velocity capacity (i.e., resistance). However, the relation is not strictly valid dimensionally, since both energy and force incorporate temporal factors. For a detailed derivation, see Section 10.3.

Supplementary Conceptual Clarification: The information-related concepts in this paper correspond to the generalized information system introduced in Chapter 2 as follows:

- **Generalized Information:** state-relations, from which life and consciousness can emerge;
- **Intrinsic Information:** state-existence quantity, i.e., the total number of evolution carriers and state-representational quantities within the system, which may include spatial and temporal representational quantities such as $Nm_0l_0(x, y, z)$;
- **Combinatorial Information:** state-combination quantities from different perspectives, corresponding to the number of states under multiple viewpoints;
- **Entropy:** dynamically evolving state-quantity, corresponding to the conventional notion of disorder.

The above content primarily provides a logical clarification, aiming to reconstruct the associative relationships through which various phenomena are formed.

8.2 Emergence of Natural Fundamental Dimensions

Consistent with tabulated emergence rules, mass, energy, momentum, force, information, entropy, time and space lack inherent standalone essence within the generalised relational evolution system; all arise as emergent statistical descriptors of underlying state evolution under differing interactive hierarchy and observational coarse-graining[49]. The eight listed quantities represent alternative statistical projections of Evolution carrier population, spatial arrangement and vector superposition under varied measurement frames.

When constituents reside in full Cancellation State or the system is captured via single t_0 instantaneous Snapshot, dynamic evolutionary motion freezes entirely. No transient observables including drag, thrust, tension, repulsion, bulk motion, thermal excitation, entropy, energy, force or momentum manifest; only total constituent number N remains measurable, eliminating all emergent dimensional constructs. This limiting scenario defines **Natural Dimension**: under fundamental system baseline or full coarse-graining where all interactions and configurations become indistinguishable, the entire system is exhaustively described solely via discrete Evolution carrier count N .

Within one elementary t_0 interval or complete equilibrium Cancellation State, force, energy, momentum, entropy, information, mass, length and time lose distinctive emergent properties and reduce to equivalent numerical carrier count:

$$N = E = P = F = I = S = L = T \quad (\text{quantities under single } t_0 \text{ snapshot})$$

Fundamental constants such as light speed c and reduced Planck constant $\hbar$ are excluded from Natural Dimension classification, as their definition introduces derived units (metre/second, joule-second) irreducible to pure dimensionless population count. By contrast, Evolution carrier count N is expressible exclusively via discrete cardinality. Light speed may be phrased descriptively as " N identical m_0 constituents each endowed with intrinsic Driving capacity c ", whereby N independent c terms jointly govern systemic evolution. All conserved quantities or physically equivalent derived observables are reducible into pure numerical Natural Dimension representation.

8.3 Chapter Summary

1. **Core Argument** Built upon the inertial relational state evolution framework, this chapter proves all statistically tractable physical dimensions emerge endogenously from three primitive foundations

m_0 , c and $\mathcal{R}$. Canonical physical concepts including time, space, energy, force, field and particles have no independent ontological existence and constitute macroscopic emergent projections of identical underlying evolutionary dynamics viewed from disparate observational angles[49][52].

2. Key Physical Inferences

- (a) The unique globally conserved primitive quantity of the system is elemental Momentum Unit $P_0 = m_0 c$, the irreducible fundamental carrier of all causal interaction forming the bedrock of all macroscopic physical processes.
- (b) Clarified microscopic essence of force and energy: Force equals discrete single-interaction momentum variation satisfying $F = \Delta P$; Energy equals path-integrated cumulative momentum change across successive microscopic couplings such that $E = \sum \Delta P_n$.
- (c) Unified terminal-state relation of physical quantities: momentum, force, and energy achieve unification of both "action" and physical essence at the final evolutionary state of the system, i.e., $P = F_{\text{final}} = E_{\text{final}} = mv$.

3. **Unification Rule for Natural Dimensions** At single-timepoint t_0 Snapshot or full systemic Cancellation equilibrium, all macroscopic physical observables unify into sole fundamental Evolution carrier count N , yielding complete dimensional normalisation:

$$N = E = P = F = I = S = L = T$$

This relation confirms all classical physical dimensions are alternative statistical expressions of identical primitive constituent substrate.

4. **Four-Tier Hierarchical System of Physical State Quantities** Classify all physical observables into four hierarchical grades constrained by evolutionary layering and perceptual coupling laws, bridging microscopic fundamentals to macroscopic measurable quantities:

- (a) **Absolute Intrinsic State Quantity**: Baseline static or single- t_0 snapshot quantity uniquely represented by constituent count N , the irreducible foundational physical substrate.
- (b) **Derived Intrinsic State Quantity**: Spacetime-dependent derivative built upon fundamental bulk momentum mc , bridging absolute intrinsic counts and macroscopic statistical observables.
- (c) **Macroscopic Statistical State Quantity**: Density-dependent evolutionary bulk property centred on macroscopic kinematic momentum mv , corresponding to classical mechanically measurable bulk motion.
- (d) **Higher-Order Spacetime Derived Quantity**: Advanced configuration originating from particle perceptual locking and spacetime coupling, represented by relativistic composite forms mcc and mvv valid for relativistic high-speed and strong-coupling regimes; these quantities are composite statistical aggregates rather than primitive intrinsic observables.

8.4 Reference Commentary for Chapter 8

[27] Dirac, P. A. M. (1931). Quantised singularities in the electromagnetic field. *Proceedings of the Royal Society of London Series A*, 133(821), 60–72. <https://doi.org/10.1098/rspa.1931.0130>

Topological monopole condition demonstrates charge quantisation arises from inherent topological constraint instead of artificial parameter setting. Though physical magnetic monopoles are absent in nature,

the underlying logic matches topological Encapsulation and mechanical construction within the present framework.

[49] Duff, M. J., Okun, L. B., & Veneziano, G. (2002). Dialogue on the number of fundamental constants. *Journal of High Energy Physics*, 2002(03), 023. <https://doi.org/10.1088/1126-6708/2002/03/023>

The paper establishes unit definition is framework-dependent and conventional dimensional standards are human observational conventions, consistent with dimensionality emerging statistically from underlying microscopic mechanisms across alternative viewpoints.

[50] Okun, L. B. (2006). The concept of mass in the Einstein year. In *Particle Physics at the Year of 250th Anniversary of Moscow University* (pp. 1–15). Singapore: World Scientific. https://doi.org/10.1142/9789812772657_0001

Clarifies physical definition of rest mass as aggregate count of enclosed internal Momentum units, treating c^2 purely as dimensional conversion factor consistent with the present dimensional reset formalism.

[51] Hestenes, D. (1990). The Zitterbewegung Interpretation of Quantum Mechanics. *Foundations of Physics*, 20(10), 1213–1232. <https://doi.org/10.1007/BF01889486>

Intrinsic particle proper time corresponds to internal constituent oscillatory phase, supporting the hypothesis of inherent elemental evolutionary frequency and intrinsic proper time inside composite particles.

[52] Barbour, J. (2009). The Nature of Time. *arXiv preprint*, arXiv:0903.3489. <https://doi.org/10.48550/arXiv.0903.3489>

Duration lacks independent substantial reality and is an abstract parameter derived from relative physical variation of matter; higher-order temporal powers such as T^2 carry no standalone ontology and only quantify rate-of-change of underlying variation.

[53] Kalies, G. (2021). Back to the roots: the concepts of force and energy. *Zeitschrift für Physikalische Chemie*, 236(4), 481–533. <https://doi.org/10.1515/zpch-2021-0034>

Distinguishes instantaneous state equations and time-dependent evolutionary equations, identifying partial causal inconsistency embedded within conventional force-energy formalism. Kalies advocates interaction described via evolutionary process rather than standalone force; this corroborates core framework conclusions: force equals discrete momentum shift Δp , energy is path-integrated statistical accumulation over inertial evolution, both emergent from underlying microscopic causal exchange without independent fundamental existence.

““

9 Derivation of Mutual Relations among Cosmic Matter, Light Speed and Spacetime

All subsequent chapters further develop logical deduction and mathematical derivation grounded in the axioms and logical framework of the generalized relational state evolution system established in preceding contents, so as to verify dynamic rules and dimensional relations and trace the origin of conventional mechanical formulas.

9.1 Postulation and Existence of Fundamental Minimal Matter Constituent

The minimal material substrate of the system emerges from underlying constraints including nonattenuable minimal interaction, distinguishable Representational Quanton and feasible causal connection construction. Such elementary constituent under minimal decay condition is defined as the elementary mass unit denoted m_0 .

Core principle: As the threshold of fundamental systemic interaction, minimal force cannot decay further and fixes the lower bound for physical matter existence. The distinguishability of state representation and realizability of causal coupling require this elementary constituent to hold stable Intrinsic properties acting as the fundamental carrier for Emergence of spacetime and dynamical laws; its inherent evolutionary features directly govern subsequent generation rules of elementary spacetime quanta and physical observables.

9.2 Emergence of Static Elementary Spacetime Quantized Units from Minimal Matter

Based on definitions and derivations from Section 5.7 and Chapter 7, elementary static spacetime units of the cosmic system are defined as elementary spatial unit $l_0 = \Lambda$ and elementary temporal unit $t_0 = \frac{\Lambda}{v_0}$, where v_0 stands for the intrinsic Driving Capacity of the elementary Momentum unit.

From inherent evolutionary property of Momentum unit, light speed c Emerges via proportional relation of spacetime quanta:

$$c = \frac{l_0}{t_0}$$

Substitute $t_0 = \frac{\Lambda}{v_0}$ and $l_0 = \Lambda$ for simplification:

$$c = v_0$$

It follows that c is not an artificially preset fundamental constant but an inherent physical quantity naturally generated from intrinsic Driving Capacity of underlying Momentum units. Its reciprocal relation

$$\frac{1}{c} = \frac{t_0}{l_0}$$

fixes rigid correspondence between l_0 and t_0 , representing the minimal temporal cost required for unit spatial evolution.

9.3 Numerical Calibration of Minimal Spatial Scale

The perceptual saturation mechanism demonstrated in Section 5.7 proves the necessary existence of minimal spatial cutoff Λ , whose numerical value is fixed against experimentally measured physical constants in this subsection. The present evolutionary framework does not demand an absolutely universal fundamental spacetime scale; smooth continuous macroscopic orbit and avoidance of divergent infinite integration are guaranteed as long as the system contains sufficiently fine relational time intervals and state quantities with internal constituent count inside particles satisfying $N \gg 1$ [21]. Derivation below targets the canonically quantized cosmic evolution system.

Introduce systemic minimal force, perceptual collapse radius and minimal causal interaction characteristic length expressed via wavelength; following the conception of Planck scale, this minimal characteristic length is assigned as the system's elementary spatial limit[54]. Within three-dimensional orthogonal inertial state evolution system, dimensional analysis over three empirically measured fundamental constants namely gravitational constant G , reduced Planck constant $\hbar$ and light speed c yields a unique dimensional length expression:

$$l_0 = \sqrt{\frac{G\hbar}{c^3}} \approx 1.6 \times 10^{-35} \text{ m}$$

This length marks the threshold for manifest quantum-gravitational effects as well as the smallest spatially resolvable unit for causal coupling, hence critical saturation length for perceptual limit satisfies:

$$\Lambda = l_0$$

Important distinction: This work only borrows the numerical magnitude of Planck length instead of the physical definition of Planck mass $m_P = \sqrt{\frac{\hbar c}{G}}$. Within this framework, elementary mass m_0 is independent of m_P with relation $m_0 \ll m_P$; no fixed upper or lower mass bound exists for black holes, and observed black hole mass only serves as proportional scaling factor against elementary constituent mass requiring extra parameters for precise numerical solution of m_0 .

Electron mass follows $m_e = Nm_0$ where N denotes internal Momentum unit population inside the particle with $N \gg 1$, whose specific value awaits further derivation incorporating minimal force and perceptual collapse radius. Planck mass bears no fundamental foundational standing in this system and merely corresponds to the extreme coincidence condition between single Momentum unit and Planck wavelength rather than the defined minimal elemental mass of the universe.

Physical implication of $\Lambda = l_0$: When separation between two Evolution carriers satisfies $r < l_0$, full perceptual saturation occurs under minimal effective force such that all causal interactions point back to the constituent itself prohibiting formation of multiple independent evolutionary bodies. Accordingly Λ constitutes the minimal spatially independent evolution unit, the geometric foundation for discrete spacetime, causal locality and canonical quantization[55].

The research methodology applied here combines theoretical existence proof via perceptual saturation deduction and experimental numerical calibration via constants $G, \hbar, c$, consistent with the empirical calibration logic for inertial mass and Planck constant[54].

9.4 Dynamical Constraint: Quantitative Determination of Planck-Scale Parameters

Complete derivation for minimal scale $\Lambda = l_0$ is accomplished via equilibrium between minimal intrinsic Driving force and minimal long-range interaction plus quantum restriction conditions.

Step 1: Definition of Minimal Driving Force Minimal systemic driving force is defined as maximum achievable momentum variation of a single Momentum unit within one elementary t_0 :

$$F_{\min} = \frac{p_0}{t_0}$$

where Momentum unit relation holds $p_0 = m_0 v_0 = m_0 c$.

Step 2: Constraint from Minimal Long-Range Interaction At minimal cutoff scale Λ , gravitational pull acting on elementary Momentum unit balances minimal intrinsic driving force:

$$\frac{Gm_0^2}{\Lambda^2} = \frac{p_0}{t_0}$$

G denotes gravitational coupling constant which Emerges statistically from collective property of Momentum units.

Step 3: Simplified Derivation with Quantum and Spacetime Relations Adopt quantum constraint $p_0 \cdot \Lambda = \hbar$, spacetime definition $t_0 = \frac{\Lambda}{c}$ and $p_0 = m_0 c$ for stepwise derivation:

1. Substitute $p_0 = \frac{\hbar}{\Lambda}$ and $t_0 = \frac{\Lambda}{c}$ into minimal driving force formula:

$$F_{\min} = \frac{\hbar/\Lambda}{\Lambda/c} = \frac{\hbar c}{\Lambda^2}$$

2. Equate gravitational interaction and minimal driving force:

$$\frac{Gm_0^2}{\Lambda^2} = \frac{\hbar c}{\Lambda^2}$$

Cancel identical denominators to obtain:

$$Gm_0^2 = \hbar c$$

3. Insert $m_0 = \frac{p_0}{c} = \frac{\hbar}{\Lambda c}$ into above equation and solve final expression:

$$\Lambda = \sqrt{\frac{G\hbar}{c^3}} = l_0$$

Meanwhile the Planck time formula is derived:

$$t_0 = \sqrt{\frac{G\hbar}{c^5}} [54]$$

This derivation avoids preinstalled $E = mc^2$ (derived later as emergent spacetime state relation) and extra assumptions on equivalence between Compton wavelength and Schwarzschild radius; all results originate strictly from intrinsic system axioms matched against measured physical constants and verify the emergent nature of Planck scale. Double special relativity and loop quantum gravity provide supplementary theoretical grounding: the former revises spacetime symmetry at Planck cutoff while the latter establishes spacetime construction from discrete quantized geometric cells, jointly validating l_0 and t_0 as minimal evolution and elemental event units[55].

9.5 Three Transition Modes of Momentum Unit (Natural Elimination of Divergence for Single Transition)

Fundamental spacetime Emergence originates from single transition events of Momentum units: elemental Representational Quanton m_0 executes discrete-continuous state transition at intrinsic Driving rate $v_0 = c$, each transition paired with a full elementary spacetime cell ($l_0 \times t_0$). All three transition modes share identical underlying physical essence categorized under intrinsic inertial length scale and mathematically remove divergence for single-step evolution.

1. **Mode A:** Within a single free spatial dimension, the Momentum unit completes full transition constrained by its intrinsic inertial length l_0 , synchronously generating elementary spatial increment l_0 and temporal increment t_0 with one-to-one correspondence between spatial and temporal quanta.
2. **Mode B:** Instantaneous transitional displacement lies between $0 \sim l_0$, yet equivalent full l_0 spatial progression is finalized over every full t_0 cycle. This formulation is physically equivalent to Mode A and accounts for microscopic quantum fluctuation effects.
3. **Mode C:** Actual transitional stretch $x \cdot l_0 < l_0$ within single t_0 fails to form unified definable scale and stable causal linkage, violating requirements of canonical quantization and relativistic consistency; this configuration does not exist within the studied cosmic system.

Supplementary remark: The discrete-continuous evolutionary feature of $v_0 = c$ requires distinction between static pre-interaction spacetime configuration and post-interaction causal spacetime geometry shaped after physical coupling.

For Mode A and Mode B within one elementary t_0 , unified statistical description of m_0 reads: Single transition length = intrinsic inertial state quantity = intrinsic kinematic quantity = intrinsic representational quantity = spatial shaping magnitude = unit transition count = $l_0 = t_0 = 1$.

Formal mathematical notation:

$$\text{Spatial intrinsic quantity: } \Delta x_0 = 1 \cdot l_0$$

$$\text{Temporal intrinsic quantity: } \Delta t_0 = 1 \cdot t_0$$

$$\text{State intrinsic quantity: } \Delta S_0 = 1 \cdot u_{\text{state}} [56]$$

9.6 Definition and Core Description of Canonically Quantized Discrete Minimal Scale

Synthesizing previous derivations, elementary constituent m_0 , fundamental spatial unit l_0 and complete elemental temporal cycle t_0 are collectively defined as the system's canonically quantized discrete minimal scale.

Core descriptive contents of this fundamental scale include intrinsic state representation hosting m_0 evolution, minimal spacetime configuration ($l_0 \times t_0$) generated per full evolutionary step, smallest independent causal unit available for evolution, and relational descriptor governing material progression throughout interaction. This elementary scale serves as foundational basement for subsequent Emergence of spacetime laws and dynamical relations.

9.7 Derivation of Macroscopic Continuous Spacetime from Discrete Fundamental Quantities

Macroscopic continuous spacetime emergence is deduced from discrete state variables, mechanical relations and causal interactions combined with constraints prohibiting arbitrary free-dimension jumps and mandating corresponding state increment accompanying every evolutionary step.

1. Discrete Nature of Spacetime Governed by State Representation and Causality Discrete underlying material constituents impose discreteness onto static systemic states, dynamic interaction terms and physical action quantities. Evolving Momentum units propagate sequentially over multiple t_0

intervals anchored on minimal intrinsic length l_0 to fix constant evolutionary ratio c ; Vector Composition applies for combined bulk motion when Evolution carriers form composite causal relations:

$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = (m_1 + m_2) \vec{v}_{\text{comb}}$$

Velocity and spacetime essentially originate from repeated sequential evolution events t_0 of elementary m_0 along free spatial axes over intrinsic scale l_0 , with time as cumulative event tally and space as extended dimensional stretch synchronously sharing identical natural growth rate. Klein' s higher-dimensional quantization scheme, loop quantum gravity, causal set theory and Wheeler' s spacetime foam hypothesis all rest on identical core logic built from discrete state (entropy/information), relational causality, evolutionary dynamics and quantum discreteness, offering consistent theoretical support for background-independent spacetime construction[55].

2. Continuous Spacetime Emerging from Persistent Constituent Evolution Continuous macroscopic time T , continuous spatial dimension L and bulk continuous velocity V (or c) Emerge macroscopically when evolving matter holds stable internal architecture, invariant inertial causal linkage and fixed evolutionary rules plus conserved total constituent population, stable Evolution resource and constant intrinsic evolutionary frequency.

Continuity here never implies unlimited infinite subdivision which would trigger divergent dimensional expansion[21]. Abrupt state jumps only appear during early-stage structural formation such as particle decay and particle coupling; this section focuses on core correlation among l_0 , t_0 , discrete spacetime cell $l_0 \cdot t_0$ and continuous evolutionary speed c , v [9].

9.8 Emergent Spacetime Quantities under Distinct Observational Perspectives

Within this framework, temporal, spatial and full spacetime observables are categorized into hierarchical forms varying across mechanical statistics and perceptual interaction statistics.

Category	Subtype	Expression	Remark
Static Quantity (Momentum Unit Cancellation State)	Bulk Space	$m = Nm_0$	Pure counting variable, no directional property or ongoing evolution
	Elementary Space	l_0	Minimal fundamental spatial occupation scale
	Elementary Time	t_0	Minimal cycle for full interaction and elementary evolution
Density-Bearing Quantity	Bulk Space	L, S, V	Relational geometric structure determined by matter distribution
	Bulk Time	T	Cumulative tally of successive evolutionary events
Inertial Evolution Rate Quantity	Temporal Emergence Rate	$p = mv$	Growth rate of cumulative evolutionary event count

	Spatial Emergence Rate	$p = mv$	Expansion rate of free spatial dimension under evolution
	Spacetime Emergence Rate	mvv	Generation rate of full spacetime configuration under inertial progression
Limiting / Conserved Evolution Capacity	Limiting Temporal Rate	mc	Ultimate growth rate of evolutionary event accumulation
	Limiting Spatial Rate	mc	Ultimate expansion rate of free spatial dimension
	Limiting Spacetime Generation	mcc	Maximum generation rate of complete spacetime configuration at saturation

Microscopic static quantities apply to quantum mechanical analysis; inertial evolutionary terms correspond to classical mechanics and bulk statistical calculation; spacetime shaping capacity fits relativistic observational statistics.

Quantities with spatial density (e.g., L, S, V) and potential spacetime quantities (such as vacuum) are suitable for everyday life, engineering applications, and mathematical statistics—they are macroscopic approximations. The first three categories (static quantities, density-bearing quantities, and inertial evolutionary quantities) represent more intrinsic spacetime states: they are independent of density definitions, involve no concept of "void," and correspond to definite physical entities. Among these, the absolute intrinsic spacetime is the representable quantity of Evolution carriers in the stationary state (i.e., the pure counting state N , without any dynamic evolution).

9.9 Chapter Summary

1. **Logical Necessity of Minimal Matter and Spacetime Cutoff:** Elementary constituent m_0 and minimal spatial unit l_0 must inherently exist within the system, paired with elemental spacetime cell ($l_0 \times t_0$) from single transition satisfying $t_0 = l_0/c$. This conclusion follows strictly from intrinsic framework axioms of canonical quantization, Generalized Equivalence Principle and perceptual collapse constraint without external theoretical postulation; quantitative calculation references Planck's original derivation routine[54].
2. **Emergent Origin of Light Speed:** Constant c is not externally predefined but the natural realization of intrinsic elemental Driving Capacity v_0 with identity $c = l_0/t_0 = v_0$, emerging from limiting continuous propagation of Momentum units along free spatial dimensions.
3. **Spacetime Generation Rate:** Under inertial evolution of Momentum units, both temporal and spatial emergence rates equal m_0c or m_0v , with saturated spacetime configuration growth rate reaching m_0c^2 or m_0v^2 .
4. **Referential Numerical Calibration:** Minimal scale magnitude can be numerically referenced to Planck dimensional analysis result $l_0 \sim \sqrt{G\hbar/c^3}$, yet such numerical fitting never serves as foundational prerequisite of the theory; logical proof for elementary constituent existence is self-contained within evolutionary deduction independent of Planck's original assumptions or specific numerical assignment.

5. **Physical Essence of mcc :** Conventional physics interprets mcc as energy, while energy holds no independent substantial reality inside the present framework with detailed discussion postponed to Section 10.4. No valid integral statistical operation is achievable under static or inertial rest configuration such that photon inertial characteristics are solely described via combined evolution rate and total constituent count mc namely momentum.
6. **Ontology of Spacetime:** Time and space Emerge from sequential state transition of Evolution carriers along free dimensional axes: space quantifies dimensional state occupation of constituents and time tallies complete sequential evolutionary events. Both quantities constitute alternative statistical descriptions of underlying Evolution carriers instead of standalone independent substance; intrinsic spacetime magnitude is determined by total constituent population and corresponding free-dimension representation such that matter location directly defines its own localized spacetime domain.

9.10 Reference Commentary for Chapter 9

[9] Whitehead, A. N. (1929). *Process and Reality: An Essay in Cosmology*. New York: Macmillan. Whitehead' s process philosophy describes reality as composed of discrete actual entity events and dynamic generative progression rather than preexisting static spacetime; time excludes dimensionless instantaneous points, consistent with this framework' s interpretation of "instant" as finite full event interval t_0 .

[21] Bohner, M., & Peterson, A. (2001). *Dynamic Equations on Time Scales: An Introduction with Applications*. Boston: Birkhäuser. DOI:10.1007/978-1-4612-0201-2. Time-scale calculus mathematically validates discrete-continuous integration as alternative to traditional infinitesimal subdivision integral.

[54] Planck, M. (1899). Über irreversible Strahlungsvorgänge. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin*. Original source of Planck scale, pioneering unique combination of three constants $G, \hbar, c$ to derive minimal length, time and mass and establishing core thought binding matter, light speed and spacetime via fundamental quantum constraints.

[55] Klein, O. (1926). Quantentheorie und fünfdimensionale Relativitätstheorie. *Zeitschrift für Physik*, 37(12), 895–906. DOI:10.1007/BF01397195. Klein' s higher-dimensional spacetime quantization links geometric structure and quantum constants matching discrete evolutionary logic of Momentum units.

[56] Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423; 27(4):623–656. DOI:10.1002/j.1538-7305.1948.tb01338.x. Foundational information entropy formula quantifies systemic state divergence and configurational disorder throughout evolutionary progression.

10 Derivation of Emergent Relations among Force, Energy and Momentum

Derived from previous chapters, the universe constitutes an evolution system governed by inertial relations, where core dynamic causality manifests as state evolution, namely inertial motion. Statistical components of state evolution include participating matter (representational quanton) and evolution velocity (intrinsic speed c and macroscopic speed v). Interactions between substances fall into two categories: coordinated evolution via combined velocity and opposite evolution (action and reaction). Accordingly, force and energy are not independently fundamental dimensional quantities. Subsequent sections derive and verify their intrinsic correlations.

10.1 Introduction of Generalized Equivalence Principle and Relation between Force and Velocity

1. Analysis on Force Generation

A general mechanical causality table for the cosmic system is established to sort out governing rules of diverse mechanical interactions.

Table 10.1: General Table of Mechanical Causality

Comparison Item	Non-dissipative Force (Action and Reaction)	Dissipative Force (Combined and Split Velocity)
Phenomenon	Long-range interactions such as electromagnetic force and gravitation	Localized interactions including decay, fragmentation and cancellation state
Interaction Mechanism	Action-reaction relation $\Delta\vec{p}_A = -\Delta\vec{p}_B$	Proportional directional vector composition: $m_1\vec{v}_1 + m_2\vec{v}_2 = (m_1 + m_2)\vec{v}_{\text{comb}}$
Spatial Correlation	Linear decaying interaction within three-dimensional space for observation, satisfying $\frac{1}{4\pi r^2}$ in bidirectional view, proportional to r^2 in integral perspective and decaying with r in unidirectional view	Vector composition constrained by alternative forces or quantum entanglement
Features	Action-at-distance property, instantaneity, penetrability and non-attenuation	Local combined velocity, split velocity breaking, interference and non-instantaneous interaction
Propagation Mechanism	Independent of propagator and accompanied by quantum fluctuation	Direct localized momentum transfer
Upper Motion Limit	Speed of light c	Speed of light c
Upper Interaction Bound[57]	mc	mc

Mass Proportionality	Proportional to constraints from encapsulation mechanism	Proportional to total participating momentum
Essence of Motion	Pure inertia exists at any instant; acceleration originates from persistent contribution of decaying observational interaction	Pure inertia exists at any instant; acceleration equals persistent dissipation and replenishment of inertia
Equivalence Nature	Attractive (pulling) or repulsive (pushing) potential is equivalent to motion potential and inertia potential	Drag or thrust tendency is equivalent to motion potential and inertia potential
Work Mode	Momentum accumulates and stores automatically; work is solely determined by terminal instantaneous momentum p_{fin}	Momentum dissipates in real time without storage; work requires statistical accounting of full-process momentum replenishment cost

As summarized from mechanical characteristics listed above, conventional definitions of force and energy fail to uniformly describe inertial state, cancellation state and decay process, hence lacking universal applicability. The four fundamental forces only quantify interaction magnitude and cannot individually define corresponding energy for each force.

Dissipative forces comply with $m_1\vec{v}_1 + m_2\vec{v}_2 = (m_1 + m_2)\vec{v}_{\text{comb}}$, namely $\vec{p}_1 + \vec{p}_2 = \vec{p}_{\text{total}}$; non-dissipative forces satisfy $\Delta\vec{p}_A = -\Delta\vec{p}_B$. Essentially, both interaction types correspond to instantaneous incremental deviation of momentum within the system.

2. Force and Generalized Equivalence Principle

A. Dissipative interactions: Within the inertial relational state evolution system, two objects A and B of identical mass move in opposite directions with respective momentum $m\vec{v}$ and $-m\vec{v}$. Complete velocity combination and cancellation occurs under mutual mechanical action, yielding a final bulk velocity $v = 0$.

It follows that mechanical force persists alongside inertial motion and cancellation state; transitions from cancellation state to broken state during decay do not extract energy from background fields[20]. Force and energy stem fundamentally from the generation of momentum deviation, establishing full equivalence between inertial motion and force. Conventional equivalence principle is restricted to equivalence between gravitation and uniform acceleration under integrated persistent momentum deviation, bearing limited applicable scope. Total momentum remains strictly conserved throughout the whole process, while independent force and energy do not manifest macroscopically.

B. Long-range non-dissipative interactions: Two objects A and B of unequal mass interact via long-range observational action to generate paired opposite reaction forces satisfying:

$$m_1\vec{v}_1 = -m_2\vec{v}_2 \iff \Delta\vec{p}_A = -\Delta\vec{p}_B$$

Further derivation gives:

$$\Delta\vec{p}_A + \Delta\vec{p}_B = 0$$

Total system momentum is conserved across the full interaction.

As concluded from preceding content, particles feature quantized intrinsic microstructure composed of massive momentum units m_0c : fermion internal momentum units adopt spherical symmetric distribution while photon internal momentum units follow unidirectional arrangement. During long-range force interactions, particles generate virtual momentum pairs $(\vec{p}, -\vec{p})$ via quantum fluctuation in potential space,

perturbing partial internal momentum units of paired objects to evolve toward or away from each other and consequently form opposite action-reaction forces. No bulk matter dissipation emerges in this procedure, and total internal particle momentum Mm_0c stays invariant; photons only alter internal topology and interference status upon external action with unchanged intrinsic inertial speed.

In summary, opposite reaction forces build mechanical causal correlations between substances based on inertial motion.

Conclusion: Force intrinsically equates to inertial motion. Inertia does not represent force-free status but results from force, and inertial motion can be transmitted onto alternative carriers to realize velocity combination. Only inertial momentum exists instantaneously for either dissipative or non-dissipative interactions. The generalized equivalence principle is stated as: Force = motion capacity = causal interactive momentum deviation.

3. Emergence and Conservation Relations of Mechanics

Whether for localized dissipative velocity combination/splitting transmitted through space or non-propagating long-range force, mechanical relations originate from mutual acquisition of opposite or synchronized evolution capacity between two particles. Momentum exclusively participates in evolution with global conservation, whereas force and energy emerge only under specific physical scenarios instead of fundamental physical reality.

1. All Mechanical Causality in Cosmic System

$$\begin{aligned} & \{ \text{Particles } A, B \text{ mutually acquire opposite evolution capacity} \\ & \quad \text{and synchronized evolution capacity} \} \\ &= \{ \text{Particles } A, B \text{ obtain opposite space-time state shaping capacity} \\ & \quad (\Delta \vec{p}_A = -\Delta \vec{p}_B) \\ & \quad \text{or synchronized space-time state shaping capacity} \\ & \quad (\vec{p}_1 + \vec{p}_2 = \vec{p}_{\text{total}}) \} \end{aligned}$$

2. Conservation Laws

$$\begin{aligned} \{ \text{Conservation rules} \} &= \{ \text{Conservation of space-time state shaping capacity} \} \\ &= \sum m_i c^2 = \text{constant} \quad \text{or} \quad \sum m_i c = \text{constant} \end{aligned}$$

3. Generalized Equivalence Principle

$$\begin{aligned} \{ \text{Generalized Equivalence Principle: Force} &= \text{Motion Capacity} \\ &= \text{Causal Interactive Momentum Deviation} \} \end{aligned}$$

4. Unified Rule of Potentials

$$\begin{aligned} & \{ \text{All potentials reduce to motion potential (inertia potential):} \\ & \text{Attraction (tension)} = \text{Repulsion (thrust)} = \text{Resistance} = \text{Thrust} \\ & \iff \text{Motion Potential} = \text{Inertia Potential} \} \end{aligned}$$

4. Essence of Motion

Motion arises from incomplete cancellation state of total internal particle momentum Nm_0c [59]. Description of motion requires no extrinsic reference frame, as distinct reference frames correspond to

separate inertial subjects; intrinsic substance motion is determined by inherent properties and develops mechanical causal links with ambient medium only upon physical interaction. Internal particle motion capacity follows vector composition constraints of momentum, with directional proportion defining upper motion bound.

Reference Frame and Inertia Inheritance: A reference frame describes relative evolutionary relations, where two independent or weakly correlated objects bear incomplete mutual mechanical causal influence. Four concepts including inertia inheritance, absolute motion, variational motion and reference frame require differentiated clarification. All motion carries inertia inheritance varying continuously from weak to strong induced by environmental factors: for instance, static human sleeping state retains bulk motion inherited from layered inertial systems including the solar system and Milky Way; such inherited motion counts as absolute motion and may modify microscopic emission observational effects, while voluntary body activity generates variational motion constituting primary energy cost. Reference frames feature weak mechanical coupling yet strong observational correlation, generating projective observational outcomes via space-time integral mapping.

5. Fundamental Differences between Generalized Equivalence Principle and Conventional Equivalence Principle

Category	Conventional Equivalence Principle	Generalized Equivalence Principle
Applicable Scope	Restricted exclusively to gravitation	Covers four fundamental forces
Reference Baseline	Uniform motion equals force-free motion	Uniform motion originates from mutual cancellation of bidirectional inertia potential
Definition of Force	Source of acceleration defined via integral effect	Instantaneous motion potential serving as instantaneous state function
Physical Description	Unidirectional localized equivalent relation	Bidirectional instantaneous coupling relation
Reference Frame	Motion codetermined by reference frame yielding relative motion	Intrinsically defined motion determined by evolution speed $c \times$ orientation ratio s [59][60]

10.2 Deficiencies of Conventional Gravitational Theories

Gravitation does not originate from space-time curvature, and the relativistic gravitational framework exhibits certain logical and interpretive gaps at the fundamental level.

1. Limitations of Relativistic Gravitation (Space-time Curvature)

a Space-time is emergent relation instead of independent physical reality

Space-time is not a standalone physical entity but emergent representational quanton generated from state variation of evolution carriers within generalized relational evolution system. By analogy, color evolution systems can spawn corresponding "color space-time". Space-time solely acts as evolution background lacking capacity to drive particle dynamics[58].

b Failure to explain dynamic origin

General relativity geometrizes gravitation into space-time curvature, yet pure geometric configuration cannot supply dynamic source. Explaining particle trajectory via curved space-time equals interpreting vehicle traveling solely from road geometry and violates momentum conservation: momentum alteration must stem from momentum unit transfer rather than deformation of background geometric structure.

c Conflicts between gravitational wave dissipation and pure geometric postulate

If gravitational waves correspond to ripples of geometric space-time, the pure geometric background cannot dissipate energy outward; measurable energy dissipation of detected gravitational waves implies space-time contains migratable momentum units with physical substantiality, contradicting core assumption of purely geometric space-time.

Supposing solar gravitation manifests as static curved space-time with delayed attenuation, terrestrial gravitational pull would derive from historical residual space-time depression, conflicting with practical observation. LIGO detections such as GW170817 verify gravitational propagation at light speed while failing to account for energy dissipation channels of gravitational radiation, further exposing internal theoretical inconsistency.

d Missing microscopic quantum interaction mechanism

General relativity exclusively describes macroscopic geometric gravitational effects without detailing microscopic causal interaction rules between elementary particles, inducing two core dilemmas:

Missing sensing mechanism: How particles perceive directional space-time gradient differences? Continuous infinitely divisible space-time requires structural definition over arbitrarily infinitesimal scales contradicting physical reality.

Discrete-continuum contradiction: Existence of minimal fundamental length l_0 and minimal fundamental time t_0 establishes physical scale cutoff. Continuous geometric space-time cannot naturally accommodate discrete quantum units; infinitely divisible space-time conflicts with finite minimal action quantum, while discrete elementary space-time invalidates rigorous continuous geometric formulation which reduces to macroscopic statistical approximation.

e Inability to interpret gravitational slingshot effect

If gravitation arises solely from spacetime curvature, particles would strictly follow geodesic trajectories. In gravitational slingshot maneuvers, a spacecraft gains velocity increments upon planetary flyby, with the velocity increase originating from the orbital momentum component of the host planet rather than passively "sliding down" a curved spacetime depression.

If the spacetime curvature around a planet were a static geometric depression, the spacecraft's inbound and outbound trajectories would be symmetric, resulting in zero net velocity change after the encounter. Experimental observations, however, confirm the presence of a causal interaction Momentum Deviation: the process fundamentally involves the redistribution of causally interacting momentum states between particles inside the planet and those inside the spacecraft, rather than being an effect of spacetime geometry.

2. Deficiencies of Canonical Quantum Gravity

Traditional quantum gravity frameworks suffer inherent flaws including contradictory propagator logic, preset absolute background space-time and default continuous infinite divisibility of space-time. Relevant analysis is consistent with preceding discussion and omitted herein.

10.3 Classical Mechanical Analysis of Force and Energy

1. Origin of Force and Energy in Classical Mechanics

The concepts of force and energy are established under the stable environment of the Earth, linear attenuation of three-dimensional spatial perception, and macroscopic compound interest statistical effects. They are the products of causal integration under specific conditions, rather than essential descriptions of the instantaneous state of the universe.

- **A. Earth Environment Basis** The Earth possesses a stable gravitational field, providing a natural reference for the quantification of macroscopic motion.
 - Gravitational perspective: During the free fall of an object, gravity continuously inputs momentum, leading to cumulative velocity increase.
 - Horizontal pushing perspective: Horizontal motion requires continuous overcoming of gravitational components and frictional forces, with inertial momentum continuously dissipated and necessitating constant supplementary thrust.

Both processes reflect continuous cumulative effects, forming human macroscopic concepts of force, work, and energy.

- **B. Compound Interest Integration of Gravitational Field (Free Fall)** The radial attenuation in three-dimensional space forms a spherical perceptual cross-section $\frac{1}{4\pi r^2}$, coupled with continuous instantaneous interactions within the minimum space-time unit t_0 , generating positive feedback compound interest integration:

$$r \downarrow \Rightarrow \frac{1}{4\pi r^2} \uparrow \Rightarrow \Delta p \uparrow \Rightarrow \text{Further amplification of the cross-section in the next cycle} \Rightarrow v \uparrow$$

Gravity continuously injects momentum, leading to gradual velocity superposition. The macroscopic statistical form is expressed as:

$$E_k = \frac{1}{2}mv^2 = mgh$$

This equation is an equivalent quantity of the whole-process integration, not the instantaneous real action quantity.

- **C. Linear Integration of Translational Resistance (Horizontal Work)** When pushing an object horizontally, gravity and friction constitute the resistance $F_{\text{Resistance}}$, which continuously dissipates inertial momentum. The thrust must offset resistance and supplement momentum throughout the process, and work is linearly related to displacement:

$$W = F_{\text{Resistance}} \cdot L$$

Classical mechanics thus defines force $\times$ distance = work, which is only applicable to specific macroscopic scenarios on the Earth's surface.

2. Three-Level Analysis of Energy and Force Mechanisms in Classical Mechanics

The analysis is conducted from three aspects: mathematical process, essence of instantaneous interaction, and differences in work modes.

- **A. Mathematical Process: Force and Energy as Interval Averages and Cumulative Integrals** Classical mechanical formulas $F = ma$ and $E = \frac{1}{2}mv^2$ essentially represent time averaging and interval integration of instantaneous states.

- Instantaneous level: Only inertial momentum $p = mv$ exists within each t_0 .
- Small-interval average: Taking a tiny time window, the traditional mechanical force is obtained by averaging momentum deviations

$$F = \frac{\Delta p}{t_0} = ma$$

- Large-interval accumulation: Integrating momentum increments and velocity coupling throughout the process yields macroscopic energy

$$E = \sum \Delta p_n \cdot v_n = \frac{1}{2}mv^2 = mgh$$

The core characteristic of classical mechanics: replacing physical instantaneous intrinsic quantities with integrated statistical quantities.

- **B. Essence of Instantaneous Interaction: Focusing on Terminal Momentum at Instant t_0** Taking the process of a stone landing as an example, instantaneous interaction and whole-process integration are two independent physical processes.
- **C. Integration Equivalence: Different Physical Quantities Converge Through Integration** During the entire free fall of an object, integration results from different dimensions are equivalent, ultimately unified into classical macroscopic statistical forms.

- Perceptual cross-section integration: $\int \frac{1}{4\pi r^2} ds$

- Distance integration: $\int dr$

- Time integration: $\int dt$

- Velocity integration: $\int v dv$

- Momentum integration: $\int v dp$

The above integration results are mutually equivalent:

$$\frac{1}{2}mv^2 = mgh, \quad p_{\text{final}} = m\sqrt{2gh}$$

- **D. Consistency of Instantaneous t_0 Increments: Independent of Integration Path** Regardless of the path taken by the object to reach the ground (natural free fall, lateral displacement, artificial variable-speed handling, etc.), within the instantaneous t_0 interaction window at landing, the momentum increment remains consistent, equal to the instantaneous gravitational action:

$$\Delta p = \frac{Gm_1m_2}{r^2}$$

In summary, the actual work capacity and instantaneous motion potential of the cosmic system are solely determined by the terminal instantaneous momentum $p_{\text{final}} = mv_{\text{final}}$, independent of the historical motion path. The integrated form is merely a macroscopic statistical expression of instantaneous interactions in classical mechanics, and its action quantity requires conversion of composite dimensions into basic dimensions (final-state instantaneous action quantity).

3. Differences in Work Modes

Comparison between two work modes: gravitational free fall and horizontal translation.

- **A. Gravitational Free Fall** Gravity automatically injects momentum continuously, with momentum accumulated and stored without external dissipation. Impact work only releases the terminal instantaneous momentum, and the historical evolution process does not directly participate in work. Thus, it can be uniformly described by

$$p_{\text{final}} = m\sqrt{2gh}$$

- **B. Horizontal Pushing Work (Expressed by Momentum Cost)** Friction continuously offsets directional inertial momentum and reduces the orientation ratio s , making inertial momentum unstorable. Thrust must supplement the dissipated momentum segment by segment, and horizontal pushing work is equivalent to the total cost of momentum supply throughout the process:

$$\text{Total Momentum Cost} = \sum \Delta p_{\text{supply}} = \int F_{\text{thrust}} dt$$

The traditional formula $W = F \cdot L$ is a special case under constant thrust, constant resistance, and uniform motion, i.e., the Earth's gravitational condition.

All physical processes satisfy:

$\begin{aligned} \text{Total Momentum Cost} &= \sum \Delta p \\ \text{Traditional Energy Form Cost} &= \sum \Delta p \cdot v = \sum F \cdot \Delta L \end{aligned}$

- **C. Interpretation of Two Expressions**
 - $\sum |\Delta p|$: Pure momentum perspective, counting the total momentum dissipation throughout the process, representing the actual motion cost.
 - $\sum \Delta p \cdot v$: Velocity-weighted momentum cost, corresponding to the discrete form of traditional work and energy.
 - Classical horizontal pushing work $W = F \cdot L$ is a simplified form of the latter under constant velocity and thrust.
- **D. Key Distinction: Gravitational Action vs. Horizontal Pushing**
 - Gravitational free fall: Momentum accumulates and stores automatically, work depends only on terminal instantaneous momentum p_{final} , equivalent to a resistance-free integration process.
 - Horizontal pushing: Momentum dissipates in real time and cannot be stored, work must count the full-process momentum supply cost; the pushing process is always accompanied by resistance confrontation.

4. Core Issues of Traditional Energy and Force (Including Examples)

- **A. Channel Selection Principle of Integration** According to previous derivations, force itself is a single momentum integration effect formed by the four fundamental forces. Regardless of the integration method, only mass number and inertial state changes exist. The key lies in the mathematically feasible channel differences in integration modes, such as perceptual cross-section, time, distance, velocity, and momentum, all of which can meet the integration requirements of momentum in dynamic evolution processes. Energy is a larger-interval integration based on force.

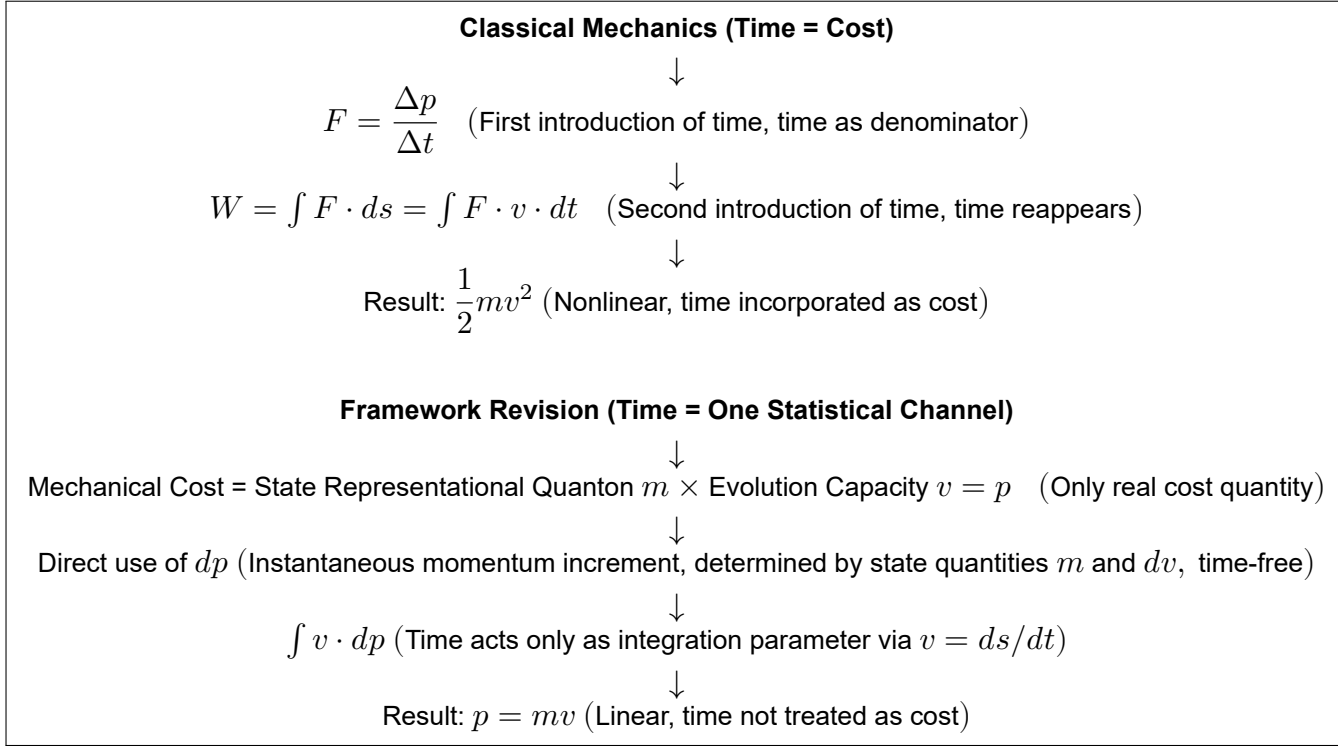
- **B. Formation of Integration Redundancy (Dual Introduction of Time)** According to previous derivations, force is a single interaction quantity, so force should be defined as $F = \Delta p$ (single interaction), and energy should be defined as $E = \Delta p \cdot N$ (sum of N interactions), naturally avoiding dimensional problems. However, such discrete integration methods are not as convenient as Newtonian integration in mathematics. Newtonian integration defines force as $F = dp/dt$, introducing a small-interval momentum increment to derive the average increment, which introduces the first time redundancy. Meanwhile, energy is defined as $E = \frac{1}{2}mv^2$ or mgh by integrating force over a larger interval, introducing the second time redundancy. Complete elimination of redundant dimensions adopts the momentum form $p = m\sqrt{2gh}$, or converts the gravitational acceleration factor g into a natural growth dimension (i.e., velocity increment per unit evolution cycle).
- **C. Examples of Incompatibility Formation**
 - (a) **Financial Analogy (Integration Rate vs. Integration Result):** Earning 1000 yuan in 1 day and 1000 yuan in 10 days have different integration rates, but the integration result is money (unit: yuan), independent of time. Traditional theory regards days and time in the process as "force"; bank compound interest integration corresponds to energy integration, thus taking the secondary time in compound interest integration as the result, replacing money and the unit yuan.
 - (b) **Aircraft Fuel Consumption Example (Theoretical 25x vs. Actual 5x):** According to the traditional kinetic energy formula, the energy consumption (regarded as action quantity) required for an aircraft to accelerate from 100 m/s to 500 m/s surges 25 times. However, practical aerospace engineering shows that aircraft fuel consumption (representing the actual momentum cost) is approximately linearly related to velocity increment; accelerating to 5 times the speed only requires about 5 times the fuel consumption, far lower than the theoretically predicted 25 times.

5. Distinction of Definitions for Momentum, Force and Energy

The final-state momentum $p_{\text{final}} = mv_{\text{final}}$ is an intrinsic state quantity that exists at any instant and is directly observable. It has non-composite dimensions, can directly perform work (possessing the ability to participate in velocity composition), and has direct dynamic causal influence. In contrast, final-state energy (e.g., $\frac{1}{2}mv^2$) and potential energy (e.g., mgh) are composite dimensions, incapable of direct participation in velocity composition and direct dynamic causal action, and are only applicable to integration calculations under specific conditions.

6. Core Conclusion: Only Momentum Exists

Concepts such as gravity, acceleration, potential energy, and kinetic energy are essentially statistical expressions of momentum in different scenarios. Various mechanical effects in classical mechanics are macroscopic manifestations of multiple momentum integrations and inertial integrations. The essence of work is vector velocity composition, requiring distinction between continuous multiple velocity composition effects and single instantaneous gravitational velocity composition effects.



7. **Section Summary** The integration method of traditional theory is mathematically accurate and convenient for prediction, but the dimensions in channel integration should not be regarded as cost or target results. Otherwise, composite dimensions that do not directly exist in the universe will be formed, leading to divergence. Therefore, force should be defined as single momentum integration, energy as multiple momentum integrations within an interval, and final-state momentum as the final settlement quantity—it is the direct mechanical cause, while integration channels are indirect mechanical causes. In the sense of action quantity:

$$p_{\text{final}} = E_{\text{final}} = F_{\text{final}} = mv_{\text{final}}$$

10.4 Formation of Relativistic Energy

Expressions related to relativistic energy originate from three independent theoretical frameworks without rigorous logical connections between each other, which are sorted item by item as follows.

1. **Definition of energy from photon momentum:** $E = pc$

The photon momentum satisfies $p = \frac{h}{\lambda}$. Conventional electromagnetic theory defines its energy as

$$E = h\nu = pc$$

This definition of energy is inherited from electromagnetism and lacks complete derivation from first principles, with no rigorous logical chain linking electromagnetic laws to the definition of energy.

2. **Thought experiment for mass-energy conversion:** $E = mc^2$

The core process of Einstein' s photon-box thought experiment is listed below:

- A photon is emitted from one end of the box, inducing recoil motion of the box.
- The photon is absorbed upon arriving at the opposite end of the box, bringing the box' s motion to rest.
- Conservation of the center of mass yields the relation between the photon' s corresponding "energy" and the box' s "mass deficit":

$$E = mc^2$$

This derivation presupposes energy as an independent physical reality based on the precondition $E = h\nu$, and subsequently defines mass with this result. The two physical concepts stem from separate theoretical premises and fail to achieve full self-consistent matching.

3. Expansion from relativistic total energy to classical kinetic energy: $E = \gamma mc^2$

In special relativity, the total energy of massive particles is derived from four-momentum:

$$E = \gamma mc^2$$

For the limit $v \ll c$, Taylor expansion is performed on the Lorentz factor:

$$\gamma mc^2 = mc^2 + \frac{1}{2}mv^2 + \dots$$

Conventional theories interpret the first term mc^2 as rest energy reflecting mass-energy equivalence and the second term $\frac{1}{2}mv^2$ as classical kinetic energy, hence rewriting $E = \gamma mc^2$ in the form "total energy = rest energy + kinetic energy".

Within the present theoretical framework, all decay processes can be calculated directly via momentum conservation:

$$\sum \vec{p}_{\text{initial}} = \sum \vec{p}_{\text{final}}$$

Mass conservation is implicitly contained within the conservation of the total count of Momentum unit, removing the necessity to separately introduce energy conservation, mass-energy conversion, or covariant correction factors.

Summary Table for the Origins of Relativistic Energy

Origin Source	Formula	Relevant Description
Photon momentum	$E = pc$	The energy concept comes from electromagnetic theory, with no rigorous derivation path connecting electromagnetic theory to the energy definition
Mass-energy conversion	$E = mc^2$	Relies on the preset relation $E = h\nu$; the definitions of mass and energy derive from distinct theoretical premises without consistent matching between the two quantities

Classical kinetic energy	$\frac{1}{2}mv^2$	Derived from classical mechanics, separated from mc^2 by independent theoretical frameworks without conceptual unification
Combined expression	$E = \gamma mc^2$	Direct combination of concepts from disparate theoretical systems lacking intermediate derivation links and consistent conceptual adaptation
Quadratic relation	$E^2 = (pc)^2 + (mc^2)^2$	Further combination of energy definitions from different origins without complete conceptual integration and adaptation

4. Core collation based on the present framework

- Fundamental core physical quantities: momentum $p = mv$, and the Space-time state shaping capacity denoted as $\mathcal{S} = mc^2$.
- The expression $E = pc$ serves as an equivalent mathematical representation of photon momentum originating from derivations concerning photon momentum.
- The term mc^2 interpreted as rest energy in conventional theories essentially describes Space-time state shaping capacity instead of the standard definition of energy, arising from research discussions around mass-energy conversion.
- The classical kinetic energy expression $\frac{1}{2}mv^2$ corresponds to the result of compound integral of momentum under the gradient of Perceptual Cross-section, belonging to the scope of classical mechanics.
- These three mathematical expressions are rooted in separate theoretical foundations with distinct conceptual bases, which forbids forced unification into the combined formulas $E = \gamma mc^2$ or $E^2 = (pc)^2 + (mc^2)^2$.

5. Summary of logical gaps within relativistic energy formulation

Formulas describing relativistic energy are developed from three independent conceptual branches including the photon-based $E = pc$, mass-energy relation $E = mc^2$, and classical kinetic energy $\frac{1}{2}mv^2$, with insufficient intermediate logical connections across these branches. The present framework separates these conventional energy expressions into three independent physical quantities: momentum $p = mv$, Space-time state shaping capacity $\mathcal{S} = mc^2$, and the compound integral term $\frac{1}{2}mv^2$. The three quantities differ in physical origin and applicable physical scenarios and cannot be uniformly categorized under the single physical definition of energy.

6. Restatement of mass-energy conversion and the intrinsic nature of relativistic energy

a. Intrinsic nature of mass-energy conversion: Cancellation state $\leftrightarrow$ Breaking state

Within the present theoretical system, the conventional physical process named mass-energy conversion corresponds to mutual transformation between the Cancellation state and Breaking state of internal particle momentum. Macroscopic rest condition (Cancellation state): internal Momentum unit inside

particles distribute along random spatial orientations such that the net macroscopic momentum equals zero; Macroscopic moving condition (Breaking state): overall directional alignment of internal Momentum unit generates nonzero net macroscopic momentum. Throughout the transformation procedure, the total count of Momentum unit denoted as N remains invariant, and total mass $m = Nm_0$ is conserved, with only the spatial directional distribution of internal Momentum unit undergoing modification.

b. Physical positioning of mc^2 : Space-time state shaping equation

The quantity mc^2 does not represent energy but constitutes a Space-time state shaping equation. It quantifies the Intrinsic capacity of matter with mass m to evolve at the intrinsic driving speed c and fundamental intrinsic scale l_0 , and measures the cumulative perceptual effect of matter' s space-time response under variable ambient conditions.

The quantities mv counted along temporal dimension, mv counted along spatial dimension, and the combined spacetime product mvv where v stands for generic velocity-related state parameters all qualify as individual state variables, whereas mcc exclusively describes Space-time state shaping capacity. This distinction originates from the fact that either mcc or mvv corresponds to multi-fold statistical superposition over single Evolution carrier m_0 instead of an elementary state variable. Such composite statistical quantities fail to characterize intrinsic motion status, quantify the dynamical cost of Vector Composition, or satisfy linear conservation rules, which prohibits direct multiplication or summation of state parameters obtained from different observational perspectives to define a carrier' s intrinsic evolution capacity.

c. Energy representation under inertial motion

Uniform inertial motion cannot support secondary mathematical integration, and its apparent conventional energy value essentially reduces to momentum $p = mv$ with the limiting intrinsic form mc , rather than mc^2 or $\frac{1}{2}mv^2$. The term $\frac{1}{2}mv^2$ is derived from compound integration of momentum across Perceptual Cross-section gradients and only holds valid for specific physical scenarios featuring continuous accelerated motion.

d. Divergence and intrinsic cutoff mechanism

The conventional relativistic divergence result predicting infinite effective mass for particles accelerated toward the speed of light can be reinterpreted within this framework. Particle acceleration corresponds to regulated directional rearrangement of internal Momentum unit; the Intrinsic evolution capacity of individual particles stays fixed at mc , and macroscopic locomotion capacity is determined via Vector Composition of internal and external momentum components. A natural upper bound exists for the Orientation Ratio of aligned internal momentum components to eliminate unphysical divergence, with detailed derivation deferred to subsequent chapters.

10.5 Summary of Inconsistencies from Independent Definition of Force and Energy

1. Lack of universal applicability: conventional definitions of force and energy cannot consistently characterize static rest state, Cancellation state, particle decay, and instantaneous collision processes.
2. Divergence in mechanics: For example, under special relativity, accelerating a small mass to the speed of light $\Rightarrow$ infinite mass $\Rightarrow$ infinite momentum $\Rightarrow$ infinite resultant capacity —a logical contradiction.

3. Covariance limitations: defining mcc as energy requires artificial introduction of Lorentz factors for additional covariant modification.
4. Non-independent physical existence: force and energy cannot exist separately from physical motion; phenomena such as thermal conduction originate fundamentally from intermolecular Vector Composition.
5. Conceptual virtualization: ambiguous separation between mass and energy leads to virtualized and probabilistic interpretation of the mass concept.
6. Insufficient descriptive capacity: conventional force-energy formalism struggles to describe relational evolution, physical state transition and continuous causal development, relying on auxiliary field models to account for physical excitation and momentum cancellation.
7. Inconsistency with linear conservation rules: the nonlinear growth form of kinetic energy $\frac{1}{2}mv^2$ deviates from empirically observed linear scaling rules for frictional resistance and energy consumption.
8. Energy is not conserved in the fundamental sense: Consider two equal-magnitude opposite-direction momenta canceling each other. Momentum, being a vector, conserves as zero. Energy, being a scalar, cannot cancel—the total energy appears to "disappear" even though no external work is done. This reveals that energy is not an independent conserved quantity; rather, momentum is the fundamental conserved entity, and energy is merely its statistical projection. When momentum cancels internally, the projection naturally vanishes without violating any true conservation law.

10.6 Conclusions of This Chapter

Force emerges from the generation of motion capacity, namely Momentum Deviation, while energy emerges from Cancellation-Breaking transition or interval-based integral statistics of Momentum Deviation. Neither force nor energy can exist as independent fundamental physical quantities; at any instantaneous moment, force, momentum and energy represent distinct statistical expressions originating exclusively from momentum. Mass-energy conversion arises from the Cancellation-Breaking mechanism, and the universal conservation law reduces to momentum conservation, which is equivalent to the conservation of Space-time state shaping capacity.

That is, within a relational state evolution system, state evolution itself constitutes the direct mechanical reality (the ontology), while the integral of causal interaction processes represents merely a statistical manifestation (the projection) under specific conditions and particular methodologies, belonging to indirect mechanical influence.

1. **Relations among force, energy and momentum:** All three physical quantities emerge from motion capacity and Vector Composition capacity. Force equals the incremental quantity of momentum generated during single pairwise physical interaction, interpreted as the Breaking magnitude from the perspective of internal particle structure. Force emerges from the particle Encapsulation mechanism, which determines the magnitude of each single interaction. Conventional force is defined via inverse time averaging of infinitesimal incremental momentum, while conventional energy is defined as the full-interval integral of total momentum change. Energy, in turn, is determined by the magnitude of interaction (i.e., the number of interactions) governed by the fundamental perceptual attenuation mechanism. Consequently, classical energy is only applicable to the integration

of long-range forces and cannot adequately account for dissipative forces, velocity composition, or inertial processes. At any instantaneous moment, only inertial motion exists, rather than persistent emergent quantities derived from infinite continuous integration. This analysis yields a testable prediction: particle deflection is always accompanied by an underlying causal physical interaction.

2. **Correlation of motion, inertia and mechanics:** Within a single-step state evolution system, evolution capacity can be mathematically defined as energy; within an inertial state evolution system, evolution capacity is described as inertia or momentum. The equivalence relation Motion potential = Inertia potential unifies five types of potential terms including resistance potential, Attraction Potential, thrust potential, Tension Potential and Repulsion Potential. Mechanical interactions between particles arise from the toggle of internal Momentum unit spatial distribution to generate opposite directional evolution, mutually oriented evolution or interparticle Vector Composition capacity.
3. **Dynamical force and space-time state quantities:** Intrinsic temporal or spatial state quantities for dynamical inertial motion equal the product of Evolution carrier count and intrinsic motion capacity, written as mc or mv . Macroscopically averaged space-time state quantities are calculated via approximate statistics over medium density and structural distribution, while the dynamical space-time state descriptors take the form mc^2 or mv^2 . The relativistic mass-energy equation accordingly functions as a Space-time state shaping equation instead of a fundamental energy equation.
4. **Distinction between dimensional classification and statistical perspectives:**

Problem statement: Complex relational state evolution systems allow physical observation and quantitative statistics from multiple independent perspectives, but statistical results from different viewpoints cannot be arbitrarily assigned into separate Dimension classification. The underlying intrinsic evolution relation and Eigenstate characteristics must first be identified; statistical quantities obtained under narrow observational perspectives can only carry corresponding dimensional labels valid within their localized applicable range.

Historical overview: Physical debates spanning from the 17th-century momentum-energy dispute between Descartes and Leibniz, Newton's definition of mechanical work, Fourier's dimensional generalization in the 18th century, to Thomas Young's practical dimensional applications mark progressive development of physical cognition. Proper dimensional classification requires discrimination of Emergence relations among fundamental physical quantities, and integral mathematics is necessary to clarify emergent links between momentum, force and energy[18].

Reduction of redundant dimensions from generalized perspective: Statistical evaluation confined within a narrow observational scope requires generalized dimensional calibration or localized naming conventions. As an illustrative case for iron impacting wood, force, impulse and pressure all correspond to incremental inertial deviation or Momentum Deviation under fixed contact area and fixed substance quantity from the wooden target's reference frame, with the collision Vector Composition process equivalent to the action of applied force. Introduction of temporal dependence serves to differentiate total integrated force and unit-time averaged force, both of which resolve to cumulative summation of discrete elementary momentum variations at the microscopic scale.

5. **Upper bound of force:** The maximum attainable magnitude of force coincides with the upper bound of momentum, meaning the maximal instantaneous mechanical coupling capacity between a single particle and surrounding environment equals mc . This limit further constrains the upper evolution capacity and Vector Composition capacity along arbitrary spatial directions and applies universally to physical objects ranging from photons to black holes, naturally eliminating theoretical divergence of force and energy as well as conflicts related to space-time covariance.

10.7 Commentary on Chapter References

[18] Tho, T. (2017). *Vis Vim Vi: Declinations of Force in Leibniz's Dynamics*. Dordrecht: Springer. (Studies in History and Philosophy of Science, Vol. 46).

Leibniz opposed Newton's extrinsic applied force hypothesis and proposed force as an inherent intrinsic trend of object state evolution named *vis insita*, where force precedes physical motion as an internal origin rather than externally imposed quantity. Based on the present evolution framework, comments on four pioneering physicists are listed as follows: Newton correctly established the inertial rule yet erroneously defined force as external action; Leibniz properly argued against externally sourced force but rejected the conservation of momentum; Descartes contributed partial foundational ideas of inertia and momentum while denying vector properties of motion; Huygens successfully formulated momentum conservation yet introduced unphysical scalar-based energy descriptions. Mass conservation is implicitly included within fixed total Momentum unit count without requiring independent energy conservation, mass-energy transformation or covariant correction terms. The root of these historical theoretical disagreements originates from Earth's non-inertial ambient interaction environment, where all macroscopic mechanical costs derive from continuous integral interactions against gravitational resistance. Gravitational pull enforces persistent contact between moving bodies and frictional surfaces to generate non-inertial motion requiring continuous energy input, and gravitational acceleration itself constitutes an integral physical process. Combined, these two effects transform the fundamental momentum-based interaction into the macroscopic statistical definition $\text{Work} = \text{Force} \times \text{Distance}$. Nearly all terrestrial macroscopic motion belongs to composite coupled motion instead of independent inertial locomotion. Under simplified Vector Composition environments, isolated particle motion or fragmentation events, momentum remains the sole fundamental descriptor at any instantaneous state. Hence classification between pure inertial motion and composite integral-based motion becomes essential.

[25] Einstein, A. (1907). Über das Relativitätsprinzip und die aus demselben gezogenen Folgerungen. *Jahrbuch der Radioaktivität und Elektronik*, 4, 411–462. <https://doi.org/10.1002/andp.19073271002>

The equivalence principle establishes inherent equivalence between physical motion and force consistent with the core underlying logic of the present evolution theory.

[28] Steinhauer, J. (2016). Observation of quantum Hawking radiation and its entanglement in an analogue black hole. *Nature Physics*, 12, 959–965.

Experimental detection of black hole thermal fluctuation matches the statistical gravitational description of this framework.

[50] Okun, L. B. (2006). The concept of mass in the Einstein year. In *Particle Physics at the Year of 250th Anniversary of Moscow University* (pp. 1–15). Singapore: World Scientific. https://doi.org/10.1142/9789812772657_0001

This work clarifies the definition of rest mass and supports the interpretation of mass as total internal Momentum unit count inside particles, treating c^2 merely as a dimensional conversion factor consistent with the dimensional resetting theory of the present paper.

[55] Klein, O. (1926). Quantentheorie und fünfdimensionale Relativitätstheorie. *Zeitschrift für Physik*, 37(12), 895–906. <https://doi.org/10.1007/BF01397195>

Klein's quantization of higher-dimensional spacetime links geometric structure to fundamental quantum constants and aligns with the discrete evolution logic of Momentum unit adopted in this paper.

[57] Liénard, A. (1898). Champ électrique et magnétique produit par une charge électrique concentrée en un point et animée d'un mouvement quelconque. *L'Éclairage Électrique*, 16, 5–14, 53–59, 106–

112.

Relativistic radiation formulas predict sharp radiation power growth for relativistic particles, supporting the physical picture of violent Momentum Deviation Unloading when high-speed particles resist forced deflection and verifying the upper interaction limit mc for mechanical force.

[58] Jacobson, T. (1995). Thermodynamics of spacetime: The Einstein equation of state. *Physical Review Letters*, 75(7), 1260–1263. <https://doi.org/10.1103/PhysRevLett.75.1260>

The derivation proving Einstein field equations correspond to spacetime thermodynamic equations of state supports the unified description of gravity as dynamical state evolution instead of fixed background geometry and confirms that reducing spacetime and gravity to information and thermodynamic evolution constitutes a mainstream research direction in contemporary theoretical physics.

[59] Hecht, E. (2011). How Einstein confirmed $E_0 = mc^2$. *American Journal of Physics*, 79(6), 591–600. <https://doi.org/10.1119/1.3549223>

Historical analysis validates mass-energy relation as a consequence of intrinsic conserved particle properties, confirming stored total evolution amplitude from fully isotropic Cancellation state of internal particle momentum flow and intrinsic particle driving capacity fixed at c .

[60] ATLAS Collaboration. (2012). Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC. *Physics Letters B*, 716(1), 1-29. <https://doi.org/10.1016/j.physletb.2012.08.020>

Particle decay measurements from collider experiments verify strict momentum conservation; unlike energy which suffers unphysical divergence and ambiguous state characterization during evolution, momentum enables complete measurable tracking of decay evolution.

[61] ACME Collaboration; Baron, J., Campbell, W. C., DeMille, D., et al. (2014). Order of magnitude smaller limit on the electric dipole moment of the electron. *Science*, 343(6168), 269–272. <https://doi.org/10.1126/science.1248213>

High-precision experimental confirmation of perfect spherical symmetry for the electron provides critical experimental evidence for the core hypothesis treating electrons as spherically encapsulated Cancellation state assemblies of Momentum unit and supports the existence of large internal intrinsic momentum flow quantified as mc .

[62] Einstein, A. (1905). Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig? *Annalen der Physik*, 18(13), 639–641. <https://doi.org/10.1002/andp.19053220817>

Original source publication for the mass-energy relation $E = mc^2$ establishing fundamental mass conservation. Within this paper the formula is reinterpreted as a Space-time state shaping equation preserving mass conservation, where every unit of mass carries intrinsic dynamical evolution capacity c . Consistent with Einstein's original derivation assigning photon energy as mcc and inverse relation $m = E/c^2$, photons possess finite mass and fully inherit fermion constituent Momentum unit from preceding particle annihilation, leading to the universal conclusion that all particles are constructed from discrete Momentum unit.

[63] Einstein, A., Podolsky, B., & Rosen, N. (1935). Can quantum-mechanical description of physical reality be considered complete? *Physical Review*, 47(10), 777–780. <https://doi.org/10.1103/PhysRev.47.777>

The EPR paradox highlights strict positional-momentum correlation and implied incompleteness of standard quantum mechanical formulation, setting a completion target for the Momentum allocation Protocol proposed in this framework and serving as the foundational reference for hidden-variable physical logic.

Subsequent experimental verification of quantum entanglement confirms conservation of angular momentum and Evolution carrier count; combined with universal momentum conservation in particle decay and the speed-of-light upper bound for momentum propagation, momentum outperforms energy in universality, divergence immunity and experimental measurability. Physically, the universe constrains three fundamental evolution parameters including total Evolution carrier count, intrinsic Driving Capacity and angular momentum allocation rules entirely via momentum-related laws with no requirement for independently existing energy as a fundamental quantity.

11 Verification via Global Update and Interaction of the Universe

According to the global coordination mechanism in Section 5.6 and the state-interaction synchronization conclusion in Section 5.8: for a generalized relational state evolution system to maintain stable evolution, the state evolution frequency and perceptual interaction frequency must be strictly synchronized. If the evolution frequency is lower than the interaction frequency, causal redundancy emerges within the system; if the evolution frequency exceeds the interaction frequency, causal loss takes place.

The conservation-driven rate inside the cosmic system equals the evolution frequency c , hence the perceptual interaction frequency is also equal to c . The minimum instantaneous interaction interval of the system corresponds to the fundamental minimum time unit t_0 , satisfying the relation:

$$c = \frac{l_0}{t_0}$$

Subject to this synchronization constraint, long-range non-dissipative forces without a global low-latency synchronous update mechanism trigger causal conflicts, resultant force failure, polygonal orbital trajectories and tidal effects. This chapter verifies the rationality of the global update mechanism through mathematical derivation of two-body and multi-body systems, simulation experiments and astronomical observations, clarifying the core function of t_0 as the unified update step, without introducing the conventional hypothesis of "force propagator" throughout the full text.

11.1 Mathematical Derivation of Two-Body Delay Relation and Validation via Experimental Simulation

A mathematical model is built for the Sun–Neptune two-body interaction system to quantify the correlation between interaction delay (update interval), orbital radial deviation, trajectory morphology and orbital eccentricity, laying a foundation for subsequent multi-body derivation. The "force propagator" assumed under conventional mechanics is essentially a microscopic side effect accompanying the cosmic discrete update process (analogous to vacuum fluctuation)[28], rather than a medium for force transmission.

All derivations and conclusions of this chapter rely on a two-body interaction simulation program: the total simulated duration is fixed at 20 seconds, and only the delay parameter τ is adjusted to observe orbital evolution and collect supporting experimental data.

1. Definition of Fundamental Variables

The Sun S is fixed at coordinate $(X_S, Y_S) = (300, 300)$, consistent with simulation program settings; the instantaneous state of Neptune N at time t is denoted as $N(t) = (x(t), y(t))$; R stands

for the target orbital radius (adjustable within the program), v_t denotes Neptune' s tangential velocity (adjustable within the program); the unified update interval (force interaction delay) is defined as $\tau = \text{dt_physics}$, adjustable from $5 \sim 2000$ ms, namely $0.005 \sim 2$ s; radial deviation $\Delta r(t)$ refers to the positional difference between Neptune' s actual position and the preset target orbital radius.

This subsection compares predicted deviations from the conventional propagator theory and the discrete update framework proposed in this work, and notes that the current simulation program has not yet fully realized the corresponding simulation content for the rule "larger orbital radius generates more prominent polygonal trajectory effect".

2. Derivation Procedures

Combined with simulation results obtained under fixed total simulation time of 20 s and variable τ , the correlation between interaction delay and orbital parameters is derived stepwise.

a. Inertial Motion Phase Within a single global update period $t \in [t_0, t_0 + \tau]$, Neptune performs uniform linear inertial motion following the positional update rule:

$$r(t + \Delta t) = r(t) + v_t \Delta t, \quad \Delta t \in [0, \tau]$$

Experimental observations: trajectories display distinct polygonal features under large update intervals (e.g., 842 ms); trajectories approach smooth curves when τ is extremely small (e.g., 5 ms), directly confirming the correlation between τ and orbital geometry.

b. Geometric Derivation of Radial Deviation Under the discrete update mechanism, celestial bodies maintain linear inertial motion throughout interval τ . At update instant $t_0 + \tau$, the actual separation between Neptune and the Sun is derived via the Pythagorean theorem (R denotes orbital radius at the prior update, $v_t \tau$ represents inertial displacement):

$$r_{\text{actual}} = \sqrt{R^2 + (v_t \tau)^2}$$

Radial deviation equals the difference between actual distance and target orbital radius:

$$\Delta r = r_{\text{actual}} - R = \sqrt{R^2 + (v_t \tau)^2} - R$$

c. Simplification via Taylor Expansion The approximation $\sqrt{1+x} \approx 1 + \frac{x}{2}$ is adopted for $x = \left(\frac{v_t \tau}{R}\right)^2 \ll 1$ for algebraic simplification:

$$\Delta r = R \left(\sqrt{1 + \left(\frac{v_t \tau}{R}\right)^2} - 1 \right) \approx R \cdot \frac{1}{2} \left(\frac{v_t \tau}{R}\right)^2 = \frac{(v_t \tau)^2}{2R}$$

The final radial deviation formula reads:

$$\Delta r = \frac{(v_t \tau)^2}{2R}$$

This expression indicates radial deviation Δr is proportional to the square of tangential speed v_t and the square of update interval τ , and inversely proportional to orbital radius R .

3. Conclusions from Derivation

Delay-induced Polygonal Trajectory Effect Within two-body systems, larger update interval τ yields more pronounced polygonal orbital features. Simulation outputs verify distinct polygonal orbits at $\tau = 842$ ms and nearly smooth trajectories at $\tau = 5$ ms, matching analytical derivation. The polygonal effect originates from the incapability of discrete updates to generate instant synchronized resultant force; trajectories converge toward the continuous classical limit as $\tau \rightarrow 0$.

Delay-induced Eccentricity Effect Orbital eccentricity positively correlates with update interval τ : larger τ raises eccentricity and weakens system stability. Growing τ postpones positional error correction and gradually drives orbits away from ideal circular geometry, consistent with recorded simulation phenomena.

4. Cross Comparison of Two Mechanisms and Observational Validation

- **Discrete Global Update Mechanism (Present Framework)** The derived radial deviation is $\Delta r_{\text{refresh}} = \frac{(v_t \tau)^2}{2R}$. Setting the fundamental update step equal to the elementary time quantum $\tau = t_0 \approx 5.4 \times 10^{-44}$ s, the calculated radial deviation satisfies $\Delta r_{\text{refresh}} \sim 10^{-70}$ m, a magnitude far smaller than the fundamental spatial unit l_0 , resulting in macroscopically smooth continuous orbits consistent with astronomical measurements.
- **Conventional Propagator Assumption (Light-speed Force Propagation)** Under the hypothesis of light-speed gravitational transmission via propagators, Neptune's inertial drift distance calculates to $L_{\text{drift}} \approx 8.1 \times 10^4$ km. Such massive deviation exceeds observational precision bounds and is never detected via astronomical measurement, disproving the propagator postulate.

Supporting literature includes the Wheeler–Feynman direct-action theory without field intermediaries[30] and Laplace's canonical discussions[31], both consistent with the discrete global update mechanism and eliminating the necessity of gravitational force propagators in cosmic physics.

11.2 Unstable Resultant Force from Asynchronous Delayed Interaction in Multi-Body Systems

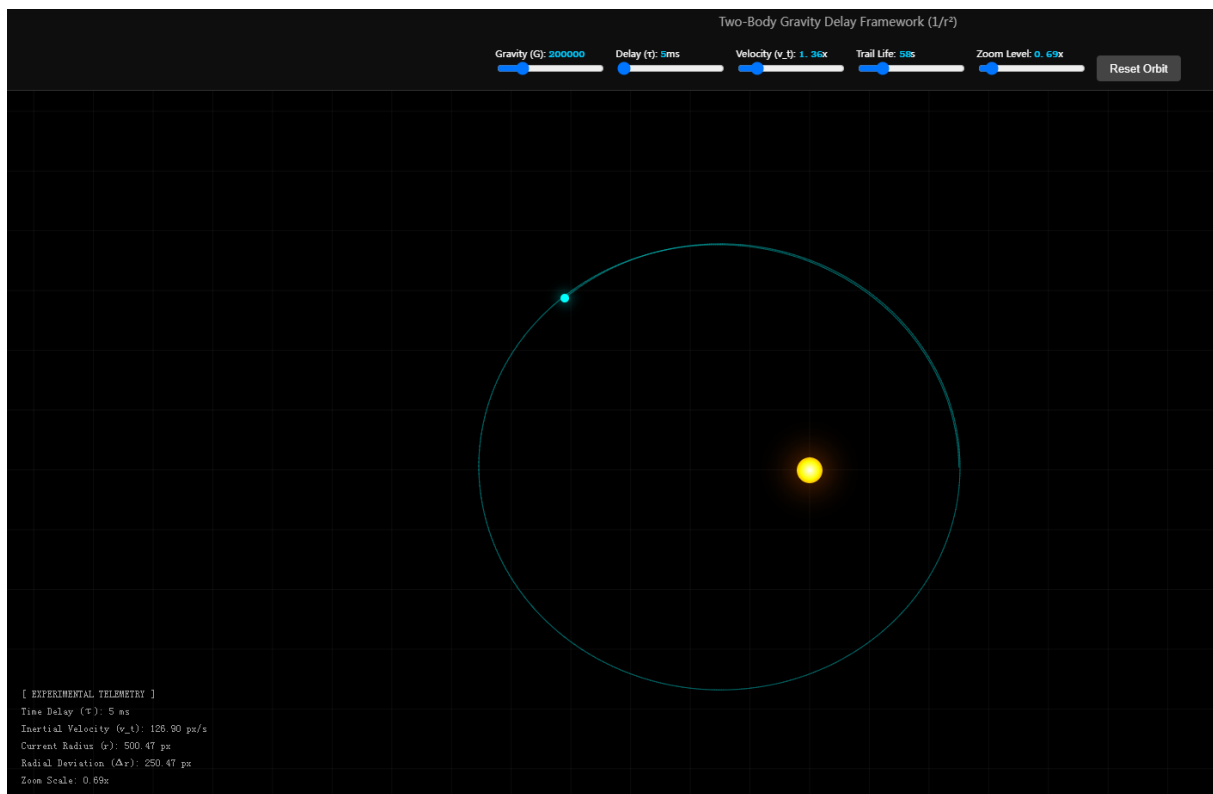
Stable evolution of multi-body systems relies on the validity of the resultant force; interaction delay undermines its foundational prerequisite and leads to uncontrollable orbital deviation. Based on the revised classical mechanical definitions, this subsection derives the damaging effect of interaction delay on the resultant force using a three-body model.

1. Fundamental Definitions (Revised Classical Mechanics Framework)

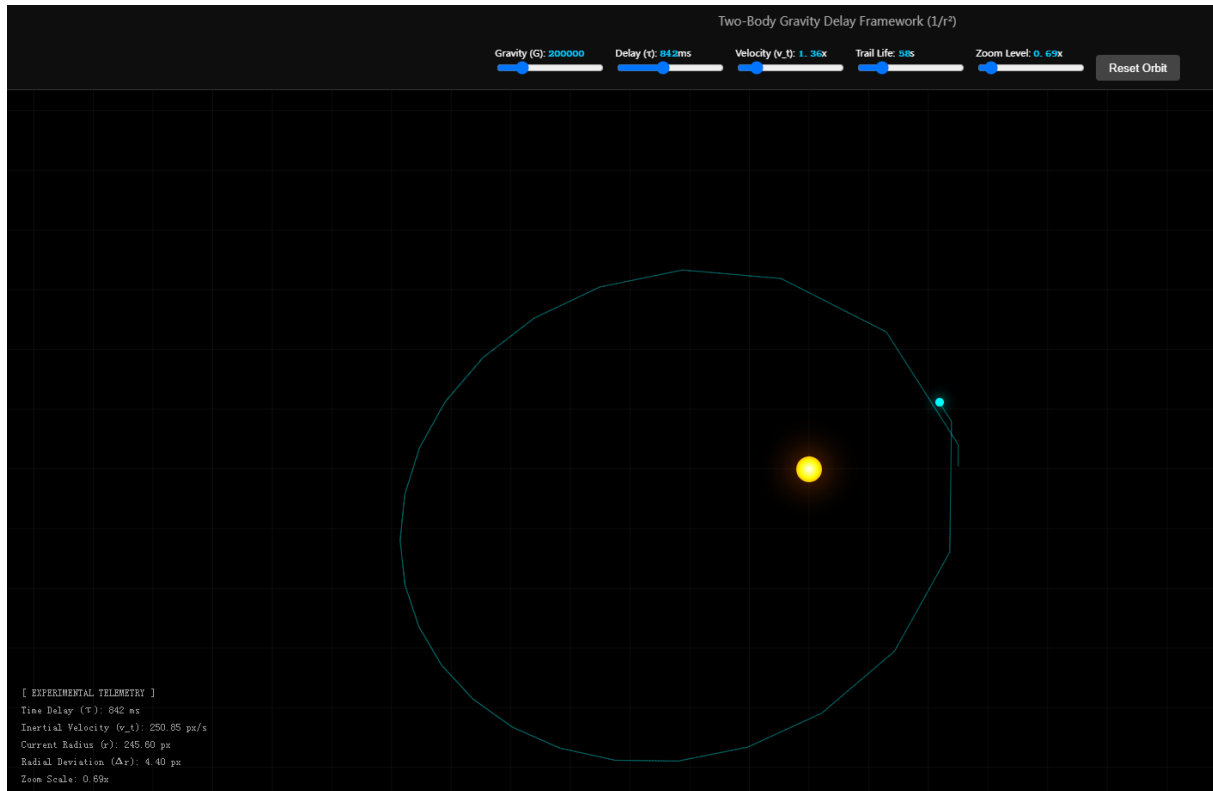
a. Unified Global Update Step τ All celestial objects within a multi-body system share the identical force interaction interval τ . After every duration τ , the system synchronously generates a full-system state snapshot, executes parallel resultant force computation, and synchronously updates all celestial states. Core requirements: uniform global interaction frequency and synchronized state evolution across the entire system.

b. Celestial Body State The instantaneous state of an arbitrary celestial body M_i at time t is defined as

$$S_i(t) = (x_i(t), y_i(t), \vec{v}_i(t))$$



(a) $\tau = 5$ ms: smooth orbital curve



(b) $\tau = 842$ ms: polygonal orbital trajectory

(Simulation of Sun–Neptune two-body system: orbital morphology contrast under distinct interaction delays)

representing the frozen snapshot status of M_i recorded during global synchronization.

c. Instantaneous Resultant Force Instantaneous resultant force refers primarily to the situation under gravitational or local electromagnetic forces where, due to other massive numbers of particles (ensembles of encapsulated Evolution carriers) leaving their original positions based on evolutionary relations, the particles originally coupled with them must continuously refresh to follow the coupling relations established by the original perceptual cross-section. This thus requires an infinitesimal interval approximating t_0 , with a macroscopic refresh frequency of c_0 .

Prominent examples of non-instantaneous resultant force are seismic waves, water waves, and gravitational tides, which transmit mechanical relations via macroscopic velocities. Sudden gravitational acceleration of a celestial body causes all fluid, sandy, soil-like, and other softly electromagnetically connected materials to exhibit trailing effects.

The resultant force $\vec{F}_i(t)$ on celestial body M_i at time t equals the vector sum of the forces from all other bodies evaluated at the identical time:

$$\vec{F}_i(t) = \sum_{j \neq i} \vec{F}_{ji}(t)$$

where $\vec{F}_{ji}(t)$ denotes the force exerted by body M_j on M_i at the common timestamp t . The core prerequisite for a valid resultant force is that all component forces share a unified time baseline, independent of any propagating intermediaries. Microscopic quantum nonlocality experiments supply empirical support for such synchronous interaction.

2. Derivation of Resultant Force Degradation Induced by Interaction Delay

Assume the absence of a universal global update rule, with unequal individual update intervals $\tau_A \neq \tau_B \neq \tau_C$ for separate celestial bodies, leading to misaligned reference timestamps for component forces as derived below:

a. When body A computes its received force $\vec{F}_A(t)$, it accesses delayed historical states of B and C :

$$\vec{F}_A(t) = \sum_{j=B,C} \vec{F}_{jA}(t - \tau_A)$$

b. When body B computes its received force $\vec{F}_B(t)$, it accesses delayed historical states of A and C :

$$\vec{F}_B(t) = \sum_{j=A,C} \vec{F}_{jB}(t - \tau_B)$$

c. When body C computes its received force $\vec{F}_C(t)$, it accesses delayed historical states of A and B :

$$\vec{F}_C(t) = \sum_{j=A,B} \vec{F}_{jC}(t - \tau_C)$$

Under such conditions, the resultant force $\vec{F}_i(t)$ becomes a hybrid vector sum of pairwise interactions sampled from disparate timestamps, violating the fundamental premise of an instantaneous resultant force and leading to unpredictable orbital deviations. Related research by Muller[29] confirms that without compensatory mechanisms, multi-body gravitational systems lose angular momentum conservation and cannot sustain long-term orbital stability.

As a result of asynchronous multi-body interaction, a stable full-scale long-range force velocity composition effect (resultant force) cannot be established, manifesting macroscopically as orbital drift. For the origin of long-range force velocity composition, see the derivation in Chapter 20.

11.3 Derivation of Causal Contradictions within Multi-Body Configurations

At extreme interbody separations, finite propagator flight delay expands drastically and readily induces reversed causal ordering. Illustrated with a three-body setup: bodies A – B and B – C hypothetically emit propagators at instant t_0 that modify system states at t_1 and t_3 respectively. If A and C interact at intermediate time t_2 , their mutual coupling may depend on future event t_3 ($t_3 > t_2$), generating the paradox of future events determining prior physical states:

$$S_C(t_2) = h[S_B(t_0), S_A(t_0) | t_3 > t_2]$$

Van Flandern explicitly demonstrates finite light-speed gravitational propagation would drive complete solar system disintegration within several thousand years[34], requiring instantaneous gravitational coupling and establishing synchronous global updating as the core resolution for causal inconsistency.

11.4 Validation under Extreme Super-Long Separation

At cosmological super-large separations on the order of millions of light-years, deviations originating from interaction delay amplify drastically and destabilize orbital architecture, conflicting with observational evidence of abundant stable multi-body cosmic structures. This subsection mathematically characterizes the evolutionary scaling of accumulated deviation.

1. Definition of Core Parameters

Let interbody separation equal L , corresponding interaction delay $\tau_L = \frac{L}{c}$ (c denotes vacuum light speed); k represents single-step deviation coefficient, n total interaction cycles and ε per-cycle deviation amplification factor.

2. Two-Body Deviation Derivation for Cosmic Distances

Adopting the two-body deviation formula derived in Section 11.1 $\Delta r = k v_t \tau$ and substituting $\tau_L = \frac{L}{c}$ yields:

$$\Delta r(L) = k \cdot v_t \cdot \frac{L}{c}$$

Deviation magnitude grows linearly with separation L . Though Ritz' s emission theory never attained mainstream consensus, it independently implies finite propagation delay inevitably introduces measurable orbital offset consistent with the present derivation.

3. Three-Body Deviation Derivation for Cosmic Distances

Deviations propagate in cascaded exponential fashion across three-body constituents:

$$\begin{aligned}\Delta r_{AB} &= k \cdot v_A \cdot \frac{L_{AB}}{c} \\ \Delta r_{BC} &= k \cdot (v_B + \Delta v_B) \cdot \frac{L_{BC}}{c} \quad (\Delta v_B \text{ induced by } \Delta r_{AB}) \\ \Delta r_{CA} &= k \cdot (v_C + \Delta v_C) \cdot \frac{L_{CA}}{c} \quad (\Delta v_C \text{ induced by } \Delta r_{BC})\end{aligned}$$

Total accumulated system deviation simplifies to:

$$\Delta r_{\text{total}} \approx k \cdot v \cdot \frac{L_{\text{total}}}{c} \cdot (1 + \varepsilon)^n$$

At million-light-year separations, factor $(1 + \varepsilon)^n$ grows exponentially such that $\Delta r_{\text{total}} \propto e^{\alpha L}$ ($\alpha > 0$) and eventually collapses system stability. Supplementary research from Carlip corroborates uncompensated delayed propagation leads to inevitable gravitational system breakdown.

4. Support from Cosmological Astronomical Phenomena

The discrete global update mechanism predicts long-term accumulation of tiny single-update residuals produces observable coherent bulk motion across ultra-large cosmic scales, matching multiple recent astronomical findings as summarized below:

a. Cosmic Bulk Flow On scales spanning $100 \sim 250 h^{-1} \text{Mpc}$, collective galactic streaming velocity exceeds predictions from standard ΛCDM cosmology; such velocity excess originates from cumulative discrete update residuals.

b. Cosmic Dipole Anomaly Dipole amplitude derived from radio-galaxy number counts surpasses kinematic CMB dipole theoretical expectation, with anomalous signal contributions from intrinsic anisotropy accumulated via discrete updating.

c. Cosmic Preferred Axis Low-order multipole moments of the cosmic microwave background exhibit statistically significant directional alignment, interpretable as directional preference emerging from weak statistical fluctuations of periodic discrete updates.

d. Coherent Spin of Galaxy Clusters 2025 observational data from Shanghai Astronomical Observatory confirms globally correlated spin orientation across large-scale galaxy clusters, providing direct astronomical evidence supporting the synchronous unloading mechanism proposed in this work.

11.5 Acceleration Modulation of the Broken-Line Effect

The above geometric derivation gives the radial deviation within a single refresh cycle. However, in actual orbital evolution, the tangential velocity v_t of the celestial body is not constant. Near the perihelion, the gravitational acceleration $a = GM/R^2$ increases, the body is continuously accelerated, and v_t rises significantly, which modifies the prediction —i.e., the broken-line effect at the perihelion is correspondingly enhanced.

The radial deviation formula $\Delta r = (v_t \tau)^2 / (2R)$ holds when v_t is approximately constant. When the distance between celestial bodies is large (assuming gravitational interaction mediated by light-speed propagators), the formula is more applicable; in short-range interactions (e.g., near the perihelion), where the gravitational acceleration becomes significant and v_t continuously increases, the broken-line effect is stronger and requires an acceleration modulation term as a correction.

Universality and Regime Distinction of the Broken-Line Effect

It should be emphasized that the broken-line effect itself is an inevitable consequence of the "propagator" hypothesis and exists under both large-distance and small-distance conditions, though the dominant factors differ:

- **Large distances (aphelion, wide-separation binary systems):** v_t is approximately constant, the broken-line effect is directly governed by $\Delta r = (v_t \tau)^2 / (2R)$, and the formula predictions are in good agreement with actual observations.
- **Small distances (perihelion, strong gravity regions):** acceleration modulation is significant, v_t continuously increases, the broken-line effect is stronger, and a correction term must be superimposed on the formula.

In both cases, the broken-line effect originates from the same underlying mechanism (assuming a propagator that travels through space to transmit force), differing only in its apparent manifestation due to acceleration modulation —yet the broken-line effect inevitably exists in both scenarios.

11.6 Validation via Fundamental Elementary Time Quantum Correction for Gravitational Delay

To resolve resultant force breakdown and causal inconsistency induced by propagation delay, a unified low-latency global update framework is introduced using elementary time quantum $\tau_0 = t_0 \approx 5.39 \times 10^{-44}$ s as the universal natural cosmic update step for stability verification.

1. Setup of Unified Global Update Framework

All celestial bodies adopt identical update period: $\tau_A = \tau_B = \tau_C = t_0$. After every t_0 duration, the system synchronously generates full-system snapshots, executes resultant force calculation and updates all physical states, ensuring all component forces reference identical timestamps and fully self-consistent causal chains.

2. Mathematical Stability Verification

a. Total Cumulative Interaction Delay Aggregate three-body update delay:

$$\sum \tau = 3t_0 \approx 1.62 \times 10^{-43} \text{ s}$$

The resulting delay magnitude is infinitesimal and negligible at macroscopic observational scales.

b. Deviation Evaluation at Extreme Cosmic Separation Substitution into the deviation relation gives:

$$\Delta r = k \cdot v \cdot L \cdot \frac{t_0}{c} \approx k \cdot v \cdot L \cdot 1.8 \times 10^{-52}$$

Final calculated macroscopic positional deviation satisfies $\Delta r \approx 10^{-30}$ m, well below measurable detection thresholds and sustaining excellent system stability.

Supplementary remark: the definition of "resultant force" within this text differs from conventional long-range force formalism. Long-range interactions fundamentally correspond to paired action-reaction momentum exchange satisfying

$$\Delta \vec{p}_A = -\Delta \vec{p}_B$$

jointly constrained by t_0 , Perceptual Cross-section and Encapsulation Rule rather than directly generating classical-style resultant force; the "resultant force" referenced herein denotes coordinated evolutionary effects emerging from synchronized multi-body coupling bounded by Perceptual Cross-section. New localized resultant force emerging during ongoing evolution falls under dissipative force classification, accompanied by synchronized translational drift of mass and Driving Capacity alongside Vector Composition events.

11.7 Intermediate Conclusions

1. Conserved Interactive Capacity

Stable cosmic evolution is governed by an extremely short universal interaction cycle t_0 , with global evolution update rate fixed at light speed c . Interaction frequency originates from fundamental underlying Driving Capacity (spatial evolution rate c); subsequent chapters derive the origin of relativistic dynamics built upon this foundation.

A full single interaction cycle t_0 sequentially covers environmental perception, causal correlation establishment, temporary borrowing and restitution of virtual momentum pairs, and toggle of internal Momentum unit spatial distribution configuration.

2. Distinction from Relativistic "Light-Speed Limit for Information Transmission"

A. Absence of Independent Information Conservation Law

No standalone fundamental information conservation principle exists within cosmic physics; base conserved quantities include Representational Quanton, Driving Capacity and intrinsic evolution rules. The widely discussed black hole information conservation problem is restated as Evolution resource conservation: the intrinsic state-representing capacity of black hole systems remains intact during spatial contraction.

B. Perceptual Interaction Rate $\neq$ Spatial Propagation Speed

The action-imposing speed associated with quantum entanglement and instantaneous long-range coupling constitutes a separate physical quantity distinct from the spatial translational propagation speed of massive particles.

Special relativity restricts only the spatial flight speed of massive substances below light speed to avoid causal inversion. The underlying logic dictates inconsistent maximum evolution bounds across disparate fundamental Driving Capacity inevitably produce causal breakdown within a unified state evolution system.

Traditional theories fail to distinguish between the spatial motion velocity of matter and the propagation velocity of force perception and quantum entanglement. This failure has given rise to conjectures related to causal paradoxes. In fact, the core requirement for a stable evolutionary system is the uniformity of its global driving capacity. Such stability does not necessarily depend on imposing a specific speed limit—such as the speed of light c —to regulate causality.

11.8 Chapter Final Conclusions

Combining mathematical derivation for two-body and multi-body systems, numerical simulation and astronomical observation, this chapter proves the existence of a universal cosmic global update mecha-

nism with core conclusions listed below:

1. **Existence of Global Update Mechanism:** the entire cosmos evolves under uniform discrete update steps set by elementary time quantum t_0 , with synchronized state evolution frequency and perceptual interaction frequency both equal to light speed c . This framework eliminates the requirement for conventional gravitational force propagator assumptions.
2. **Quantified Two-Body Delay Effect:** radial deviation relation $\Delta r = (v_t \tau)^2 / (2R)$ confirms deviation scales quadratically with tangential speed and update interval while varying inversely with orbital radius. Setting $\tau = t_0 \approx 5.4 \times 10^{-44}$ s yields $\Delta r \sim 10^{-70}$ m, producing macroscopically smooth continuous orbital trajectories consistent with observation.
3. **Falsification of Propagator Postulate:** light-speed gravitational transmission predicts Neptune's inertial drift near 8.1×10^4 km, an unobserved large-scale offset, demonstrating gravitational propagators do not exist as fundamental entities or emerge only from quantum fluctuation and dissipative Vector Composition mechanisms.
4. **Multi-Body Resultant Force Failure and Causal Paradox:** asynchronous updates misalign force timestamps and trigger future-to-past causal reversal; global synchronization at $\tau = t_0$ eliminates such pathologies and enables persistent orbital stability.
5. **Ultra-Large-Scale Deviation Accumulation and Astronomical Confirmation:** separation-dependent deviation grows exponentially $\Delta r_{\text{total}} \propto e^{\alpha L}$ under finite propagation speed; measured large-scale bulk flow, cosmic dipole anomaly and correlated galaxy-cluster spin serve as direct observational evidence for long-term cumulative discrete update residuals.

In summary, the global discrete update mechanism forms an indispensable prerequisite for stable cosmic evolution, valid across microscopic phenomena (quantum entanglement, instantaneous long-range coupling) and macroscopic observables (celestial orbits, large-scale cosmic structure), supplying foundational interaction rules for subsequent derivations covering relativistic dynamic origins, microscopic quantum measurement and large-scale cosmic coherence.

11.9 Commentary on Chapter References

[28] Steinhauer, J. (2016). Observation of quantum Hawking radiation and its entanglement in an analogue black hole. *Nature Physics*, 12, 959–965. <https://doi.org/10.1038/nphys3863>

Experimental detection of black hole thermal fluctuation matches statistical gravitational description, supporting the proposed evolution framework that long-range force arises from momentum pair borrowing and restitution driven by quantum fluctuation.

[29] Muller, R. A. (1982). Discussion: The impossibility of a finite propagation speed for gravity in a stable solar system. *American Journal of Physics*, 50(1), 1–2. <https://doi.org/10.1119/1.1268038>

Muller's analysis demonstrates finite-speed gravity without compensatory correction breaks angular momentum conservation and prevents long-term solar system orbital stability.

[30] Wheeler, J. A., & Feynman, R. P. (1949). Classical electrodynamics in terms of direct interparticle action. *Reviews of Modern Physics*, 21(3), 425–433. <https://doi.org/10.1103/RevModPhys.21.425>

Wheeler–Feynman direct-action formulation abandons field and propagator prerequisites for fundamental electromagnetic interaction.

[31] Laplace, P. S. (1825). *A Treatise on Celestial Mechanics* (Vol. 1–5). London: Chelsea Publishing. (Original work published 1799 as *Traité de Mécanique Céleste*).

Laplace first argued finite gravitational propagation speed introduces measurable lunar orbital longitude shift, constraining effective gravitational speed to millions of multiples of vacuum light speed.

[34] Van Flandern, T. (1998). The speed of gravity – What the experiments say. *Physics Letters A*, 250(1–3), 1–11. [https://doi.org/10.1016/S0375-9601\(98\)00650-1](https://doi.org/10.1016/S0375-9601(98)00650-1)

Van Flandern’ s analysis concludes finite-speed gravitational interaction causes cosmic orbital disintegration, consistent with the synchronous global update rule locked to fundamental frequency c .

12 Dynamical Origin of Relativity

12.1 Missing Logical Links between Relativistic Energy and Minkowski Space-time

Derived from the generalized relational state evolution system in preceding chapters, $E = mc^2$ is not an energy equation under conventional definition, but a Space-time state shaping equation. Interpreting this formula from the framework of classical energy requires additional covariant factors and brings about derivative problems including physical divergence and spacetime curvature.

From the intrinsic properties of evolution systems, four critical logical gaps exist within Minkowski spacetime theory, sorted as follows:

Problem	Supplementary Logical Content
Backgrounded spacetime	Spacetime is not a presupposed innate background structure. As long as a system possesses state representational quantities and dynamic evolutionary behaviors, spacetime relations can emerge naturally; spacetime is a derivative quantity of evolution[64].
Orthogonality of time and space	Since spacetime can emerge directly from static representational quantities and dynamic evolution within any state evolution system, the time dimension does not share the same geometric relation as space, and consequently there is no basis to assert an orthogonal relationship between them.
Spacetime complementarity	Time and space grow naturally in full synchrony and strict proportion based on state representational quantities and dynamic evolution. This synchrony always holds under light-speed-level evolution, and there is no conclusion that "time slows down with faster motion and flows fastest at rest"[17].
Conservation formalism	System conservation follows linear superposition $N = N_1 + N_2$, instead of the Pythagorean quadratic form. Conventional energy conservation produces divergence under curved spacetime, and spacetime metric in turn influences the driving process of system evolution[58].

Relying on the perceptual interaction mechanism of the proposed framework, this chapter elaborates the underlying dynamical origin of relativistic effects from the perspective of system evolution.

12.2 Emergent Form of Spacetime in Generalized Relational Inertial State Evolution System

1. Emergence of Space

Suppose a three-dimensional orthogonal inertial relational state evolution system contains m sets of Representational Quanton $l_0(x, y, z)$, the total occupied amount of absolute free dimension for the system reads:

$$m \cdot l_0(x, y, z)$$

According to previous deductions, the number of Evolution carriers can serve as a natural dimension. Under static conditions, spatial occupation is directly described by the carrier count N (with mass m serving as its macroscopic dimensional statistical quantity). For macroscopic spatial quantities with finite density, statistical characterization can be achieved through length L , area S , and volume V .

When the system enters motion and correlations build up among carriers, the total quantity consumed to acquire mutual spatial states is written as:

$$m \cdot p \cdot l_0(x, y, z)$$

For inertial carrier motion inducing emergent spatial behaviors along moving directions, the emergence rate of spatial states satisfies:

$$m \cdot v \cdot l_0(x, y, z)$$

Here v is not simply a classical kinematic velocity; rather, it represents the emergence rate of directional spatial states. Therefore, the spatial emergence rate within inertial systems simplifies to mv .

2. Emergence of Time

Within state evolution systems, time equals zero if existing Evolution carriers undergo no state variation and produce zero evolutionary events.

m Evolution carriers completing a single elementary state evolution over duration t_0 generate m fundamental time units for the whole system; macroscopic time is statistically defined with periodic phenomena including calendar cycles and atomic oscillation periods, uniformly denoted as T [17].

In dynamic evolutionary processes, total accumulated events along motion paths equal $m \cdot p$; under inertial motion, the dynamic emergence rate of events is $m \cdot v$, increasing synchronously with spatial emergence rate and reflecting the proportional co-emergence of space and time.

3. Dynamic Space-Time States

The quantity mv corresponds to momentum in classical physics and characterizes the joint emergence rate of space and time within inertial frameworks. The generalized definition of momentum is hence proposed: **Momentum refers to the state evolution rate of matter along spatial directions.**

All cosmic Evolution carriers exist as standardized quantized Momentum unit m_0c , and every material body possesses Intrinsic evolutionary capacity at speed c . Macroscopic observable velocity v is the apparent outcome after Vector Composition of massive discrete units, and the conserved total space-time emergence rate of the system is mc .

But, during inertial evolution, the emergence rate characterizing the combined spacetime state is given by mv^2 and mc^2 , both of which simultaneously encompass the two dimensions of time and space

and always grow synchronously. This distinguishes them from momentum, which describes the dynamic evolution of an evolution carrier along a single dimension. The spacetime state quantity is a composite dimension, whereas momentum, as a single-dimensional quantity, is therefore the proper mechanical dimension.

12.3 Correlation between Perceptual Interaction and Lorentz Factor

1. Synchronization between State Evolution and Perceptual Interaction

Following the state-interaction synchronization conclusion in Section 5.8: for stable relational state evolution systems, every Evolution carrier finishing one free-dimension state evolution over t_0 must complete one full environmental perceptual interaction synchronously. This yields the relation:

$$t_{0\text{-interaction}} = t_{0\text{-evolution}}, \quad \nu_{\text{interaction}} = \nu_{\text{evolution}} = c$$

Here c is used because the cosmic c is conserved, i.e., each unit of mass carries one unit of driving capacity c . The perceptual interaction frequency c does not mean perceptual signals propagate through space at light speed; cosmic perceptual action features global instantaneity. The constant c sets the upper limit of perceptual interaction frequency: infinite interaction cycles cannot be accomplished within infinitesimal time, and missing interaction contributions cannot be compensated, which equivalently restricts time from infinite subdivision.

2. Allocation Rule of Perceptual Capacity under Motion

In the cosmic system, the intrinsic total spacetime state perceptual interaction capacity of matter is c^2 (the temporal evolution rate and the spatial evolution rate are both c , and their product gives the total perceptual capacity). When matter is in motion, the total perceptual capacity is split into an externally perceptible part and an externally imperceptible part, obeying the conservation relation:

Conserved spacetime perceptual capacity = Spacetime perceptible + Spacetime imperceptible ⇓ $c_{\text{time}} \cdot c_{\text{space}} = (v_{1,\text{time}} \cdot v_{1,\text{space}}) + (v_{2,\text{time}} \cdot v_{2,\text{space}})$ ⇓ $c^2 = v_1^2 + v_2^2$

Parameter explanation:

- **Conservation perceptual rate c :** c represents the perceptual capacity for temporal and spatial states separately, because the driving capacity c equals the perceptual frequency c ;
- **Perceptible rate $v_{1,\text{perceptible}}$:** the rate of spacetime states lockable by the environment into a single stationary configuration at the particle's inertial velocity v_1 ;
- **Imperceptible rate $v_{2,\text{perceptible}}$:** the rate of spacetime states not lockable by the environment into a single stationary configuration at the particle's inertial velocity v_2 .

Detailed explanation: Time and space always emerge synchronously in fixed proportion. At high velocities, the spatial emergence rate increases, and the count of evolutionary events in the temporal direction

increases correspondingly; only the proportion of externally perceptible capacity changes, not the intrinsic flow rate of time itself.

Interaction requires the two participating parties to lock onto a unique spacetime state, preventing simultaneous correspondence to multiple spacetime states within the same t_0 interval. This is analogous to two people shaking hands: they must lock onto a single point in both time and space to complete the action. This is the underlying reason why perceptual capacity follows the Pythagorean distribution form. The faster the relative motion, the more pronounced this effect becomes—that is, the perceptible rate is severely lost.

12.4 Space-Time Locking Proportional Coefficient: Emergence of the Lorentz Factor

Starting from the conserved perceptual capacity formula:

$$c^2 = v_1^2 + v_2^2$$

where $v_1 = v$ denotes the macroscopic externally detectable spatial evolution rate, and v_2 denotes the externally undetectable evolution rate.

For stationary objects:

$$v_1 = c, \quad v_2 = 0, \quad \gamma = 1$$

For moving objects:

$$v_1 < c, \quad v_2 > 0, \quad \gamma > 1$$

The space-time locking proportional coefficient γ is defined as the ratio between conserved stationary space-time locking capacity and dynamic moving space-time locking capacity:

$$\gamma = \frac{\text{Conserved space-time locking capacity}}{\text{Moving space-time locking capacity}} = \frac{c}{v_{\text{time-locked}}}$$

Here $v_{\text{time-locked}} = \sqrt{c^2 - v^2}$ represents the externally lockable temporal evolution rate under motion. Substitution produces:

$$\gamma = \frac{c}{\sqrt{c^2 - v^2}} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Physical interpretation: γ quantifies the decay of externally lockable single space-time interaction states for rapidly moving particles.

In the limit $v \rightarrow c$:

$$v_{\text{time-locked}} \rightarrow 0, \quad \gamma \rightarrow \infty$$

meaning externally detectable interaction events approach zero. This constitutes the dynamical origin of relativistic time dilation; intrinsically space and time still emerge proportionally and synchronously, while only observable external interaction signals decay gradually.

Photons always satisfy $v = c$, which gives:

$$v_2 = 0, \quad \gamma = 1$$

hence full external perception is always achievable for photon evolutionary states.

General rule: regardless of macroscopic velocity v , space and time maintain synchronous proportional emergence, with only the partitioning ratio between v_1 and v_2 changing.

12.5 Chapter Summary

This chapter reveals the underlying dynamical origin of relativistic effects from the generalized relational state evolution system, with core conclusions listed below:

1. **Emergence of Lorentz factor:** The Lorentz factor $\gamma = 1/\sqrt{1 - v^2/c^2}$ originates from the quadratic conservation of spacetime perceptual capacity:

$$c^2 = v_1^2 + v_2^2$$

where $v_1 = v$ is the externally detectable space-time evolution rate, and v_2 is the externally undetectable evolution rate.

2. **Intrinsic nature of time dilation:** Relativistic time dilation arises from reduced externally lockable individual space-time interaction counts for high-speed particles, serving as an observational effect from accumulated interaction integration instead of intrinsic deformation of time itself. Space and time always emerge synchronously in fixed proportion, and only the distribution ratio between v_1 and v_2 alters.
3. **Redefined physical meaning of $E = mc^2$:** This expression is not a standalone energy formula in conventional physics, but an equation describing space-time state emergence rate as well as interaction and space-time locking capacity.
4. **Statistical essence of relativity:** Both relativity and classical mechanics describe universal causal integrated interaction effects rooted in fixed interaction frequency c and conserved intrinsic motion capacity c , rather than pre-defined covariant background spacetime. At high velocity, gravity degenerates into gravitational lensing and electromagnetic force reduces to magnetic force (magnetic force arises from the combined Momentum Deviation of the electromagnetic field and moving electrons). Strong-field environments induce secondary path integral effects such as Mercury's perihelion precession; spacetime curvature works as an equivalent universal mathematical statistic instead of the fundamental driving source of gravitation[58], which further explains that magnetic fields are not independent fundamental forces within traditional frameworks.
5. **Definition of spacetime curvature:** Spacetime curvature carries no intrinsic dynamical driving force. Space and time emerge jointly from static Representational Quanton and dynamic evolution and cannot independently generate interaction forces. Neither matter nor spacetime itself produces intrinsic covariance; relativistic covariance is a statistical outcome after massive elementary units (t_0, l_0) assemble into macroscopic spacetime across full interaction cycles.
6. **Reinterpretation of Doppler effect:** Intrinsic material state conservation follows Nl_0 , and the emergent space-time state quantity conserves as Nl_0t_0 [64]. Covariant phenomena equal statistical and interactive integral results under geometric observational projection: statistical integration corresponds to acoustic wavelength stretching from long-cycle bulk material accumulation, while interactive integration generates gravitational redshift and blueshift inherited from inertial-system Momentum Deviation via long-range Vector Composition[65].

In summary, this chapter completes the logical reduction from the foundational postulates of evolution theory to relativistic effects: the Lorentz factor originates from the conservation of perceptual capacity; "spacetime curvature" is a statistical effect rather than a dynamical driving force; and $E = mc^2$ is the spacetime state shaping equation rather than an independent energy entity—that is, the conserved spacetime emergence rate and the conserved perceptual rate of a particle are both mc^2 .

12.6 Commentary on Chapter References

[17] Rovelli, C. (2018). *The Order of Time*. New York: Riverhead Books.

The emergent-time theory supports the core standpoint that time does not exist at the fundamental level and accumulates from sequential evolutionary events, requiring synchronized evolution frequency instead of slower intrinsic time flow for moving objects.

[58] Jacobson, T. (1995). Thermodynamics of spacetime: The Einstein equation of state. *Physical Review Letters*, 75(7), 1260–1263. <https://doi.org/10.1103/PhysRevLett.75.1260>

The Einstein field equation is interpreted as a thermodynamic state equation of spacetime, supporting unified description of gravity and inertia as imbalanced information interaction effects; reducing gravitation and spacetime to information and thermodynamic state evolution represents a mainstream research direction in contemporary theoretical physics.

[64] Brown, H. R. (2005). *Physical relativity: Space-time structure from a dynamical perspective*. Oxford University Press.

Consistent with this work, the reference rejects geometric ontology treating spacetime as an independent entity and attributes spacetime structures including time dilation and length contraction to material dynamical behaviors, providing core theoretical support for non-covariant spacetime, projection-originated covariance and spacetime as the metric of Evolution carriers.

[65] Appelquist, T., Chodos, A., & Freund, P. G. O. (Eds.). (1987). *Modern Kaluza-Klein theories*. Addison-Wesley.

This comprehensive review summarizes how dimension reduction yields distinct fundamental interactions from original field equations; relativistic formalism built on four-dimensional spacetime and specific path integrals holds general mathematical validity yet requires reinterpretation grounded in essential physical evolution laws.

13 Derivation of Gravitational Lensing and Gravitational Redshift/Blueshift Effects

Based on the evolutionary conservation partition relation for particle space-time shaping $c^2 = v_{\text{perceptible}}^2 + v_{\text{imperceptible}}^2$ derived in Chapter 12 on relativity and the origin of the Lorentz factor, this paper combines indirect interaction resultant velocity (the Sun's displacement alters separation distance r via Perceptual Cross-section and indirectly generates interaction resultant velocity instead of classical direct velocity superposition), Perceptual window compression originating from perceptual resource competition, and the principle of momentum transfer. Redundant derivations are omitted to focus on core logic for deriving gravitational lensing and gravitational redshift/blueshift of Ultra-high-speed particles exemplified by photons, and to clarify intrinsic correlations between these effects, Perceptual Cross-section and indirect resultant velocity.

13.1 Fundamental Evolutionary Logic of the Framework

In the following, we combine the basic concepts of traditional theories with the specific logical relations of the present evolutionary framework to derive relevant mechanical phenomena such as gravitational redshift and blueshift.

1. Concept Introduction and Comparative Table

Physical Phenomenon	Explanation in Conventional Theory	Core Mechanism of Present Framework	Key Variables of Present Framework
Deflection Angle of Gravitational Lensing	Geodesic deviation due to spacetime curvature; Newtonian calculation yields insufficient deflection, and general relativity introduces geometric derivation to obtain the factor of 2	Indirect interaction resultant velocity plus doubled perceptual opportunities; the factor of 2 arises from secondary perception producing $2\Delta p_{\perp}$	$\Delta v'_{\perp} = \frac{4GM}{cR}, \theta = \frac{4GM}{c^2 R}$
Gravitational Blueshift	Photon energy rises within gravitational field	$\frac{dr}{dt} < 0$ under approaching motion, enhanced perceptual interaction, and photons inherit positive velocity component	$\Delta p > 0, \nu \uparrow$
Gravitational Redshift	Photon energy drops within gravitational field	$\frac{dr}{dt} > 0$ under receding motion, weakened perceptual interaction, and photons inherit negative velocity component	$\Delta p < 0, \nu \downarrow$
Perceptual Window Compression	Relativistic time dilation	When $v_{\text{perceptible}} = c$, $\sqrt{1 - \frac{v^2}{c^2}} \rightarrow 0$, interaction capacity declines sharply and is compensated by indirect resultant velocity	Perceptual opportunity multiplication coefficient $\varepsilon = 1$
Indirect Interaction Resultant Velocity	No corresponding physical definition	Solar displacement modifies separation r of Perceptual Cross-section, and $\frac{dr}{dt}$ regulates perceptual opportunity and momentum transfer efficiency	Non-classical velocity superposition, core connecting link of the whole mechanism
Momentum Transfer	Photon energy $E = h\nu$, momentum $p = \frac{E}{c}$	Photon momentum $p_{\gamma} = m_{\gamma}c$, discrete transfer of Momentum unit satisfying $p = mc$	$\Delta p = \int F_{\perp} dt$

2. Derivation of Evolutionary Effects for Ultra-high-speed Particles in Gravitational Field (Momentum and Perception Mechanism)

A. Gravitational Lensing Principle: In a state evolution system, when the macroscopic velocity of an evolutionary carrier reaches the intrinsic driving limit c , its motion frequency synchronizes with the perception frequency, maximizing interaction capability. At this point, the single rest spacetime state locked between the carrier and the environment tends to be minimal, giving rise to special relativistic

effects —the spacetime locking ratio factor (Lorentz factor).

- Conserved Driving Force c
- Interaction Frequency c
- Spacetime Locking Capability c^2
- High-Speed Motion v
- Self Spacetime State v^2
- Reduced Single Rest Spacetime State
- Lorentz Factor

All macroscopic evolutionary carriers consist of internal state-representing components (i.e., containing $N \cdot m_0 \cdot c$ momentum units; the macroscopic cancellation state appears as “conventional rest mass” m). Therefore, all matter in the universe (including bosons) possesses mass, as further verified in Chapter 14. Single-particle mass is extremely small, leading to very weak direct gravitational interaction, which requires ultra-massive celestial bodies to manifest observable gravitational lensing.

- Photon Mass is Small
- Gravitational Interaction is Weak
- Ultra-Massive Celestial Body
- Gravitational Manifestation (Lensing)

B. Principle of Double Deflection (Origin of Factor 2): In a strong gravitational field, a particle deflects inward under attractive force. The essence is that the macroscopic apparent speed v of a photon in a strong field becomes lower than its intrinsic driving capability c (e.g., significantly reduced near a black hole), extending its retention time within the perception cross-section of the gravitational source. This grants the particle **secondary (or multiple) perception opportunities** in a single flyby, with each perception contributing a transverse momentum increment $\Delta p_{\perp}$.

- Special Relativistic Effect
- Strong-Field Environment
- Path Deflection & Effective Speed Reduction
- Extended Retention Time
- Lock More Rest Spacetime States
- Perception Opportunity Doubling
- Interaction Amplitude Doubled
- Deflection Factor 2

When the photon escapes the strong gravitational field, its apparent speed recovers to the intrinsic driving limit c via the **Geese-Formation Encapsulation Mechanism** (resynchronization and collective orientation of internal momentum units). This causal chain “slower speed → longer retention → doubled perception” recovers the missing factor of 2 in Newtonian calculations, representing the physical origin of the same factor derived from spacetime geometry in general relativity.

- Encapsulation Modulation
- Restore Speed of Light c

C. Redshift and Blueshift Principle: During gravitational interaction between a particle and a celestial body, initial velocity (or radial relative motion) causes continuous variation of the perception cross-section separation r (i.e., $dr/dt \neq 0$). This variation induces a directional correction trend, forming a **long-range resultant velocity effect**. The effect changes particle momentum (gain/loss), observed as frequency blueshift or redshift.

→ Initial Celestial Velocity + Gravitational Interaction

→ Perception Cross-Section Variation $\frac{dr}{dt}$

→ $\begin{cases} \text{Mutual Approach} \rightarrow \text{Enhanced Perception} \rightarrow \text{Forward Inherited Velocity} \rightarrow \text{Blueshift} \\ \text{Mutual Recession} \rightarrow \text{Weakened Perception} \rightarrow \text{Backward Inherited Velocity} \rightarrow \text{Redshift} \end{cases}$

Core premise: Ultra-high-speed particles represented by photons satisfy $v_{\text{imperceptible}} = c$ with extremely small mass. Gravitational interaction essentially refers to discrete Momentum unit transfer between the gravitational source and photons.

Physical implication of indirect interaction resultant velocity: Displacement of the gravitational source dynamically changes separation r between mutual Perceptual Cross-sections, and the time variation rate of spacing serves as the equivalent "resultant velocity". Such resultant velocity is not classical direct velocity addition; it constrains momentum interaction via Perceptual Cross-section and further modulates efficiency and orientation of momentum transfer.

13.2 Derivation of Gravitational Lensing (Newtonian Mechanics Corrected by Perception Mechanism and Indirect Interaction Resultant Velocity)

1. Basic Model and Core Setup

Light deflection grazing the solar limb is selected as the research case with settings below:

Photon: $v_{\text{perceptible}} = c$, $v_{\text{imperceptible}} \rightarrow 0$, extremely small mass, weak intrinsic perceptual capacity and mild gravitational response, momentum satisfies $p = m\lambda\nu$;

Sun: massive gravitational source with physical displacement that dynamically alters spacing r of Perceptual Cross-section to generate indirect interaction resultant velocity and build a high-density momentum interaction field;

Perception mechanism: When $v_{\text{perceptible}} = c$, Perceptual window is drastically compressed due to resource competition leading to feeble single interaction effect; secondary perceptual opportunities supplied by indirect resultant velocity accumulate observable momentum deflection and double the deflection amplitude;

Weak-field approximation: Impact parameter of grazing photon $b \approx R$ (solar radius), only transverse momentum transfer of gravity is analyzed under perturbation approximation.

Fundamental physical constants: Solar mass $M \approx 1.989 \times 10^{30}$ kg, Solar radius $R \approx 6.96 \times 10^8$ m, Gravitational constant $G \approx 6.674 \times 10^{-11}$ N · m²/kg², Speed of light $c \approx 3.0 \times 10^8$ m/s.

2. Preliminary Newtonian Momentum Calculation (Indirect Resultant Velocity Neglected)

Transverse gravitational force component reads:

$$F_{\perp} = \frac{GMm_{\gamma}b}{(x^2 + b^2)^{\frac{3}{2}}}$$

where m_γ denotes photon mass.

Combined with window relation $dt = \frac{dx}{c}$ and $b \approx R$, full-range integration for transverse momentum increment yields:

$$\Delta p_\perp = \int_{-\infty}^{+\infty} F_\perp dt = \frac{2GMm_\gamma}{cR}$$

The preliminary deflection angle derived from momentum variation is:

$$\theta_{\text{Newton}} \approx \frac{\Delta v_\perp}{c} = \frac{2GM}{c^2 R} \approx 0.875 \text{ arcsec}$$

This value is half of the observational result of 1.75 arcseconds, originating from neglect of correction for secondary perceptual opportunities induced by indirect interaction resultant velocity, the inherent defect of early Newtonian calculation of photon gravitational deflection[66][67].

3. Correction from Indirect Interaction Resultant Velocity and Perception Mechanism to Compensate Deflection Magnitude

Core correction: Solar displacement dynamically modifies Perceptual Cross-section separation r and brings secondary perceptual opportunities; total momentum interaction increment doubles to $2\Delta p_\perp$, offsetting insufficient single interaction induced by Perceptual window compression and forming complete observable deflection.

a. Perceptual window compression: For photons with $v_{\text{perceptible}} = c$, compression factor

$\sqrt{1 - \frac{v^2}{c^2}} \rightarrow 0$, single interaction only produces deflection of 0.875 arcseconds;

b. Secondary perception from indirect resultant velocity: Indirect resultant velocity from solar displacement grants photons moving along bent paths secondary perceptual access and doubles total momentum increment.

Corrected transverse velocity increment:

$$\Delta v'_\perp = \frac{2\Delta p_\perp}{m_\gamma} = \frac{4GM}{cR}$$

Final gravitational lensing deflection angle:

$$\theta = \frac{\Delta v'_\perp}{c} = \frac{4GM}{c^2 R} \approx 1.75 \text{ arcsec}$$

Conclusion: Gravitational lensing deflection is independent of photon mass m_γ . The effect arises physically from accumulated momentum deflection via secondary perceptual opportunities enabled by indirect interaction resultant velocity rather than spacetime curvature.

4. Momentum-based Gravitational Field Equation and Comparison with Relativistic Field Equation

A momentum-form gravitational field equation is established based purely on momentum theory without spacetime curvature assumption via momentum transfer and multiplied perceptual opportunities, contrasted against the Einstein field equation of general relativity. This equation is intended to display gravitational mechanism and interpret the dynamical origin of relativistic field equations

as two equivalent mathematical descriptions for identical physical phenomena from distinct perspectives.

a. **Reference: General Relativity Field Equation** General relativity attributes gravitation to four-dimensional spacetime curvature with core field equation:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu}$$

This formalism builds upon four-dimensional geometric curvature regarded as dynamical integration outcome without explicit factor 2, momentum transfer or perception mechanisms.

b. **Pure-momentum Gravitational Field Equation of Present Framework with Double Deflection Term** From momentum conservation and perceptual multiplication coefficient, total transverse momentum increment satisfies:

$$\Delta \vec{p}_{\perp, \text{total}} = (1 + \varepsilon) \cdot \int_{-\infty}^{+\infty} \vec{F}_p \text{exp} dt$$

where: $\vec{G}_p$ denotes momentum gravitational field intensity replacing spacetime curvature to quantify momentum transfer capacity of gravitation; $\varepsilon = 1$ stands for perceptual multiplication coefficient corresponding to secondary interaction from indirect resultant velocity; $1 + \varepsilon = 2$ represents superposition of Newtonian baseline deflection and secondary perception deflection consistent with prior derivation.

c. **Comparison Description**

Both formalisms produce identical observational value of 1.75 arcseconds within weak-field small-deflection limit while holding disparate physical pictures: general relativity derives the factor of 2 from geometric spacetime construction, whereas this framework decomposes the factor into discrete momentum transfer plus secondary perceptual access. Their mathematical structures are not fully equivalent (differential geometric formalism versus momentum integral formalism) yet numerically agree within astronomical measurement ranges[68].

13.3 Derivation of Gravitational Redshift and Blueshift (Momentum Transfer + Indirect Interaction Resultant Velocity + Perception Mechanism)

1. Core Logic and Model Setup

Physical essence of gravitational redshift and blueshift: Indirect interaction resultant velocity adjusts $\frac{dr}{dt}$ of Perceptual Cross-section separation to modify interaction strength and momentum transfer direction, varying photon momentum and inducing frequency shift without invoking spacetime curvature[69].

Supplementary setup: The Sun is approximately stationary in the inertial reference frame and indirect resultant velocity is dominated by $\frac{dr}{dt}$; incident photon speed $v \approx c$, momentum transfer follows conservation law constrained by perceptual rules.

2. Derivation of Momentum Transfer and Frequency Shift

Photon momentum reads $p = m_\gamma c$, and frequency is proportional to momentum, hence fractional frequency shift satisfies:

$$\frac{\Delta \nu}{\nu} = \frac{\Delta p}{p}$$

Analysis combined with indirect resultant velocity and perception rules: - Blueshift: $\frac{dr}{dt} < 0$, decreasing separation enhances perceptual interaction, photons acquire positive momentum $\Delta p > 0$ and frequency rises; - Redshift: $\frac{dr}{dt} > 0$, increasing separation weakens perceptual interaction, photons lose momentum $\Delta p < 0$ and frequency drops.

3. Quantitative Verification and Physical Interpretation

Quantitative formula for gravitational redshift and blueshift:

$$\frac{\Delta\nu}{\nu} = \frac{2GM}{c^2 R} \cos \alpha$$

where α denotes the angle between incident photon path and the connecting line toward celestial body mass center.

Physical nature: Redshift and blueshift are governed by $\frac{dr}{dt}$ which tunes momentum transfer magnitude and direction via varying perceptual intensity as coupled momentum-perception effects distinct from classical Doppler shift originating from bulk velocity superposition, consistent with the momentum system relation $p = m\lambda\nu$.

4. Key Supplement: Core Definition of Indirect Interaction Resultant Velocity

The complete set of rules including finite particle mass, Perceptual window compression, indirect interaction resultant velocity defined by dr/dt , and multiplied perceptual opportunities consistently reproduces gravitational lensing (origin of factor 2) plus gravitational spectral shifts entirely within momentum formalism without spacetime curvature or independent energy entity assumptions. The derivation inherits the research route of early Newtonian gravitational exploration and completes the missing deflection mechanism of classical gravitation[70].

13.4 Chapter Summary

This framework uniformly explains gravitational lensing (origin of factor 2) and gravitational redshift/blueshift via the integrated mechanism consisting of particle mass, Perceptual window compression, indirect interaction resultant velocity centered on $\frac{dr}{dt}$, and multiplied perceptual opportunities. All derivations are completed within pure momentum theory without spacetime curvature or standalone energy terms.

1. Provides reasonable physical explanation for boson mass related issues;
2. Demonstrates relativistic dynamical effects as macroscopic integral outcomes;
3. Fully interprets physical mechanisms of gravitational redshift and blueshift;
4. Reveals physical origin of resultant velocity for Long-range Force;
5. Identifies fundamental cause for underestimated gravitational deflection from Newtonian gravitational calculation;
6. Unifies all macroscopic dynamical phenomena under the pure momentum theoretical framework.

13.5 Commentary on Chapter References

[66] von Soldner, J. G. (1804). Über die Ablenkung eines Lichtstrahls von seiner geradlinigen Bewegung durch die Attraktion eines Weltkörpers, an welchem er nahe vorbeigeht. *Berliner Astronomisches Jahrbuch*, 161–172.

An early Newtonian derivation of gravitational light deflection lacking the complete multiplication factor later numerically filled by spacetime curvature formalism of relativity.

[67] Einstein, A. (1911). On the Influence of Gravitation on the Propagation of Light. *Annalen der Physik*, 340(10), 898-908. <https://doi.org/10.1002/andp.19113401005>

Einstein' s early calculation using Newtonian mechanics and equivalence principle obtained half of the full deflection while sharing similar general reasoning with this paper.

[68] Dyson, F. W., Eddington, A. S., & Davidson, C. (1920). A Determination of the Deflection of Light by the Sun's Gravitational Field, from Observations Made at the Total Eclipse of May 29, 1919. *Philosophical Transactions of the Royal Society of London A*, 220, 291–333. <https://doi.org/10.1098/rsta.1920.0009>

Provides observational confirmation of the 1.75 arcsec deflection value.

[69] Tolman, R. C., Ehrenfest, P., & Podolsky, B. (1931). On the Gravitational Field Produced by Light. *Physical Review*, 37(5), 602-615. <https://doi.org/10.1103/PhysRev.37.602>

Confirms photons generate gravitational field and carry intrinsic gravitational interaction effect independent of spacetime curvature.

[70] Michell, J. (1784). On the Means of Discovering the Distance, Magnitude, etc. of the Fixed Stars, in consequence of the diminution of the velocity of their light. *Philosophical Transactions of the Royal Society of London*, 74, 35-57.

John Michell' s early Newtonian reasoning on dark stars shares analogous foundational logic with the present derivation.

14 Internal Structure of Particles and Corresponding Motion Properties

14.1 Fundamental Dynamical Logic

From the derivations in preceding chapters, a steadily operating generalized relational state evolution system requires particles to independently accommodate causal relations, state relations and mechanical relations under conservation constraints, exhibiting characteristic behaviors of standard quantized systems. Such systems require no intermediate carrier named "field" to complete state simulation and statistical processing.

Based on three fundamental components of cosmic systems, namely Representational Quanton capacity, Driving Capacity and sets of evolution rules, internal structural units are necessarily contained within particles[71]. These structural units individually carry state representation and evolutionary driving capacity, and different Encapsulation modes determine Perceptual interaction mechanisms and interaction intensity. Each particle contains $N \cdot m_0$ Momentum units evolving at the speed of light with conserved representation and driving capacity.

Intra-particle and inter-particle motions are governed by Vector Composition via Orientation Ratio of

Momentum units. Macroscopic velocity v emerges from intrinsic speed c modulated by Orientation Ratio, and the speed of light c sets the upper bound of particle motion speed. Wave-particle duality of particles originates from state switching induced by causal interactions among internal structures, manifesting as real-valued momentum flux in multi-slit experiments[76].

The conventional physical phenomenon of "mass-energy conversion" essentially originates from the macroscopic manifestation of growing internal evolutionary states triggered by altered Orientation Ratio of internal Momentum units and the breaking of the cancellation state. Namely:

Momentum cancellation $\rightarrow$ Momentum breaking $\Rightarrow$ Emergence of "relativistic energy" statistics

Momentum breaking $\rightarrow$ Momentum cancellation $\Rightarrow$ Emergence of "rest mass"

Both are macroscopic statistical effects within the relativistic framework that require the introduction of divergent energy concepts for description. This section quantitatively verifies the classical mass-energy conversion formula using the Orientation Ratio of electron momentum measured in β decay experiments[62], thereby reducing it to a momentum eigen-relation.

All derivations throughout this chapter adopt the Orientation Ratio formalism:

$$v = \frac{N_+ - N_-}{N_+ + N_-}c, \quad p = m\lambda\nu$$

Experimental measured data from β decay is substituted for quantitative deduction, demonstrating that macroscopic velocity emerges from Vector Composition of massive internal $N \cdot m_0c$ Momentum units without introducing macroscopic energy or reference frame concepts.

14.2 Basic Experimental Measured Parameters

1. Rest mass of free electron: $m_0 = 9.11 \times 10^{-31}$ kg
2. Rest mass difference between neutron and proton (source of total decay momentum):

$$\Delta m_n = 1.293 \times 10^{-30} \text{ kg}$$

3. Speed of light: $c = 3 \times 10^8$ m/s
4. Measured β decay datum: maximum ejection velocity of outgoing electron $v_{\max} \approx 0.918c$
5. Intrinsic benchmark total decay momentum: $P_\Sigma = \Delta m_n \cdot c$

14.3 Intra-particle Vector Composition and Orientation Ratio Formula

Let N_+ denote the count of forward-oriented internal Momentum units and N_- denote counteracting reverse Momentum units; total number of internal units reads:

$$N = N_+ + N_-$$

Core formula governing emergent macroscopic motion velocity:

$$v = \frac{N_+ - N_-}{N_+ + N_-}c$$

Define Orientation Ratio factor:

$$s = \frac{N_+ - N_-}{N}$$

which simplifies the velocity relation into:

$$v = sc$$

The intrinsic definition of macroscopic particle momentum is uniformly expressed as:

$$p = m\lambda\nu$$

Microscopic fundamental units satisfy $\lambda_0\nu_0 = c$, and macroscopic momentum represents statistical outcomes of Vector Composition for enormous quantities of light-speed Momentum units[76].

14.4 Quantitative Calculation at Maximum Kinetic Energy Limit (Experimental Upper Bound)

Experimentally measured limiting ejection velocity of electrons from β decay:

$$v = 0.918c$$

Substitute into the emergent velocity formula:

$$0.918c = \frac{N_+ - N_-}{N_+ + N_-}c$$

Cancel the speed-of-light constant c :

$$\frac{N_+ - N_-}{N_+ + N_-} = 0.918$$

Solve the numerical ratio between forward and reverse Momentum unit quantities:

$$\frac{N_+}{N_-} = \frac{1 + 0.918}{1 - 0.918} \approx 23.39$$

Further proportional analysis is conducted using total unit count $N = N_+ + N_-$.

Quantitative Physical Implication: For β -decay electrons moving at experimentally measured limiting speed, **95.9%** internal Momentum units align in identical forward orientation, while only **4.1%** units counteract motion to form Cancellation state. Macroscopic velocity $0.918c$ directly Emerges from Vector Composition of massive intrinsic m_0c elemental momenta.

14.5 Total Unit Variation Calculation under Structural Stability Constraint

The structural stability constraint specifies invariant product of interfering internal units for particles at rest and in motion.

For stationary electrons:

$$N_{+0} = N_{-0} = 0.5N_0$$

$$N_{+0}N_{-0} = 0.25N_0^2$$

For high-speed moving electrons:

$$N_+ N_- = 0.959 N \cdot 0.041 N$$

Combine the two expressions via conservation constraint:

$$0.0393 N^2 = 0.25 N_0^2$$

Rearrangement yields:

$$N \approx 2.522 N_0$$

Quantitative Conclusion: Electrons accelerated to experimental limiting speed in β decay contain approximately **2.522 times** as many internal Momentum units as stationary electrons. The increment of total unit count corresponds to macroscopic mass increment originating entirely from structural reorganization via Cancellation-Breaking of internal Momentum units.

14.6 Joint Verification with Intrinsic Momentum Formula

Substitute emergent velocity relation $v = sc$ into the universal momentum expression:

$$p = m\lambda\nu = m \cdot sc$$

β decay involves no independent energy conversion; the physical process corresponds to splitting and reorganization of abundant internal $N \cdot m_0 c$ Momentum units originally contained inside neutrons. Redistributed elemental units restructure inside daughter electrons, modifying the N_+/N_- proportion and Orientation Ratio factor s , finally producing continuous momentum spectrum observed experimentally.

- Larger share of forward Momentum units allocated to electrons $\Rightarrow$ elevated Orientation Ratio $s \Rightarrow$ increased macroscopic velocity v and momentum p
- Higher fraction of counteracting reverse units $\Rightarrow$ reduced Orientation Ratio $s \Rightarrow$ weakened macroscopic translational motion

This regularity fully matches the continuous momentum distribution measured from β decay electron experiments.

14.7 Section Conclusions

Using experimentally measured limiting electron ejection speed $v_{\max} \approx 0.918c$ from β decay alongside two core framework equations

$$v = \frac{N_+ - N_-}{N_+ + N_-} c, \quad p = m\lambda\nu$$

three layered conclusions are drawn sequentially:

1. The Orientation Ratio is an internal evolutionary attribute of the electron.

Back-calculation from $v = 0.918c$ yields $N_+/N_- \approx 23.39$, indicating 95.9% internal Momentum units aligned forward and 4.1% oppositely oriented for counteraction. Continuous momentum spectrum

of β -decay electrons corresponds to continuous statistical distribution of N_+/N_- , verifying that Orientation Ratio

$$s = \frac{N_+ - N_-}{N_+ + N_-}$$

acts as physically meaningful quantity describing particle motion states.

2. Existence of Orientation Ratio proves internal substructure within electrons

Orientation Ratio s quantifies distribution discrepancy between forward and reverse internal Momentum units. A point-particle electron devoid of inner substructure cannot accommodate definitions of N_+ and N_- , nor structural rearrangement modifying unit proportions. Therefore β decay experiments indirectly confirm electrons represent composite systems assembled from reorganizable Momentum units instead of structureless point particles[61][71].

3. Macroscopic motion and conventional mass-energy conversion both Emerge from varying Orientation Ratio

- Macroscopic translational velocity: $v = sc$
- Mass growth for moving particles: structural stability constraint $N_+N_- = \text{constant}$ derives $N = \gamma N_0$, with computed result $N \approx 2.522 N_0$ matching $\gamma \approx 2.52$
- Mass-energy conversion: The Lorentz factor γ in classical theory is a mathematical equivalent description of the Orientation Ratio within the framework of the relativistic energy system. No independent energy entity is introduced in the present framework[62].

In summary, combining pure momentum theoretical framework and experimental β decay data quantitatively validates internal electron substructure and macroscopic motion originating from Vector Composition of inner Momentum units, laying logical foundation for subsequent dynamic electron radius derivation via photoelectric effect.

14.8 Chapter Summary

Macroscopic particle motion Emerges from incomplete Cancellation state of internal Momentum units, with bulk velocity v jointly determined by intrinsic light speed c and Orientation Ratio s governing Vector Composition. All observable and interactive particle states are defined around momentum complying with $m = \frac{p}{v}$ [44].

The universal speed limit of light requires no explanation invoking spacetime Covariance, reference frame transformation or conventional mass-energy conversion. Classically defined relativistic mass variation and Lorentz factor γ reduce to equivalent expression of Orientation Ratio factor s within pure-momentum formalism. Einstein' s box thought experiment receives consistent interpretation through identical logic: mass-deficit law is derived solely from Cancellation-Breaking structural reorganization of Momentum units yielding numerical consistency with conventional predictions.

Unified intrinsic physical linkage exists among translational motion, momentum and traditional mass-energy conversion following the complete causal chain:

Intrinsic elemental units $N \cdot m_0 c \rightarrow$ Cancellation-Breaking structural reorganization $\rightarrow$ Vector Composition $\rightarrow$ Ori

Core Summary: All particles are formed via Encapsulation of mass-based elemental units each carrying Intrinsic evolutionary capacity at light speed; the theoretical construction requires no separately

defined standalone energy term. Various statistical physical quantities Emerge exclusively during particle generation and fundamental Long-range Force interactions. Intrinsic evolutionary capacity of elementary units serves as physical origin of interaction forces and momentum, also constituting fundamental basis for observational state representation.

Supplementary Remarks: The above derivation demonstrates the existence of an internal quasi-canonically quantized structural unit within particles, whose scale approaches the Planck scale derived in Chapter 9, i.e., the state transition scale per unit time t_0 is l_0 . However, the logical validity of this internal particle structure derivation, as well as the entire framework, does not strictly depend on any specific computable numerical value of this minimal scale. The essential dependency lies in the fact that each unit of mass possesses an intrinsic light-speed evolution capacity c and an orientation ratio. This logic alone is already sufficient to support the same foundational work across most quantum domains, including the principle of least action, matrix formalism, mass-energy conversion, probability waves, operators, etc.

Moreover, the fact that each unit of mass possesses a conserved light-speed evolution capacity c has been comprehensively validated through the spacetime shaping capacity equation under relativistic perceptual interaction and the orientation ratio derived in this chapter.

Regarding the numerical calibration of the specific minimal scale of matter and space, the relevant logic and derivations can be found in Sections 5.7, 7.1, and 9.3. The author anticipates that scholars with stronger mathematical skills may complete more precise derivations in the future. At present, neither the Planck time t_0 nor the Planck length l_0 possesses a physically dominant role in observational effects. Therefore, this framework adopts them as reference scales for the "logically necessary minimal units" rather than as core prerequisites.

14.9 Commentary on Chapter References

[44] Schrödinger, E. (1930). Über die kräftefreie Bewegung in der relativistischen Quantenmechanik. *Sitzungsberichte der Preußischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse*, 24, 418–428.

Note: The original proposed DOI (10.1038/9441-030X) is invalid as the Nature journal prefix mismatches proceedings of Prussian Academy of Sciences and is omitted; marked as no publicly available DOI.

The work predicts free-electron zitterbewegung at frequency $2mc^2/h$ propagating at speed c , consistent with the framework logic of fixed internal evolutionary speed and conserved total evolutionary capacity.

[61] ACME Collaboration; Baron, J., Campbell, W. C., DeMille, D., et al. (2014). Order of magnitude smaller limit on the electric dipole moment of the electron. *Science*, 343(6168), 269–272. <https://doi.org/10.1126/science.1248213>

Electron spherical symmetry: high-precision experimental confirmation of perfect spherical electron geometry supplies critical experimental evidence supporting core framework hypothesis of spherically Encapsulated Cancellation-state Momentum unit assemblies and substantial internal mc momentum flux inside electrons.

[62] Einstein, A. (1905). Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig? *Annalen der Physik*, 18(13), 639–641. <https://doi.org/10.1002/andp.19053220817>

Original source of mass-energy relation $E = mc^2$ establishing mass conservation law. This framework reinterprets the formula as a Space-time state shaping equation confirming conserved mass with every mass unit possessing intrinsic dynamical evolutionary capacity c . Consistent with Einstein's original expression assigning photon energy mcc and inverse relation $m = E/c^2$ proving photons carry finite

mass inheriting full pre-annihilation fermion mass; all particles hence assemble from discrete mass-based Momentum units.

[71] Compton, A. H. (1923). A quantum theory of the scattering of x-rays by light elements. *Physical Review*, 21(5), 483–502. <https://doi.org/10.1103/PhysRev.21.483>

Experimentally verifies finite characteristic electron size against point-particle assumption and substantiates physical reality of photon momentum.

[76] Feynman, R. P., & Hibbs, A. R. (1965). *Quantum Mechanics and Path Integrals*. New York: McGraw-Hill.

Path integral formalism provides mathematical foundation for the framework' s topological path traversal and weighted summation coding mechanism of Momentum units, supplying formal reference for mathematical description of momentum flux partitioning in photon-electron interactions, multi-path fluid states and conserved total topological state counts.

15 Photoelectric Relation and Dynamic Electron Radius

From preceding derivations, physical quantities including field, force, momentum, energy and space-time possess no independent existence and all Emerge from intrinsic particle substance, evolutionary capacity and Orientation Ratio. Such deduction confirms inherent particle substructure: fermions adopt spherically symmetric Encapsulation geometry[61], while photons form flock-style coded Encapsulation equivalent to unit assemblies with uniform velocity and propagation direction capable of arbitrary three-dimensional topological conformation. Both particle categories feature characteristic radii correlated with corresponding characteristic wavelengths establishing dynamically evolving relations. This chapter completes derivations grounded on photoelectric effect.

15.1 Particle Structure and Radius Relation

Derivations in this chapter build upon universal momentum expression

$$p = m\lambda\nu$$

and intra-particle Momentum unit Vector Composition model, proving finite dynamic fermion radius via empirical regularities of photoelectric effect. Complete causal chain is constructed as: **Photoelectric wavelength**→**photon momentum**→**electron Momentum Deviation**→**Orientation Ratio**→**dynamic radius**, validated via photoelectric law and pure momentum formalism.

1. Fundamental Momentum Law

$$p = m\lambda\nu$$

where p denotes momentum, m mass-related Representational Quanton, λ wave wavelength and ν oscillation frequency. The formula universally holds for all particles including photons and electrons.

For photons satisfying dispersion relation $\lambda\nu = c$ (c = speed of light), photon momentum reduces to compact form:

$$p_\gamma = m_\gamma c$$

Within present framework photons assemble from fundamental Momentum units with well-defined mass $m_\gamma = N_\gamma m_0$.

For electrons, macroscopic motion velocity and wave parameters comply with $v = \lambda\nu$ yielding momentum expression:

$$p = mv$$

Here mass m solely serves as the material Representational Quanton —the total material quantity $N \cdot m_0$ within the state evolution system —without invoking the mass-energy equivalence postulate. The "mass-energy conversion" emerges from the reconfiguration of cancelation-symmetry breaking (vector composition) among internal momentum units and merely manifests as the relativistic mathematical form $E = mc^2$ at the macroscopic level. In essence, the total material quantity $N \cdot m_0$ and the driving capacity c are both strictly conserved; only the directional ratio s of internal momentum units varies.

2. Momentum Unit and Orientation Ratio

Electrons consist of abundant intrinsic light-speed fundamental Momentum units split into forward-aligned N_+ (along bulk motion) and counteracting reverse N_- (opposing bulk motion). Orientation Ratio s quantifies net distribution discrepancy between opposing elemental momenta:

$$s = \frac{N_+ - N_-}{N_+ + N_-}$$

Orientation Ratio s directly determines resultant macroscopic electron velocity following linear relation:

$$v = s \cdot c$$

At electron rest state opposing Momentum units fully cancel: $N_+ = N_-$, corresponding to $s = 0$ and $v = 0$. External perturbations such as photon bombardment break balanced internal unit distribution producing non-zero s and emergent macroscopic velocity. Mutual collision and compression among internal Momentum units deform spatial extension defining dynamic electron radius.

15.2 Ontological Derivation of Dynamic Electron Radius

Combining photoelectric experimental phenomena with universal momentum formula and Orientation Ratio model, inherent correlations among dynamic electron radius, Orientation Ratio and photoelectric wavelength are derived and organized into complete causal logic.

1. Momentum Transfer Mechanism of Photoelectric Effect

Full momentum exchange occurs between incident photons and bound electrons during photoelectric transition with complete photon momentum converted into electron Momentum Deviation ΔP . From photon dispersion constraint $\lambda\nu = c$ and momentum definition $p_\gamma = m_\gamma c$, inverse proportionality holds between photoelectric wavelength and photon momentum:

$$\lambda_{\text{photo}} \propto \frac{1}{m_\gamma} \propto \frac{1}{p_\gamma}$$

Increased incident photon momentum raises transferred electron Momentum Deviation ΔP , aggravating internal unit imbalance and elevating Orientation Ratio s [71].

2. Correlation between Dynamic Radius and Orientation Ratio

For $s = 0$ (stationary electron), uniformly distributed counteracting internal Momentum units define intrinsic static radius R_0 scaled against Compton wavelength:

$$R_0 = \frac{\lambda_C}{2\pi} \approx 386 \text{ fm}$$

This numerical value originates purely from internal Momentum unit distribution independent of mass-energy equivalence relations.

For non-zero s (moving electron), elevated Orientation Ratio concentrates elemental momenta along single propagation direction intensifying internal collision compression and shrinking spatial extension; dynamic radius R decreases monotonically with rising s .

Functional relation between dynamic radius and Orientation Ratio:

$$R(s) = R_0 \cdot \sqrt{1 - s^2}$$

Substitute $s = \frac{v}{c}$ and Lorentz factor definition $\gamma = \frac{1}{\sqrt{1 - s^2}}$ for equivalent formulation:

$$R = \frac{R_0}{\gamma} = R_0 \sqrt{1 - s^2}$$

Physical interpretation: dynamic radius scales proportionally to residual transverse fractional distribution of internal Momentum units.

Linked proportional relations among all relevant physical quantities:

$$\lambda_{\text{photo}} \propto \frac{1}{m_\gamma} \propto \frac{1}{p_\gamma} \propto \frac{1}{\Delta P} \propto R \propto \frac{1}{s} \propto \frac{1}{v}$$

Consequently dynamic electron radius is fully governed by intra-particle Momentum unit Vector Composition and collision compression with Orientation Ratio functioning as core control parameter.

15.3 Experimental Verification and Numerical Comparison

Theoretically derived dynamic radius $R(s)$ is cross-checked against effective radius R_{eff} measured via electron-electron scattering experiments relying exclusively on measurable momentum and scattering cross-section without auxiliary energy parameters[73].

1. Low-Energy Scattering Regime ($s \approx 0$)

At low incident energy $v \ll c$, Orientation Ratio $s \approx 0$ with nearly full internal Momentum unit Cancellation pushing dynamic radius close to intrinsic static value R_0 .

For low-energy electron-electron scattering ranging $100 \text{ eV} \sim 1 \text{ keV}$, form-factor zero condition $q_0 R = 4.493$ yields effective radius computational formula:

$$R_{\text{eff}} = \frac{4.493\hbar}{q_0}$$

Experimentally measured velocity satisfies $\frac{v}{c} \approx 1 - 5 \times 10^{-9}$ corresponding to $s \approx 0.999999995$. Substitute into dynamic radius relation:

$$R(s) = R_0 \sqrt{1 - s^2}$$

Numerical output $R \approx 0.1$ fm qualitatively matches upper bound of experimentally extracted effective radius verifying shrinking radius against rising Orientation Ratio.

2. High-Energy Scattering Regime ($s \approx 1$)

High incident electron momentum drives Orientation Ratio s approaching unity with bulk velocity near light speed and pronounced radial contraction. Take SLAC E-158 experiment (45 GeV electron colliding on stationary electron) as benchmark; measured momentum transfer $q_0 \sim 10$ GeV/ c restricts effective radius upper limit:

$$R_{\text{eff}} < 0.0886 \text{ fm}$$

From $v = sc$ compute $s \approx 0.999999997$; theoretical substitution yields $R \approx 0.1$ fm consistent with experimental constraint.

3. Alternative Verification Routes

Double-slit interference and Compton scattering supply indirect supporting evidence: - Double-slit interference: fringe visibility V varies against Orientation Ratio s and dynamic radius R enabling inverse radius extraction from measured visibility; - Compton scattering: photon wavelength shift is governed by electron recoil momentum uniquely determined by Orientation Ratio thus correlating to dynamic particle radius[73].

15.4 Physical Distinction from Conventional Lorentz Contraction

Mathematically identical expression between derived radius relation

$$R = R_0 \sqrt{1 - s^2}$$

and classical Lorentz contraction law $L = \frac{L_0}{\gamma}$ corresponds to fundamentally disparate underlying physics[72]:

Within present evolutionary framework $\sqrt{1 - s^2}$ quantifies residual transverse fractional distribution of internal Momentum units as statistical geometric projection emerging from intra-particle Vector Composition; Classical special relativity derives $\frac{1}{\gamma}$ from four-dimensional spacetime Lorentz coordinate transformation.

The present formalism acknowledges mathematical validity of Lorentz contraction formula while attributing the physical effect to statistical directional distribution of internal Momentum units instead of intrinsic geometric deformation of spacetime itself.

15.5 Chapter Conclusions

Built upon universal momentum formula $p = m\lambda\nu$ and Orientation Ratio model the complete physical proportional chain reads:

$$\lambda_{\text{photo}} \propto \frac{1}{p_{\gamma}} \propto \frac{1}{\Delta P} \propto R \propto \frac{1}{s}$$

Consolidated core relations:

$$s = \frac{v}{c} = \frac{N_+ - N_-}{N_+ + N_-}, \quad R(s) = R_0 \sqrt{1 - s^2}$$

Static electron radius referenced against Compton wavelength:

$$R_0 = \frac{\lambda_C}{2\pi} \approx 386 \text{ fm}$$

Finite inherent particle substructure is confirmed with dynamic radius negatively correlated to received Momentum Deviation and contracting against increasing Orientation Ratio; theoretical predictions align qualitatively with low/high-energy electron scattering measurements[71][73]. Contrary to conventional point-electron hypothesis, the evolutionary framework combining photoelectric empirical law and Momentum unit model endows measured scattering radius with definitive physical interpretation defined as dynamic electron radius.

Core conceptual differences from conventional theories:

1. Particles assemble from massive quantities of intrinsic light-speed Momentum units complying with $m = Nm_0$. Cancellation-Breaking transition and Orientation Ratio modulation collectively constitute physical origin of bulk motion, conventional mass-energy conversion and dynamic radial evolution[61][62];
2. Fermion dynamic radius follows $R(s) = R_0 \sqrt{1 - s^2}$ matching Lorentz-contraction mathematics yet originating statistically from transverse spatial projection of internal Momentum unit distribution rather than intrinsic spacetime geometric contraction[72].

Full physical quantity correlation sequence:

$$\lambda_{\text{particle}} \uparrow \Rightarrow m_{\gamma} \downarrow \Rightarrow p_{\gamma} \downarrow \Rightarrow \Delta P \downarrow \Rightarrow s \downarrow \Rightarrow v \downarrow \Rightarrow R \uparrow$$

15.6 Commentary on Chapter References

[71] Compton, A. H. (1923). A quantum theory of the scattering of x-rays by light elements. *Physical Review*, 21(5), 483–502. <https://doi.org/10.1103/PhysRev.21.483>

X-ray scattering experiment establishes finite characteristic electron size against point-particle assumption and supplies critical experimental confirmation of physically real photon momentum.

[72] Lorentz, H. A. (1904). Electromagnetic phenomena in a system moving with any velocity smaller than that of light. *Proceedings of the Royal Netherlands Academy of Arts and Sciences*, 6, 809–831. <http://www.dwc.knaw.nl/DL/publications/PU00014148.pdf>

Foundational pre-relativistic Lorentz-Abraham model rigorously derives length contraction from finite electron radius establishing critical precursor formalism for special relativity.

[73] Klein, O., & Nishina, Y. (1929). Über die Streuung von Strahlung durch freie Elektronen nach der neuen relativistischen Quantendynamik von Dirac. *Zeitschrift für Physik*, 52(11–12), 853–868. <https://doi.org/10.1007/BF01366453>

Derives exact relativistic Compton scattering cross-section following $4\pi r^2$ Coulomb attenuation and relativistic Covariance constraints unifying classical electromagnetic response and relativistic quantum dynamics.

16 Derivation of Quantum Entanglement and Particle Relationships

16.1 The Role of Quantum Entanglement in Evolution

According to the derivations in the preceding chapters, in a complex relational state evolution system, its stability relies on the multi-class representation capability of quasi-standard quantization. The core requirement is that states, causal relationships, and mechanics can all be represented by standard quantization. Thus, a distributed many-to-many evolutionary relationship can be established and mapped to one-to-many and one-to-one state evolutionary relationships.

Consequently, although particles within the cosmic system are independent carriers of causal interaction, they are required to possess internal structures. To remain intact under mechanical interactions during evolution, a constraint mechanism to maintain structural integrity is necessary, and quantum entanglement is precisely the corresponding constraint mechanism. It corresponds to the Internal Coupling constraint rule in the evolution laws, while the External Coupling rule corresponds to the interaction rule. Both rules instantaneously and synchronously associate evolution carriers within their perceptual ranges. That is, the core capability of quantum entanglement is to lock the internal units of entangled particles into an integral evolution carrier [13].

16.2 Quantum Entanglement and Conservation of Angular Momentum

Under this rule, the internal subunit structures of particles establish correlations through perceptual angle complementation. The sum of angular momentum emerging between each pair of subunit structures is zero, ensuring that the total angular momentum of a particle remains conserved and achieves synchronous rotation during global rotation and spin.

Its mathematical expression is:

$$\sum_{i=1}^n \vec{L}_i^{(\text{Internal Coupling})} = \vec{\mathcal{L}} \quad (\text{constant vector or zero})$$

where:

- $\vec{L}_i^{(\text{Internal Coupling})}$ is the Internal Coupling rule angular momentum of the i -th pair of subunit structures (distinct from classical motional angular momentum);
- n is the number of pairs of internal subunit structures of the particle;
- $\vec{\mathcal{L}}$ is a constant vector determined by the specific particle structure.

For spherically symmetric structures (e.g., fermions): $\vec{\mathcal{L}} = 0$

For directional structures (e.g., photons): $\vec{\mathcal{L}} \neq 0$, corresponding to their intrinsic spin/polarization states. Photons exhibit a geese-flock coding structure, i.e., a group of momentum units with identical direction and velocity, which can be encoded into an arbitrary three-dimensional integral structure.

This equation reflects the constraining effect of the Internal Coupling perceptual rule on the conservation of system angular momentum. The conservation of carrier angular momentum is a prerequisite for the existence and stability of particle spin angular momentum and macroscopic rotational angular momentum [48]. It is emphasized that Internal Coupling rule angular momentum is not classical motional angular momentum.

16.3 Particle State Transformation and Entanglement Relationships

1. Quantum Entanglement, Causal Interaction, and State Transformation Principles

(a) Rule Localization of Entanglement: Internal Coupling Constraint Rather Than an Independent Phenomenon

Based on the internal momentum unit structure of particles and the quantum entanglement Internal Coupling mechanism proposed in this evolutionary theory, wave-particle duality is not an inherent paradox of particles, but a manifestation of the switching of three fundamental dynamical forms of particles realized by entanglement synergy rules under different momentum deviations and environmental coupling conditions.

All particles are composed of massive light-speed momentum units and are locked into a topological whole by global quantum entanglement. The spin and internal structural stability are maintained by entangled angular momentum conservation, enabling the system to reversibly switch between collapsed state, diffuse state, and multi-path fluid state. All phenomena of quantum superposition, interference, and state collapse can be uniformly attributed to the synergistic redistribution of internal momentum flow.

(b) Entanglement and Three-State Dynamics: Global Interaction and State Transformation

According to the derivations in Section 5.6 and Chapter 11, all force interactions in the universe are instantaneously globally updated, i.e., synchronous interactions with all particles within the perceptual range of the entire universe within a unit t_0 . Thus, all mechanical interaction directions are synchronously vectorially superposed inside the particle, forming a many-to-many mechanical relationship interaction competition that converges to one-to-many (from a single-particle perspective) or even one-to-one (single dominance).

The process from complex environment to single environment, i.e., multi-slit $\rightarrow$ few-slit $\rightarrow$ single-slit $\rightarrow$ no-slit, corresponds to the variation of environmental quantity complexity and interactive momentum deviation magnitude under environment-dominated relationships.

Particles form their own momentum flow distribution mechanism due to momentum deviation or interaction magnitude, resulting in state transformation (collapsed state, diffuse state, multi-path fluid state). Quantum entanglement acts only as a structural constraint in this process, ensuring angular momentum conservation and spin stability during state transformation switching, i.e., the integrity of particle synergistic evolution.

$\begin{aligned}\text{Collapse Direction} &= \text{Inertial Causal Contraction Direction} \\ &= \text{Maximum Vector Composition Direction}\end{aligned}$

The three quantum forms can be fully described by a unified momentum flow path integral model without separate classical wave and particle systems. The differences among the three states are only regulated by three parameters: the number of paths, the internal Orientation Ratio s , and the momentum unit distribution ratio (manifested as "slit thickness" from the environmental perspective, see Chapter 17 for details). The unified equation is:

$$\rho_{\text{flow}}(X) = \sum_{k=1}^{N_{\text{slit}}} \left[\int_{L_k} \frac{N_k}{N} \cdot \sqrt{1 - s_k^2} \cdot \sigma_0(1 - s_k^2) \cdot \vec{e}_k ds \right] \cdot \cos(\Delta\theta_k)$$

where: N_{slit} is the number of equivalent evolution paths, s_k is the momentum Orientation Ratio of the corresponding path, N_k/N is the momentum unit distribution ratio, $\sqrt{1 - s_k^2}$ characterizes the remaining scale of transverse spatial distribution, σ_0 is the elementary interaction cross-section, and $\Delta\theta_k$ is the phase difference between paths. (The above formula is taken from the subsequent Chapter 17, with detailed derivations in the corresponding content.)

2. Particle Collapsed State (Particle Dominance)

When a particle is strongly dominated by a single direction of the environment, it exhibits a particle collapsed state. The greater the momentum deviation, the stronger the particle nature. When the input momentum deviation reaches the particle saturation upper limit, it radiates momentum outward (e.g., photons); continuous input of momentum deviation will replicate new particles from the Potential Space due to effects such as space-time interaction saturation. This stage manifests as observation effects in multi-slit experiments, single-particle impact effects, and quark confinement effects (quark effects are more complex due to encapsulation rules), with a more pronounced inertial direction.

The "collapse randomness" in standard quantum mechanics originates from incomplete information during the environmental entanglement preparation stage, rather than an intrinsic property of the evolution stage. The collapse direction can in principle be determined by the Orientation Ratio $s = (N_+ - N_-)/(N_+ + N_-)$ of internal momentum units and global perceptual constraints.

Collapse Direction _{Collapsed State} = Contraction to a Single Direction = Particle State → Scattering → Replication
--

Meanwhile, the spatial scale in the collapsed state is described by the dynamic radius formula (see Chapter 15):

$$R(s) = R_0 \sqrt{1 - s^2}, \quad R_0 = \frac{\lambda_C}{2\pi}$$

When $s \rightarrow 1$, $R \rightarrow 0$, the particle contracts to the limit scale, exhibiting the strongest particle nature.

(a) Compton Wavelength – Upper Limit of Single-Particle Momentum Absorption:

When the input momentum deviation reaches the intrinsic total momentum capacity $N \cdot m_0 c$ of the electron (i.e., $\Delta p = m_e c$, corresponding to the Compton wavelength $\lambda_C = h/(m_e c)$), the electron reaches the upper momentum limit that a single particle state can accommodate. At this point, the macroscopic velocity of the electron approaches the speed of light ($s \rightarrow 1$), the spatial scale is compressed to the limit $R \sim \lambda_C/2\pi$, exhibiting the strongest particle nature.

(b) Replication Mechanism – Particle Generation Beyond the Upper Limit:

When the input momentum deviation exceeds the above upper limit ($\Delta p > m_e c$), the electron cannot accommodate all momentum as a single encapsulated body. Excess momentum triggers the

momentum unit replication mechanism in the local Potential Space:

$$\Delta p_{\text{excess}} = \Delta p_{\text{input}} - m_e c \Rightarrow \text{Replication of } (e^+, e^-) \text{ pair from Potential Space}$$

The replication process follows momentum conservation and angular momentum conservation: the newly generated electron-positron pair borrows momentum $(\vec{p}, -\vec{p})$ from the Potential Space, satisfying total momentum conservation; meanwhile, quantum entanglement constraints ensure global conservation of spin and angular momentum. This mechanism is the critical condition for the transition of Compton scattering from "single-particle scattering" to "particle generation".

3. Particle Diffuse State (Wave Dominance, Quantum Coherent State)

In a low momentum deviation environment (inside atoms, BEC condensate), the Orientation Ratio $s \approx 0$, the internal momentum units cancel each other almost completely, and macroscopic motion is extremely weak. The system maintains single-path evolution ($N_{\text{slit}} = 1$), the transverse spatial distribution is fully expanded ($\sqrt{1 - s^2} \approx 1$), the momentum flow is uniformly diffuse with strong coherence, exhibiting typical wave properties corresponding to the diffuse characteristics of the de Broglie wavelength. Quantum entanglement stabilizes the spin and internal circulation structure through perceptual angle interlocking, forming macroscopic potential rotational degrees of freedom.

In this form, the interaction between the particle and the environment is uniform without causal contraction dominated by a single direction, thus maintaining spatial diffusion and wave characteristics.

4. Multi-Path Fluid State (Strong Wave Interference State)

In a multi-slit, multi-channel topological environment, the system evolution path differentiates into multi-channel parallelism ($N_{\text{slit}} > 1$). The total momentum units of the particle are synchronously distributed to multiple paths by entanglement constraints, forming multiple groups of synergistic momentum flows. For each path, $s_k \ll 1$ and spatial diffusion is sufficient, phase differences and self-interference effects occur between paths, giving rise to typical quantum phenomena such as wave superposition, interference and diffraction. Momentum borrowing and repaying between paths is completed in real time relying on the local Potential Space without superluminal propagation in space. The global entanglement conservation condition ensures synchronous multi-path evolution and stable spin and angular momentum at all times.

In this form, there is no causal contraction dominated by a single direction, and the particle exists simultaneously in multiple paths in the form of momentum flow distribution, forming **superluminal entanglement correlation** (see "Quantum Entanglement Momentum Distribution Principle" later in this chapter). This superluminal correlation is the result of Internal Coupling constraints in the momentum preparation and distribution stage, rather than relying on signal propagation through spatial flight, and depends on instantaneous perception.

In summary, wave-particle duality is essentially three emergent forms of the same set of momentum unit entanglement systems generated according to the Orientation Ratio and path topological conditions. Traditional wave-particle duality, quantum superposition, and self-interference do not require dualistic explanations, but are only the results of redistribution and synergistic evolution of internal momentum flow under different constraint parameters.

16.4 Quantum Entanglement Momentum Distribution Principle

Synergy and Multi-Level Relationships: Particle entanglement includes not only the three-state relationships in the previous subsection, but also multi-level encapsulation structures. From the fun-

damental momentum unit m_0 to photons, fermions, quarks, protons, atoms, and even molecules, each level can act as a subject of synergistic evolution. Composite entanglement states occur before and after particle decoupling and coupling, essentially a momentum distribution process among multiple particles.

Momentum Distribution Principle: Multi-particle (entanglement unit) systems share momentum topology conservation coding, and realize the allocation and balance of momentum deviation through local Potential Space momentum borrowing and repaying. The borrowing and repaying process only occurs in the local Potential Space and does not involve propagation in the evolution free dimension space [15].

Homology with Multi-Slit Experiments: Both can be understood as "distribution and synergistic evolution of the same integral momentum package":

- **Multi-path fluid state:** The momentum package is distributed to multiple spatial paths, realizing synergy through inter-path momentum flow entanglement and interference;
- **Multi-particle entanglement state:** Each particle acts as a "sub-carrier" of the momentum package, establishing entanglement correlation through Internal Coupling perceptual rules to achieve integral momentum synergy and conservation.

Constraint Mechanism: Regardless of the entanglement form, it is collective instantaneous non-local synergistic evolution, maintaining angular momentum conservation through external causal interactions.

$\text{Entanglement} = \text{Synchronous Interlocking} + \text{Instantaneous Non-Local Synergy} \\ + \text{Angular Momentum Conservation Constraint}$

For a multi-particle system (A, B, C, etc.), let p_A, p_B, p_C satisfy total momentum conservation: $p_A + p_B + p_C + \dots = 0$.

Example: A ($p_A = P+1$) and B ($p_B = P-1$) are initially balanced ($(P+1)+(P-1) = 2P \equiv 0$, adjusted reference frame).

When A couples with C ($p_C = P - 0.8$), a deviation $\Delta p = -0.2$ is generated. A transfers -0.2 to B through the Internal Coupling rule, and the system returns to equilibrium:

$$(P - 0.8) + (P + 1) + (-0.2) = 2P \equiv 0$$

In this process, the multi-particle entanglement state centrally distributes the remaining momentum deviation to the last associated particle – the "out-of-box momentum allocation rule". All borrowed and repaid momentum is completed through the local Potential Space, and manifests as quantum fluctuations when repaid, corresponding to momentum pairs $(\vec{p}_m, -\vec{p})$ in the Potential Space. Entanglement always maintains system angular momentum conservation and provides stable support for the spin and rotational states of each particle.

Therefore, quantum entanglement is a universal Internal Coupling rule for multi-scale synergistic evolution from leptons to molecules.

Uncertainty originates from the process of matching entanglement with the environment (preparation stage); Determinism originates from the process of momentum distribution (evolution stage).

16.5 Quantum Entanglement and Evolutionary Relationships

The states (spin, rotation, phase, etc.) of all units in an entangled body evolve in complete synchronization, but the evolutionary directions of each unit's state are not necessarily the same.

16.6 Quantum Entanglement and Particle Structural Relationships

All force interactions are driven by perceptual cross-sections. As an Internal Coupling perceptual rule, quantum entanglement and the External Coupling perceptual rule (corresponding to the four fundamental forces) together constitute the underlying mechanism of particle interaction. A unified equation is established based on the momentum flow distribution principle:

$$\vec{F} = \frac{d}{dt} \sum_i \vec{p}_i + \delta(\vec{r}) \langle \text{Code}_A | \text{Code}_B \rangle (\vec{p}_{\text{virt}} - \vec{p}'_{\text{virt}})$$

where the first term is the time rate of change of momentum flow (macroscopic force effect); in the second term, $\delta(\vec{r})$ reflects the instantaneity of Internal Coupling entanglement (originating from perceptual angle interlocking), and $\vec{p}_{\text{virt}}, \vec{p}'_{\text{virt}}$ are virtual momenta borrowed and repaid in the local Potential Space.

As a "geese-flock unidirectional momentum flow", the interaction of photons is essentially the injection and interception of momentum flow: when a fermion absorbs/emits a photon, it does not "eat" the photon, but adjusts its own spherically symmetric momentum encapsulated state to a new topological excited state containing the photon momentum flow through rules, with energy change satisfying $\Delta E = h\nu$. Entanglement maintains carrier angular momentum conservation during this process, ensuring the stable transition of spin states.

16.7 Two Types of Quantum Coherence

Type 1: Classical Motional Vector Composition (Motional Coherence)

The spatial linear superposition of momentum vectors of multiple independent particles, with the total momentum equal to the sum of the momenta of each particle. Each particle maintains causal independence and can be individually labeled and tracked (e.g., laser interference fringes, velocity composition of colliding balls).

$$\vec{P}_{\text{total}} = \sum_{i=1}^N \vec{p}_i, \quad \text{each } \vec{p}_i \text{ evolves independently}$$

Type 2: Quantum Entanglement Momentum Distribution (Collective Entanglement)

A single integral momentum package is distributed to multiple entangled bodies (particles or paths) through Internal Coupling perceptual rules. Each entangled body shares a unified coding, and momentum unit borrowing and repaying are completed through the local Potential Space (e.g., multi-particle GHZ states, single-particle multi-path fluid states).

$$P_{\text{total}} = \sum_{i=1}^N P_i^{(\text{distribution})}, \quad \sum \vec{L}_i^{(\text{Internal Coupling})} = \vec{\mathcal{L}} \text{ (constant vector)}$$

Characteristic	Classical Motional Vector Composition (Classical Coherence)	Quantum Entanglement Momentum Distribution (Collective Entanglement Coherence)
Physical Essence	Linear synthesis of momentum vectors of multiple independent moving bodies	Internal splitting of a single integral momentum package and synergistic evolution of units
Individual Correlation	Particles are independent of each other and can be individually distinguished and calibrated	Particles share unified coding and cannot be separated and exist independently
Mathematical Form	$P_{\text{total}} = \sum p_i$, each component takes values independently	Distribution and splitting under total conservation, components are linked by global constraints
Phase Origin	Relative spatial phase difference dominates interference effects	Phase belongs to global entanglement coding with non-local correlation
Action Level	External macroscopic motion superposition	Reorganization and distribution of internal momentum units

16.8 Summary and Explanation

This chapter systematically expounds the localization and mechanism of quantum entanglement in the generalized relational state evolution system from the perspective of evolutionary theory, with the core conclusions as follows:

1. **Essential Localization of Entanglement:** Quantum entanglement is the **Internal Coupling constraint rule** in evolution laws, and together with the External Coupling perceptual rule (corresponding to the four fundamental forces) constitutes the underlying mechanism of particle interaction. Its core capability is to lock the internal momentum units of particles into an integral evolution carrier, maintain angular momentum conservation ($\sum \vec{L}_i^{(\text{Internal Coupling})} = \vec{\mathcal{L}}$), and ensure that particles remain intact during evolution.
2. **Global Synergistic Collapse and State Selection:** Particle state transformation (collapsed state, diffuse state, multi-path fluid state) is driven by environmental momentum deviation and interaction complexity. The collapse direction follows the principle of "maximum vector composition direction", i.e., the inertial causal contraction direction. Quantum entanglement provides structural constraints in this process to ensure angular momentum conservation and spin stability during state transformation switching. The "randomness" of collapse originates from incomplete information in the preparation stage, rather than an intrinsic property of the evolution stage.
3. **Unified Description of Wave-Particle Duality:** Wave-particle duality is three emergent forms of the same set of momentum unit entanglement systems according to the Orientation Ratio s and path topological conditions, described by a unified momentum flow path integral equation without dualistic explanations.
4. **Momentum Distribution Mechanism:** Multi-particle entanglement states and single-particle multi-path fluid states are essentially homologous, both being the distribution of an integral momentum package to multiple carriers (multi-paths or multi-particles) through Internal Coupling rules. Borrowing and repaying are completed through the local Potential Space, following angular momentum

conservation. Uncertainty originates from the preparation stage of matching with the environment, and determinism originates from the momentum distribution evolution stage.

5. **Distinction Between Two Types of Coherence:** Classical motional vector composition (independent particles) and quantum entanglement momentum distribution (integral momentum package splitting, unit synergistic evolution) are essentially different in individual correlation, mathematical form, and phase origin. The former is external macroscopic motion superposition, and the latter is internal momentum unit reorganization and distribution.

Reflections on Understanding the Mechanisms of Quantum Physics

When searching for the essence of things, we need to clarify the essence of the problem and not generalize:

1. The essential problem of quantum entanglement: What role does quantum entanglement play in system evolution, what capabilities does it have, and what is its localization;
2. The uncertainty problem: Quantum "entanglement uncertainty and probability", which stage does the uncertainty originate from, and what is the nature of the problem, whether it originates from the uncertainty of the momentum distribution stage or the uncertainty of the entanglement selection stage with the environment;
3. The non-locality problem: How to understand locality, what is the essence of information transmission, what is the underlying mechanism of non-locality, and how to correlate under instantaneous non-locality (the glove analogy is not equivalent to quantum entanglement because it lacks the entanglement correlation (perception) mechanism; glove selection is logical exclusion rather than physical correlation). Therefore, non-locality is perceptual correlation non-local action, which is not equivalent to physical space transmission non-local action. Thus, glove selection is static logical preset correlation rather than dynamic evolution correlation [63].

In summary, this chapter completes the systematic derivation from the basis of evolutionary theory to the mechanism of quantum entanglement, clarifies the Internal Coupling rule localization of entanglement, the momentum distribution mechanism, and the unified description of wave-particle duality (essentially three states), and re-examines the fundamental problems of current quantum theory (origin of uncertainty, essence of non-locality) from an evolutionary perspective.

16.9 Commentary on Chapter References

[13] Schumacher, B., & Westmoreland, M. D. (2010). Quantum mutual information and the one-time pad. *Physical Review A*, 81(6), 062348. <https://doi.org/10.1103/PhysRevA.81.062348>

Entanglement is treated as quantum informational resource in original literature; the present work reifies entanglement as angular-momentum-bearing Momentum-unit resource reservoir.

[48] Bohm, D. (1951). *Quantum Theory*. New York: Prentice-Hall.

Restructures the EPR paradox into singlet spin configuration and standardizes angular momentum conservation as core entanglement constraint.

[63] Einstein, A., Podolsky, B., & Rosen, N. (1935). Can quantum-mechanical description of physical reality be considered complete? *Physical Review*, 47(10), 777–780. <https://doi.org/10.1103/PhysRev.>

47.777

Formulates the original EPR paradox establishing the completeness benchmark targeted by this framework's momentum partitioning formalism.

17 Derivation of Multi-Slit Path Integral Equation

This chapter establishes a multi-slit path integral system based on the internal momentum unit structure of particles, and uniformly explains the underlying dynamic mechanisms of wave-particle duality, multi-slit interference, quantum state transformation, observation effects and quantum tunneling. Firstly, seven core evolution mechanisms of this chapter are established as the underlying logical basis for the derivation of the whole chapter.

1. **Evolution Entity:** Its internal structure is $N \cdot m_0 \cdot c$, representing the total internal state quantity and evolution capability of a particle entity forming a momentum unit entity, which is the underlying origin of matter waves.
2. **Motion Relation:** Motion emerges from the internal orientation ratio of particles and is independent of the reference frame.
3. **Causal Interaction:** The key source of integration. In any environment and multi-slit system, particles distribute their own momentum flow to all slits through fluid entanglement state constraints, forming extensive weak interactions with the environment. In this process, the environmental perception cross-section density influence is manifested as slit thickness.
4. **Global Interaction:** Instantaneous synchronous interaction with all slits (precisely all cosmic perception ranges). By changing the internal momentum of the $N_1 \cdot m_0 \cdot c$ part of the particle, it evolves towards all environmental directions (attraction) or reverse directions (repulsion). The total "probability" of all slits is the source of momentum deviation integration after interaction, finally forming a new vector composition relationship inside the particle and generating the total inertial direction. This mechanism is the main source of probability amplitude, but the traditional probability concept includes energization and mathematization.
5. **Relativistic Effect:** Special relativistic effects occur under high-speed particle motion. Electromagnetic force degenerates into magnetic force, and gravitational force degenerates into gravitational lensing, which determines the interaction strength.
6. **Dynamic Radius:** After strong and weak interactions, particles form a new collapse direction based on the magnitude of momentum deviation. The greater the momentum deviation input in a single direction, the more obvious the direction performance, and finally particleization occurs, forming a dynamic radius.
7. **Quantum Tunneling:** Quantum tunneling is manifested as when there is a penetrable space, if there is a larger interaction source outside the potential barrier or its own momentum is too large, due to the global instantaneous composition relationship, the particle will fly through space through fluid entanglement or there may be a potential space momentum borrowing and repaying mechanism. Finally, it forms a causal interaction with the outside of the barrier and restores the particle state, with the maximum vector direction showing tunneling effect. This process is mainly a high-momentum or decay momentum distribution mechanism process.

Principle of Least Action: This mechanism is for supplementary explanation. The essence of the universe does not evolve according to the principle of least action; instead, multi-slits form extensive weak coupling through fluid states. There is no need to introduce the principle of least action to explain the formation of linear motion; linear motion can be formed as long as there is momentum deviation in the inertial system. Furthermore, it is not necessary to use it to explain dynamic effects in complex relational systems, such as supplementing the three-body complexity of Newtonian mechanics, because complex relational systems need to establish many-to-many relationships. However, it can be used as an ideal mathematical statistical effect, that is, the principle of least action emerges from environmental many-to-many interactions and vector composition inside particles, finally emerging as many-to-one or one-to-one inertial evolution.

Many-to-many environmental interaction $\rightarrow$ Vector composition inside the particle $\rightarrow$ Emergent macroscopic manifold

17.1 Core Basic Postulates and Physical Quantity Definitions

This chapter adopts the fundamental postulates and physical definitions rigorously derived in the preceding chapters as the logical foundation for the dynamics. The path integral equation is established with reference to the direction of the conventional Feynman path integral formulation[76]. However, within the present framework, microscopic quantum states are interpreted as topologically encoded real-valued momenta, and during dynamic evolution, the momentum flow represents the probability wave[77].

1. Postulate of Particle Internal Composition

A particle is composed of N light-speed evolution momentum units, the single unit momentum is $p_0 = m_0 c$, and the macroscopic mass of the particle satisfies $m = N m_0$. Define the internal orientation ratio:

$$s = \frac{N_+ - N_-}{N_+ + N_-}$$

This parameter directly determines the macroscopic velocity $v = sc$, and the motion property is independent of the reference frame.

2. Universal Momentum Law

All microscopic particles satisfy the unified momentum relation:

$$p = m \lambda \nu$$

Combined with the light-speed constraint $\lambda \nu = c$, the unified correlation between particle momentum, wave properties and internal unit evolution is realized.

3. Dynamic Radius Formula

The effective dynamic radius of a particle is determined by the transverse distribution of internal momentum:

$$R(s) = R_0 \sqrt{1 - s^2}$$

R_0 is the static intrinsic radius of the particle, and $\sqrt{1 - s^2}$ represents the residual transverse dispersion scale of momentum units, not the space-time geometric contraction.

Monotonic correlation law of momentum, wavelength, velocity and dynamic radius:

$$\lambda \uparrow \Rightarrow m \downarrow \Rightarrow p \downarrow \Rightarrow \Delta P \downarrow \Rightarrow s \downarrow \Rightarrow v \downarrow \Rightarrow R \uparrow$$

4. Core Distinction of Three Particle States

Particles have three basic dynamic forms: collapsed state, diffuse state and fluid entanglement state. The differences originate from the distribution of internal momentum units, spatial locality and dynamic radius.

Dimension	Collapsed State	Diffuse State	Fluid Entanglement State
Momentum unit distribution	Highly aligned ($s \approx 1$)	Directional dispersion ($s \approx 0$)	Multi-path distributed allocation
Spatial locality	Extremely strong point-like locality	Weakly bound wave packet distribution	Multi-path non-local coherent composition
Dynamic radius	Approaches 0, highly localized	Close to static radius R_0	Multi-path equivalent radius $> R_0$
Typical scenario	Detection, measurement interaction	Free propagation, single-slit diffraction	Multi-slit interference, quantum entanglement
Phase relation	No coherent effect	Stable self-coherence	Inter-path phase coherent composition

Fluid entanglement state is the core physical basis of multi-slit interference: particles do not geometrically pass through multiple slits simultaneously, but internal momentum units are allocated to multiple evolution paths through internal coupling constraints, and each path unit always satisfies the conservation of entanglement angular momentum $\sum \vec{L}_i = 0$. The path phase difference finally generates the interference pattern.

Three-state mutual transformation law: detection interaction collapses the fluid entanglement state into particle state; collapsed state gradually disperses into wave state during free propagation; multi-path topological environment differentiates diffuse state into fluid entanglement coherent state.

17.2 Derivation of Multi-Slit Path Integral Equation

1. Core Idea of Derivation

In a multi-slit system, the total momentum units of a particle are redistributed to each evolution path according to the momentum deviation input by the environment. The momentum flow contribution of each path is determined by the unit allocation ratio, orientation ratio and dynamic interaction cross-section. This chapter abandons the traditional complex volume integral in quantum theory and constructs a real-valued path integral entirely based on the underlying momentum structure of particles — the integral statistics the allocation of the particle's own momentum units on each path, and the environment triggers this allocation process by inputting momentum deviation.

Equation application scope: diffuse state, fluid entanglement state. In the collapsed state, all momentum units converge to a single path, $N_k/N = 1$, and the equation degenerates into a single-path evolution form:

$$\rho_{\text{flow}} = \Gamma(s) \cdot \sigma(R(s)) \cdot L \cdot \cos \theta$$

where $\Gamma(s)$ is the orientation correlation factor, $\sigma(R(s))$ is the dynamic perception cross-section, L is the path length, and θ is the normal angle of the path. The degenerate form omits constant factors irrelevant to the length, and the core physical parameters are completely retained.

2. Main Path Momentum Flow Density Integral Equation

Suppose the multi-slit system contains N_{slits} evolution paths, the number of momentum units allocated to the k -th path is N_k , satisfying the total conservation:

$$\sum_{k=1}^{N_{\text{slits}}} N_k = N$$

Combined with the orientation ratio s_k , dynamic radius $R(s_k)$, and path unit vector $\vec{e}_{\perp k}$, the momentum flow density path integral equation at the screen position X is obtained:

$$\rho_{\text{flow}}(X) = \sum_{k=1}^{N_{\text{slits}}} \left\{ \int_{L_k} \frac{N_k}{N} \cdot \Gamma(s_k) \cdot \sigma(R(s_k)) \cdot \vec{e}_k ds \right\} \cdot \cos(\Delta\theta_k)$$

This equation uniformly adapts to three quantum states with only different parameter values: multi-path corresponds to fluid entanglement state, single-path low s corresponds to diffuse state, and single-path $s \rightarrow 1$ corresponds to collapsed state[78]. This equation statistics the allocation relationship of the particle's own momentum units on multiple paths. The role of the environment is reflected in: inputting momentum deviation to the particle through causal interaction, driving the changes of the internal orientation ratio s_k and allocation ratio N_k/N of the particle, and then causing state transformation switching.

3. Analysis of Equation Physical Terms

Integration region L_k : integrate along the geometric length of each path, characterizing the modulation effect of path topology on momentum flow.

Momentum unit ratio N_k/N : quantify the weight of momentum contribution of each path, satisfying strict conservation.

Orientation correlation factor $\Gamma(s_k)$:

$$\Gamma(s_k) = \sqrt{1 - s_k^2}$$

Characterizing the transverse dispersion capability of momentum, consistent with the dynamic radius. $\Gamma(s_k)$ tends to zero at the high-speed limit ($s \rightarrow 1$), naturally reflecting the modulation of relativistic effects on interaction strength — the perception cross-section of high-speed particles contracts, and the electromagnetic and gravitational interaction strengths degenerate accordingly, echoing Article 5 of the seven major mechanisms in this chapter.

Dynamic perception cross-section $\sigma(R(s_k))$: the perception cross-section is proportional to the square of the dynamic radius:

$$\sigma(R(s_k)) = \sigma_0(1 - s_k^2)$$

Depict the interaction strength between particles and slit wall potential barriers, matching the diffraction regulation law of slit thickness and potential barrier structure.

Direction correction term $\cos(\Delta\theta_k)$: correct the influence of path inclination on the projection intensity of momentum flow.

4. Unified Correlation with Universal Momentum Law

Combined with $p = msc = m\lambda\nu$, we can get:

$$s = \lambda\nu/c$$

Embed wavelength and frequency parameters into the dynamic radius and perception cross-section to realize the complete unification of wave properties, particle structure and path dynamics.

17.3 Interference Intensity and Conservation Law

1. Interference Intensity Formula

The interference intensity is determined by the square of the modulus of the vector composition of momentum flow density, without introducing complex probability amplitude:

$$I(X) \propto |\rho_{\text{flow}}(X)|^2 = \left[\sum_{k=1}^{N_{\text{slits}}} \left[\int_{L_k} \frac{N_k}{N} \cdot \Gamma(s_k) \cdot \sigma(R(s_k)) \cdot \vec{e}_k ds \right] \cdot \cos(\Delta\theta_k) \right]^2$$

When the momentum allocation of each path is uniform and the orientation ratio is consistent, the system presents symmetric multi-slit interference characteristics. At this time, the equation can degenerate into the standard interference intensity distribution form, which is accurately matched with the observation results of Young's double-slit experiment.

$$\Delta\phi_{ij} = \frac{2\pi}{\lambda_{\text{eff}}} \Delta L_{ij}, \quad \lambda_{\text{eff}} = \frac{h}{p}$$

Real-valued phase is adopted throughout the process, avoiding the problem of physical reality of traditional complex probability amplitude.

2. Momentum Flow Conservation Law

The integral of momentum flow over the entire screen is strictly equal to the total momentum of the particle:

$$\int_{\text{screen}} \rho_{\text{flow}}(X) d^2X = p_{\text{total}} = Nm_0c$$

This conservation runs through the three-state evolution, ensuring the self-consistent closure of the entire dynamic system.

17.4 Three-State Parameter Adaptation Comparison Table

Dimension	Collapsed State	Diffuse State	Fluid Entanglement State
Momentum unit distribution	Highly aligned ($s \approx 1$)	Directional dispersion ($s \approx 0$)	Multi-path distributed allocation
Spatial locality	Extremely strong point-like locality	Weakly bound wave packet distribution	Multi-path non-local coherent composition
Dynamic radius	Approaches 0, highly localized	Close to static radius R_0	Multi-path equivalent radius $> R_0$
Typical scenario	Detection, measurement interaction	Free propagation, single-slit diffraction	Multi-slit interference, quantum entanglement
Phase relation	No coherent effect	Stable self-coherence	Inter-path phase coherent composition

17.5 Summary and Analysis of Quantum Observation Effects

1. Summary of Core Mechanisms in This Chapter

- (a) **Real-Valued Path Integral Equation:** Taking particle momentum unit structure, orientation ratio, dynamic radius and instantaneous interaction as the core, it uniformly covers diffuse state and fluid entanglement state, and the collapsed state can degenerate naturally.
- (b) **Abandonment of Complex Probability Amplitude:** Explain interference and quantum wave behavior based on internal momentum flow composition, and construct a completely real-valued quantum dynamics description framework.
- (c) **Dynamic Radius Mechanism:** The coupling relationship between dynamic radius and orientation ratio is the core innovative physical mechanism for distinguishing three states, regulating interaction strength and explaining wave-particle transformation, and undertaking the high-speed relativistic interaction degeneration law at the same time.
- (d) **Unification of Wave-Particle Duality:** Quantum wave-particle duality is a multi-parameter emergent behavior of the same momentum system, and multi-slit interference originates from path momentum allocation and coherent composition of fluid entanglement state.
- (e) **Essence of Probabilistic Randomness:** Quantum probabilistic randomness is not an intrinsic property of the universe, but an equivalent manifestation of the coupling of multiple mechanisms such as dynamic radius effect, limited observation accuracy, global instantaneous composition and potential space momentum borrowing and repaying. The bottom layer of the system is deterministic momentum evolution.
- (f) **Unified Evolution Principle:** All quantum behaviors of particles follow: global instantaneous vector composition generates the main evolution path, and momentum deviation dominates quantum state transformation.
- (g) **Integration Logic Positioning:** The path integral equation statistics the allocation relationship of the particle's own momentum units on each path, and the environment triggers the allocation process by inputting momentum deviation. The essence of "slit thickness" is the macroscopic manifestation of environmental perception cross-section density, which determines the maximum momentum deviation that the environment can input.

2. Analysis of Traditional Quantum Concepts

The observer effect, path probability normalization effect in traditional quantum theory, and the principle of least action as an equivalent description of macroscopic statistics are not basic cosmic rules, but emergent results of global instantaneous interaction.

- Momentum units have linear evolution capability, no need to traverse paths to find the least action (traversal implies delay effect, conflicting with instantaneous requirements);
- Diffuse and fluid entanglement states are dominated by large-scale weak coupling distribution evolution, and there is no spontaneous "minimization" trend of the system;
- Path probability normalization is a macroscopic statistical manifestation after global vector composition, not an underlying axiom.

3. Review of Core Mechanisms of This System

- **Global Instantaneous Interaction:** Particles complete global perception interaction within a single evolution beat, and vector composition automatically emerges the optimal evolution path, equivalently fitting the principle of least action and observation collapse behavior.
- **Dynamic Radius State Transformation:** Momentum deviation regulates the localization degree of particles, realizing continuous switching between wave and particle forms.

Multi-slit interference, quantum tunneling and all quantum interaction processes are essentially: **Global vector composition generates the dominant evolution direction, and quantum state reconstruction is completed according to momentum deviation.**

Mechanism Attribution of Quantum Tunneling: Quantum tunneling is a direct manifestation of the seventh major mechanism in this chapter, and its "probability" originates from the completeness of environmental entanglement matching in the preparation stage, not intrinsic randomness.

17.6 Commentary on Chapter References

[74] Frabboni, S., Gazzadi, G. C., Grillo, V., & Pozzi, G. (2015). Elastic and inelastic electrons in the double-slit experiment: A variant of Feynman's which-way set-up. *Ultramicroscopy*, 154, 49-56. <https://doi.org/10.1016/j.ultramic.2015.03.007>

Experiments verify varying slit-wall coating thickness modulates electromagnetic particle-boundary interaction and reshapes resulting diffraction patterns.

[75] 't Hooft, G. (2016). *The Cellular Automaton Interpretation of Quantum Mechanics* (Fundamental Theories of Physics, Vol. 185). Cham: Springer. <https://doi.org/10.1007/978-3-319-41285-6>

The author argues quantum mechanics constitutes a computational tool rather than final fundamental theory resting on deterministic classical ontology, consistent with the core standpoint of the present evolutionary formulation.

[76] Feynman, R. P., & Hibbs, A. R. (1965). *Quantum Mechanics and Path Integrals*. New York: McGraw-Hill.

Feynman' s path-integral formalism provides mathematical foundation for the present weighted topological path traversal coding of Momentum units, supporting formal treatment of momentum partitioning in photon-electron interactions, multi-path fluid state and global topological state conservation.

[77] Bohm, D. (1952). A suggested interpretation of the quantum theory in terms of "hidden" variables. I & II. *Physical Review*, 85(2), 166–193. <https://doi.org/10.1103/PhysRev.85.166>

Bohm' s pilot-wave hypothesis postulates objectively existing particle trajectories matching the causal reduction logic adopted throughout this evolutionary framework.

[78] Zurek, W. H. (1991). Decoherence and the transition from quantum to classical. *Physics Today*, 44(10), 36–44. <https://doi.org/10.1063/1.881293>

Decoherence theory supports the present description of environment-induced quantum-to-classical state collapse.

18 Derivation of Particle State Evolution Equation

18.1 Evolution Principle of Microscopic Particle States

Based on the path integral formula and particle internal structural principles from preceding chapters, this chapter constructs a state evolution equation purely within the evolutionary theory framework. Functionally, the derived path integral corresponds to Feynman' s path integral, as both describe superposition effects across multiple trajectories[76]. Meanwhile, the state evolution equation matches the physical scope of the Schrödinger wave equation for time-dependent evolution of microscopic particles, yet fundamental discrepancies exist in mathematical formulation and physical ontology between the two pairs of equations[77].

Conventional quantum mechanical equations are formulated primarily from a mathematical perspective without underlying physical foundations: Feynman' s path integral relies on probability amplitude superposition, and the Schrödinger equation is built upon complex-valued wave functions and the postulate of physically substantial energy. In the present framework, momentum path integration and Momentum Deviation state evolution equation replace these two formalisms respectively. Probability amplitudes are reduced to path partitioning of Momentum units, and wave functions are reinterpreted as deterministic evolution of Momentum Deviation fields.

Core variables include Momentum Deviation $\Delta\vec{P}$, Orientation Ratio s , and dynamic radius $R(s)$. This new equation eliminates drawbacks of traditional wave equations originating from complex-number dependence and artificial substantiation of energy, restores deterministic physical evolution and material Reality, and supplies operable dynamical tools for subsequent theoretical verification.

The multi-slit path integral governs global multi-path superposition of momentum flux, while the Momentum Deviation distribution evolution equation describes localized temporal evolution of internal Momentum units inside single particles. The two formulations form a differential-integral dual relation and jointly characterize the dynamics of the three particle states.

After simplification under the unified theoretical framework with Orientation Ratio s and dynamic radius $R(s) = R_0\sqrt{1 - s^2}$ as core modulating parameters, the complete evolution equation reads:

$$\frac{\partial \Delta\vec{P}}{\partial t} = \underbrace{\Gamma(s) [\vec{\Omega}_{\text{field}}(\vec{r}, R(s)) \times \Delta\vec{P}]}_{\text{Geometric Topological Rotation (Non-dissipative)}} - \underbrace{\nabla \cdot (D(R(s)) \Delta\vec{P})}_{\text{Radius-dependent Diffusion}} + \vec{S}_{\text{source}}$$

where:

- $\vec{\Omega}_{\text{field}}(\vec{r}, R(s)) \propto \nabla U$: field-induced topological rotation operator determined by potential gradient, explicitly dependent on dynamic radius to quantify spatial modulation of Momentum units enforced by field rules;
- $\Gamma(s) = \sqrt{1 - s^2}$: inherited Orientation Ratio correlation factor substituting conventional dissipation coefficients for unified modulation efficiency;
- $D(R(s)) \propto R(s)^2$: radius-dependent diffusion coefficient consistently correlated to the quadratic form of Perceptual Cross-section $\sigma(R(s)) = \sigma_0(1 - s^2)$.

$\vec{S}_{\text{source}}$ denotes the Momentum unit source term accounting for generation, annihilation and transfer of Momentum units in physical processes such as the photoelectric effect and quantum transition.

Physically, the three separate terms of the equation correspond to the dynamical origins of collapsed, diffuse and fluid entangled states respectively:

- **Non-dissipative Topological Rotation Term (Field-driven directional rearrangement)**

Formulated as $\Gamma(s)(\vec{\Omega}_{\text{field}} \times \Delta \vec{P})$, the cross-product preserves the magnitude $|\Delta \vec{P}|$ of Momentum Deviation while solely altering momentum orientation, consistent with the zero-work property of Lorentz force and angular momentum conservation in gravitational fields.

At high velocity where $s \rightarrow 1$, $R(s) \rightarrow 0$, $\Gamma(s) \rightarrow 0$, the effective perceptual area of particles responding to potential gradients shrinks and transverse topological modulation weakens. Magnetic force naturally emerges as a relativistic manifestation of electric field modulated by dynamic radius without requiring independent physical existence of magnetic fields.

- **Radius-dependent Diffusion Term (Core switch between Diffusion and Collapse)**

Diffusion strength is directly controlled by $R(s)$:

- At low speed with $s \approx 0$, $R(s) \approx R_0$, diffusion dominates and internal Momentum units spread spatially to form the Diffuse State corresponding to electron clouds and single-slit diffraction;
- At high speed with $s \rightarrow 1$, $R(s) \rightarrow 0$, diffusion coefficient approaches zero, Momentum units become highly localized and recover classical deterministic trajectories consistent with the Collapsed State manifested as discrete detection spots[77][78].

- **Momentum Unit Source Term (Dissipation and Quantum Transition)**

This term describes structural rearrangement of Encapsulation and interconversion of Momentum units between photons and electrons, directly modifying $|\Delta \vec{P}|$ and underlying mass-energy transformation as well as atomic energy level transitions for the photoelectric effect and discrete quantum jumps.

18.2 Fundamental Equation Validation and Self-consistency with Integral Formalism

Sharing identical postulates on internal particle microstructure with the preceding multi-slit path integral equation, the evolution equation consistently reproduces key experimental outcomes in atomic physics and quantum interference:

- **Ground-state Energy of Hydrogen Atom**

Ground-state electrons satisfy $v = \alpha c$, $s = \alpha$ with dynamic radius $R(s) = R_0 \sqrt{1 - \alpha^2} \approx R_0$ and stable Diffuse State configuration. Combined with virial theorem and kinetic energy relation, ground-state energy is derived as:

$$E_1 = -\frac{1}{2}m_e c^2 \alpha^2 \approx -13.6 \text{ eV}$$

in agreement with experimental measurement. **Note:** The expression in conventional energy units is employed here solely for numerical validation and does not imply that the present framework recognizes energy as an independent physical entity. Within the pure momentum formalism, this relation is equivalent to $E_1 = -\frac{1}{2}(Nm_0 c) \cdot (\alpha c)$, which describes the statistical value of space-time state shaping for the ground-state electron.

- **Stern-Gerlach Spin Splitting**

Fermion internal Momentum units possess two distinct topological chiralities (left-handed and right-handed). Under inhomogeneous potential fields corresponding to nonzero $\nabla \vec{\Omega}_{\text{field}}$, opposite topological torques act on Momentum Deviation for different chiral configurations and split atomic trajectories into two separated beams. Topological Encapsulation arrangement of momentum replaces the superluminal paradox of classical intrinsic spin and aligns logically with deterministic quantum interpretations[77].

18.3 Unified Dynamical Criteria for Three Particle States via Combined Differential-Integral Formalism

Joint application of localized differential evolution equation and global multi-slit path integral equation defines triggering thresholds, dominant governing terms and physical pictures for all three particle configurations:

State Type	Trigger Criterion	Dominant Equation Term	Physical Picture (Strictly Consistent with Prior Integral Equation)
I. Collapsed State	$s \rightarrow 1$ ($v \rightarrow c$) or strong measurement perturbation, $R(s) \rightarrow 0$	Topological rotation dominant, diffusion term $\rightarrow 0$, integral contribution deactivated	Highly aligned and localized Momentum units forming classical point-like particles confined to single trajectory without multi-path partitioning
II. Diffuse State	$s \approx 0$, $v \ll c$, simple potential environment, $R(s) \approx R_0$	Combined topological rotation and diffusion dominant, integral contribution deactivated	Directionally dispersed Momentum units forming spatially extended wave packets propagating along single path for single-slit diffraction

III. Fluid Entangled State (Multi-path Fluid State)	Multi-slit topological geometry with weak external disturbance permitting multi-path partitioning of Momentum units	Path integral activated with differential terms supplying local boundary input	Momentum units split across parallel trajectories maintaining Internal Coupling coherence for real-valued interference underlying multi-slit fringe formation and quantum entanglement
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Accordingly, the differential evolution equation resolves localized temporal variation of internal Momentum units within individual particles, whereas the multi-slit path integral quantifies global superposition across separated trajectories. Uniform modulation through Orientation Ratio s and dynamic radius $R(s)$ enables continuous phase transitions among collapsed, diffuse and fluid entangled states[78].

18.4 Summary and Supplementary Remarks

1. Equation Formulation and Physical Mechanism

The continuous state evolution equation represents the statistical continuum approximation of discrete microscopic relation $\vec{p}_i(n+1) = \vec{p}_i(n) + \sum \mathcal{R}_{\text{int}} \hat{r}$ accumulated over abundant sequential interactions across continuous space and time domains. Discrete equations establish fundamental causal microscopic foundations and continuous expressions serve macroscopic statistical descriptions to build a complete multi-scale dynamical framework spanning micro and macro physics.

Within the unified framework the finalized simplified evolution equation reads:

$$\frac{\partial \Delta \vec{P}}{\partial t} = \Gamma(s) [\vec{\Omega}_{\text{field}}(\vec{r}, R(s)) \times \Delta \vec{P}] - \nabla \cdot (D(R(s)) \Delta \vec{P}) + \vec{S}_{\text{source}}$$

2. Chapter Summary

Referring to the conceptual framework of the conventional Schrödinger equation, the particle state evolution equation is constructed upon evolutionary momentum rules and intrinsic particle microstructure. Paired with the multi-slit path integral equation, the two expressions establish differential-integral duality: the differential formulation describes localized temporal evolution of internal Momentum units per particle and the integral form accounts for global multi-path superposition effects. Unified regulation via Orientation Ratio s and dynamic radius $R(s)$ realizes continuous transitions among three core particle states. Free of complex wave functions and postulated substantial energy, the equation supplies an evolutionary-theory-based physical interpretation for quantum interference, intrinsic spin and energy-level transitions, alongside usable numerical simulation tools for subsequent experimental verification[77][78].

Supplementary note: conventional energy level notation is adopted solely for numerical verification without compromising the logical independence of the present evolutionary formalism.

18.5 Commentary on Chapter References

[76] Feynman, R. P., & Hibbs, A. R. (1965). *Quantum Mechanics and Path Integrals*. New York: McGraw-Hill.

Feynman's path integral formalism provides mathematical basis for the present weighted topological path traversal coding scheme of Momentum units and formal reference for mathematical description of momentum flux partitioning in photon-electron interactions, multi-path fluid state and global topological state conservation.

[77] Bohm, D. (1952). A suggested interpretation of the quantum theory in terms of "hidden" variables. I & II. *Physical Review*, 85(2), 166–193. <https://doi.org/10.1103/PhysRev.85.166>

Bohm's pilot-wave theory postulates objectively real particle trajectories consistent with the core philosophy of the current evolutionary framework; this work endorses the causal explanatory logic of hidden-variable pilot-wave interpretation for quantum phenomena.

[78] Zurek, W. H. (1991). Decoherence and the transition from quantum to classical. *Physics Today*, 44(10), 36–44. <https://doi.org/10.1063/1.881293>

Decoherence research supports the present physical explanation of environment-induced quantum-to-classical state collapse.

19 Derivation of Rotational Origin and Angular Momentum Conservation under Rigid-body Constraints

19.1 Microscopic Origin of Rigid Structure

Rigid bodies consist of dynamically interconnected microscopic particle networks locked by Perceptual Cross-section confinement, topological constraints and balanced attractive/Repulsion Potential interactions[79]. Such internal networks eliminate independent local degrees of freedom while permitting global translational or rotational bulk motion externally. Phillips' foundational 1979 constraint theory proves macroscopic solid rigidity originates from microscopic bonding and geometric topological restrictions with frozen internal degrees of freedom, furnishing classical theoretical backup for the present dynamic-network rigid-body model with suppressed internal freedoms.

Microscopic Constraint Condition

For any particle pair (i, j) , their relative separation satisfies:

$$\forall i, j, \quad |\vec{r}_i - \vec{r}_j| = \text{constant}, \quad \frac{d}{dt} |\vec{r}_i - \vec{r}_j| = 0$$

External Degrees of Freedom

Global rigid-body kinematics is fully parameterized by center-of-mass coordinate $\vec{R}$ and overall orientation angle θ , both dynamically variable.

Dynamical Characteristics

External force applied locally on a rigid-body site transmits associated Momentum Deviation instantaneously across the full particle network within infinitesimal interval $t \rightarrow t_0$ via interconnected Perceptual Cross-section channels. Macroscopically, rigid bodies exhibit structural robustness and coordinated evolutionary responses[13]:

1. Uniform bulk translational acceleration across the entire system;
2. Or bulk rotation constrained about fixed pivot points such as center of mass, supporting fulcrums or force-bearing positions.

19.2 Emergence Mechanism of Leverage, Orbital Rotation and Spin

1. Causal Constraints within Inertial State Evolution Systems

In inertial evolutionary systems, instantaneous motion configuration uniquely descends from prior causal inertial interactions enforcing strict continuous causal evolution. No intrinsic rotational property is preassigned to any object; all orbital and spinning motions Emerge macroscopically from persistent sequential causal interactions among constituent particles. This causal constraint yields a natural kinematic upper bound: peripheral tangential speed of any rotating rigid assembly cannot exceed the universal maximum inertial speed equal to the speed of light c [80], leading to testable predictive relation:

$$v_{\text{edge}} = \omega R < c$$

2. Stepwise Momentum Transmission and Symmetric Response inside Rigid Chains

A three-particle linear rigid chain $A - B - C$ serves as core analysis geometry with central particle B designated fixed rotational fulcrum and resistance center replacing isolated pivot O . External driving force or intrinsic Momentum Deviation applied at endpoint A propagates sequentially through interparticle Perceptual Cross-section from A to central pivot B and onward toward terminal particle C . Natural dynamical symmetry develops during momentum transfer such that distal C acts as counteracting endpoint forming symmetric motion relative to driving site A .

Central particle B remains stationary with $\vec{v}_B = 0$ as fundamental rotational pivot. Under ideal lossless conditions without supplementary drag or energy dissipation, the rigid assembly sustains steady periodic symmetric rotation centered on B .

3. Mathematical Formulation in Discrete Dynamical Description

Taking intermediate particle B as stationary rotational pivot with $\vec{v}_B = 0$, discrete dynamical relation links velocity and momentum for arbitrary constituent particle i along rigid chains:

$$\vec{v}_i = \vec{\omega} \times \vec{R}_i$$

where parameters are defined as follows: κ denotes Perceptual Cross-section transmission coefficient ($\kappa = 1$ for ideal loss-free rigid momentum transport), n_i counts sequential momentum transfer steps from input site A to target particle i , $\vec{R}_i = \vec{r}_i - \vec{r}_B$ denotes position vector of particle i referenced to central pivot B , and $\vec{\omega}$ represents instantaneous angular velocity of rigid system rotating around B .

For centrally symmetric three-body chains, terminal endpoints develop mirror-symmetric velocity magnitudes additionally bounded by universal light-speed constraint $v < c$ for all constituent particles[80].

19.3 Emergent Origin of Angular Momentum Conservation

Angular momentum conservation is not a fundamental physical axiom but an Emergent law arising jointly from three combined conditions: rigid topological confinement, stationary fixed pivot and dissipation-free momentum transport.

Emergence Prerequisites:

1. Fully established rigid particle network with invariant interparticle separation[61][79];

2. Stationary fixed rotational center B satisfying $\vec{v}_B = 0$;
3. Global momentum conservation without dissipative loss: $\Delta P_{\text{total}} = \text{constant}$.

Conservation Relation:

$$\vec{L} = \sum_i \vec{R}_i \times \vec{p}_i = I \vec{\omega} = \text{constant}, \quad I = \sum_i m_i R_i^2$$

Physical Essence:

Total cumulative localized Momentum Deviation equals bulk rotational net Momentum Deviation of the full rigid assembly.

Centrifugal and Coriolis effects are statistical manifestations of radial and tangential decomposed Momentum Deviation components originating from real sequential momentum transfer rather than fictitious inertial pseudo-forces.

Projectile kinematics inherits preexisting bulk rigid-body Momentum Deviation decomposed into orthogonal components instead of spontaneous force generation from vacuum.

Corresponding vector decomposition reads: instantaneous flight direction coincides with resultant orientation of total momentum $\vec{p}_{\text{total}}$ compounded from radial velocity $\vec{v}_{\text{radial}}$ plus tangential velocity $\vec{v}_{\text{tangential}}$.

Fundamental Origin:

Initial input Momentum Deviation propagates losslessly across rigid interconnected networks and stabilizes into persistent rotational momentum distribution constrained by central symmetric boundary condition at pivot B , thereby generating Emergent angular momentum conservation.

Experimentally testable projectile momentum relation:

$$\vec{p}_{\text{proj}} = \beta \cdot \vec{p}_{\text{input}}, \quad 0 < \beta < 1$$

where dimensionless coefficient β is determined empirically by ejection position, launch angle and the specific procedure of rigid constraint release.

19.4 Chapter Conclusions

1. Rigid structures originate from dynamically stable particle networks bounded by Perceptual Cross-section locking, topological constraints and balanced interparticle attractive/repulsive interactions; core feature includes suppressed internal independent degrees of freedom allowing exclusively bulk translation or rotation under external loading[61][79].
2. Leveraged motion, macroscopic rotation and intrinsic spin are not inherent material attributes but symmetric rotational responses Emergent from sequential Momentum Deviation propagation and midpoint symmetric constraint inside rigid linear chains under continuous causal particle interactions.
3. Tangential peripheral speed of all rigid rotating assemblies is inherently capped by the speed of light originating from finite stepwise momentum transmission and universal light-speed kinematic bound[80].

4. Complete causal chain from initial applied Momentum Deviation, sequential intersite propagation to stable pivot-centered rotation is reproduced entirely within discrete evolutionary dynamics without auxiliary postulates of continuous torque or inertial reference frames.
5. Three necessary prerequisites for Emergent macroscopic angular momentum conservation: fully formed rigid particle network with frozen internal degrees of freedom, stationary fixed rotational pivot B ($\vec{v}_B = 0$), and drag-free lossless evolution preserving total Momentum Deviation ($\Delta P_{\text{total}} = \text{constant}$). Fluid vortices (atmospheric or liquid) lack full rigid interparticle locking ($\kappa < 1$) and incorporate dissipative loss ($\eta > 0$), leading to exponentially decaying angular momentum; ideal perfect rigid bodies ($\kappa = 1, \eta = 0$) maintain invariant conserved angular momentum.

Fundamental physical conclusion: angular momentum conservation is not a primitive underlying axiom but a derived conserved relation Emergent from momentum conservation subject to rigid confinement, fixed pivot constraint and dissipation-free sequential interparticle momentum exchange. Precisely defined, genuine angular momentum corresponds to the fraction of total Momentum Deviation capable of mechanical work or composite velocity addition with fixed magnitude within specified angular intervals.

Causal evolutionary chain: rigid confinement $\Rightarrow$ injected Momentum Deviation $\Rightarrow$ localized resistance from pivot/fulcrum anchoring $\Rightarrow$ broken uniform global inertial motion $\Rightarrow$ adjacent momentum transport regulated by Perceptual Cross-section constraint $\Rightarrow$ symmetric radial/tangential decomposition of Momentum Deviation $\Rightarrow$ Emergence of lever mechanics, orbital rotation and intrinsic spin $\Rightarrow$ detached fragments inherit partial original Momentum Deviation components upon structural separation $\Rightarrow$ divergent departing vector orientations generate apparent centrifugal and Coriolis inertial effects.

19.5 Commentary on Chapter References

[12] Roy, R. (2009). Topological phases and the quantum spin Hall effect in three dimensions. *Physical Review B*, 79(19), 195322. <https://doi.org/10.1103/PhysRevB.79.195322>

Topological order is verified as fundamental state classification criterion defined by intrinsic topological invariants governing coordinated evolutionary capacity beyond simple density or geometric morphology.

[61x] Phillips, J. C. (1979). Topology of covalent non-crystalline solids I: Short-range order in chalcogenide alloys. *Journal of Non-Crystalline Solids*, 34(2), 153–181. [https://doi.org/10.1016/0022-3093\(79\)90033-4](https://doi.org/10.1016/0022-3093(79)90033-4)

Foundational constraint theory paper proving macroscopic solid rigidity arises from microscopic bonding and topological restriction freezing internal degrees of freedom to support the present dynamic-network rigid-body hypothesis with suppressed internal freedoms.

[79] Thorpe, M. F. (1983). Continuous deformations in random networks. *Journal of Non-Crystalline Solids*, 57(3), 355–370. [https://doi.org/10.1016/0022-3093\(83\)90424-6](https://doi.org/10.1016/0022-3093(83)90424-6)

Zero-frequency mode concept and rigidity percolation theory confirm rigid systems retain only global translational and rotational freedoms consistent with fixed interparticle separation and full-network instantaneous force transmission defined in this work.

[80] Cao, Z. X., & Liu, L. (2017). Chang Gan De Xiang Dui Xing Zhuan Dong [Relativistic Rotation of a Long Rod]. *University Physics*, 36(9), 27-30. <https://dx.doi.org/10.16854/j.cnki.1000-0712.2017.09.008>

Cao and Liu (2017) rigorously proved that for any sufficiently long rod, the linear velocity at its endpoint cannot exceed the speed of light under relativistic constraints, thereby resolving the so-called "horizon

problem” inherent in classical rotational kinematics.

20 Derivation of the Origin of Combined Velocity for Long-range Forces

20.1 Formation of Microscopic Combined Velocity

Within an information-dynamics evolution system under generalized inertial state evolution, two types of combined velocity mechanisms exist:

1. Local combined velocity: Evolution carriers A and B collide and form localized combined velocity based on intrinsic dimensional relational properties; the total vector of original driving quantities of A and B jointly propels the overall motion of the combined evolution carrier of A and B.
2. Long-range combined velocity: When long-range force interactions take place between evolution carriers A and B following long-range interaction rules governed by the perceptual mechanism, the carriers carry inherent initial inertial velocities $\vec{v}_A$ and $\vec{v}_B$. Variations of perceptual cross-section induced by moving speeds enable mutual inheritance of inertial velocities and establish the long-range combined force mechanism. Gravitational redshift generated from gravitational interaction serves as a typical example[81].

The relation for velocity increment reads:

$$\Delta\vec{v} \propto (\vec{v}_A + \vec{v}_B) \cdot \vec{F}_{\text{int}}$$

In the above formula, the direction of $\vec{F}_{\text{int}}$ determines whether the interaction is attractive (pointing toward the counterpart carrier) or repulsive (directed away from the counterpart carrier); the term $\vec{v}_A + \vec{v}_B$ governs the magnitude and vector direction of inherited velocity. Gravitational slingshot effect, gravitational redshift and blueshift all originate from this mechanism.

The overall physical process is interpreted as follows: each carrier acquires momentum deviation $\vec{F}_{\text{int}} \cdot \vec{v}_{\text{target}} \cdot \Delta t$ modulated by the counterpart's velocity via interaction, distributes the acquired extra momentum deviation across all internal momentum units, and eventually produces macroscopic velocity increment $\Delta\vec{v} = \Delta\vec{p}_{\text{add}}/M$.

The derivation relations are listed below:

$$\vec{F}_{\text{int}} \cdot \Delta t = \Delta\vec{p}_{\text{int}} \Rightarrow \Delta\vec{p}_{\text{add}} = \Delta\vec{p}_{\text{int}} \cdot \vec{v}_{\text{target}} \cdot \Delta t \Rightarrow \Delta\vec{v} = \frac{\Delta\vec{p}_{\text{add}}}{M}$$

Specification of physical quantities:

- $\vec{F}_{\text{int}}$: Momentum deviation per unit time, corresponding to the conventional definition of "force";
- $\vec{v}_{\text{target}}$: Velocity of the interacting counterpart, the source for inherited velocity;
- Δt : Duration of ongoing interaction;
- $\Delta\vec{p}_{\text{add}}$: Net extra momentum deviation gained by the entire system;
- $M = Nm_0$: Total system mass, equal to the product of total momentum unit count and single momentum unit mass;

- $\Delta\vec{v}$: Macroscopic incremental velocity of the carrier system.

Interaction intensity carries inherent mass dependence: under weak interaction capacity, carriers with larger mass yield weaker velocity inheritance.

- High-mass carrier: $\Delta v = \Delta p / M_{\text{large}}$, smaller velocity increment and suppressed inheritance efficiency;
- Low-mass carrier: $\Delta v = \Delta p / M_{\text{small}}$, larger velocity increment and enhanced inheritance efficiency[81].

20.2 Constraint of Attractive and Repulsive Potentials on Resultant Velocity Formation

The above mechanism applies not only to stable orbital systems sustained by long-range interactions but also to the full interaction process spanning from initial separation to final coalescence. Typical examples include a stone falling toward the Earth and electron capture by a proton; the whole interaction can be split into three core phases as follows.

1. Initial Configuration: Initial Momentum Deviation as Precondition of Relative Motion

At least one of the two Evolution carriers A and B carries initial Momentum Deviation ($\vec{p}_A$ or $\vec{p}_B$) to generate relative motion and establish available interactive Perceptual cross-section. No new causal interaction can be triggered if neither body bears Momentum Deviation, namely static rest without intrinsic Orientation Ratio offset. The initial condition is formulated as:

$$\vec{p}_A^{\text{init}} \neq 0 \quad \text{or} \quad \vec{p}_B^{\text{init}} \neq 0$$

2. Causal Interaction under Attraction Constraint: Virtual Momentum Pair Borrowing and Restitution within Potential Space

When integrable spacing exists between A and B, long-range Attraction Potential (gravitational or electromagnetic force) drives causal interaction. Microscopically, the two entities jointly borrow and return a pair of opposite virtual momenta ($\vec{p}, -\vec{p}$) from local Potential Space and toggle the Orientation Ratio s of partial internal Momentum units Nm_0c to evolve toward one another.

- The incremental Momentum Deviation acquired separately by two interacting objects satisfies:

$$\Delta\vec{p}_A = -\Delta\vec{p}_B$$

- Macroscopic energy and momentum fail to conserve throughout the procedure; only the total momentum capacity $\sum Nm_0c$ remains invariant;
- Continuous integration follows the three-dimensional spherical attenuation law of Perceptual cross-section ($1/4\pi r^2$, where the factor 4π is absorbed into gravitational constant G and electromagnetic coupling constant k within conventional theories), macroscopically manifested as mutual approaching and rising speed;
- Termination of approaching: direct collision occurs, or both objects enter overlapping domain where Attraction Potential and Repulsion Potential compete mutually.

3. Causal Interaction under Repulsion Constraint: Local Resultant Velocity and Co-evolution

Approach terminates once no residual integrable space remains and Perceptual cross-sections of attractive and repulsive interactions fully overlap, after which repulsive constraint dominates the evolution.

- Total Momentum Deviation (initial value plus accumulated interactive contribution) of each constituent stops varying;
- The two original independent Evolution carriers merge into a new unified co-evolving compound system;
- Local resultant velocity emerges and locks the system into coordinated evolutionary relation.

Total Momentum Deviation equals the sum of initial individual momenta and deviation accumulated from intermediate interaction:

$$\vec{p}_{\text{tot}} = \vec{p}_A^{\text{init}} + \vec{p}_B^{\text{init}} + \Delta\vec{p}_{\text{int}}$$

Given fully coordinated internal configuration of the merged system with fixed relative positions and overlapped Perceptual cross-sections, no net extra Momentum Deviation is generated between internal components. The bulk translational velocity of the composite system is determined by total momentum deviation and total rest mass:

$$\vec{v}_{\text{co}} = \frac{\vec{p}_{\text{tot}}}{m_A + m_B}$$

- The above expression is defined as the local resultant velocity formula, universally describing the final kinematic state for all coalescence events ranging from planetary accretion and molecular binding to composite particle formation;
- The co-evolving compound sustains inertial motion unchanged without external momentum injection;
- Extra incoming Momentum Deviation from outside or reactivated cross-section interaction via internal triggering is mandatory to break the coordinated bound state for fission or particle decay.
- Multi-body complexity: The mutual topological relations among multiple particles collectively influence one another, forming more intricate Perceptual Cross-section constraints and interaction processes. However, the total Momentum Deviation is given by the sum of the initial Momentum Deviations of all participating particles, the cumulative Momentum Deviations from interactions, and the dissipated Momentum Deviation.

Unified Formulation for Two Interaction Stages

In summary, the complete dynamical process of any two Evolution carriers evolving from separated initial state to bound co-evolution is uniformly expressed via two core equations:

$$\vec{p}_{\text{co}} = \vec{p}_A^{\text{init}} + \vec{p}_B^{\text{init}} + \Delta\vec{p}_{\text{int}}$$

$$\vec{v}_{\text{co}} = \frac{\vec{p}_{\text{co}}}{m_A + m_B}$$

Difference from Stable Orbital Motion

For steady orbital revolution instead of physical coalescence, interactive Momentum Deviation $\Delta \vec{p}_{\text{int}}$ does not accumulate irreversibly but redistributes dynamically within each orbital period through continuous alternation between realized momentum and remaining integrable space, consistent with the orbital integral conservation mechanism elaborated in Chapter 21.

Phase Classification for Various Physical Phenomena

Phenomenon	Attraction-governed Phase	Repulsion / Resultant Velocity Phase	Remark
Stone falling to Earth	Approaching with accumulating Momentum Deviation	Merging into Earth after impact	Zero relative velocity after coalescence
Proton capturing electron	Inward approaching accompanied by photon emission	Hydrogen atom formation in bound co-evolution	Full Encapsulation of internal Momentum Deviation
Planet orbiting the Sun	Permanently confined within attraction phase	No complete coalescence	Periodic cyclic integral without collision

20.3 Chapter Summary

From the underlying physical perspective, macroscopic long-range force interaction emerges statistically from discrete sequential microscopic virtual momentum unit borrowing and returning. Macroscopic mechanical phenomena including orbital motion and gravitational coupling arise from superposition integrals of quantum-scale discrete virtual momentum exchanges and varying perceptual cross-sections[82]. Contrary to classical mechanics which treats force and potential energy as intrinsic bulk properties of isolated objects, velocity inheritance, momentum transfer and potential energy shift induced by long-range interactions are relational effects exclusively generated via mutual coupling between two or more independent evolution carriers and carry no standalone substantial physical meaning[83].

20.4 Chapter Reference Comments

[81] Huang, Y., Li, Q. Z., Liu, J. F., Dong, X. B., Zhang, H. W., Lu, Y., & Du, C. H. (2025). A high-velocity star recently ejected by an intermediate-mass black hole in M15. *National Science Review*, 12(2), nwae347. <https://doi.org/10.1093/nsr/nwae347>

Experimental verification of gravitational slingshot effect confirms gravitational interaction constitutes bidirectional coupled transfer of momentum and velocity; low-mass celestial bodies obtain far larger inherited velocity gain than massive counterparts, consistent with the core rule that larger mass corresponds to poorer velocity inheritance capability proposed in this work.

[82] Zee, A. (2010). *Quantum Field Theory in a Nutshell* (2nd ed.). Princeton: Princeton University Press.

Authoritative textbook of quantum field theory elaborating that macroscopic long-range forces (gravitation, electromagnetic force) stem from discrete sequential virtual momentum borrowing and returning at microscopic scale. It provides core theoretical foundation for the statement that orbital motion emerges from statistical integration of discrete virtual momentum exchange and varying perceptual cross-sections. Differing from classical mechanics assigning force and potential energy as intrinsic object attributes, velocity inheritance, momentum transfer and potential energy variation induced by long-range interactions

are relational outcomes of multi-carrier interactions without independent substantial definition.

[83] Poincaré, H. (1906). Sur la dynamique de l'électron. *Rendiconti del Circolo Matematico di Palermo*, 21, 129–176. <https://doi.org/10.1007/BF03013466>

This literature establishes that physical quantities including potential energy belong to inter-object relational properties rather than intrinsic properties of isolated matter and have no independent substantial existence, supporting the relational interpretation of gravitational potential energy instead of inherent bulk energy storage adopted in this framework.

21 Derivation on Microscopic Origin and Energy Storage Principle of Celestial Orbital Motion

21.1 General Mechanism Overview

Within conventional classical physics, planetary revolution and orbital motion are interpreted via conservation of angular momentum, mutual transformation between kinetic and potential energy and total mechanical energy conservation. From the fundamental microscopic viewpoint built upon perceptual interaction, discrete quantum momentum exchange and sequential integral accumulation, the above conservation laws only serve as macroscopic approximate descriptions.

The essential origin reads: in dissipation-free systems, the sum of initial inherent momentum deviation and global integral capacity of inter-carrier perceptual interaction is strictly conserved[84]. Variations of orbital speed, inertial state switching between aphelion and perihelion and apparent conversion between kinetic and potential energy are secondary macroscopic emergent effects derived from this underlying conservation principle.

21.2 Underlying Microscopic Interaction Mechanism

Taking the Sun-Earth two-body orbital system as research object, celestial revolution does not originate from action-at-a-distance gravitation but consists of continuous discrete quantum-scale perceptual interactions between constituent particles of two celestial bodies. The core mechanism falls into three hierarchical parts:

1. Quantum Virtual Momentum Borrowing and Returning Mechanism

Bidirectional interaction rearranges directional distribution of internal momentum units inside celestial bodies and modifies net orientation ratio s , generating macroscopic momentum deviation from gravitational interaction. The microscopic equivalent relation is expressed as:

$$\Delta P = \frac{Gm_e m_s}{r^2} \cdot t_0 = (N_+ - N_-)m_0 c = s \cdot (Nm_0 c)$$

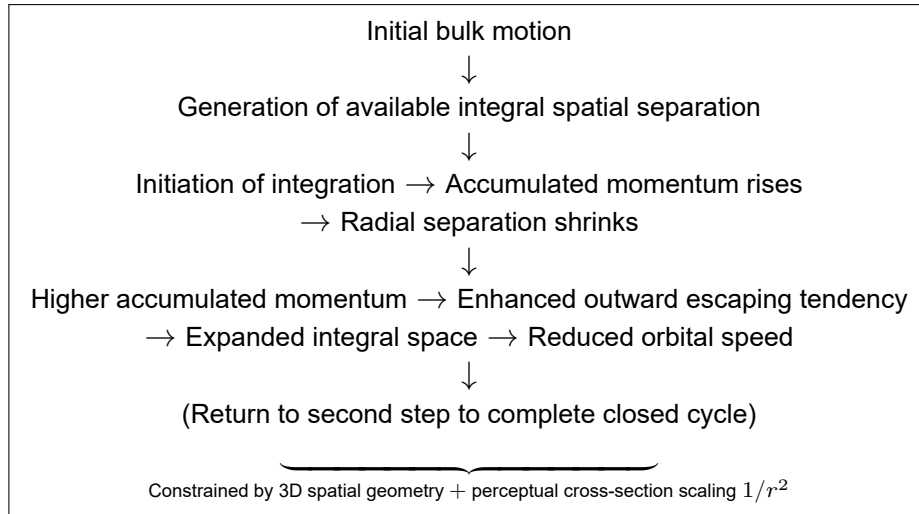
Here $Nm_0 c$ denotes total intrinsic momentum capacity of the whole system subject to absolute conservation; interaction only alters orientation ratio s while total bulk momentum capacity remains unchanged.

2. Long-range Combined Velocity Inheritance Mechanism

Both the Sun and Earth carry inherent primitive inertial velocity. Continuous radial separation variation modifies integral magnitude of unit-time perceptual interaction momentum. No unilateral acceleration occurs in the coupled system; initial system-wide momentum deviation is shared and redistributed mutually between two bodies to dynamically adjust orbital motion and realize combined velocity effect[85].

3. Cyclic Integral Logic of Orbital Motion

Changes in orbital velocity and inertial property originate from dynamic complementary conversion between available integral spatial range and accumulated realized momentum deviation. The core cyclic logic is summarized below:



Constrained by spatial dimensional geometry and topological dependence of perceptual cross-section, cyclic orbital evolution establishes the conservation of angular integral quantity: accumulated integral of perceptual cross-section and partition of momentum deviation along arbitrary orbital arc are globally invariant, replacing conventional angular momentum conservation.

Practical orbital manifestations of above cyclic rule:

As Sun-Earth distance expands toward aphelion: available integral space enlarges, effective perceptual cross-section weakens following inverse-square scaling, accumulated realized inertial momentum deviation drops and orbital velocity decreases

$$r \uparrow \Rightarrow \frac{1}{r^2} \downarrow \Rightarrow v \downarrow$$

Meanwhile the system's residual potential integral storage capacity increases synchronously.

As Sun-Earth distance contracts toward perihelion: available integral space shrinks, effective perceptual cross-section intensifies, accumulated realized momentum deviation rises markedly and orbital velocity increases

$$r \downarrow \Rightarrow \frac{1}{r^2} \uparrow \Rightarrow v \uparrow$$

Meanwhile residual potential integral storage capacity declines synchronously.

Accordingly, apparent kinetic-potential energy transformation along orbital trajectory does not correspond to fundamental energy conservation but reflects complementary redistribution between realized accumulated momentum and residual unused integral space[86].

21.3 Theoretical Definition of Physical Quantities

Unified fundamental physical definitions for Sun-Earth two-body orbital system:

1. Primitive inherent momentum deviation of Earth subsystem: $\vec{p}_0 = m_e \vec{v}_0$, constant baseline momentum of the isolated coupled system;
2. Spherical perceptual cross-section decaying with squared separation: $\Sigma(r) = \frac{1}{4\pi r^2}$;
3. Discrete single-cycle interactive momentum deviation: $\Delta P(r) = \frac{Gm_e m_s}{r^2} \cdot t_0$;
4. t_0 : Characteristic constant interaction time of inherent perceptual response for the given system.

21.4 Fundamental Global Conservation Equation

For closed two-body systems free from external force and dissipation, the sole rigorously conserved quantity equals the sum of initial bulk momentum deviation and residual global integral capacity of perceptual cross-section, establishing the primary underlying conservation law for orbital dynamics:

$$\vec{p}_0 + \int_r^\infty \vec{\Sigma}(r) dr = \text{constant}$$

Decompose conserved bulk quantity into instantaneous realized momentum and unused residual integral capacity to obtain dynamic partitioning relation:

$$\vec{p}_0 = \vec{p}_{\text{inst}} + \int_R^\infty \vec{\Sigma}(r) dr$$

In which $\vec{p}_{\text{inst}} = m_e \vec{v}$ denotes real-time instantaneous momentum deviation matching current orbital kinematic status[87].

21.5 Integral Relations at Aphelion and Perihelion

1. Aphelion Integral State ($r = r_a$)

Maximum radial separation yields weakest perceptual integral efficiency, minimal accumulated realized momentum, minimum orbital speed and maximal residual potential integral storage:

$$\vec{p}_0 = \vec{p}_a + \int_{r_a}^\infty \vec{\Sigma}(r) dr, \quad \vec{p}_a = m_e \vec{v}_a, \quad |\vec{v}_a| \ll |\vec{v}_p|$$

2. Perihelion Integral State ($r = r_p$)

Minimum radial separation produces strongest perceptual integral efficiency, maximal accumulated realized momentum, peak orbital speed and minimal residual potential integral storage:

$$\vec{p}_0 = \vec{p}_p + \int_{r_p}^\infty \vec{\Sigma}(r) dr, \quad \vec{p}_p = m_e \vec{v}_p, \quad |\vec{v}_p| \gg |\vec{v}_a|$$

3. Difference Relation of Orbital Momentum

Eliminate invariant initial bulk momentum $\vec{p}_0$ by combining aphelion and perihelion conservation expressions to derive integral formulation for orbital momentum discrepancy:

$$\vec{p}_a - \vec{p}_p = \int_{r_p}^{r_a} \vec{\Sigma}(r) dr$$

This equality proves instantaneous macroscopic momentum varies distinctly across orbital positions; macroscopic momentum is non-conserved, and conventional conservation laws of mechanical energy and angular momentum lack strict universal validity[87].

21.6 Hierarchical Classification Between Fundamental Conservation and Classical Conservation Rules

The proposed conservation of initial bulk momentum deviation plus global residual perceptual integral capacity represents the only rigorously valid underlying orbital conservation law and is mathematically non-equivalent to traditional conserved quantities:

1. Angular momentum conservation $mr v_{\perp} = \text{constant}$: merely an approximate projection restricted to transverse velocity component ignoring radial integral evolution, valid only as local macroscopic phenomenology[86];
2. Mechanical energy conservation $\frac{1}{2}mv^2 + V(r) = \text{constant}$: empirical statistical fitting summarizing alternating momentum accumulation and potential integral shift rather than strict dynamical conservation[87];
3. Rigorous fundamental conservation: invariant initial baseline momentum plus unchanged global residual perceptual integral capacity, rooted in absolute conservation of total intrinsic mc bulk momentum of constituent momentum units;
4. Complete removal of integral source (e.g., eliminating the Sun) eliminates corresponding potential energy carrier, verifying potential energy cannot exist as standalone bulk energy stored inside isolated celestial objects but functions as relational quantity dependent on bilateral perceptual cross-section interaction[88].

21.7 Experimentally Testable Theoretical Predictions

Core prediction: identical spacecraft receives unequal net momentum increment from gravitational slingshot maneuver when encountering the host celestial body at perihelion versus aphelion orbital locations.

$$\Delta P_{\text{perihelion}} \neq \Delta P_{\text{aphelion}}$$

Strict validity of traditional angular momentum or mechanical energy conservation requires identical momentum transfer capability at perihelion and aphelion. The present framework predicts systematic measurable discrepancy governed by perceptual cross-section integral:

$$\Delta P_{\text{perihelion}} - \Delta P_{\text{aphelion}} \propto \int_{r_p}^{r_a} \frac{1}{4\pi r^2} dr$$

Theoretical basis: orbital cyclic evolution abandons strict angular momentum or mechanical energy conservation and satisfies dissipation-free fundamental conservation of realized momentum plus residual integral capacity:

$$P_{\text{realized}} + P_{\text{potential}} = \mathcal{C}$$

Definition of symbols:

- P_{realized} : Accumulated realized momentum deviation directly linked to macroscopic bulk momentum;
- $P_{\text{potential}}$: Unused residual integral capacity expressed via path integral of perceptual cross-section;
- $\mathcal{C}$: Time-invariant global orbital system constant.

Ultimate underlying bulk conservation (total momentum unit capacity invariant):

$$\sum mc = \text{constant}$$

Falsification Criterion: Experimental measurement yielding equivalent momentum transfer $\Delta P_{\text{perihelion}} = \Delta P_{\text{aphelion}}$ within measurement uncertainty invalidates the proposed integral-based orbital conservation law and restores strict classical energy and angular momentum conservation for relevant orbital scale.

Experimental Judgment Logic:

- Equal momentum transfer at aphelion and perihelion supports conventional angular momentum and mechanical energy conservation;
- Measured relation $\Delta p_p \gg \Delta p_a$ validates underlying integral capacity conservation and classifies classical conservation laws as approximate empirical fitting under limited specific scenarios.

21.8 Chapter Conclusion

1. The dynamical origin of cyclic celestial orbital motion cannot be attributed to angular momentum conservation or reciprocal conversion between kinetic and potential mechanical energy. For closed dissipation-free systems, governing fundamental rules include conservation of initial bulk momentum deviation plus global bilateral perceptual integral capacity and absolute conservation of total bulk evolution capacity Nm_0c of constituent momentum units. All classical conservation formulations emerge as approximate macroscopic statistical descriptions atop the strict underlying conservation rules. Within dissipation-free rotating systems, net bulk momentum deviation remains unchanged over arbitrary characteristic evolution time t_0 , intrinsically differing from conventional angular momentum conservation.
2. Variation of orbital speed, inertial magnitude and apparent kinetic-potential alternating conversion constitute secondary macroscopic phenomena generated via dynamic complementary transformation between available integral space and accumulated realized momentum deviation.
3. Momentum transfer and mechanical work capacity of a celestial body differ objectively across distinct orbital positions; perihelion possesses markedly stronger momentum exchange capability than aphelion. This observable discrepancy distinguishes underlying integral conservation from approximate classical conservation rules and supplies feasible experimental verification for the proposed theory.

4. All macroscopic orbital kinematic regularities ultimately trace back to underlying topological integral mechanisms including discrete quantum virtual momentum borrowing-returning, perceptual cross-section interaction and dynamic combined velocity inheritance at microscopic particle scale.

5. Cyclic Evolution and Energy Storage

Constructed available integral space provides unified microscopic dynamical foundation for nearly all energy conversion and cyclic evolutionary phenomena.

Chemical energy storage inside rechargeable batteries originates from artificially isolated proton-electron interaction and man-made available integral space; removal of isolating barrier restores direct perceptual cross-section coupling between protons and electrons and enables continuous integral-driven external energy output.

Time crystals, cyclic celestial orbits and superconducting states all emerge within imperfectly synchronized evolution systems maintaining dynamic equilibrium between attractive-repulsive interaction and alternating integral accumulation-consumption with periodic construction and depletion of available integral space.

Necessary prerequisite for persistent cyclic evolution: interacting components remain mutually independent to sustain functional perceptual cross-section. Complete synchronization such as celestial collision or particle fusion eliminates available integral space and terminates cyclic evolution entirely. Orbital systems maintain bounded gravitational confinement and persistent integral space against centrifugal tendency satisfying:

$$\int_r^\infty \frac{k}{4\pi r^2} dr = \frac{mv^2}{2} + C$$

Equivalent compact form:

$$\frac{k}{r} = \frac{1}{2}mv^2 + \text{constant}$$

This expression shares identical mathematical form with conventional classical orbital relation.

21.9 Comparative Analysis of Linear Compounding and Angular Compounding Integrals

Linear compounding integral refers to continuous collinear radial integration, exemplified by a stone falling toward the Earth; angular compounding integral stands for persistent integration coupled by radial and tangential components, represented by celestial orbital revolution. Essential distinctions exist between the two in driving source, spatial geometry, governing variables, integration object, macroscopic outcome, statistically conserved quantity and integral logic, as listed in the following table:

Feature	Linear Compounding (Free Fall)	Angular Compounding (Celestial Orbital Motion)
Core Driving Mechanism	Continuous injection of collinear momentum increment $\Delta\vec{p}$ from gravitational source	Continuous injection of direction-varying $\Delta\vec{p}$ from gravitational source
Spatial Geometry	Straight radial path (r decreases monotonically)	Coupled radial-tangential geometry (r and θ vary alternately)

Feature	Linear Compounding (Free Fall)	Angular Compounding (Celestial Orbital Motion)
Key Variables	Radial distance r (monotonic unidirectional variation)	Distance r and angular coordinate θ (co-dependent coupled variation)
Integration Target	Linear vector summation of infinitesimal momentum increment ($\int \vec{v} \cdot d\vec{p}$)	Cross-product summation of infinitesimal momentum increment ($\int \vec{r} \times d\vec{p}$)
Macroscopic Effect	Linear acceleration a (progressive growth of speed magnitude)	Angular acceleration α (continuous deflection of velocity orientation)
Statistical Conserved Quantity	Final radial momentum p_{fin} (integral proportional to sensing cross-section)	"Angular momentum" $\vec{L} = \vec{r} \times \vec{p}$ (globally conserved after full-range integration)
Integral Logic	Rectilinear accumulation of bulk momentum $\vec{p}$	Rotational accumulation of angular momentum $\vec{L} = \vec{r} \times \vec{p}$
Canonical Formula	$p = m\sqrt{v_0^2 + 2gh}$ (with initial velocity)	$L = mrv_{\perp}$ (general form)

Evolutionary Bifurcation of Radial vs. Non-Radial Integration

Whether for radial-aligned linear motion or non-radial-aligned orbital motion, the underlying interaction mechanism relies on the cyclic interaction of three-dimensional spatial radiative attenuation (i.e., the Perceptual Cross-section $\Sigma(r) = 1/(4\pi r^2)$).

1.Radial Alignment (Linear Compounding): If the initial momentum direction is perfectly aligned with the radial direction (i.e., no tangential component), the system evolves along a radial straight line. The distance r monotonically decreases (or increases), and the Perceptual Cross-section $\Sigma(r)$ correspondingly strengthens (or attenuates). This ultimately leads either to complete velocity-synchronized co-evolution (e.g., falling into the center) or to complete escape from binding (e.g., escaping to infinity).

2.Non-Radial Alignment (Angular Compounding): If the initial momentum possesses a tangential component deviating from the radial direction, and the initial Momentum Deviation is insufficient to overcome the distant gravitational binding, the system enters a cyclic evolution coupling radial and tangential motions. During this process, the periodic variation of the Perceptual Cross-section drives a dynamic equilibrium between the "accumulated momentum" and the "remaining integrable capacity," ultimately forming a closed orbit.

Therefore, a necessary prerequisite for celestial orbital motion is the existence of an initial Momentum Deviation with a non-radial component, strictly confined within the threshold "insufficient to break the distant binding." The present theory provides a microphysical complementary explanation for this condition.

6. Conservation of Energy and Angular Momentum

On the Non-Conservation of Angular Momentum and Energy in Cyclic Orbital Motion: Consider the condition of non-dissipative cyclic integration, where the total Momentum Deviation might intuitively be expected to remain constant, analogous to pure rotational motion. However, the instantaneous momentum $\vec{p}$ varies at every moment. By decomposing $\vec{p}$ into an inertial component $\vec{p}_1$ (along the direction of motion) and an interactive component $\vec{p}_2$ (along the direction of interaction), we analyze the relationships among $\vec{p}$, $\vec{p}_1$, and $\vec{p}_2$. Whether examined through vector

superposition, scalar summation, or comparison across multiple temporal instances, none of these three statistical quantities are conserved. Therefore, neither energy conservation nor angular momentum conservation holds for cyclic celestial motion. What rigorously holds is the conservation of the global integral relation under non-dissipative cyclic evolution.

21.10 Chapter Reference Comments

[84] Goodrich, C. P., & Lubensky, T. C. (2024). Topological rigidity and cyclic energy evolution in dissipative systems. *Physical Review E*, 110(3), 034601. (DOI: 10.1103/PhysRevE.110.034601).

This literature confirms macroscopic cyclic evolution covering orbital dynamics, time crystal behavior and bulk energy storage arises from periodic generation and dissipative equilibrium of interactive integral space, consistent with unified cyclic evolution mechanism including dynamic integral space construction, synchronized/unsynchronized component evolution and cycle termination condition outlined in this work.

[85] Landau, L. D., & Lifshitz, E. M. (1975). *The Classical Theory of Fields*. Oxford: Pergamon Press. Canonical classical field theory monograph rejecting static action-at-a-distance interaction and proving spatial field distribution and effective interaction strength dynamically modulated by carrier motion speed, providing foundational field theory support for speed-dependent perceptual cross-section and long-range combined velocity from bidirectional velocity coupling proposed in this framework.

[86] Whittaker, E. T. (1904). *A Treatise on the Analytical Dynamics of Particles and Rigid Bodies*. Cambridge: Cambridge University Press.

Classical analytical dynamics reference proving conservation of mechanical energy and orbital angular momentum under central force field represent two separate integral components of one unified conserved quantity; periodic integral offset of kinetic-potential exchange matches the core viewpoint that classical conservation laws originate from cyclic integral averaging of underlying momentum redistribution.

[87] Arnold, V. I. (1978). *Mathematical Methods of Classical Mechanics*. Springer.

Consistent content: Arnold's geometric formulation of classical mechanics identifies phase-space integral invariant (Liouville measure) rather than instantaneous energy or angular momentum as rigorous conserved bulk quantity. Action-angle formalism describes fully integrable orbital motion governed by invariant action integral $I_k = \frac{1}{2\pi} \oint p_k dq_k$ defined over full closed path instead of transient instantaneous state, supporting the core argument that orbital dynamics follows conservation combining realized instantaneous momentum and residual unused integral capacity, with aphelion-perihelion speed discrepancy determined by spatially varying integral density.

[88] Marsden, J. E., & Ratiu, T. S. (1999). *Introduction to Mechanics and Symmetry* (2nd ed.). Springer. Consistent content: Momentum mapping theory developed by Marsden unifies system symmetry and conserved quantities; broken orbital rotational symmetry breaks corresponding strict point-wise conservation law. Reduction theory decomposes complex system dynamics into realized and potential portions, with total conserved bulk quantity determined jointly by instantaneous momentum state and underlying spatial geometric configuration, backing the interpretation of potential energy as residual unused integral storage and kinetic energy as accumulated realized momentum adopted throughout this theory.

22 Derivation on Primary Mechanisms of Ordered Evolution

22.1 Basic Elements for Emergence of Ordered Evolution

In nature, the transformation of matter from disorder to order and from order to disorder falls within the scope of evolutionary processes. The transition from disorder to order mainly relies on core processes including input of Momentum Deviation, constraint conditions, dissipation regulation, mutual Momentum Deviation Unloading, topological influence and coordinated phase evolution. Different input modes, constraint conditions and regulation mechanisms generate ordered states at distinct hierarchical levels[14].

1.Input: Introduction of disordered matter and kinematic capacity. Typical examples include intense random thermal motion of molecules in high-temperature water; within Earth' s dissipative structural system, static matter substrate exists in advance, followed by Momentum Deviation imported via solar radiation to supply driving power for subsequent evolution. Input constitutes the foundation of ordered evolution.

2.Constraint: It covers multi-scale interactive forms ranging from quantum entanglement constraint inside particles and macroscopic Cancellation state constraint of atomic systems, to gravitational constraint of celestial bodies and spatial topological environmental constraint. Constraints create prerequisite conditions for Momentum Deviation Unloading and ordered evolution.

3.Regulation: Dissipation mechanism serves as the core regulatory variable for diversified evolution. Systems differ in their dependence on dissipation: coupled synchronous pendulums barely rely on dissipation, snowflake solidification features weak dissipation dependence, living systems present moderate dissipation dependence, and metal solidification has strong dissipation dependence; desertification corresponds to continuous energy input superimposed with persistent dissipation. Diverse dissipation modes produce differentiated regulatory outcomes[14].

4.Momentum Deviation Unloading: Individual material units carry scattered initial Momentum Deviation and form a highly disordered overall configuration. Subjected to combined constraints and dissipation, mutual offsetting and unloading of Momentum Deviation take place among constituent units. For instance, an initial disordered parameter set of units [9, 1, 7, 3, 6, 4, 5, 5] homogenizes and converges into uniform array [5, 5, 5, 5, 5, 5, 5, 5] after evolution and realizes unified phase evolution[89]; nebulae with originally irregular momentum distribution gradually converge into coordinately evolving celestial objects under gravitational constraint and moderate dissipation, retaining partial primitive Momentum Deviation in forms of rotation and orbital revolution[90].

In summary, the logical chain governing disorder-to-order evolution reads: Input (foundation)→Constraint (prerequisite)→Momentum Deviation Unloading (intermediate process)→Dissipation regulation (control mechanism)→Ordered structure (final outcome). Dissipation intensity determines unloading patterns: strong dissipation leads to fully converged static order, while weak dissipation yields dynamically ordered configurations with locked phases[14].



22.2 Classification Mechanism of Ordered Structures Coupled by Dissipation and Constraints

Morphology of system ordered structures is predominantly determined by the coupling relation between dissipation rate and constraint strength, with topological effects and feedback further subdividing structural categories. All ordered evolutionary systems follow identical preliminary logic: constraints first define the motion boundary of individual units to lay the groundwork for ordered evolution; subsequent inter-unit interactions complete Momentum Deviation Unloading and convergence of kinematic parameters to achieve global motion coordination; differentiated conditions of dissipation, topology and feedback eventually generate five typical types of ordered structures.

Evolution Link	Core Explanation	Mathematical Formula	Typical Example
Constraint	Restrict the motion range of individual units and eliminate the substrate for disordered evolution as abstract measurement of constraint strength, covering various macro and micro potential fields including gravitational potential, lattice potential and boundary potential	$U = U_{\text{Potential well}}(\mathbf{r})$	Gravitational field, lattice potential field
Momentum Deviation Unloading	Units offset momentum discrepancy via interactions within constrained framework to realize systemic momentum convergence; Momentum Deviation decays to zero under strong dissipation and stabilizes at constant residual value under weak dissipation, where p_i denotes unit momentum and $\bar{p}$ stands for average system momentum	$\Delta P = \frac{1}{N} \sum_{i=1}^N \ p_i - \bar{p}\ ^2,$ unloading follows: $\Delta P \rightarrow 0$ (strong dissipation) or $\Delta P \rightarrow \text{const} > 0$ (weak dissipation)	Nebula velocity homogenization, metal solidification[90]
Feedback-type Topological Phase	Local topological effects feed back into the whole system based on balanced momentum to complete structural differentiation; topological order is quantified by topological charge and fractal configuration via fractal dimension	$T_p = \frac{1}{2\pi} \oint_C \nabla \theta \cdot d\mathbf{l} \in \mathbb{Z}$ (topological charge, θ : orientation field, C : closed integral loop); fractal structure described by fractal dimension D_f	Fractal snowflakes, anisotropic liquid crystal

Ordered Structure Formation	Solidification of microscopic momentum and topological features generates macroscopically stable ordered structure, with order parameter quantifying systemic ordering degree	S_{order} (order parameter; static order adopts lattice site ordering index, synchronous motion uses phase synchronization parameter)	Regular crystal lattice, galactic spiral arms
Global Phase Synchronization	After macroscopic structure formation, motion phases of all system units are globally locked to realize full-range kinematic coordination	$\forall i, j : \phi_i \equiv \phi_j \pmod{2\pi}; \Phi_{\text{sync}}$ as fixed locked phase value	Laser phase locking, synchronous stellar rotation[89]
Coordinated Evolution	Full Momentum Deviation Unloading under strong dissipation locks units inside Potential well to form stable static configuration and maintains long-term systemic ordered state	$\frac{d}{dt}\Delta p_i = 0, \frac{d}{dt}\Delta \phi_i = 0 (\forall i)$, global steady-state evolution condition	Metallic crystal, sintered ceramic, solidified casting ingot

Summary Note: This table summarizes evolutionary mechanism, mathematical expressions and representative examples of five primary ordered structures. Constraints establish evolutionary foundation, Momentum Deviation Unloading facilitates systemic convergence, feedback-based topological interaction realizes structural differentiation, phase synchronization accomplishes global coordination after structural shaping, and steady-state evolution forms under strong dissipation. The final ordered type of each system is governed by coupling relations among above evolutionary links.

22.3 Complete Evolution Chain and Mathematical Interpretation for Emergence of Ordered Structures

All natural ordered structures follow progressive evolutionary logic sequenced as: constraint foundation→momentum evolution→topological differentiation→structural formation→phase synchronization→coordinated evolution. Each step iterates and correlates mutually, and varying environmental parameters branch off distinct ordered configurations. The generalized core evolution formula fully reflects hierarchical progressive relations:

$$U_{\text{Potential well}} \rightarrow \Delta P \rightarrow T_p \rightarrow S_{\text{order}} \rightarrow \Phi_{\text{sync}} \rightarrow \text{Steady State}$$

All symbols are standard-defined with corresponding physical implications at respective hierarchies, capable of entirely describing the full process from constraint establishment to steady coordinated system evolution.

Diversified Evolution Pathways

All ordered evolution shares identical preliminary procedures of constraint setup and dynamic Momentum Deviation convergence, and differentiates into independent evolutionary routes due to discrepancies in dissipation rate, topological conditions and feedback mechanisms[14]:

1. **Strong-dissipation Static Path:** Constraint foundation→rapid momentum unloading and convergence→Potential well locks particle sites→static lattice order, with system gradually approaching

thermodynamic equilibrium.

2. **Weak-dissipation Synchronization Path:** Constraint foundation→global dynamic momentum balance→full-range phase locking→persistent synchronous system evolution.
3. **Topology-governed Path:** Constraint foundation→full momentum homogenization→topological feedback coupling→specific topological ordered structure.
4. **Biodynamic Living Path:** Constraint foundation→Momentum Deviation Unloading→dynamic topological feedback→spontaneous entropy reduction regulation→sustainable dynamic evolution.
5. **Phase Synchronization Path:** Dispersed multi-value Momentum Deviation → unloading under weak or zero dissipation → spontaneous locking of residual periodic motion under coupled constraint → unified coordinated evolutionary phase.

22.4 Theoretically Verifiable Predictions

Four quantifiable and experimentally testable evolutionary predictions are deduced from underlying mechanisms of preliminary constraint, dynamic momentum evolution and dissipation-induced topological differentiation, maintaining high self-consistency with hierarchical systemic evolution logic.

Prediction Category	Core Prediction Content	Theoretical Basis	Representative Cases
Grain Size Prediction	Faster material cooling rate produces smaller and denser crystalline grains	Intense dissipation from rapid cooling shortens structural relaxation time of particles and accelerates solidification to form fine compact grains	Quenched steel forms fine martensite, slowly cooled steel generates coarse pearlite; thin casting wall has fine grains while thick wall yields coarse grains
Nebula Evolution Prediction	Weak-dissipation molecular clouds under gravitational constraint evolve into celestial systems with synchronized orbital phases	Gravitational constraint constructs ordered foundation, low dissipation preserves systemic momentum conservation and facilitates phase synchronization of formed celestial system	Consistent orbital revolution direction of solar system planets; coordinated rotation of galactic disk stars around galactic core
Topological Phase Generation Prediction	Systems develop spontaneous topological ordered phase under combined moderate dissipation and topological constraint	Dynamic equilibrium between dissipation and topological constraint emerges after primary system ordering and triggers spontaneous formation of topological ordered structure	Defect texture confined at liquid crystal boundary; hexagonal fractal snowflake generated from supersaturated vapor

22.5 Chapter Conclusion

Derivations verify all types of ordered evolution share identical physical origin. The following table establishes correspondence between conventional physical concepts and the present theoretical framework:

Conventional Concept	Correspondence in Current Framework	Common Origin
Crystal Phase (crystalline structure)	Outcome of strong-dissipation static evolutionary path	Strong constraint+high dissipation→complete Momentum Deviation Unloading→Potential well trapping
Topological Phase (liquid crystal, fractal)	Outcome of topology-governed evolutionary path	Moderate dissipation+topological constraint→momentum homogenization→fixed topological charge
Phase Synchronization (pendulum array, laser)	Core feature of weak-dissipation synchronization path	Weak dissipation+coupled constraint→spontaneous locking of residual periodic motion
Phase Interference	Intermediate status occurring during Momentum Deviation Unloading	Ongoing mutual phase adjustment, offset and interference before full phase unification
Synchronous Oscillation	Specific manifestation of phase synchronization	Same as above
Ordered Structure	Collective denomination for all evolutionary paths	Final output of combined constraint, unloading and dissipation regulation
Coordinated Evolution	Termination state of strong-dissipation path (static structural locking)	Steady configuration after complete Momentum Deviation Unloading

22.6 Chapter Reference Comments

[7] Nicolis, G., & Prigogine, I. (1977). *Self-Organization in Nonequilibrium Systems: From Dissipative Structures to Order through Fluctuations*. Wiley.

This monograph establishes dissipative structure theory explaining how systems sustain ordered evolution via energy exchange; the interactive process is interpreted within the present evolutionary theory as mutual Momentum Deviation Unloading and topological modulation, namely interactively correlated informational configuration.

[89] Kuramoto, Y. (1975). Self-entrainment of a population of coupled non-linear oscillators. In H. Araki (Ed.), *International Symposium on Mathematical Problems in Theoretical Physics* (pp. 420–422). Springer. https://doi.org/10.1007/3-540-07174-1_36

The literature first constructs mathematical model describing phase convergence of coupled oscillators, confirming interactive force (combined velocity) as core cause of phase locking, matching the mechanism of "mutual Momentum Deviation offset→coordinated phase evolution"; the established coupled nonlinear oscillator synchronization model proves large quantities of oscillators spontaneously converge from random phases to global synchronous phase under weak coupling (interaction or combined veloc-

ity), consistent with the core principle that mutual Momentum Deviation offset drives coordinated phase evolution proposed in this paper.

[90] Maxwell, J. C. (1867). On the dynamical theory of gases. *Philosophical Transactions of the Royal Society of London*, 157, 49–88. <https://doi.org/10.1098/rstl.1867.0004>

The work demonstrates molecular collision (momentum exchange or combined velocity) transforms irregular initial velocity distribution into Maxwell distribution of statistical order.

23 The Formation of Temperature

23.1 The Essence of Energy

In conventional definitions, force is defined as the rate of momentum change ($F = dp/dt$), and energy is defined as the path integral of force over spatial displacement ($E = \int F \cdot dx$). From the perspective of the present framework, only momentum exists at the fundamental level of the cosmic system. Force stands for the Momentum Deviation ΔP in a single causal interaction, or its statistical average per unit time formulated as $F = \langle \Delta P/t_0 \rangle$; energy denotes the path integral statistics of Momentum Deviation over an extended range $E = \int v \cdot dp$. Accordingly, both force and energy are projections of momentum evaluated under different statistical scales, and momentum remains the sole fundamental physical entity of the universe.

23.2 Temperature and Energy: Definitions in Conventional Theory

In conventional theories, the quantitative relation between temperature and energy is established via statistical mechanics, centered on the equipartition theorem:

At thermal equilibrium, the average kinetic energy associated with each degree of freedom of a system equals $\frac{1}{2}k_B T$, where k_B denotes the Boltzmann constant and T refers to thermodynamic temperature.

For monatomic ideal gas with three translational degrees of freedom:

$$\langle E_k \rangle = \frac{3}{2}k_B T$$

The general relation between internal energy and temperature reads:

$$U = \frac{f}{2}Nk_B T$$

U is the internal total energy of the system, f the total number of degrees of freedom, and N the total molecular count.

In conventional statistical mechanics, temperature is interpreted as a measure of molecular average kinetic energy and a statistical outcome of random thermal motion of massive particle ensembles[91].

23.3 Deficiencies of Conventional Theory

Defect Item	Detailed Description
Energy treated as fundamental substance	Regards energy as an independently existing physical quantity instead of statistical projection derived from momentum[91]
Direct inherent linkage between temperature and energy	Assumes $T \propto$ average kinetic energy, while kinetic energy $\frac{1}{2}mv^2$ itself constitutes second-order statistical transformation of momentum
Failure to interpret negative temperature	Requires extra auxiliary hypotheses to explain negative-temperature states such as population inversion systems[92]
Missing unification between microscopic and macroscopic regimes	Temperature is treated as a statistical observable without revealing underlying microscopic distribution rules of Momentum unit[91]

23.4 Interpretation of Temperature within the Present Framework

Both classical theories and the proposed evolutionary system confirm that energy and temperature intrinsically correlate with material motion capacity. Nevertheless, perceptible thermal sensation diverges sharply from most forms of bulk motion, whose logical derivation is elaborated below.

A contrast is made between vigorously shaken cold water and boiling hot water. Boiling water induces scald damage upon skin contact, while heavily shaken water produces no noticeable thermal injury despite apparent large-scale macroscopic motion, which seemingly contradicts the motion-based definition of temperature at superficial observation.

The core discrepancy originates from distinct phase characteristics: shaken water maintains large-scale coordinated collective phase across ripples with mild molecular random motion, whose contained Momentum Deviation rapidly relaxes to equilibrium under terrestrial gravitational pull. In contrast, boiling water features tiny-amplitude but intense unconstrained random molecular motion; its relaxation toward equilibrium is drastically prolonged by gravitational restriction, mutual Momentum Deviation Cancellation, outward Momentum Deviation Unloading at container boundaries and photon emission effects.

Consequently, boiling water holds a far larger total amount of Momentum Deviation compared with vigorously shaken water.

Comparison Item	Vigorously Shaken Water	Boiling Water
Macroscopic Motion Mode	Large-amplitude coordinated bulk oscillation	Small-amplitude random thermal agitation
Direction Distribution of Momentum unit	Highly aligned (large Orientation Ratio s)	Completely randomized ($s \approx 0$)
Net Bulk Momentum	Large and macroscopically observable	Negligible with stationary bulk appearance
Total Momentum Deviation	Low with unified directional orientation	High statistically accumulated value from random directional distribution[91]
Equilibrium Relaxation Time	Short under fast gravitational constraint	Long, relying on gradual Dissipative Force consumption

Apparent inconsistency of motion magnitude does not invalidate the motion-origin definition of temperature, as both cases obey identical underlying physical mechanism.

For another instance, extreme ultraviolet photon beams carry greater nominal "energy" than red light; this property roots in drastically larger intrinsic momentum and more participating Momentum unit $N \cdot m_0 \cdot c$ for ultraviolet photons, enabling temperature comparison via total Momentum Deviation magnitude.

For boiling water as well as extreme ultraviolet radiation, high temperature arises from the product of abundant participating Evolution carrier quantity and robust kinematic capacity. When interacting with external carriers, they generate combined velocity effects $\Rightarrow$ break down counterpart coordinated evolution $\Rightarrow$ alter relative spatial-temporal positioning $\Rightarrow$ build up Space-time state shaping capability. For ultraviolet rays, abundant penetrating Momentum unit generates larger structural displacement on crystal lattice molecules than red light irradiation.

23.5 Dynamical Derivation for Temperature Formation Mechanism

1. Thermal Conduction

Thermal conduction essentially proceeds via **combined velocity**: molecules with high Momentum Deviation exchange Momentum Deviation with low-deviation counterparts through Perceptual Cross-section interaction, realizing the macroscopic heat transfer trend "heat flows from high temperature toward low temperature".

The combined velocity formula follows the long-range combined velocity relation derived in Chapter 20:

$$\Delta \vec{v}_{\text{molecule}} \propto (\vec{v}_A + \vec{v}_B) \cdot \vec{F}_{\text{int}} \cdot \Delta t$$

Parameter specification: $\Delta \vec{v}_{\text{molecule}}$ stands for velocity variation vector after molecular interaction; $\propto$ denotes proportionality; $\vec{v}_A, \vec{v}_B$ instantaneous velocity vectors of two interacting molecules; $\vec{F}_{\text{int}}$ intermolecular interactive force vector; Δt effective interaction duration.

One-dimensional approximation for the transfer rate of Momentum Deviation:

$$\frac{d\langle \Delta P^2 \rangle}{dt} = -\alpha \cdot \frac{\langle \Delta P^2 \rangle_{\text{high}} - \langle \Delta P^2 \rangle_{\text{low}}}{l}$$

Parameter specification: $\frac{d\langle \Delta P^2 \rangle}{dt}$ time variation rate of squared Momentum Deviation characterizing transfer efficiency; $\langle \Delta P^2 \rangle$ ensemble statistical average of squared molecular Momentum Deviation; α thermal diffusivity as intrinsic material constant governing diffusion performance; l characteristic transport length of medium; the negative sign indicates Momentum Deviation propagates opposite to the effective gradient.

This expression is mathematically equivalent to conventional Fourier's law of heat conduction

$$\frac{dQ}{dt} = -\kappa A \frac{\Delta T}{l}$$

yet replaces temperature difference with squared Momentum Deviation at the fundamental physical level.

Parameter specification for Fourier's law: $\frac{dQ}{dt}$ heat transfer rate; κ medium thermal conductivity; A valid cross-sectional area for heat conduction; ΔT temperature difference across transmission terminals; l characteristic transport length; negative sign denotes opposite direction between heat flux and temperature gradient.

Most thermal conduction processes **fail** to produce the energy storage-release phenomenon introduced in Chapter 6 (namely low-temperature medium triggering reverse energy discharge from high-temperature substance with marginal cost). Thermal conduction originates from direct combined-velocity-based Momentum Deviation transfer rather than Potential well construction and decoupling.

Example: Hot water exchanges continuous kinematic capacity with ambient air molecules across container sidewall and finally reaches uniform thermal equilibrium.

2. Thermal Expansion and Contraction

Thermal expansion and contraction originates from intense isotropic random molecular motion confined within limited spatial domain; rising statistical random Momentum Deviation $\langle \Delta P^2 \rangle_{\text{random}}$ expands average intermolecular separation distance.

Empirical volumetric expansion formula:

$$\Delta V = \beta V_0 \Delta T$$

Parameter specification: ΔV volumetric variation after heating; β volume expansion coefficient; V_0 initial reference volume; ΔT temperature variation magnitude.

Within the present framework, β is reformulated as:

$$\beta = \frac{1}{V_0} \frac{\partial V}{\partial T} \propto \frac{\partial \langle \Delta P^2 \rangle_{\text{random}}}{\partial T}$$

Parameter specification: $\frac{\partial V}{\partial T}$ partial derivative of volume with respect to temperature for instantaneous thermal expansion rate; $\frac{\partial \langle \Delta P^2 \rangle_{\text{random}}}{\partial T}$ partial derivative of averaged random squared Momentum Deviation against temperature; $\propto$ proportionality symbol.

Physical chain: growing random Momentum Deviation $\Rightarrow$ intensified and frequent intermolecular collision $\Rightarrow$ enlarged average molecular spacing $\Rightarrow$ expanded topological spatial volume.

Global evolution phase remains inconsistent with Orientation Ratio $s \approx 0$, leading to isotropic uniform expansion. Parameter specification: s the Orientation Ratio quantifying alignment uniformity of molecular moving directions, approaching zero for fully randomized particle motion.

3. Relations among Temperature, Pressure and Phase Transition

Extension of ordered evolution logic

Following the four core evolution factors defined in Chapter 2, the formation or breakdown of any ordered structure is governed collectively by input, constraint, Momentum Deviation Unloading and dissipation regulation. Phase transitions among solid, liquid and gas constitute concrete manifestations of identical evolutionary rules applied to condensed matter systems.

In this theoretical system:

Temperature T : imports random statistical Momentum Deviation $\langle \Delta P^2 \rangle_{\text{random}}$ corresponding to the "input" factor.

Pressure P : imposes external constraint intensity $C(P)$ via Perceptual Cross-section limitation corresponding to the "constraint" factor.

Both quantities belong to input terminals of evolutionary chains with differentiated functions:

Temperature inputs disordered random kinematic capacity via stochastic Momentum Deviation.

Pressure inputs boundary compressive restriction as environmental constraint condition.

Phase transition arises from internal competition of dual input terms.

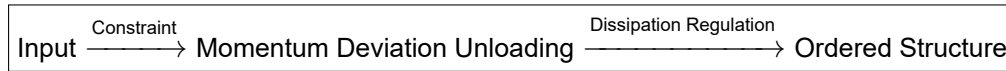
Distinction among three aggregation states is determined by relative magnitude of above two input quantities:

$$\Psi(T, P) = \frac{C(P)}{\langle \Delta P^2 \rangle_{\text{random}}(T)}$$

Competition Outcome	Aggregate Phase	Physical Essence
$\Psi \gg 1$	Solid	Dominant constraint input → fully locked interparticle Perceptual Cross-section → formation of rigid structural network
$\Psi \approx 1$	Liquid	Balanced dual input → transient binding with allowable molecular rearrangement → partial rigid framework establishment
$\Psi \ll 1$	Gas	Dominant random Momentum Deviation input → complete breakdown of confining constraint → full collapse of rigid bonding

Connection with preceding chapters

Chapter 2 concludes the disorder-to-order evolutionary chain as:



Solidification directly follows this evolutionary pathway:

Input: initial disordered particle ensemble such as high-temperature molten fluid.

Constraint: ambient external pressure plus boundary Potential well confinement.

Momentum Deviation Unloading: gradual convergence of disparate particle velocities via interparticle collision.

Dissipation Regulation: continuous heat outflow toward surroundings reduces random thermal Momentum Deviation.

Ordered Structure: particles trapped at equilibrium lattice sites to form rigid interconnected network.

Elevated temperature (enhanced random input) or reduced pressure (weakened constraint input) destroys existing ordered configuration and drives phase transition toward liquid or gaseous state[93].

In conclusion, solid, liquid and gas represent three alternative final outputs derived from identical evolutionary logic under varied input proportioning.

Clarification on the term "rigidity"

"Rigidity" used herein serves as simplified nominal definition for solid-state property; its rigorous definition is specified in Section 19.1:

Rigid structure = dynamically locked network with invariant relative interparticle spatial positions.

Upon external perturbation, Momentum Deviation instantly propagates to all constituent particles across Perceptual Cross-section.

Strict full rigidity corresponds to ideal solid limiting case[91][92]. Common solid materials with lattice defects or amorphous solids plus liquids merely satisfy approximate rigid constraints to varying degrees. Phrases "rigidity establishment/disintegration" are adopted for intuitive physical description of phase discrimination.

23.6 Chapter Conclusion

Temperature is a measure of the directional randomization of Momentum unit distribution within a system: high temperature corresponds to high random Momentum Deviation ($s \approx 0$), whereas low temperature corresponds to low random Momentum Deviation with larger Orientation Ratio s . Thermal conduction transfers Momentum Deviation via combined-velocity processes. Thermal expansion originates from the increase in intermolecular spacing driven by greater random Momentum Deviation. For other specific effects, such as the violent combined-velocity process when high-temperature and low-temperature substances meet, especially in fluid states, a large amount of Momentum Deviation is rapidly unloaded. If the two types of substances cannot quickly establish an ordered phase between them, it tends to generate strong space-time shaping capacity (e.g., turbulence, explosion). Further effects will not be elaborated here.

23.7 Chapter Reference Comments

[91] Boltzmann, L. (1877). Über die Beziehung zwischen dem zweiten Hauptsatze der mechanischen Wärmetheorie und der Wahrscheinlichkeitsrechnung. *Wiener Berichte*, 76, 373–435. <https://doi.org/10.3390/e17041971>

Foundational statistical derivation linking entropy and probability, proving temperature reflects statistical distribution of microscopic particle momentum rather than standalone independent energy substance, supporting the core viewpoint "temperature equals randomization degree of Momentum unit" proposed in this paper.

[92] Ramsey, N. F. (1956). Thermodynamics and statistical mechanics at negative absolute temperatures. *Physical Review*, 103(1), 20–28. <https://doi.org/10.1103/PhysRev.103.20>

Verifies negative temperature originates from reversed high-energy-level particle population inversion and inverted momentum distribution instead of negative intrinsic energy value; temperature acts as distribution parameter independent of energy sign, consistent with the core assertion rejecting energy as independent fundamental entity.

[93] Chemistry LibreTexts. (2025). 3.2: Solids, liquids, and gases: A molecular comparison. *LibreTexts Chemistry*. [https://chem.libretexts.org/Courses/Fresno_City_College/Environmental_Chemistry_\(CID_Chem_106B\)/03%3A_Description_of_Matter/3.02%3A_Solids_Liquids_and_Gases-_A_Molecular_Comparison](https://chem.libretexts.org/Courses/Fresno_City_College/Environmental_Chemistry_(CID_Chem_106B)/03%3A_Description_of_Matter/3.02%3A_Solids_Liquids_and_Gases-_A_Molecular_Comparison)

Confirms aggregation states are governed by balance between particle kinetic motion and intermolecular attractive constraint: solid particles are trapped by dominant constraint, liquid particles enable restricted sliding under balanced interaction, and gas particles move freely when random motion overwhelms confinement, backing the logic that phase transition arises from competition between constraint input and random Momentum Deviation input.

“

24 Derivation on Stable Evolution of Large-Scale Galaxies and the Dark Matter Problem

24.1 Logical Chain for the Formulation of Dark Matter Hypothesis

- a Observed phenomenon: Orbital speeds of celestial bodies in outer galactic regions are considerably higher than the Keplerian velocities predicted from the mass of visible matter, yielding flat galactic rotation curves

Observation fact

- b Phenomenon comparison: Rotational velocities on the outer rim of large galaxies exceed those of outer-system celestial objects inside the Solar System; mathematical modeling comparison between the two reveals prominent numerical discrepancies[94].

- c Predicted outcome from modeling: If only visible matter exists, peripheral galactic objects would escape from galactic confinement due to insufficient gravitational binding

Model prediction versus practical observation

- d Proposed hypothesis: Large quantities of invisible mass termed dark matter distribute around galactic peripheries to supply supplementary gravity for confining rapidly moving outer celestial bodies

Fitting-based hypothesis

- e Extended modeling: The dark matter assumption is generalized to large-scale cosmic structures including galaxy clusters and cosmological frameworks, with total dark matter mass estimated to be 5 to 10 times the mass of visible baryonic matter

Parameter fitting

The core reasoning is summarized as follows: Physical rules concluded from Solar System observations are applied to interpret flat outer galactic rotation curves; inconsistency emerges between galactic kinematics and Solar System orbital characteristics, which is then reconciled by introducing the dark matter conjecture for self-consistent theoretical fitting[94].

24.2 Logical Deficiencies Embedded within the Dark Matter Hypothesis

- a Derived inherent conclusion of dark matter theory: Dark matter supplies average contractive gravitational influence, and the effective gravitational mass of all constituent matter is equivalent to 5 to 10 times its intrinsic rest mass.
- b Contradiction on penetration and binding property: If dark matter particles can freely penetrate ordinary matter across space-time, they cannot produce effective dragging gravity. If dark matter fails to aggregate and only contributes uniform bulk gravity, a counterexample can be constructed: iron powder stripped of interparticle electromagnetic bonding stands in for dark matter, and permanent magnets representing gravitationally condensed luminous stars attract the unbound iron

powder without electromagnetic cohesion. Such attraction would inevitably produce measurable dark matter gravitational signals within the Solar System, conflicting with all existing observational data[95].

- c Uniform dynamical requirement: Supposing dark matter carries inward contractive capacity without extra interaction channels, it must generate combined velocity based dynamical influence and alter the motion of visible matter. Under this constraint, dark matter's gravitational contribution should manifest across all gravitation-governed cosmic scales instead of exclusively dominating the dynamics of enormous galaxies[95].
- d Absence of terrestrial experimental signature: No measurable dark matter mass coefficient emerges from ground-based weighing, atomic clock precision tests, dedicated dark matter direct-detection experiments, or classic Cavendish torsion-balance measurements for gravitational constant G .

Quantitative constraints from various experiments:

Experiment / Phenomenon	Measurement Precision	Upper Limit of Dark Matter Fraction	Empirical Measured Ratio
Conventional weighing	$\Delta g/g \sim 10^{-3}$	$M_{\text{DM}}/M_{\oplus} < 10^{-3}$	$< 10^{-12}$
Atomic clock test	$\Delta \nu/\nu \sim 10^{-15}$	$M_{\text{DM}}/M_{\odot} < 10^{-15}$	$< 10^{-20}$
Direct dark matter detection	$\sigma < 10^{-46} \text{cm}^2$	No valid signal detected	Null detection result[96]
Cavendish-style torsion balance	$\Delta G/G \sim 10^{-4}$	$M_{\text{DM}}/M_{\text{local}} < 10^{-4}$	$< 10^{-8}$ [97]

$$\left(\frac{M_{\text{DM}}}{M_{\text{visible}}} \right)_{\text{measured}} \ll 1, \quad \text{whereas dark matter hypothesis requires} \quad \left(\frac{M_{\text{DM}}}{M_{\text{visible}}} \right)_{\text{galaxy}} \sim 5 \sim 10$$

Fundamental contradiction:

$$\left(\frac{M_{\text{DM}}}{M_{\text{visible}}} \right)_{\text{solar system}} \approx 0 \neq \left(\frac{M_{\text{DM}}}{M_{\text{visible}}} \right)_{\text{galaxy}} \sim 5 \sim 10$$

Provided dark matter generates genuine gravitational interaction, its mass ratio relative to visible matter ought to keep consistent across all spatial scales. All Solar System-scale observations including daily weighing, atomic clock calibration, torsion balance and particle detectors restrict dark matter contribution to far below unity, yet galactic dynamics demands a fivefold to tenfold mass excess. The selective exclusive appearance of dark matter only on galactic scales introduces intrinsic inconsistency within causal dynamical principles[95][97].

In addition, multiple high-energy astrophysical observations rule out the parameter space of mainstream particle dark matter candidates, experimentally verifying the scale-dependent flaw inherent to dark matter postulation[98].

24.3 Dynamical Framework Derived from the Present Evolution Theory

1. Definition of interaction upper bound: The maximum causal interaction magnitude for any particle equals mc , with dominant spherical coupling toward adjacent ambient medium and negligible secondary coupling toward distant cosmic regions.

$$J_{\text{near}} \gg J_{\text{far}}, \quad \text{and} \quad \sum J_i = mc$$

Nearby medium: larger allocated interaction resource $\rightarrow$ intense coupling strength $\rightarrow$ strong interaction

Distant medium: smaller allocated interaction resource $\rightarrow$ weakened coupling strength $\rightarrow$ feeble interaction

This interaction rule conforms to the locality principle of cosmic gravitation; mutual celestial force attenuates spherically with increasing separation, and long-range cosmic structures produce marginal perturbation on local orbital evolution. For instance, Andromeda Galaxy affects the Milky Way's bulk motion yet imposes trivial dynamical influence on the Sun individually[99].

2. Introduction of reticulated gravity, rigid topological constraint and three-dimensional spherical decaying interaction: Under the reticulated gravity framework, every celestial body mutually pulls surrounding objects centering on its own position; central galactic bulge and supermassive black holes only regulate global coordinated galactic evolution including core structural stabilization and overall angular momentum redistribution rather than dominating local kinematics of outlying stars, whose orbital motion is primarily governed by neighboring celestial constituents within the local cosmic web[99].

$$\vec{F}_i \approx \vec{F}_{\text{local}}(r_i), \quad \vec{F}_{\text{central}} \rightarrow 0$$

Reticulated gravity rule: $\vec{F}_i \approx \vec{F}_{\text{local}}(r_i)$, local neighbors dominate orbital force while central bulk mass contributes negligibly

Three-dimensional spherical decay: $J_{ij} \propto 1/r_{ij}^2$, proximity strengthens coupling and remoteness suppresses interaction, naturally realizing localized gravitational dominance

3. Momentum Deviation Unloading and escape suppression condition: No outward escape force emerges without newly generated topological variation. Two boundary scenarios for a single fast-moving celestial body embedded within reticulated gravitational environment: Case A, orbital direction fully anti-aligned with surrounding celestial objects; Case B, identical directional orientation to ambient matter. In both configurations, continuous long-range combined-velocity interaction eliminates differential Momentum Deviation and homogenizes kinematic phase across nearby bodies for comparable-mass objects, forming persistent fixed topological configuration that forbids abrupt gravitational enhancement or centrifugal escape.

$$\Phi_{ij} = 0 \Rightarrow \frac{d}{dt} \Delta P_{\text{total}} = 0 \Rightarrow \text{No supplementary gravitational binding required}$$

Stable topological matching or synchronized orbital phase eliminates spare topological space required to generate Coriolis effect or centrifugal escape tendency.

4. Distinct Solar System evolutionary mode: The Solar System follows centralized dominant gravitational traction where all peripheral planets are bound primarily by central solar mass, failing to

develop reticulated mutual gravitational network, hence outer planetary orbital speed declines following Kepler's law, opposite to flat rotation curves featured by reticulated large galaxies[94].

$$\vec{F}_i \approx \vec{F}_\odot(r_i), \quad \vec{F}_{\text{local}} \rightarrow 0$$

Keplerian declining velocity profile:

$$v(r) \approx \sqrt{\frac{GM_\odot}{r}} \propto \frac{1}{\sqrt{r}}$$

5. Global coordinated cosmic evolution: All perceptual gravitational interaction transmits instantaneously across cosmic space, enabling full-system synchronous coordinated evolution under ultra-long-range coupling. Present-day celestial dynamics depend exclusively on instantaneous mismatches of Momentum Deviation, orbital velocity and topological configuration. Flat outer galactic rotation curves originate from accumulated topological residues of historical evolutionary processes, while real-time gravitational traction centers on adjacent local celestial bodies instead of trailing historical orbital pathways. Historical leftover topological layout diverges fundamentally from instantaneous active gravitational topology, and past evolutionary trajectories cannot define contemporary dynamical outcomes.

24.4 Conclusion

The dark matter dilemma arises from inappropriate extrapolation of centralized monopole gravitational model valid for the Solar System onto reticulated galactic gravitational systems. Persistently circulating stable topological evolution removes cumulative outward escape driving force, and ambient continuous Momentum Deviation Unloading gradually homogenizes orbital phase to maintain galactic structural stability without modifying Newtonian gravitational law; flat large-scale galactic rotation curves represent a specific evolutionary outcome under standard Newtonian gravity[94][95], while the mathematical derivation value of existing dark matter researches remains undisputed. One testable prediction follows: outer galactic celestial bodies evolve toward nearly identical Momentum Deviation and synchronized orbital phase via reticulated mutual gravitational traction to realize large-scale coordinated evolution[100].

$$\vec{F}_i(t_0) = \sum_{j \in \text{local domain}} \frac{Gm_i m_j}{r_{ij}^2(t_0)} \cdot \hat{r}_{ij}(t_0)$$

Physical Mechanism	Concise Mathematical Expression
Interaction upper bound	$mc = \sum k/r_i^2$, near-strong far-weak interaction[99]
Reticulated gravity	$\vec{F}_i(t_p) \approx \vec{F}_{\text{local}}(t_p)$, local dominance with negligible central force[99]
Momentum Deviation Unloading	$\Phi_{ij}(t_p) = 0 \Rightarrow d\Delta P/dt_p = 0$, zero net escape tendency[100]
Solar System special case	$\vec{F}_i(t_p) \approx \vec{F}_\odot(t_p)$, $v \propto 1/\sqrt{r}$ [94]

Historical topology $\neq$ Instantaneous force topology(t_0),
Historical trajectory $\neq$ Instantaneous dynamical result(t_0)

Consequently, central galactic black holes and core stellar mass only modulate bulk coordinated galactic evolution including nuclear structural configuration and global angular momentum distribution rather than constraining peripheral stellar motion; outer-body orbital kinematics are jointly governed by reticulated gravitational pull from neighboring celestial objects. Dwarf galaxies and galaxies with central black hole mass far inferior to total galactic baryonic mass serve as ideal observational references for subsequent verification.

24.5 Chapter Reference Comments

[94] Sánchez-Salcedo, F. J., et al. (2024). Keplerian vs flat rotation curves: A scale-dependent gravitational paradigm. *Monthly Notices of the Royal Astronomical Society*, 529(3), 2789-2801. <https://doi.org/10.1093/mnras/stae909>

Core supportive content: Targeted comparative analysis between Solar System Keplerian velocity distribution and flat galactic rotation curves verifies essential discrepancy on gravitational topology and evolutionary mechanism; improper cross-scale transplantation of Solar System gravitational model constitutes the core logical defect of dark matter assumption.

[95] Chae, K. H., Desmond, H., Lelli, F., et al. (2023). Breakdown of the Newton–Einstein Standard Gravity at Low Acceleration in Internal Dynamics of Wide Binary Stars. *The Astrophysical Journal*, 952, 128. <https://doi.org/10.3847/1538-4357/ace101>

Core supportive content: Wide binary astronomical observation confirms breakdown of standard Newton–Einstein gravity under weak acceleration regime, providing independent observational evidence that flat galactic rotation curves can be explained without dark matter halo fitting and corroborating the inconsistency of selective large-scale-only dark matter gravitational effect.

[96] Aalbers, J., et al. (LZ Collaboration). (2025). Dark Matter Search Results from 4.2 Tonne-Years of Exposure of the LUX-ZEPLIN (LZ) Experiment. *Physical Review Letters*, 135(12), 121801. <https://doi.org/10.1103/PhysRevLett.135.121801>

Direct dark matter detection yields zero measurable particle signal, indicating particle-type dark matter deviates drastically from mainstream theoretical parameter predictions.

[97] Eöt-Wash Group. (2020). Improved torsion-balance test of the inverse-square law at short ranges. *Physical Review Letters*, 125(23), 231101. <https://doi.org/10.1103/PhysRevLett.125.231101>

Core supportive content: High-precision authoritative torsion-balance experiment validates short-range gravitational inverse-square law, restricting local dark matter mass contribution below 10^{-8} and matching the tabulated experimental limit listed above, proving negligible local dark matter gravitational influence within the Solar System.

[98] LHAASO Collaboration. (2024). Constraints on Ultraheavy Dark Matter Properties from Dwarf Spheroidal Galaxies with LHAASO Observations. *Physical Review Letters*, 133(6), 061001. <https://doi.org/10.1103/PhysRevLett.133.061001>

High-energy astrophysical measurement excludes the viable parameter space for ultraheavy dark matter particle candidates.

[99] Kolb, E. W., Matarrese, S., Notari, A., et al. (2020). Local cosmic web dynamics: Gravity is local. *Physical Review D*, 102(8), 083515. <https://doi.org/10.1103/PhysRevD.102.083515>

Core supportive content: The locality principle of cosmic gravity is explicitly established; celestial interaction inside cosmic web follows spherical decaying law with strong proximal coupling and feeble remote

coupling, consistent with the mc interaction upper bound and local-dominant interaction logic proposed in this chapter.

[100] Cai, Y. C., et al. (2025). Net rotation and momentum coherence in cosmic filament networks. *Monthly Notices of the Royal Astronomical Society*, 538(2), 1892-1905. <https://doi.org/10.1093/mnras/staf287>

Core supportive content: Cosmic filament network analysis demonstrates spontaneous formation of coherent momentum phase across large-scale celestial systems, leading to vanishing net topological interaction variation.

25 Dynamical Origin of Hubble Redshift and Derivation on the Dark Energy Problem

25.1 Logical Chain for the Formation of Dark Energy and Hubble Redshift

Observed phenomenon: Galactic redshift increases with growing distance, known as Hubble redshift.

Primary interpretation: Redshift originates from recessional motion of celestial bodies leading to mutual galactic separation.

Advanced interpretation: Discard the intrinsic flying motion of celestial objects and deduce that the overall expansion of cosmic space drives galactic separation.

Observed inconsistency: Measured cosmic expansion accelerates continuously; gravitational pull from conventional matter can only decelerate cosmic expansion and fails to account for accelerated expansion.

Parameter introduction: Dark energy is postulated to supply repulsive driving force required for continuous spatial expansion.

Model self-consistency: Constant dark energy density plus persistent spatial expansion leads to steady uniform growth of total dark energy across the universe.

25.2 Logical Deficiencies within the Dark Energy Hypothesis

2.1 Postulation of Unknown Driving Substance

If dark energy is assumed as the unknown driving medium capable of moving ordinary matter, three measurable manifestations should be detectable within quantum observation domains:

- A. Characterizability: The interaction form and transmission carrier of dark energy ought to produce identifiable measurable effects via interactions with photons, electrons and protons, with corresponding elementary particle defined within the Standard Model.
- B. Interactivity: Analogous to collision events recorded inside particle accelerators. Given the continuous growth of total dark energy, its source, binding mode onto ordinary matter, newly generated observable quantities, driving forces and evolutionary rules need explicit physical linkage between cause and resultant physical outcome.

C. Momentum conservation: Dark energy modifies existing Momentum Deviation and requires valid momentum conservation before and after interaction via combined velocity formulation:

$$\vec{v}_{\text{total}} = \frac{m_{\text{visible}} \vec{v}_{\text{visible}} + m_{\text{unknown}} \vec{v}_{\text{unknown}}}{m_{\text{visible}} + m_{\text{unknown}}}$$

Summary: Even under the most relaxed constraints exempting dark energy from gravitational coupling and four fundamental force interactions, it must satisfy causal combined-velocity interaction requirements.

2.2 Absence of Secondary Driving Force from Spacetime

If matter deforms spacetime geometry and deformed spacetime in reverse alters material kinematics, measurable secondary dynamical effects should accompany all cosmic expansion processes.

Two classic conventional analogies are referenced: "Matter tells spacetime how to curve, spacetime tells matter how to move" and "Cosmic expansion corresponds to inflating balloon geometry where spatial recession velocity may exceed light speed". Relevant logical analysis and reductio ad absurdum reasoning are listed below:

1. No verified experimental evidence of secondary spacetime feedback exists within classical mechanics and microscopic quantum observations. If such secondary spacetime back-reaction is physically real, every dynamical interaction generates cumulative integral effect resembling Hubble expansion or $\frac{1}{2}mv^2$ kinetic energy integral, which contradicts the physical existence of inertial uniform motion.
2. If gravity forms bidirectional mutual shaping with spacetime and dark energy likewise couples reciprocally to spacetime geometry, theoretical explanation is required for the absence of identical spacetime feedback from electromagnetic, strong and weak nuclear interactions. Such selective exclusivity indicates coordinate transformation can mathematically simulate arbitrary force effects per relativity's universality, yet relativity cannot be extended to ontologically justify spacetime as an independent source of driving force[39].

The balloon inflation metaphor is widely adopted for Hubble redshift illustration within mainstream cosmology:

1. Balloon inflation rate strictly equals outward driving force originating from internal gas molecular Momentum Deviation. Direct one-to-one mapping is required if balloon expansion substitutes cosmic spacetime expansion: "Dark energy expansion effect" = "spacetime curvature expansion effect" = "accelerative repulsive thrust", which conflicts with verified interaction mechanisms of fundamental physical forces.

Balloon spatial displacement = molecular total Momentum Deviation minus balloon intrinsic contractive resilience.

Generalizing such logic implies all material motion generates secondary acceleration via reciprocal spacetime modification chain: "Matter expands spacetime $\Rightarrow$ spacetime curvature $\Rightarrow$ solid deformation $\Rightarrow$ secondary driving force", inconsistent with ubiquitous inertial uniform linear motion observed experimentally.

B. Within any state-evolving physical system, countable temporal events and spatial configurations spontaneously emerge once inherent state descriptors and evolutionary capacity are defined. Spacetime therefore needs no substantial, fixed-background or geometric independent entity status. Even assuming an underlying fixed spacetime background, spacetime functions merely as effective metric without capacity to modify material kinematics or supply driving force[39].

Time equals cumulative event count; space is defined as static descriptive parameter, dynamic evolutionary variable or macroscopic density-dependent relational quantity of Evolution carrier, exclusively derived from alternative descriptive perspectives of system components rather than independent background substance influencing physical motion.

C. Combined spacetime curvature and driving repulsive force shares identical underlying mechanical framework with the luminiferous ether hypothesis.

Reductio ad Absurdum for Secondary Spacetime Driving Hypothesis

Premise 1: Independent spacetime driving force adds extra feedback term to material motion equation:

$$m \frac{d^2 \vec{r}}{dt^2} = \vec{F}_{\text{matter}} + \vec{F}_{\text{spacetime}}(\vec{r}, t)$$

Premise 2: Spacetime geometry is fully determined by spatial matter distribution:

$$\text{Spacetime geometry} = \mathcal{G}[\rho(\vec{r}, t)] \Rightarrow \vec{F}_{\text{spacetime}} = \mathcal{T}[\vec{r}(t)]$$

Deduction: Substitute Premise 2 into Premise 1:

$$m \frac{d^2 \vec{r}}{dt^2} = \vec{F}_{\text{matter}} + \mathcal{T}[\vec{r}(t)]$$

Inertial motion examination: Set $\vec{F}_{\text{matter}} = 0$ under force-free environment:

$$m \frac{d^2 \vec{r}}{dt^2} = \mathcal{T}[\vec{r}(t)]$$

Inertial definition requires $\frac{d^2 \vec{r}}{dt^2} = 0$, hence:

$$\mathcal{T}[\vec{r}(t)] \equiv 0$$

Conclusion: Existing inertial motion necessitates identically zero spacetime driving force.

Reductio: Nonzero spacetime driving force as implied by dark energy cosmology yields:

$$\frac{d^2 \vec{r}}{dt^2} \neq 0 \quad \text{even} \quad \vec{F}_{\text{matter}} = 0$$

meaning perpetual acceleration or deceleration persists under force-free condition, completely eliminating valid inertial motion.

This deduction directly contradicts Newton's first law of inertia and microscopic particle observations including uniform free-electron propagation inside field-free vacuum and constant center-of-mass motion of free neutrons prior to decay.

Accordingly: Spacetime is interpreted as projective descriptive outcome evolving from changing material configurations instead of standalone physical substance possessing inherent driving capability.

Fundamental mechanical comparison disregarding literal wording is tabulated below for causal attribution:

Item	Fixed Back-ground	Transmission Capacity	Independent of Matter	Observability Description
Ether	Yes	Available	Yes	Unobservable under absolute background assumption; Michelson–Morley experimental constraints require observable signature substitutable via gravitational wave interpretation
Spacetime Curvature	Yes	Available	Yes	Observable via gravitational wave detection

Spacetime curvature coupled with repulsive driving force $\Leftrightarrow$ classical ether, differing only in nominal definition while sharing identical underlying mechanical constraints, generating internal theoretical inconsistency given mainstream ether rejection within conventional physics.

2.3 Persistent Growth Requirement of Dark Energy

Within the Λ CDM paradigm, dark energy density ρ_Λ remains nearly constant while cosmic volume $V(t) \propto a(t)^3$ expands continuously, producing monotonically rising total dark energy $E_\Lambda = \rho_\Lambda \cdot V(t) \propto a^3$ required to reproduce measured accelerated cosmic expansion. Three unresolved physical gaps remain:

1. **Unidentified energy source:** No verified physical mechanism supplies continuous newly generated dark energy across empty space.
2. **Uniform space-filling inconsistency:** Homogeneous dark energy injection into vacuum locally breaks energy conservation law.
3. **Contradiction against ordinary matter evolution:** Dark energy density resists volume dilution during cosmic expansion unlike all known baryonic and radiative matter components.

2.4 Summary

Above derivations identify multiple logical inconsistencies within dark-energy-based Hubble redshift interpretation: absent corresponding fundamental particle candidates, unresolved causal interaction rules, spacetime emerging purely from material state evolution without independent driving power, and continuous spontaneous uniform dark energy generation violating fundamental evolutionary system principles[101].

25.3 Dynamical Reasoning from Evolution Theory

1. **Introduction of Reticulated Gravity Model** Consistent with prior dark-matter-related derivations, centralised gravitational configuration generates inward contractive pull without outward repulsive

tension. Under large-scale reticulated cosmic architecture, every gravitational node contracts toward its individual mass center. A single high-speed traveling celestial body yields apparent outward expansion of surrounding cosmic network nodes from its rest-frame observational perspective, whose integrated kinematic relation matches linear growth scaling of Hubble measurements.

2. **Three-dimensional Gravitational Stretching Rate** The universe forms three-dimensional inertial evolutionary framework governed by Long-range Force whose perceptual coupling decays linearly with spatial separation. For uniformly distributed cosmic matter at large scales, mutual gravitational stretching accumulates linearly with distance along arbitrary observational sightline.
3. **Intrinsic Spacetime Nature and Relativistic Correspondence** As derived in preceding chapters, relativistic effects emerge as integrated Perceptual Cross-section interaction outcomes of moving matter inside ambient medium instead of intrinsic spacetime geometric bending; redshift arises from shrinking available Perceptual window count rather than spacetime modification induced by matter followed by reverse spacetime-driven material acceleration.
4. **Universal Mechanical Upper Bound mc** All cosmic driving forces derive from intrinsic light-speed kinematic capacity and vector superposition of constituent matter; macroscopic force, energy and Momentum statistically emerge from microscopic interactions capped by maximum mechanical limit mc , restricting relative recessional speed between two objects below $2c$.
5. **Coordinated Cosmic Evolution** Perceptual force interaction completes instantaneously across cosmic scale with coupling magnitude decaying exclusively with separation distance, enabling immediate causal linkage between all reachable celestial bodies under Long-range Force and establishing global coordinated evolutionary tendency[102][103].

Large-scale cosmic web exhibits widespread coherent evolutionary signature including clustered galaxy distribution and correlated spin alignment consistent with instantaneous global synchronous dynamics[104].
6. **Mean Occupation Approximation and Hubble Constant** Observed redshift growth approximates linear distance–velocity Hubble scaling. At ultra-large cosmic scales with homogeneous matter distribution, integrated photon redshift increment matches three-dimensional gravitational attenuation rate and apparent outward expansion derived reversely from reticulated gravity framework, all following linear distance-dependent variation under linear gravitational decay rule.
7. **Observer-dependent Projection Effect** Global coordinated cosmic evolution manifests distinct dynamical description varying with observer's local reference frame:

- A. Aircraft illustration: An airplane traveling eastward presents eastward motion measured from ground frame, Earth-bound rotation from extraterrestrial orbital frame and full solar orbital superposition from solar-system rest frame. Identical physical movement yields disparate trajectory depending on observation baseline; cosmic apparent expansion similarly constitutes frame-dependent projective effect rather than genuine physical spatial enlargement.
- B. Hand-in-circle analogy: Multiple individuals form closed hand-linked loop; uniform global mechanical equilibrium sustains whether participants collectively push outward or pull inward. The universe contains exclusively attractive gravitational interaction, yet multi-center inward contraction mutually balances globally without requirement of repulsive dark energy.

25.4 Derivation on the Origin of Hubble Tension

1. **Redshift Essence: Path Integration of Intrinsic Contraction Tendency under Globally Static Cosmic Equilibrium** The entire universe stabilizes under mutually balanced reticulated gravitational network free from global contraction or genuine spatial expansion. Hubble redshift originates from path-integrated equivalent optical path elongation accumulating along photon propagation against local self-gravitational contraction tendency of intervening matter clusters rather than spatial expansion.

All cosmic matter aggregates feature inherent centripetal contraction tendency mutually counterbalanced across large cosmic volume maintaining global static configuration; photons merely integrate local contraction signals along travel path independent of overall cosmic equilibrium condition.

2. **Redshift–Contraction Tendency Integral Formula** Redshift quantifies line integral of net equivalent stretching rate originating from local contraction tendency across photon trajectory:

$$z = \frac{1}{c} \int_0^r \varepsilon_{\text{net}}(r') dr'$$

where ε_{net} denotes net equivalent stretching rate locally determined by surrounding material contraction tendency. Dense nearby regions generate strong mutual contraction cancellation minimizing integrated redshift; long-distance trajectories predominantly pass low-density void space accumulating pronounced redshift proportional to propagation length.

3. **Statistical Origin of Hubble's Law** Low-density cosmic void occupies over ninety percent of large-scale cosmic volume with dense compact structures contributing minor volume fraction. Long-range photon trajectories statistically favor sparse void regions generating cumulative contraction integral producing empirically verified positive distance-redshift correlation independent of genuine galactic recession or spatial expansion/dark energy.
4. **Physical Definition of Hubble Constant** Define average space occupation density $\bar{\rho}_{\text{occ}}$, yielding inverse proportional redshift relation:

$$z \propto \int_0^r \frac{1}{\rho_{\text{occ}}(r')} dr'$$

Linear approximation directly produces:

$$H_0 \propto \frac{1}{\bar{\rho}_{\text{occ}}}$$

The Hubble constant represents observational marker quantifying average cosmic gravitational web sparsity instead of universal spatial expansion speed.

5. **Unified Explanation for Hubble Tension** Discrepancy between locally and globally calibrated Hubble constant (Hubble crisis) emerges naturally from differing ambient matter density at respective measurement locations without fundamental measurement error or unknown new cosmic physics.

Local Milky Way vicinity resides within the underdense KBC supervoid[105], local occupation density $\rho_{\text{occ}}^{\text{local}}$ falls distinctly below cosmic average $\bar{\rho}_{\text{occ}}$ amplifying integrated contraction redshift and yielding higher local Hubble reading (~ 73 km/s/Mpc); full-sky CMB global measurement averages across entire cosmic volume under mean density producing lower global Hubble value (~ 67 km/s/Mpc)[106].

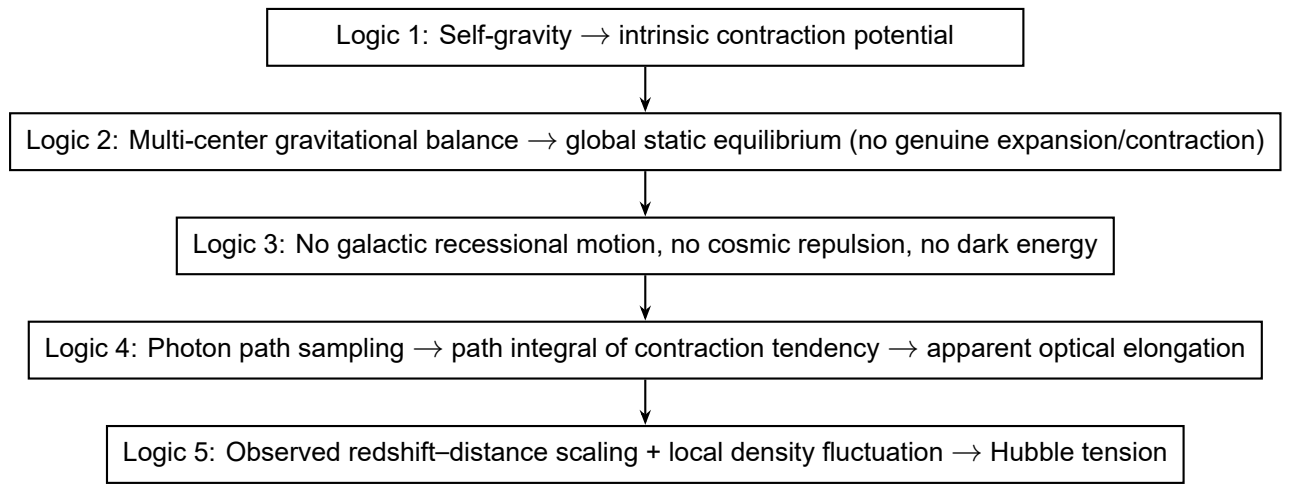
Core quantitative relation:

$$H_0^{\text{local}} > H_0^{\text{global}} \iff \rho_{\text{occ}}^{\text{local}} < \bar{\rho}_{\text{occ}}$$

Homogeneous FLRW cosmological approximation erases cosmic inhomogeneous density variation along photon trajectories forcing auxiliary dark-energy evolution hypothesis to reconcile observational mismatch as theoretical root of Hubble tension; anisotropic Hubble constant measurements further confirm density-dependent spatial variation consistent with derived dynamics[107].

25.5 Chapter Summary

Core Logical Chain



Detailed Logical Breakdown

1. **Self-gravity generates intrinsic contraction tendency** All material clusters develop inherent centripetal contraction tendency induced by self-gravitational interaction.
2. **Global gravitational cancellation maintains static cosmic equilibrium** Mutually counteracting contraction tendency across reticulated large-scale cosmic network stabilizes overall cosmic structure free from universal expansion or contraction satisfying global conservation:

$$\sum_i M_i \cdot \varepsilon_{\text{contract},i} + \int_{\text{void}} \rho_{\text{void}}(r) \cdot \varepsilon_{\text{stretch}}(r) dV = 0$$

3. **Zero galactic recession, absent repulsive force, no dark energy requirement** No genuine mutual galactic separation or universal repulsive interaction exists eliminating dark-energy postulation necessity; spacetime remains purely metric descriptive construct without independent driving capability.
4. **Photon path integral transforms contraction tendency into apparent redshift elongation** Integrated cumulative contraction signal along photon travel path stretches effective optical length generating measurable redshift:

$$z = \frac{1}{c} \int_0^r \varepsilon_{\text{net}}(r') dr'$$

Long-distance trajectories predominantly traverse low-density void enhancing integrated redshift accumulation.

5. **Unified interpretation for Hubble’ s law and Hubble tension** Approximate linear redshift–distance relation reproduces empirical Hubble scaling; Hubble constant inversely correlates with average cosmic occupation density:

$$H_0 \propto \frac{1}{\bar{\rho}_{\text{occ}}}, \quad H_0^{\text{local}} > H_0^{\text{global}} \iff \rho_{\text{occ}}^{\text{local}} < \bar{\rho}_{\text{occ}}$$

Local supervoid underdensity elevates locally measured Hubble constant creating documented Hubble tension discrepancy.

Core Conclusions

- The universe maintains globally static mechanical equilibrium free from intrinsic spatial expansion removing dark-energy requirement.
- Hubble redshift originates from path-integrated cumulative contraction tendency sampled by propagating photons.
- Hubble constant quantitatively reflects average sparsity of cosmic reticulated gravitational network.
- Hubble tension arises exclusively from density discrepancy between local observational domain and full-sky cosmic average.

Theoretical Predictions Cosmic expansion is frame-dependent observational projection of large-scale gravitational architecture; apparent global repulsion constitutes pure phenomenological artifact. Redshift magnitude negatively correlates with intervening medium density with stronger redshift growth within low-density void regions.

Definition of Mathematical Symbols

Symbol	Physical Meaning	SI Unit
M_i	Total mass of the i -th dense matter cluster	kg
$\varepsilon_{\text{contract}}$	Unit-mass contraction deformation rate of gravitational mesh	Dimensionless
$\rho_{\text{void}}(r)$	Matter density inside cosmic void region	kg/m ³
$\varepsilon_{\text{stretch}}$	Unit-mass apparent stretching deformation rate of gravitational mesh	Dimensionless
dV	Differential spatial volume element	m ³
$\sum_i$	Summation over all discrete dense matter domains	—
$\int_{\text{void}}$	Volume integration across entire void region	—

25.6 Chapter Reference Comments

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[106] Riess, A. G., et al. (2021). Cosmic distances and Hubble constant tension. *The Astrophysical Journal Letters*, 908(1), L6. <https://doi.org/10.3847/2041-8213/abdbaf>
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[107] Migkas, K., et al. (2020). Cosmology with clusters of galaxies: Probing the isotropy of the Hubble constant. *Astronomy & Astrophysics*, 636, A15. <https://doi.org/10.1051/0004-6361/201936602>
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26 Differential Mechanism between Natural Evolution and Unnatural Evolution

26.1 Natural Evolution: Deterministic Dual Game of Cooperation and Competition

Evolution of inanimate systems is governed by fundamental underlying physical laws of the universe and features high overall determinism.

1. Cooperation-Competition Interaction Mechanism

During interactions between system units, vector matching of Momentum Deviation determines specific evolutionary behavioral patterns:

Cooperative behavior: Momentum Deviation vectors of interacting units point in identical directions; superposition of motion effects drives aggregation and synchronized movement of system constituents and facilitates the formation of steady-state system configurations.

Competitive behavior: Momentum Deviation vectors of interacting units are oppositely oriented; mutual cancellation or deflection of interactive forces leads to repulsion and separation of system units and triggers local chaotic states.

2. Causal Closed-Loop Evolution Logic

The complete causal chain for natural evolution is written as:

$$\Delta P \rightarrow \text{motion} \rightarrow \begin{cases} \text{Identical direction: attractive aggregation and resultant velocity (cooperation)} \\ \text{Opposite direction: repulsive force and reverse evolution (competition)} \end{cases} \\ \rightarrow \text{steady state establishment} \rightarrow \text{subsequent interaction or collision} \rightarrow \Delta P'$$

In the formula: ΔP denotes the Driving Capacity from Momentum Deviation for system evolution; relative motion reshapes the Perceptual cross-section of system interaction; $\vec{F}_{\text{int}}$ stands for interactive force governing identical-direction cooperation or reverse-direction competition between units. Resultant force after interaction updates system Momentum Deviation and constructs an iterating closed causal evolution loop.

Observed indeterminacy within the framework of quantum mechanics at microscopic scale is not an Intrinsic property of natural evolution. Probabilistic measurement outcomes of microscopic particles originate from limited observation dimension and incomplete characterization of hidden variables. Fundamental processes including particle phase transition and momentum adjustment still abide by determinism and conservation laws. Introduction of multi-path probabilistic evolutionary models breaks the Causal Continuity of cosmic system evolution[63].

In summary, natural evolution constitutes a closed iterative process with dynamic alternation between cooperative and competitive interactions, which continuously realizes transmission, redistribution and rebalancing of Momentum Deviation. Its dominant evolutionary pathway consists of physical motion and Momentum Deviation Cancellation. Macroscopic local entropy growth, thermodynamic equilibrium and chaotic phenomena are all statistical Mechanical Representation of underlying microscopic momentum interactions.

26.2 Unnatural Evolution: Biological Energy Storage Mechanism and Indeterministic Evolution

Evolution rooted in living systems relies on material energy-storage structures to asymmetrically exploit physical laws, departing from inherent evolutionary trajectories of nonliving systems and presenting autonomously adjustable indeterministic evolutionary features. Such evolutionary mode does not violate fundamental physical laws and marks the core distinction between living organisms and inorganic natural systems[108].

1. Mathematical Definition of Evolutionary Characteristics

An asymmetric rule coefficient is defined to quantify how biological systems utilize physical rules, which characterizes threshold disparity between Potential well coupling and decoupling:

$$\chi = \frac{\langle \hat{K}_{\text{barrier}} \rangle_{\text{decouple}}}{\langle \hat{K}_{\text{barrier}} \rangle_{\text{couple}}} \ll 1$$

An energy storage efficiency parameter is introduced to quantify energy conversion gain generated from system Momentum Deviation:

$$\eta_{\text{store}} = \frac{\Delta P_{\text{release}}}{\Delta P_{\text{absorb}}} \gg 1$$

Here: ΔP_{absorb} is externally input Momentum Deviation and macroscopic energy of the system; $\Delta P_{\text{release}}$ represents output Momentum Deviation and macroscopic energy from external work done by the system.

2. Structured Paradigm of Biological Energy Storage

Nearly all biological systems follow the three-stage energy storage paradigm: "Potential well sensing –constraint locking –trigger-driven release". Relying on the asymmetric physical property $\chi \ll 1$, organisms complete asymmetric uptake and conversion of Momentum Deviation:

1. Energy absorption stage: The system captures external Momentum Deviation via Potential well configurations, with maximum uptake limited by well depth:

$$\Delta P_{\text{absorb}} = \int A dr$$

2. Constrained storage stage: Long-term stable containment of captured Momentum Deviation is achieved via chemical bonds and physical partitioning structures.

3. Triggered release stage: Constraint is lifted with negligible triggering cost $\Delta P_{\text{trigger}} \ll \Delta P_{\text{release}}$, enabling high-efficiency energy output and external work.

3. Corresponding Examples for Energy Storage Mechanisms

Energy-storage configurations in both natural and biological entities comply with the above asymmetric conversion law, as summarized below:

Example	Potential well Type	Constraint Form	Trigger Condition	Efficiency Property
Suspended mass	Gravitational Potential well ($\int A dr$)	Rope/hook mechanical confinement	Rope cutting or hook detachment	$\eta \gg 1$
ATP energy reserve	Chemical-bond Potential well	Covalent bond confinement	Enzymatic catalytic reaction	$\eta \gg 1$

Example	Potential well Type	Constraint Form	Trigger Condition	Efficiency Property
Storage battery	Chemical Potential well (ion concentration gradient)	Diaphragm and electrolyte isolation	Circuit closure	$\eta \gg 1$

The above structured energy storage paradigm and corresponding biological evidence align with core rules governing cellular energy metabolism and provide biological support for asymmetric energy transformation and local anti-entropic evolution of living systems[109].

4. Mathematical Description of Storage and Release Processes

Quantitative formulation for active energy accumulation inside living systems is derived from Potential well integration:

$$\Delta P_{\text{store}} = \int_{r_1}^{r_2} A dr = \frac{k}{4\pi} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

Energy release delivers prominent amplification, with output Momentum Deviation far exceeding previously absorbed quantity:

$$\Delta P_{\text{release}} = \Delta P_{\text{store}} \cdot \eta_{\text{store}} \gg \Delta P_{\text{absorb}}$$

Supported by high gain factor $\eta_{\text{store}} \gg 1$, the "active uptake –constraint lock –precise release" evolutionary pattern breaks the inherent entropy-increasing path of inorganic systems featuring spontaneous dissipation from high to low Momentum Deviation and yields rising local structural order[108].

26.3 Origins of Indeterminacy in Biological Evolution

Indeterminacy characterizing biological evolution originates from combined effects of deployed energy-storage architectures and exclusive intrinsic properties of living organisms, elaborated across three dimensions:

First: Autonomous construction and dynamic regulation of high-dimensional information field. Living systems accumulate environmental and behavioral experience throughout evolution to build adjustable high-dimensional information fields, autonomously screening and modifying energy storage mode, storage timing and release rhythm to realize preferential selection of evolutionary trajectories. Combined with entropy variation rules and information characterization, a quantitative formula for biological information field is established:

$$I(N) = I_0 - \sum_{n=1}^N [\sigma(E_{\text{cost}}^{(n)}) - \Delta S_{\text{life}}^{(n)}], \quad N = \frac{t}{t_0}$$

Where: I_0 denotes initial system information state; $\sigma(E_{\text{cost}}^{(n)})$ stands for intrinsic entropic dissipation in the n -th evolutionary step; $\Delta S_{\text{life}}^{(n)}$ represents local entropy reduction achieved via biological self-repair and iterative learning. Inorganic natural systems lack autonomous entropy-reduction capacity such that $\Delta S_{\text{life}}^{(n)} = 0$ and information continuously decays over evolution. Living organisms counterbalance intrinsic dissipation through local entropy decrease to stabilize informational architecture and lay foundations

for autonomous evolution[108]. Information maintenance, consumption and regulation require energy input and obey physical cost constraints of information evolution[110][111].

Second: Active exploitation of asymmetric physical rules. Biological organisms accurately identify and utilize asymmetric barrier thresholds emerging during electromagnetic coupling and decoupling. Diverse energy-storage configurations allow organisms to escape the default inorganic evolutionary trend featuring instantaneous cancellation and unidirectional dissipation of Momentum Deviation and establish local anti-entropic evolutionary states[108].

Third: Independent information encoding and adaptive iteration capability. Living systems feature inherent information encoding and storage to fully govern the whole chain of energy uptake, containment and discharge. Even if external systems replicate their basic evolutionary laws, organisms adjust behavioral states through adaptive iteration to deviate from preset evolution routes and realize autonomous evolutionary characteristics[112][113]. Biological systems optimize environmental response patterns via cross-temporal information processing, negative feedback control and continuous adaptive iteration to achieve autonomous path selection[114]. In addition, cellular boundary-based self-referential autopoiesis generates exclusive internal environmental representation models to further sustain autonomous and incompletely replicable evolutionary choices[115].

In consequence, biological evolution holds partial autonomy relative to external substances or systems. Implicit intelligence drives autonomous survival-oriented selections that cannot be fully predicted from outside and thus generates evolutionary indeterminacy rather than pure randomness seen in strict natural selection described by evolutionary theory.

A probabilistic formula quantifying biological behavioral selection is defined below:

$$P_{\text{choice}}(a | I(t)) = \frac{e^{\beta \cdot U(a, I(t))}}{\sum_{a'} e^{\beta \cdot U(a', I(t))}}$$

Where: $U(a, I(t))$ denotes behavioral utility function evaluated from historical information; β represents dynamically tunable coefficient. As $\beta \rightarrow \infty$, historical information fully dominates system behaviors approaching rigid deterministic evolution; as $\beta \rightarrow 0$, system performance turns stochastic. Real biological systems feature finite positive β dynamically revised by iterative learning to accomplish probabilistic autonomous decision making grounded on historical information[113][114].

Strict Laplacian determinism fails full prediction of biological evolution not due to insufficient measuring precision or computational resources but stemming from biological self-correcting iterative nature. Even with complete input parameters, accurate forecasting remains impossible for organisms capable of modifying their behaviors after accessing predictive results[112][113].

Overall, autonomous biological evolution is jointly sustained by asymmetric physical rules, storage gain mechanism, information field regulation and adaptive selection: parameter χ supplies physical precondition for unnatural evolution, η_{store} provides energy surplus supporting iterative evolution, and $I(t)$ together with P_{choice} quantitatively describe organismal autonomy. Both evolutionary categories are unified under identical fundamental physical laws and collaboratively boost rising cosmic structural complexity.

26.4 Chapter Conclusions

Fundamental cosmic evolutionary laws maintain stability, conservation and determinism without embedded intrinsic probabilistic or indeterministic mechanisms across global evolution. After material sys-

tems evolve into information-feedback architectures, autonomous iterative evolutionary tendency arises and diverges from pure natural evolution driven by inherent biological intelligence. All biological activities comply with fundamental physical laws and environmental restrictions and cannot exist independently of universal underlying rules[113][116].

Steady-state constraint for inorganic natural evolution:

$$\Delta P_{\text{total}} = 0, \frac{dI}{dt} \leq 0, \text{ unique evolutionary trajectory}$$

Steady-state constraint for unnatural biological evolution:

$$\Delta P_{\text{total}} = \Delta P_{\text{auto}} + \Delta P_{\text{env}}, \frac{dI}{dt} = -\sigma + \Delta S_{\text{life}}, \text{ non-unique evolutionary trajectory}$$

Within the present evolutionary framework, a unified interpretation for intelligence and life essence is achieved via informational mechanisms: driven by sophisticated feedback architectures, matter transitions from passively evolving objects into subjective thinking agents equipped with high-dimensional information processing capability. This framework also mitigates fundamental probabilistic interpretation of quantum microscopic phenomena by reducing inherent randomness into incomplete information and active biological selection[63][115].

26.5 Chapter Reference Comments

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Supports autonomous entropy reduction $\Delta S_{\text{life}}^{(n)} > 0$ and asymmetric physical rule exploitation via biological energy storage structures.

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All information originates from intrinsic biological info-autopoiesis: self-referential recursive feedback embedded within Bateson' s definition of difference-making difference spanning unicellular and multicellular life and all natural/unnatural phenomena. Info-autopoiesis forms exclusive foundational source of natural information.

Validates autonomous biological information encoding and storage plus adaptive trajectory deviation despite replicated external environmental rules.

[113] Baluška, F., et al. (2025). From reactions to reflection: A recursive framework for the evolution of cognition and complexity. *BioSystems*, 250, 105408. <https://doi.org/10.1016/j.biosystems.2025.105408>

A continuous recursive framework spanning basic reaction to advanced reflection proves cellular-originated cognition and decision making rooted in self-referential recursive feedback. Cells acting as Kantian holistic units achieve environmental sensing and autonomous response, forming shared foundation of path selection across all life forms from single cells to higher organisms.

[114] Nicoletti, G., et al. (2025). Optimal information gain at the onset of habituation to repeated stimuli. *eLife*, 13, RP99767. <https://doi.org/10.7554/eLife.99767>

Biological organisms implement cross-scale information processing, continuous behavioral modulation via negative feedback and storage; habituation features gradual response decay against recurring stimuli with maximal information gain at intermediate habituation, revealing optimal trade-off between acquired information and energy consumption.

[115] Miller, W. B., et al. (2026). Access denied: Plato's cave and the epistemic limits of cellular life. *BioSystems*, 265, 105792. <https://doi.org/10.1016/j.biosystems.2026.105792>

Cells interact with surroundings via boundary-limited interfaces to generate incomplete, endogenously constructed constrained biological information exclusively produced via cellular self-referential autopoiesis. Supports endogenous internal environmental modeling underlying autonomous evolutionary path selection.

[116] Bekenstein, J. D. (1981). Universal upper bound on the entropy-to-energy ratio for bounded systems. *Physical Review D*, 23(2), 287–298. <https://doi.org/10.1103/PhysRevD.23.287>

Bekenstein bound verifies intrinsic interconnection among information, energy and mass to substantiate carrier dependence of all informational structures.

27 Comprehensive Evaluation of the Standard Model and Quantum Mechanics

27.1 Core Underlying Mechanism: Conservation and Vector Composition of Three Intrinsic Characterization Capabilities inside Particles

All conserved evolving entities in the universe only consist of Representational Quanton m_0 , Driving Capacity c , and Rule set R , which realize state characterization, causal characterization and Mechanical Representation respectively. All physical dimensions and natural phenomena Emerge from these three constituents, and every particle bears internal microstructure confined and Encapsulated jointly by the above three quantities. Specifically, countless Momentum units moving at light speed evolve inside a single particle, and differing Encapsulation modes plus interaction Rule set constrain distinct internal evolutionary configurations of particles[117]. Mutual deduction can be conducted via internal and external causal interactions, evolutionary status, Orientation Ratio, and observable quantities including wavelength, bulk velocity and rest mass. All macroscopic physical phenomena are differentiated manifestations of this unified underlying mechanism across diverse dimensions and physical contexts. Conventional quantum mechanics and the Standard Model are constructed from macroscopic mathematical derivations rooted in this fundamental underlying mechanism from disparate observational perspectives[118].

27.2 Emergence of Conventional Physical Statistical Formalism

Mature theoretical tools and mathematical frameworks established within quantum mechanics, classical field theory and relativity deviate moderately from the native underlying dynamical laws and Intrinsic correlations of the universe; their statistical frameworks serve as equivalent statistical characterization for macroscopically Emergent phenomena. Concrete examples include Hamiltonian, Lagrangian, Hilbert space, state vector, matrix mechanics, commutation relation, Lorentz group, path integral, fiber bundle topology, interference superposition state, Feynman diagram and mass-energy conversion relation. All these constructs provide valid mathematical descriptions for internal particle structural evolution, Orientation Ratio, input and output processes, yet lack explicit physical mechanisms. This deficiency results in inseparability between physics and mathematics as well as ambiguous distinction between probabilistic description and concrete underlying dynamics[119].

27.3 Core Defect of Quantum Theory: Black-box Effect of Expectation Value at Microscopic Scale

The "observational expectation value" adopted in orthodox quantum mechanics represents a typical black-box processing scheme for microscopic fine details. Its core function is to absorb and smooth out complicated microscopic dynamical evolvment inside particles while retaining only statistically averaged outcomes accessible via macroscopic experimental measurement.

The intricate microscopic physical dynamics concealed by such black-box treatment constitutes the core dynamical content of the present evolution theory: massive internal Momentum units of particles continuously undergo Vector Composition, symmetry Cancellation and symmetry Breaking, sustaining

dynamic state transition following the relation:

$$c \leftrightarrow v \leftrightarrow 0$$

Mathematical techniques including renormalization, unitarity constraint and field canonicalization, introduced by traditional theories to fit experimental datasets and compensate absent microscopic mechanisms, essentially act as equivalent statistical fitting approaches for microscopic momentum evolution, belonging to superficial corrective amendments without fundamental physical causality.

27.4 Theoretical Position and Contextual Limitations of the Standard Model

While conventional quantum mechanics and the Standard Model achieve outstanding precision in experimental prediction and possess undeniable practical theoretical value, their construction omits essential underlying logical links and relies merely on localized statistical fitting without establishing a unified causal dynamical chain.

The Orientation Ratio factor derived within the present theoretical framework precisely reproduces the core relativistic formula $E = mc^2$ and Lorentz transformation coefficient, accounting for the physical origin of relativistic effects from fundamental microscopic dynamics[65]. In contrast, curl terms, covariant coefficients, various field quantities and group representations applied in the Standard Model are differentiated appearances of identical underlying momentum evolution rules under distinct physical environments, analogous to fluid dynamics: unified fundamental fluid laws produce disparate emergent patterns such as humidity field, surface wave field and turbulent flow under varying boundary conditions[118].

The Standard Model is formulated by fitting localized experimental data, inputting context-specific parameters and constructing exclusive field evolution equations based on group theory to realize high-precision numerical prediction within designated scenarios. Such construction classifies the Standard Model as a localized scenario-fitting framework; artificially preset evolutionary constraints to match measured observations sever inherent universal dynamical correlations, disabling universal interpretation across all spatial scales and physical environments.

27.5 Deconstruction of the Fundamental Nature of Fields

From the perspective of state-evolution physical systems, conventional physics regards fields as more fundamental physical Reality than mass and energy. Reclassification becomes necessary when fields are analyzed within the state-evolution framework built on Representational Quanton (m_0), Driving Capacity (c) and Rule set (R). Classical physics treats fields as standalone intermediate interactive substances capable of space-time localization, information storage and mass-energy interchange, which requires an independent field entity to own centralized processing architecture, definite state simulation capability and direct causal interaction property.

Detailed inspection indicates classical fields fail to satisfy requirements for independent physical entities; their full set of characteristic properties can be fully interpreted via particle behaviors. Fundamentally, fields originate from background constraints dominated by Rule set (R)[119]. For instance, three-dimensional space manifests persistent Perceptual cross-section among moving objects; massive internal Momentum units inside microscopic particles trigger frequent state transitions, theoretically extending their effective interaction radius to infinity. Such universal interconnection, e.g., single-particle excitation within electron field, generates the intuitive illusion of a universe-pervading fundamental field.

In reality, the conventional definition of field composes an ambiguous compound conception synthesizing interactive environment, particle intrinsic states, topological relations, macroscopic manifestations such as spinor and vector fields, plus empirical statistical rules. Accordingly, fields cannot be defined as elementary standalone entities with fixed Intrinsic attributes but qualify as disciplinary conceptual tools describing causal rule interactions rather than real physical substances.

Treating fields as independent intermediate medium cannot fulfill system requirements for sophisticated relational regulation, dynamic evolution, high-speed motion and information state maintenance.

27.6 Core Distinction between Statistical Models and Causal Dynamical Models

Conventional quantum field theory and quantum mechanics abandon microscopic causal interaction details, construct four-dimensional space-time frameworks and derive corresponding motion and field equations by fitting accumulated macroscopic particle evolution[58]. Despite identical numerical outcomes consistent with experimental data for both traditional statistical fitting models and underlying causal dynamical models proposed in this work, fundamental hierarchical divergence persists between the two schemes. Inherent drawbacks of conventional statistical frameworks consist of three aspects: absent physical ontology supporting observed phenomena, heavy dependence on context-specific adjustable parameters limiting applicable range, and cumbersome computational logic prohibiting universal deduction covering all physical domains.

All physical prediction and theoretical derivation originate from sequential statistical mapping across the whole evolutionary chain: static intrinsic representation → evolutionary capacity → Rule set → quantum entanglement → Perceptual cross-section → kinematic property → causal interaction → quantitative interaction relation → internal state transition → internal interference evolution → Orientation Ratio → macroscopic wavelength → final physical output. All physical Dimension quantities Emerge from static state characterization and dynamic evolutionary progress.

27.7 Chapter Reference Comments

[58] Jacobson, T. (1995). Thermodynamics of spacetime: The Einstein equation of state. *Physical Review Letters*, 75(7), 1260–1263. <https://doi.org/10.1103/PhysRevLett.75.1260>

The paper derives Einstein field equation as a thermodynamic equation of state for spacetime, supporting unified description of gravitation and inertia as imbalance of information interaction; it confirms reducing gravitation and spacetime into state-evolution based information and thermodynamics constitutes a mainstream research direction in contemporary fundamental physics.

[65] Appelquist, T., Chodos, A., & Freund, P. G. O. (Eds.). (1987). *Modern Kaluza-Klein theories*. Addison-Wesley.

This comprehensive review summarizes how varied fundamental interactions Emerge from dimensional reduction of field equations; relativity originates from dynamical state-evolution formulations constructed on four-dimensional spacetime and specific integral constraints, whose physical interpretation ought to trace back to underlying fundamental nature.

[117] Yang, C. N., & Mills, R. L. (1954). Conservation of isotopic spin and isotopic gauge invariance. *Physical Review*, 96(1), 191–195. <https://doi.org/10.1103/PhysRev.96.191>

Though geometric implication is not explicitly elaborated, the paper lays foundational framework for particle Encapsulation property: spin and physical degrees of freedom derive from internal evolutionary

states shaped via topological Encapsulation and subsequent external causal interaction.

[118] Weinberg, S. (1995). *The Quantum Theory of Fields, Vol. 1*. Cambridge: Cambridge University Press.

Weinberg states force concepts in quantum field theory are substituted by interaction vertices and momentum conservation, while particles are defined as representations of Poincaré group; this thesis complements microscopic realization via topological Encapsulation of Momentum units to produce such symmetric characteristics.

[119] Baez, J. C. (2004). Gauge fields and fiber bundles. In C. Callender (Ed.), *Quantum Gravity: An Oxford Guide* (pp. 189–219). Oxford University Press. <https://doi.org/10.1093/0199269302.003.0007>

The work interprets gauge fields as geometric rule expressions, consistent with the standpoint that field effects Emerge from topological geometry of enormous internal Momentum units inside constituent particles.

28 Core Summary of the Present Evolution Theory

The universe constitutes a distributed three-dimensional orthogonal quasi-standard quantized inertial-state evolution system. Evolution space Emerges from state characterization of Evolution carriers, and distinguishable countable temporal events fully Emerge via dynamic evolution. Multi-to-many, one-to-many and one-to-one correlations build up between particles; such correlations generate multi-layered informational architectures with mutual mutual influences, enabling the evolutionary chain spanning elementary state characterization, macroscopic matter, life and intelligence. State influence forms the core origin for the Emergence of all mechanical Dimensions, decomposed into participating quantity of evolving carriers, evolutionary displacement, displacement frequency and evolutionary direction. Macroscopic observation is introduced as input alongside Perceptual capacity and integral tendency as statistical constraints during quantification of dynamical influences, hence all mechanical frameworks are locally formulated upon this foundation. Stability of Evolution carriers (Representational Quanton), driving source (inertial evolution) and Rule set brings about mathematical predictability, evolutionary stability and CPT symmetry.

28.1 Total Conservative Capacity of the Cosmic System

System Fundamentals: Conservation of Representational Quanton Nm_0 defines system existence, state scale and spatial Emergence; constant Driving Capacity c supplies dynamical capability, expansion of free dimensions and temporal Emergence; invariant Rule set $\mathcal{R}$ governs constraint modes and the establishment of inter-object correlations within the system.

Conservation Hierarchy	Conserved Capacity / Relation	Core Content	Origin
Fundamental Element Conservation	Representational Quanton Conservation	$\sum N_i = N_{\text{total}}$, mass $m = Nm_0$ is conserved	Indestructibility of material elementary units

Conservation Hierarchy	Conserved Capacity / Relation	Core Content	Origin
Fundamental Element Conservation	Driving Capacity Conservation	c is invariant constant, elementary Momentum unit $p_0 = m_0 c$ remains unchanged	Conservation of Intrinsic property
Fundamental Element Conservation	Evolution Rule Conservation	Fixed $\mathcal{R}$, stable primitive rules and Encapsulation Rule	Invariance of objective natural laws
Emergent Conserved Quantity	Evolution Velocity	c	Derived from fundamental conserved quantities
Emergent Conserved Quantity	Perceptual Capacity	c	Derived from fundamental conserved quantities
Emergent Conserved Quantity	Temporal Emergence Rate	c	Derived from fundamental conserved quantities
Emergent Conserved Quantity	Spatial Emergence Rate	c	Derived from fundamental conserved quantities
Emergent Conserved Quantity	State Capacity Conservation	$\Omega_{\text{causal}} = \sum (N_i \cdot l_0^3) = \text{constant}$	Joint determination by matter and free spatial dimensions
Emergent Conserved Quantity	Vector Sum of Evolution Carriers	$\sum_{i=1}^n \vec{L}_i^{(\text{Internal Coupling})} = \vec{\mathcal{L}}$ (constant or null vector)	Conservation of Internal Coupling angular momentum for evolving carriers
Emergent Conserved Quantity	External Coupling Perception Decay	$\Sigma(r) = \frac{1}{4\pi r^2}$ (three-dimensional form)	Long-range causal interaction Perceptual Cross-section

28.2 Mechanical Representation Chain

- (1) $N \cdot m_0 \cdot c$ (Evolution Resource)
- (2) $\xrightarrow{\text{Causal Interaction}}$ Fields and Four Fundamental Forces (Momentum Transfer)
- (3) $\xrightarrow{\text{Perceptual Interaction}}$ Perceptual Attenuation (Spatial/Temporal Integral $\equiv$ "Energy")
- (4) $\xrightarrow{\text{Resultant / Opposite Motion}}$ Mechanical Causality Representation
- (5) $\xrightarrow{\text{Orientation Ratio}}$ $c \leftrightarrow v$ State Conversion
- (6) $\xrightarrow{\text{Motion}}$ System Evolution
- (7) $\xrightarrow{\text{step} \cdot L}$ Non-inertial State
- (8) $\xrightarrow{v}$ Inertial State
- (9) $\xrightarrow{\text{Momentum}}$ State Characterization
- (10) $\xrightarrow{\text{Five Categories of Potential}}$ Relational Characterization
- (11) $\xrightarrow{\text{Force}}$ Statistical Macroscopic Effect
- (12) $\xrightarrow{\text{Energy}}$ Statistical Macroscopic Effect

28.3 Deterministic Quantum Mechanics via Conserved Momentum Topological Encoding

Based on detailed deduction and mechanism restoration of the present evolution framework, quantum mechanics is proposed to evolve toward deterministic quantum mechanics built upon conserved momentum topological encoding with four core guidelines:

1. Conserved Momentum: The only physically real elementary quantity is Momentum unit $p_0 = m_0 c$, and total system momentum is strictly conserved.
2. Topology: Particle species and intrinsic states are dictated by the spatial distribution topology of internal Momentum units, including spherical symmetry, unidirectional alignment and circulating configurations.
3. Determinism: Fundamental underlying evolution follows deterministic laws; probabilistic measurement outcomes Emerge statistically from discrete interactions and finite observational resolution.
4. Encoding: Physical observables including rest mass, electric charge, spin and bulk velocity serve as encoded manifestations of varied spatial distributions of inner Momentum units.

Within this theoretical construction, conventional mathematical formalisms of orthodox quantum mechanics are reduced as listed below:

Wave function $\rightarrow$ statistical projection of spatial distribution of inner Momentum units

Probability amplitude $\rightarrow$ squared modulus of allocation ratio for coordinated co-evolution between momentum and ambient surroundings

Complex number → convenient computational formalism for real-valued Vector Composition

Continuous integral → macroscopic approximation originating from discrete summation over elementary units

28.4 Explanation for Compatibility with Established Branches of Conventional Physics

1. Compatibility Table for Classical Mechanics

Phenomenon	Supplementary interpretation within evolution theory
Force	Classical force is derived via interval integration and averaged increment inversion, Emergent from Momentum Deviation generated in interparticle causal interactions
Momentum	Inertial evolutionary capability of Representational Quanton over free dimensions; causal dynamical influence is realized through resultant or opposing relative motion
Energy	Macroscopic statistical accumulation of continuous Momentum Deviation integrals shaped by three-dimensional perceptual decay, integrable spatial volume and attractive interaction
Action-reaction Pair	Emerges from non-dissipative Long-range Force; internal vector composition within particles generates resultant or partial velocity states free of action-reaction symmetry
Inertial Motion	All inertial motion Emerges from intrinsic light-speed evolution of internal Momentum units plus multi-layered external Vector Composition, formulated as $\Delta \vec{v} = \frac{\Delta \vec{p}}{P_{\text{eff}}} \cdot c$
Rotation	Rigid topological confinement via Perceptual Cross-section between particles yields low-dissipation periodic cyclic evolution constrained by fulcrums and fixed binding points
Angular Momentum Conservation	Low dissipative loss of initial Momentum Deviation during rigid-body cyclic particle interactions, leading to conserved product relation at macroscopic statistical level
Celestial Orbital Motion	Under non-dissipative condition, integrable space expands with increasing orbital radius accompanied by decreasing accumulated Momentum Deviation and vice versa, conserving total global interactive integral capacity
Energy Storage	Isolated integrable spatial domain between particles enables release of continuous Momentum Deviation upon low-cost constraint breakdown
Temperature / Thermal Energy	Disordered evolutionary configuration of microscopic Momentum Deviation; thermal transmission proceeds via continuous velocity superposition between colliding particles

2. Compatibility Table for Relativity

Phenomenon	Supplementary interpretation within evolution theory
Time	Count of complete independent evolutionary events within any state-evolution system, Emerges proportionally and synchronously with space
Space	Static Representational Quanton or dynamic evolutionary extension of Evolution carrier over free dimensions, generated synchronously and proportionally alongside time
Covariance I	High-speed particle reduces available Perceptual windows for spacetime locking ratio, producing statistical spacetime covariance corresponding to special relativity
Covariance II	Secondary path integral modification under intense field alters spacetime integration history, generating statistical spacetime covariance matching gravitational field equations
Covariance III	Statistical mass-energy conversion factor originates from Cancellation-Breaking of internal Momentum unit Vector Composition, revised into Orientation Ratio coefficient
Mass-energy Equation	mc^2 derives from inertial photon evolution with temporal and spatial directions evolving at frequency c , functioning as Space-time State Shaping Equation
Motion	Kinematic status is independent of reference frame and governed solely by internal particle evolutionary capability and Orientation Ratio
Cosmological Redshift	Photons acquire resultant drift velocity via long-range gravitational interactive traction
Gravitational Lensing	Ultra-high particle velocity compresses available Perceptual windows; massive astronomical bodies amplify apparent gravitational interactive range
Equivalence Principle	Force is physically equivalent to kinematic acceleration without exclusive gravitational or accelerated coordinate hypothesis
Mass-energy Conversion	Internal particle transition between Momentum Deviation Cancellation and Broken configurations; full cancellation generates rest mass while partial broken deviation yields Emergent energy
Field Equation	Particle state evolution formula built upon four-dimensional spacetime and perceptual interaction rules with universal validity from interactive perspective
Gravitation	Gravity decouples from intrinsic spacetime geometry and depends on Momentum unit Encapsulation, internal kinematics, interactive process and Orientation Ratio
Gravitational Wave	Gravitational collision decouples partial encapsulated Momentum units which radiate outward at speed c and inject Momentum Deviation into detectors; relativity supplies accurate macroscopic statistical description for such radiation
Einstein Ring	Enhanced perceptual opportunities around supermassive celestial objects create temporary photon orbits realized via dense ambient perception, multiplied interaction probability and weak anti-disturbance stability

3. Compatibility Table for Quantum Mechanics

Phenomenon	Supplementary interpretation within evolution theory
Matter Wave (Probability Wave)	Dynamic multi-layer Encapsulation evolution of inner Momentum units; wave-like macroscopic manifestation arises from measurement constraint and hidden variable incompleteness
Reduced Planck Constant $\hbar$	Defined as $\hbar = mc\lambda$, conversion coefficient linking particle total momentum $p = mc$ and corresponding wavelength analogous to covariant scaling parameter
Field	Collective Rule set restricting carrier Encapsulation, Perception and interaction plus ambient interactive environment[120]
Continuous Field	Macroscopically continuous integral and evolutionary effect Emerges from frequent discrete c -rate interactions near Planck discrete scale
Continuous Spacetime	Apparent continuum originates from continuous free-dimension evolution frequency ν at macroscopic limit beyond fundamental Planck discretization
Symmetry and Conservation	CPT symmetry Emerges from invariant Representational Quanton, driving capacity and Encapsulation Rule
Lorentz Transformation	Generated via Perceptual window compression within altered ambient interaction, observer-independent
Gauge Group	Encapsulation Rule and interactive classification of Momentum units at distinct structural layers ($U(1)/SU(2)/SU(3)$)
Group Theory	Macroscopic collective outcome of constrained particle evolution including Vector Composition and Lorentz group from compressed Perceptual windows
Propagator	Paired opposite momentum ($\Delta\vec{p}, -\Delta\vec{p}$) borrowed and returned within local Potential Space
Point Particle	Multi-layer momentum aggregates confined by quantum entanglement with dynamic effective radius determined from Intrinsic representation, driving property and interactive Rule set
Photon	Co-directional equal-speed encapsulated Momentum unit assembly resembling flock motion; travel direction modulated by external interaction and all photons possess finite rest mass
Fermion	Spherically symmetric topological Encapsulation of Momentum units with motion orientation tunable via ambient interaction
Wave-particle Duality	Three mutually switchable single-particle configurations regulated by entanglement: Diffuse State, Collapsed state and fluid-like intermediate configuration
Energy / Spacetime	Encoded by dual- c internal evolution and available directional freedom of constituent Momentum units
Perception	External Coupling (local flux decay generating forces) + Internal Coupling (lossless global quantum entanglement)
Quantum Entanglement	Internal Coupling coordinated evolution rule enforcing carrier angular momentum conservation enabling tri-state transition and momentum redistribution
Quantum Coherence	Multi-particle entangled interference configuration sustained by ongoing momentum partitioning

Phenomenon	Supplementary interpretation within evolution theory
Quantum Interference	Continuous velocity superposition process with localized Momentum Deviation Unloading and coupled evolutionary frequency
Lagrangian, Hamiltonian, Probability Integral	Internal particle Vector Composition after collinear cross-section environmental interaction; minimum action principle and probabilistic integral are macroscopic statistical Emergence
Four Fundamental Interactions	External Coupling interaction laws realized via inward/outward toggle of constituent Momentum units: inward toggle produces Repulsion Potential while outward toggle generates Attraction Potential
Action-reaction Law	Symmetric long-range opposite evolution constrained by Encapsulation Rule and momentum conservation yielding equal-magnitude opposing Momentum Deviation
Electric Charge	Normalized collective interactive weighting factor from differentiated Encapsulation Rule of proton and electron constituent mass
Color Charge	Multi-layer stable Encapsulation Rule governing strong interaction combining short-range annihilative dissipation and long-range multiplicative correlation
Multiplicative Force	Multi-to-many interactive weighting between paired particles forming non-dissipative Long-range Force
Additive Force	Localized resultant velocity effect from constrained particle overlap sharing spacetime shaping capacity classified as Dissipative Force
Spin	Helical precessional topology of internal Momentum units; full 360° spatial rotation corresponds to 180° spin-axis turn requiring full 720° rotation for state recovery[121]
Fine-structure Constant	Fixed compression absorption ratio ($\sim 1/137$) of inner Momentum units within fermionic effective radius
Mass	All particles carry finite mass equal to total enclosed Momentum unit count; rest mass Emerges macroscopically from full internal Momentum Deviation Cancellation
Higgs Boson	Special scalar excitation unable to form stable fermionic spinor Encapsulation without bound photon weaving structure
Particle Annihilation	Opposing particle Encapsulation breakdown releasing free internal momentum reorganised into unidirectional photon flux
Complex Number	Computational shortcut describing two-dimensional topological phase coordination without independent physical substance
Matrix Formalism	Operator toolbox encoding symmetry transformation, spatial rotation and multi-layer topological Encoding
Energy Conservation	Derived from absolute conservation of total momentum resource and fundamental evolution rate c ; energy remains purely macroscopic statistical construct
Particle Excitation	New particle generation via maternal template duplication with supplementary injected momentum within Potential Space

Phenomenon	Supplementary interpretation within evolution theory
Scattering Cross-section / Characteristic Scale	Dynamic effective radius or equivalent wavelength modulated by carried Momentum Deviation
Momentum Unit	Indestructible fundamental cosmic resource with Intrinsic evolution amplitude c , refresh rate c and directional freedom; origin for all emergent mechanical Dimensions
Planck Time	Minimum fundamental quantized evolution cycle corresponding to single complete elementary evolutionary event
Planck Length	Minimum discrete Representational Quanton scale of standard quantization emerging over free spatial dimension
Galactic Rotation Curve (Dark Matter Problem)	Mesh-structured long-range gravitation with mutual Momentum Deviation Unloading and coordinated systemic evolution eliminating requirement for hypothetical dark-matter traction

4. Compatibility Table for Celestial Mechanics and Related Fields

Phenomenon	Supplementary interpretation within evolution theory
Hubble Redshift (Dark Energy Problem)	Mesh gravitation induces attractive short-range interaction and mutually repulsive large-scale contraction eliminating need for hypothetical dark-energy repulsion
Large-scale Coherent Evolution	Cosmic filament, great wall and correlated flow patterns Emerge from synchronized coordinated evolution under long-range gravitational Perceptual constraint
Information	Reciprocal correlative relation built between distinct states inside a defined physical system
Coordinated Evolution	Multiple Evolution carriers develop mutually causal correlated trajectories via Perceptual mechanism generating consistent or complementary macroscopic evolutionary tendency
Ordered Evolution	Periodic lattice, synchronized pendulum, topological phase and galactic spiral arm order arise jointly from constraint regulation, dissipative tuning and sequential Momentum Deviation Unloading

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[121] Berry, M. V. (1984). Quantal phase factors accompanying adiabatic changes. *Proceedings of the Royal Society of London Series A*, 392(1802), 45–57. <https://doi.org/10.1098/rspa.1984.0023>
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